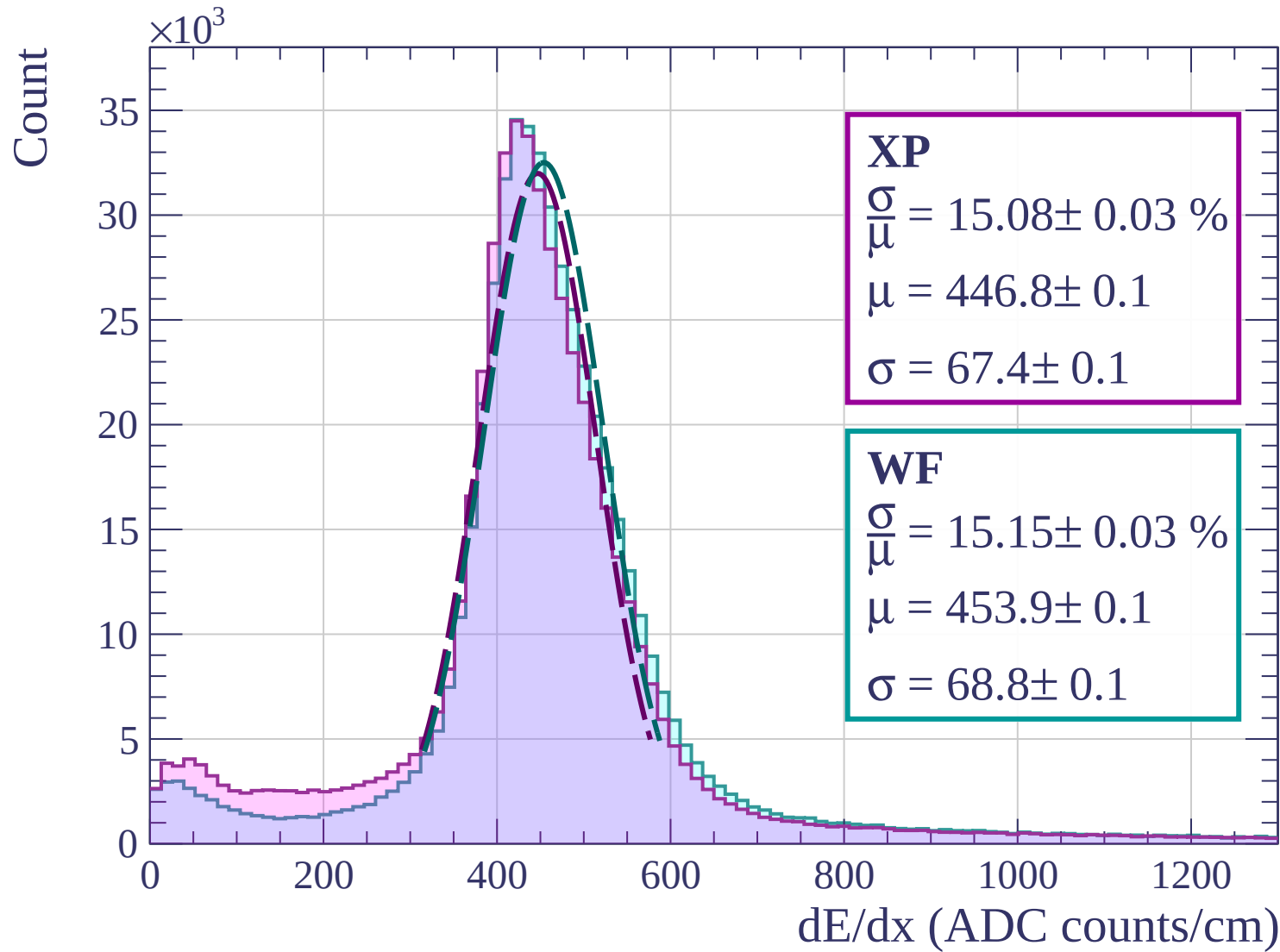
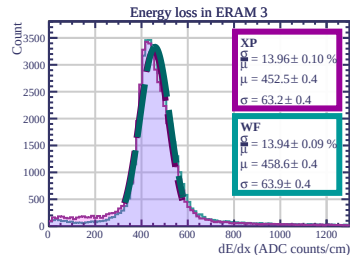
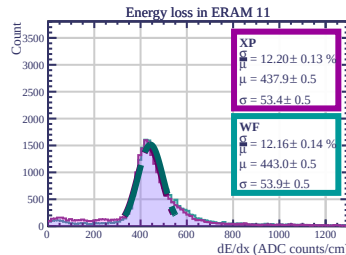
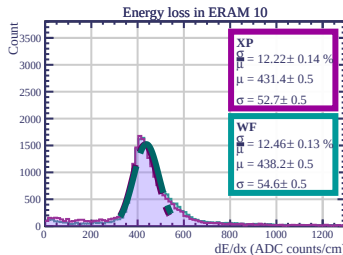
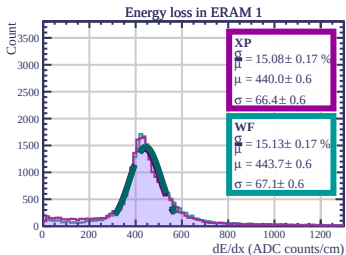
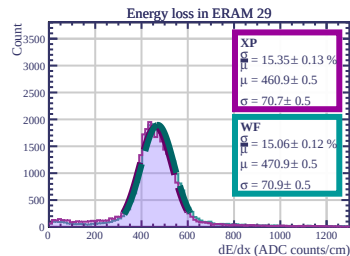
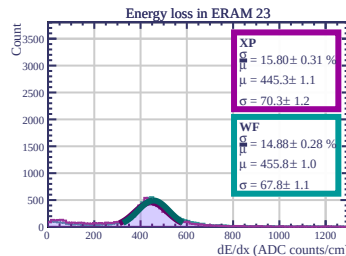
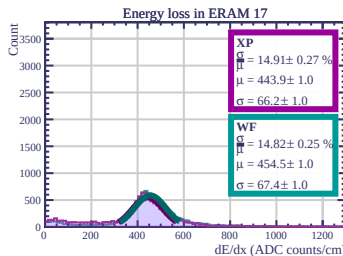
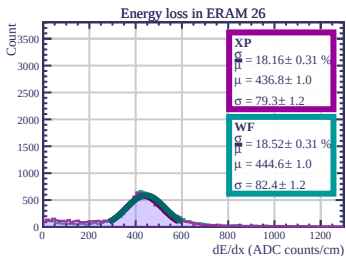
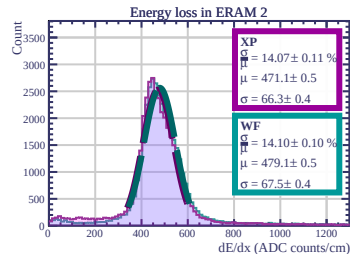
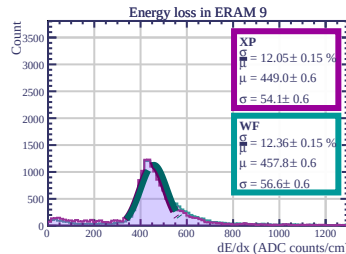
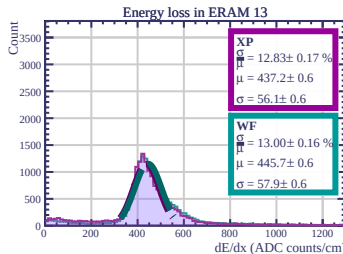
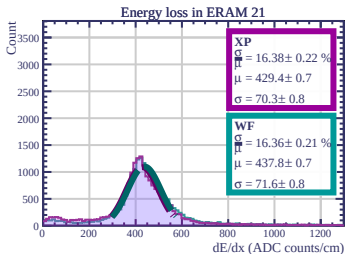
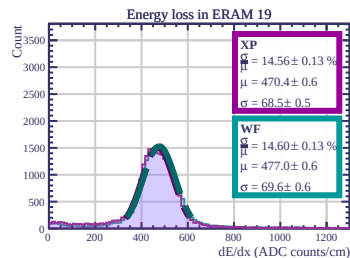
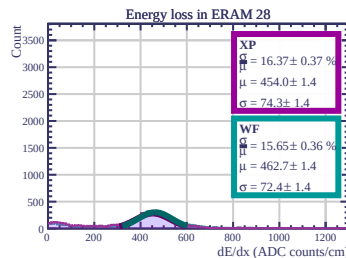
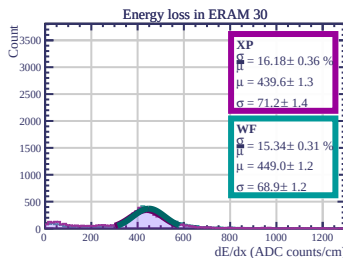
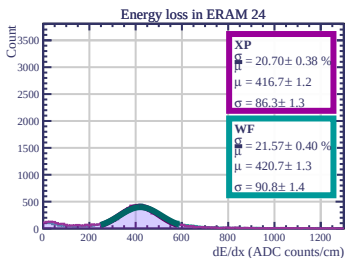
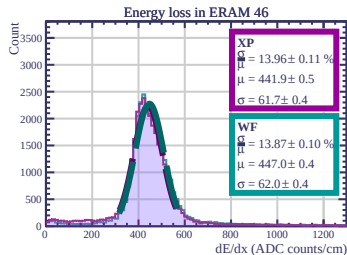
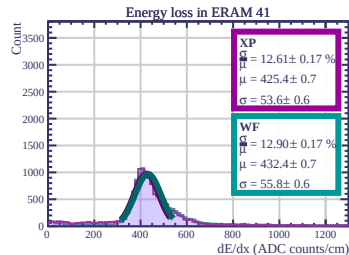
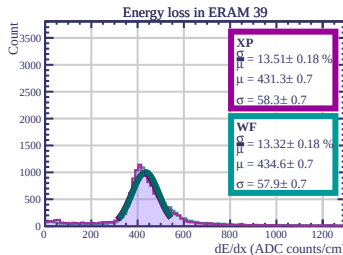
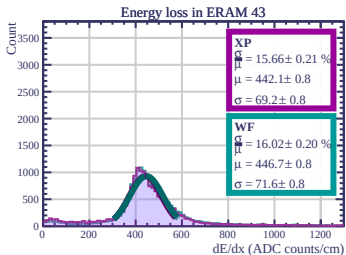
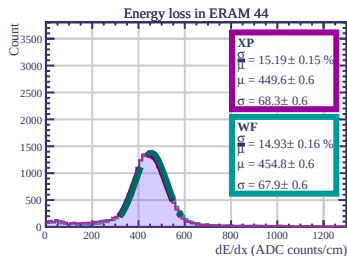
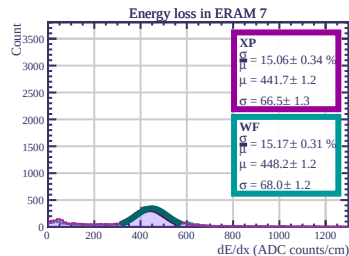
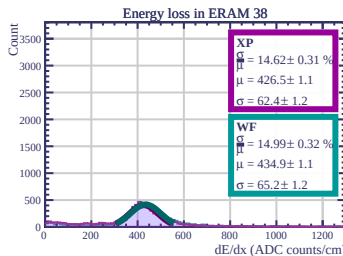
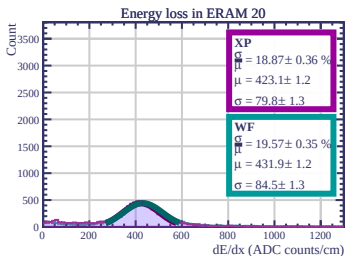
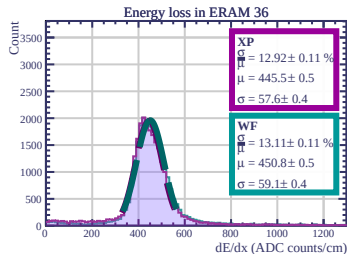
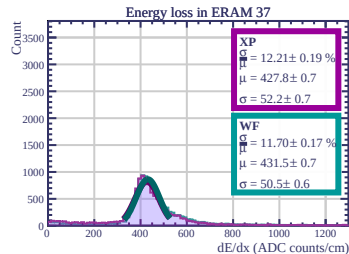
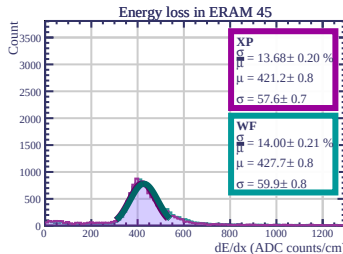
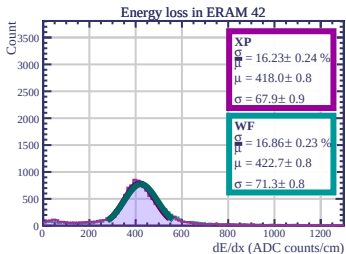
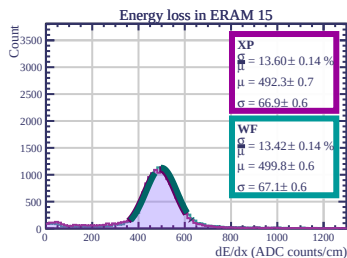
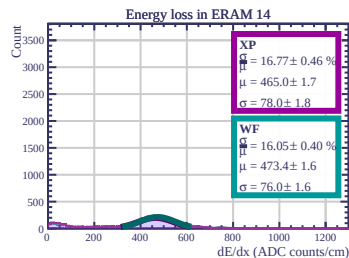
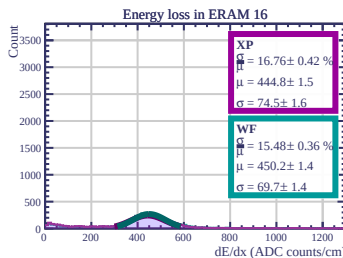
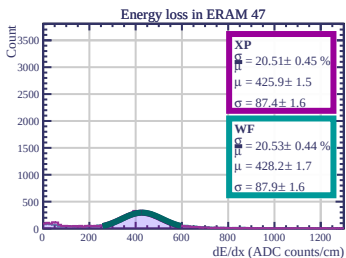
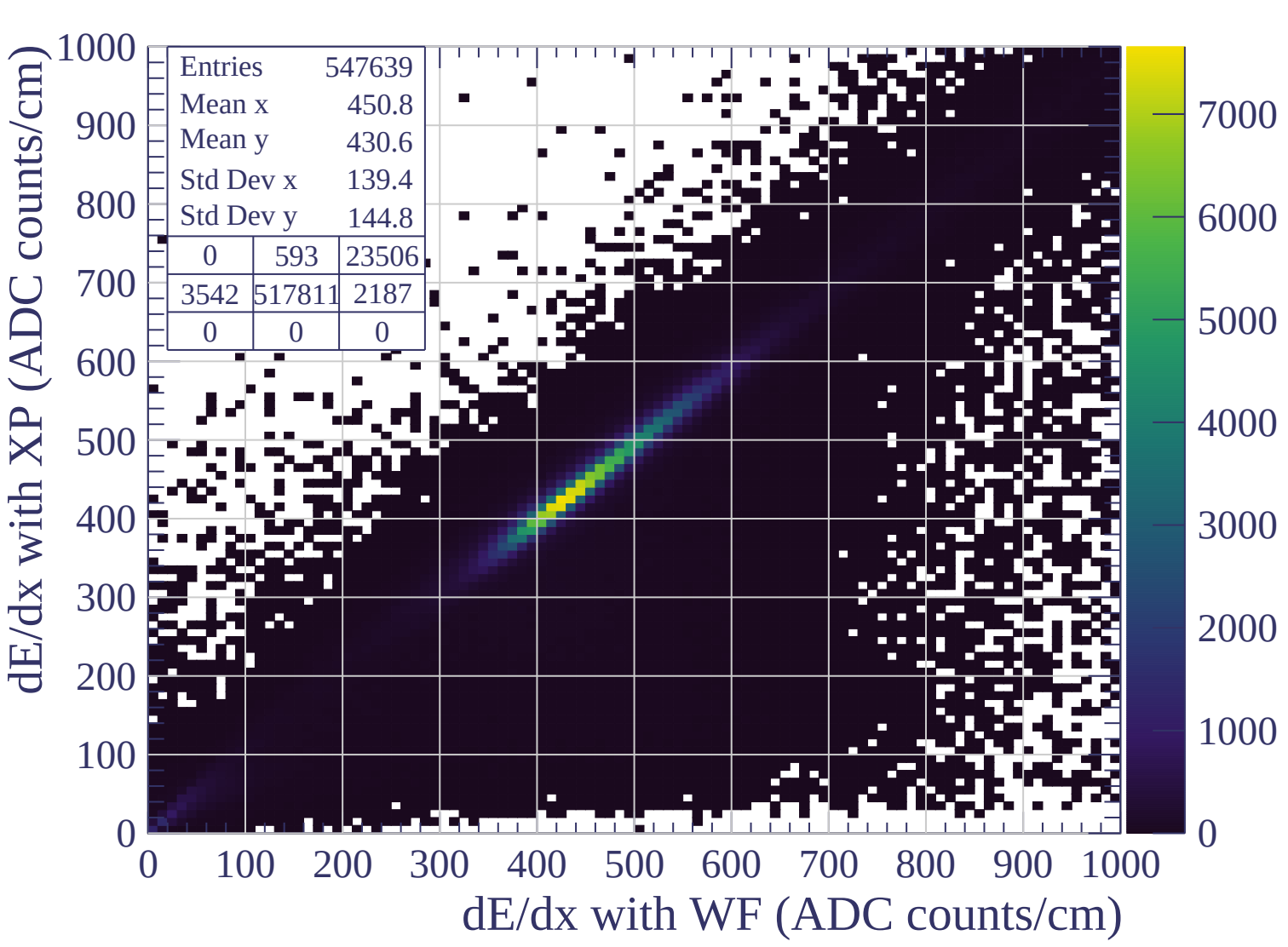
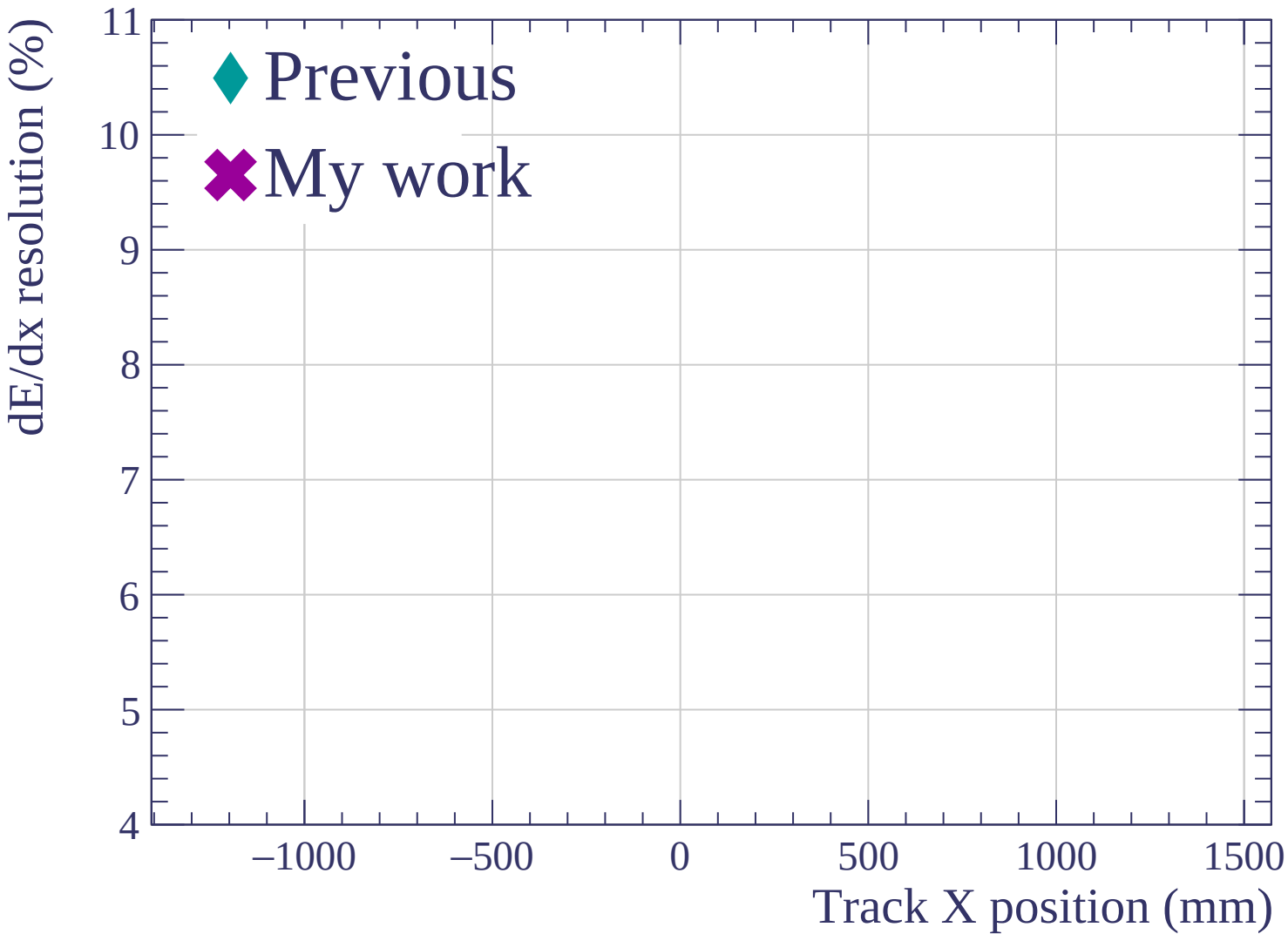


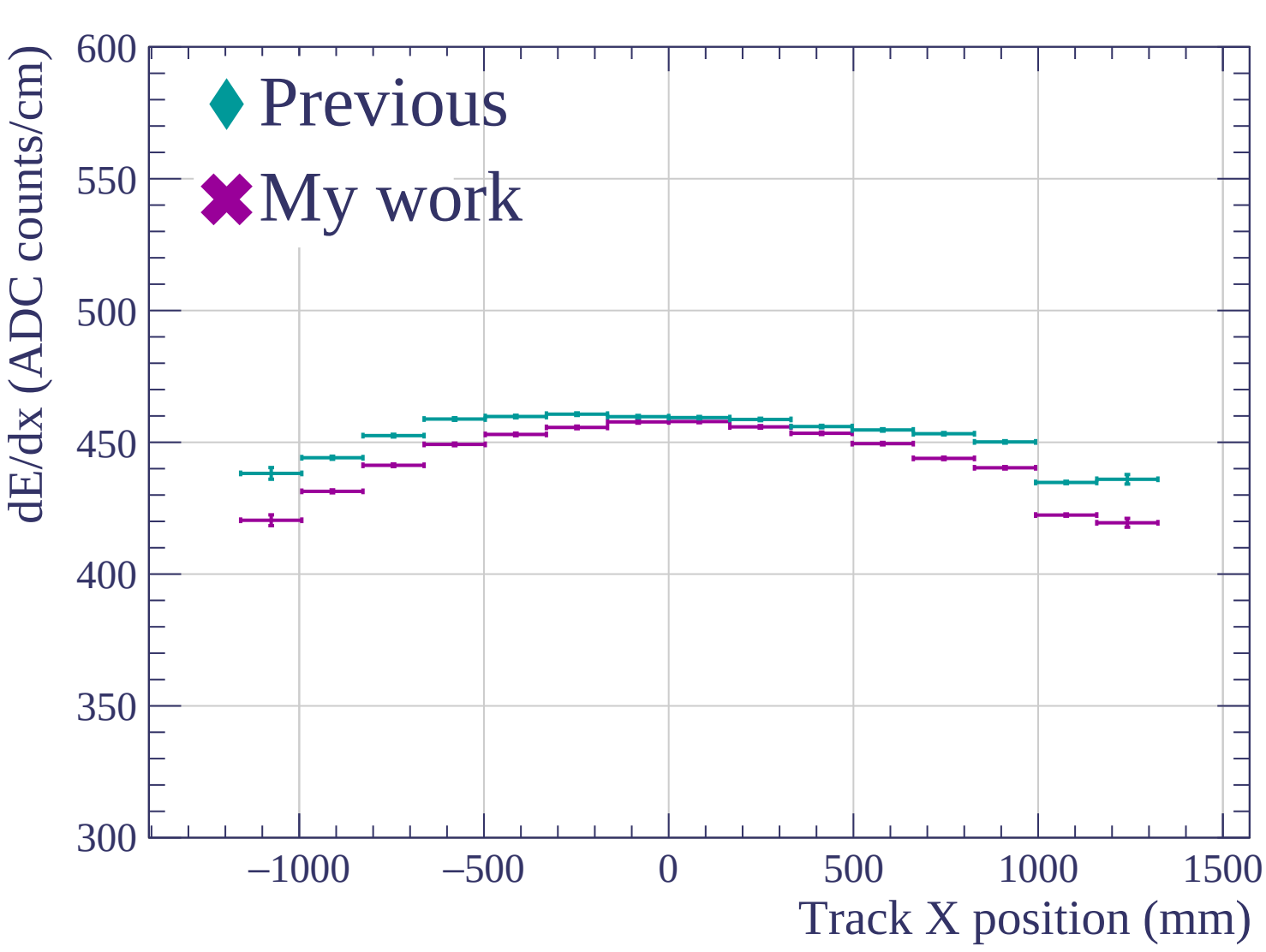


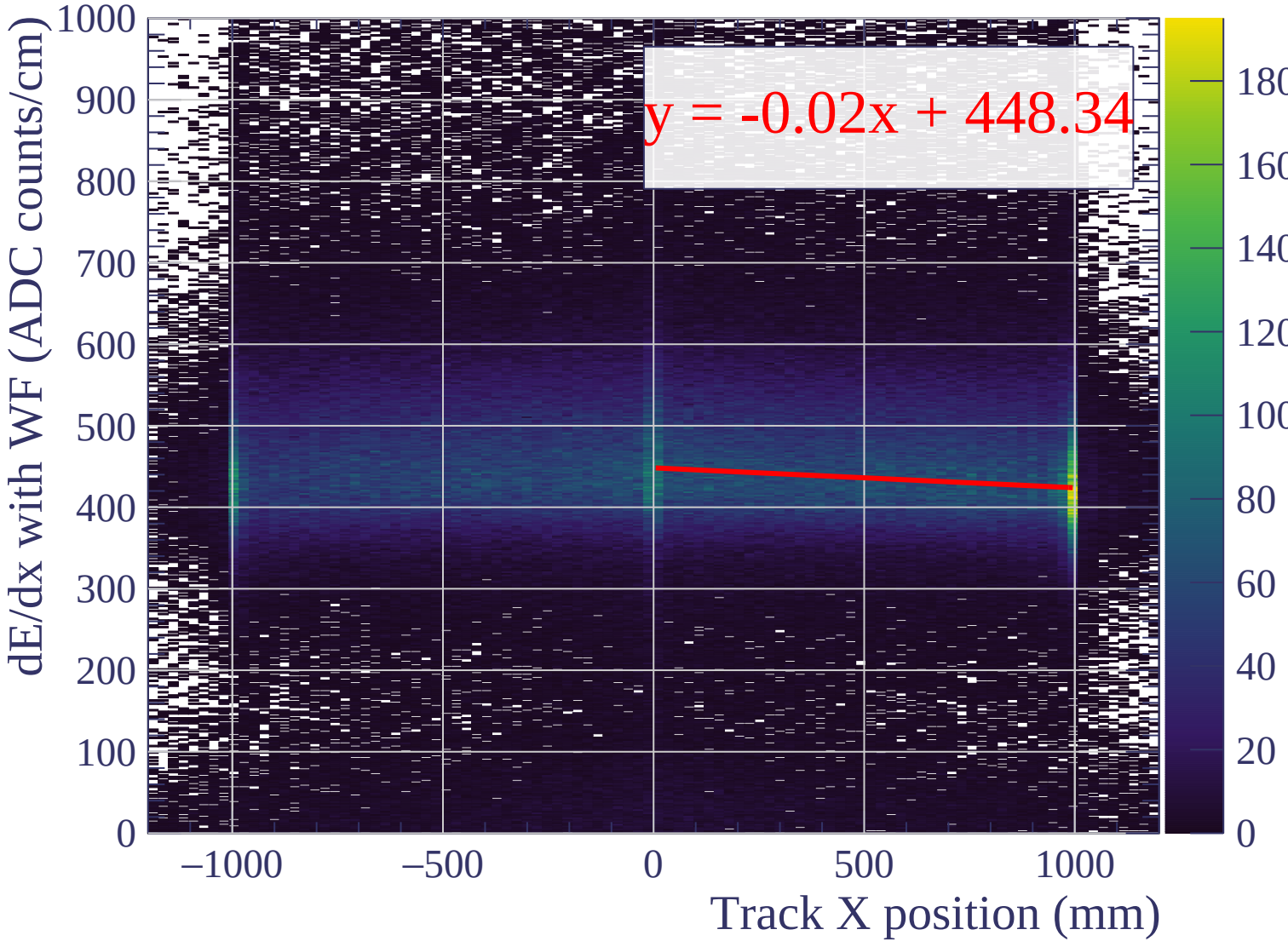


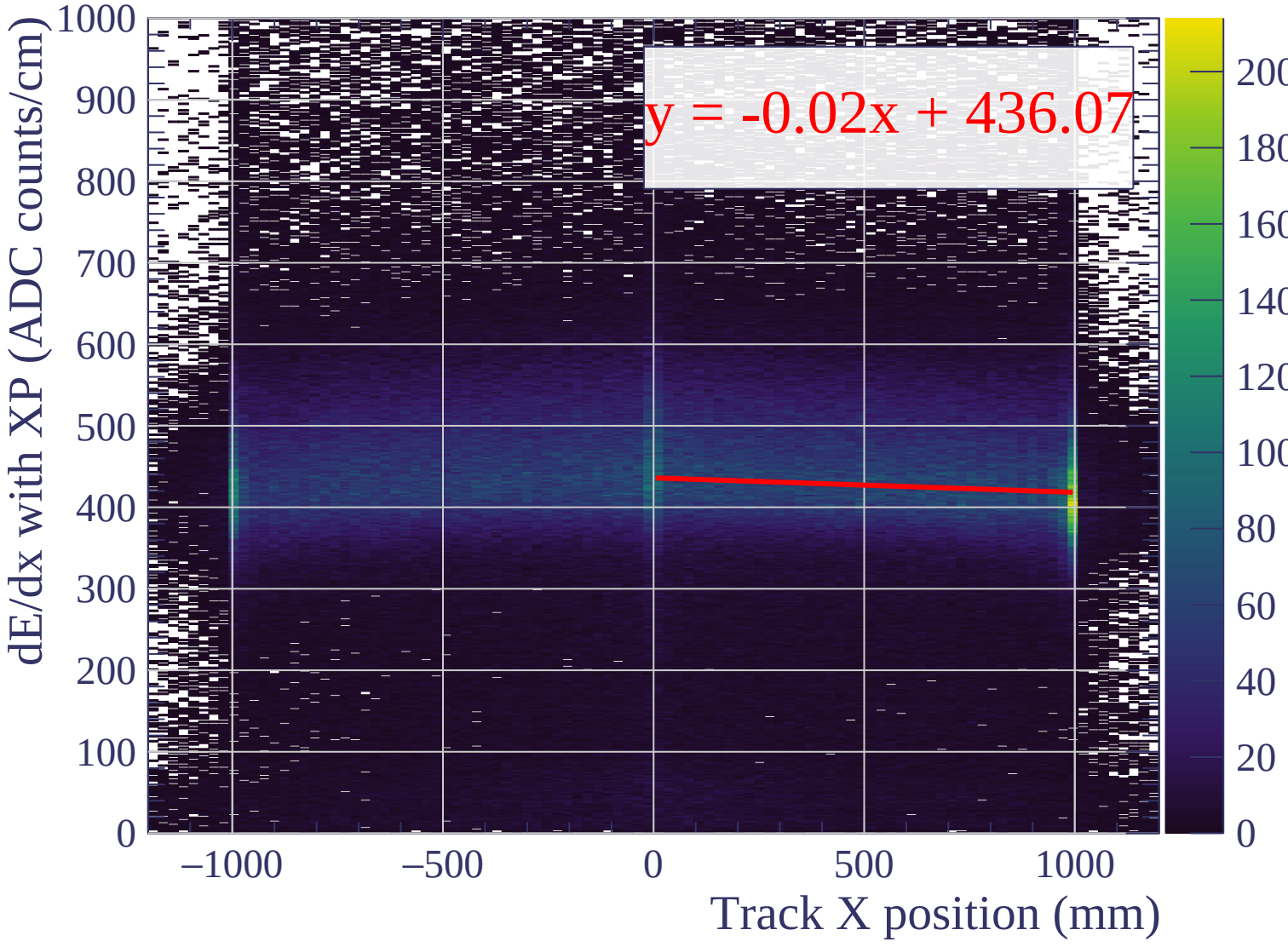


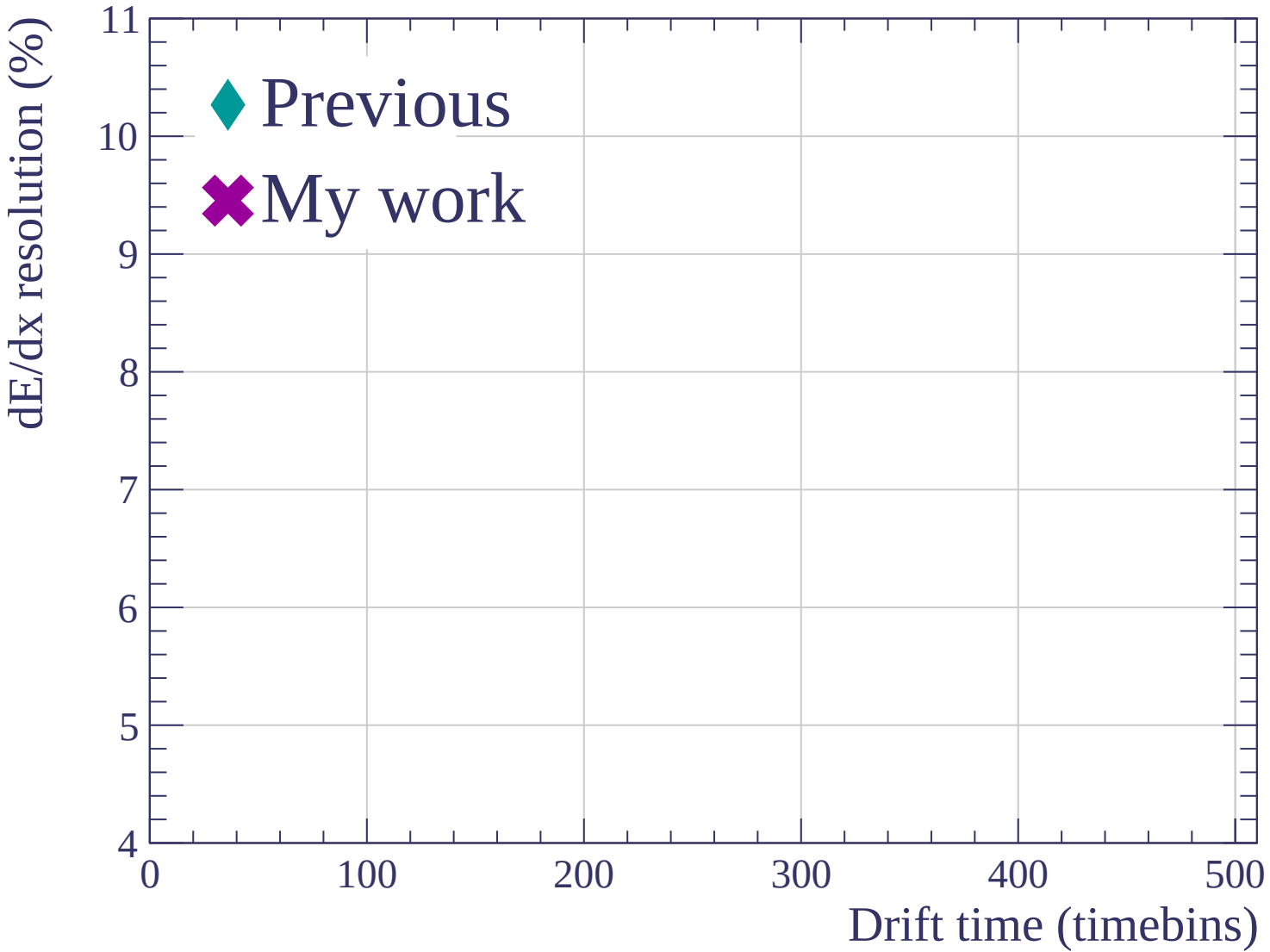


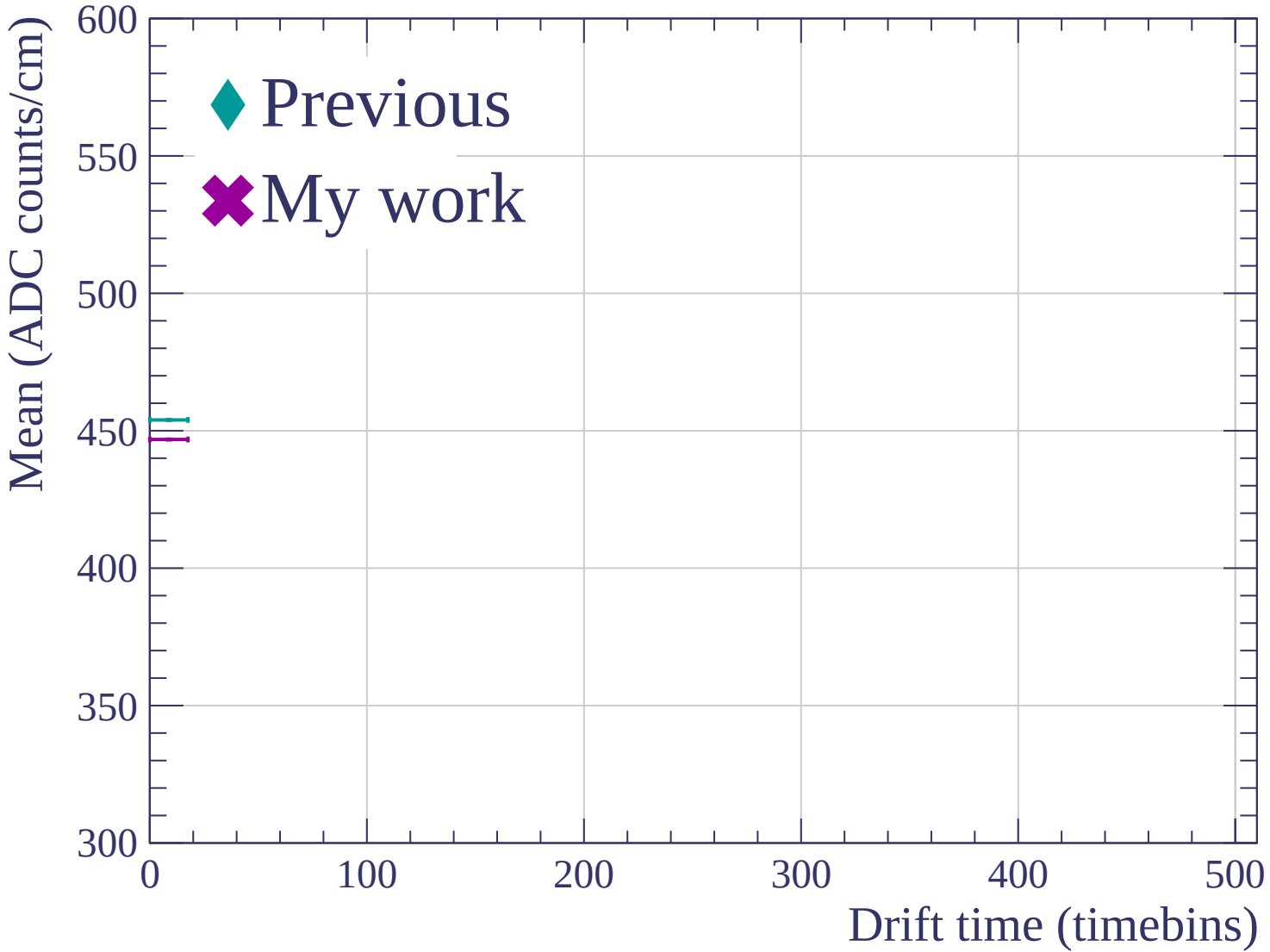


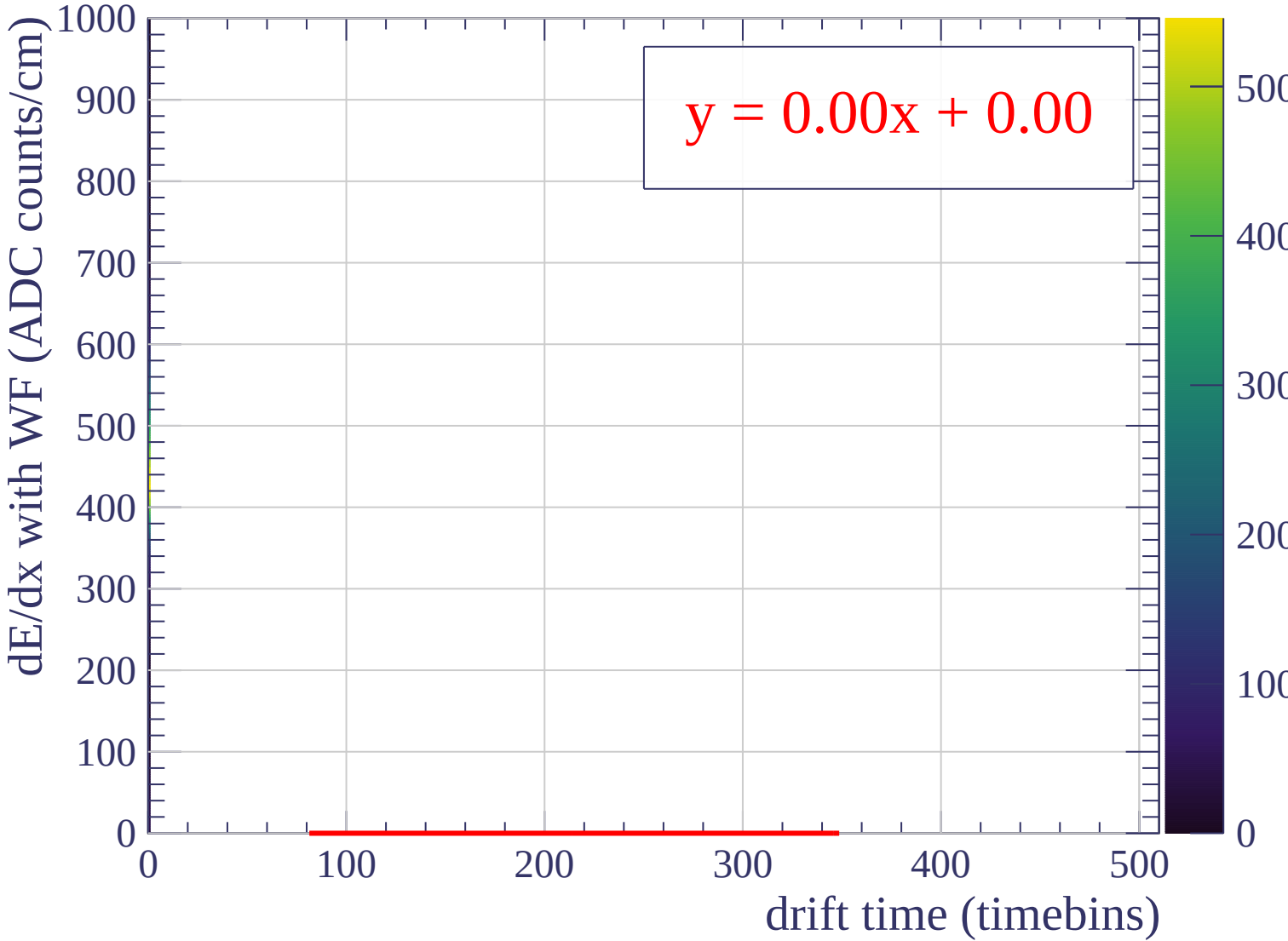


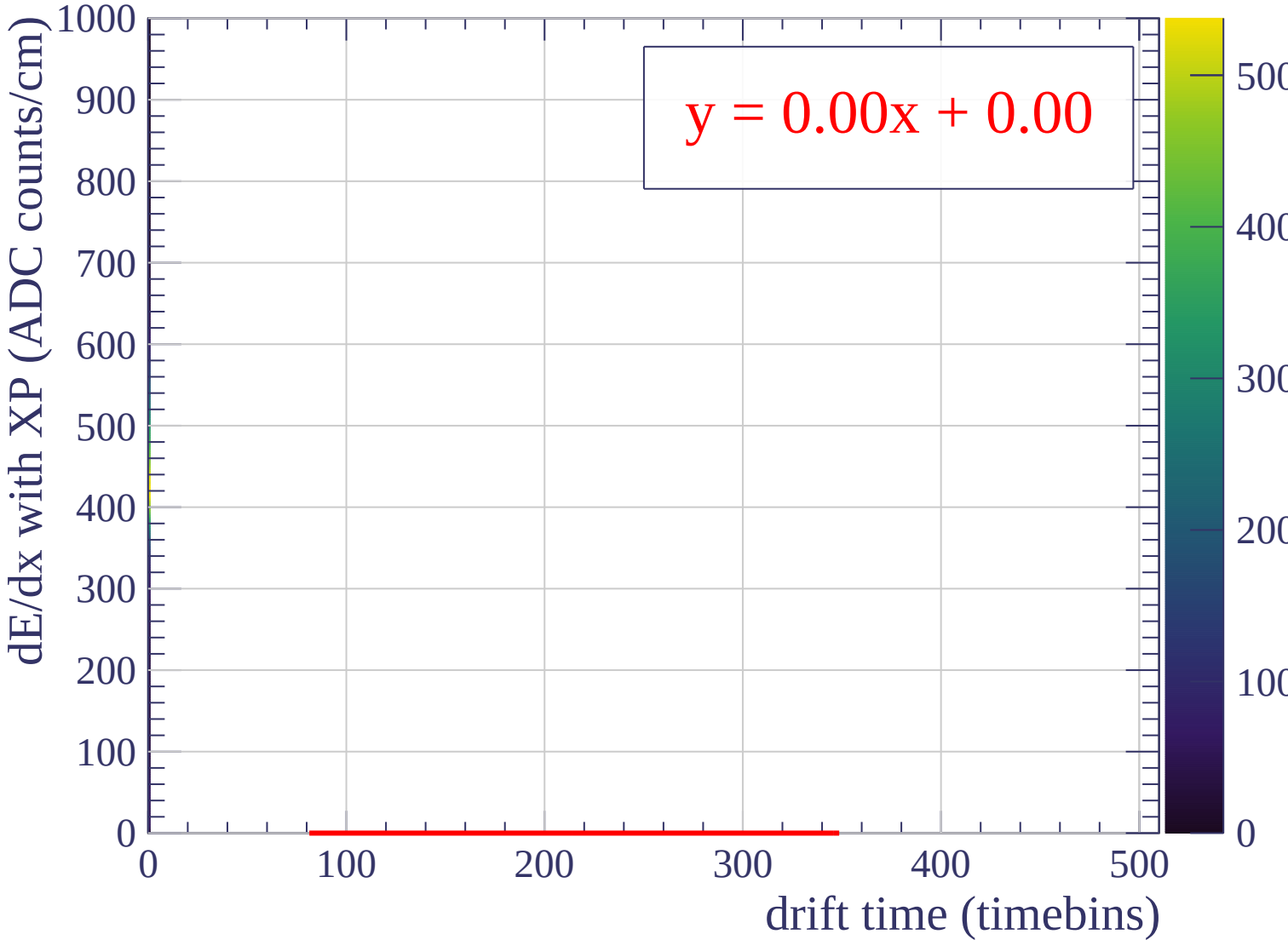


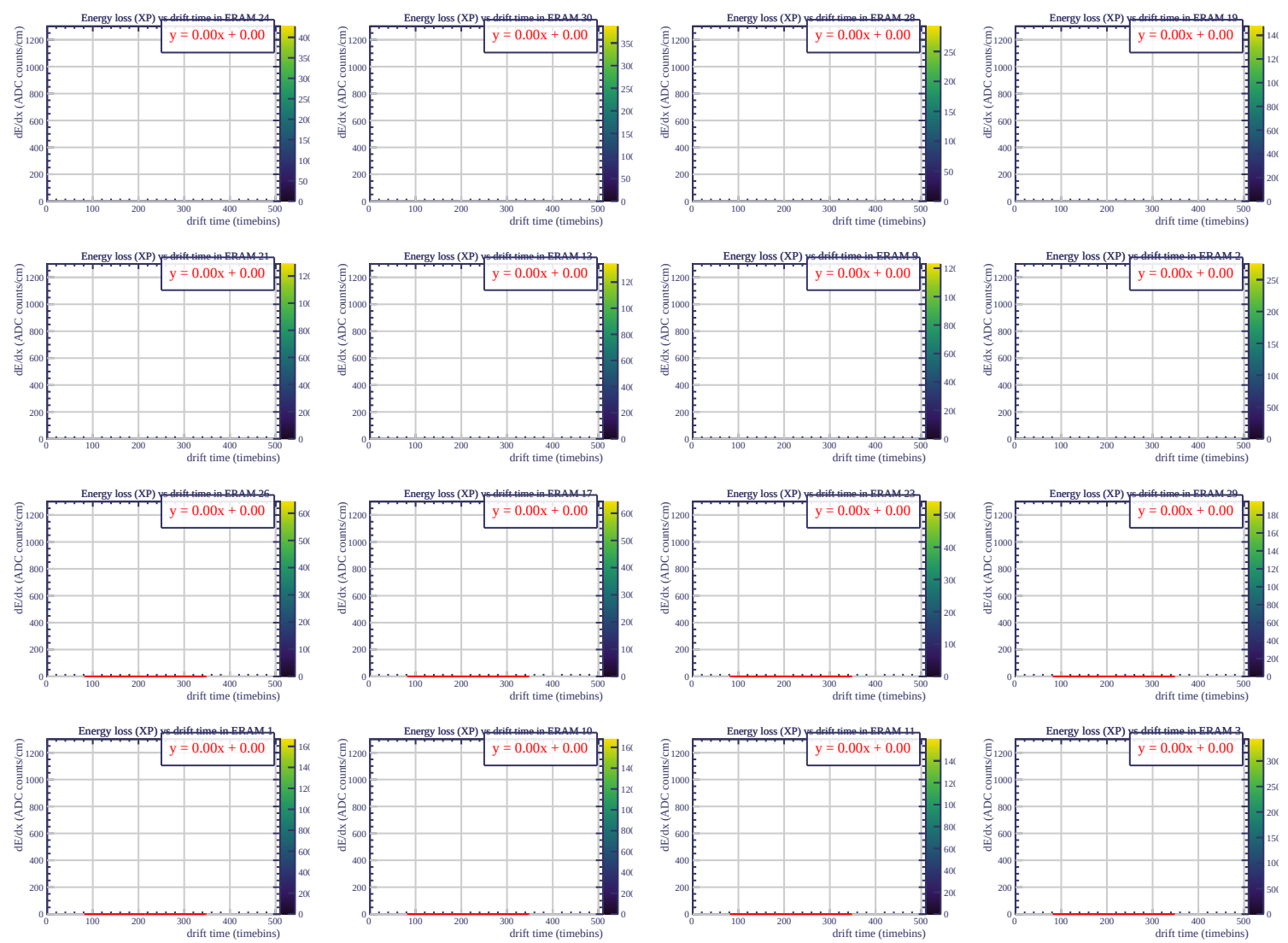


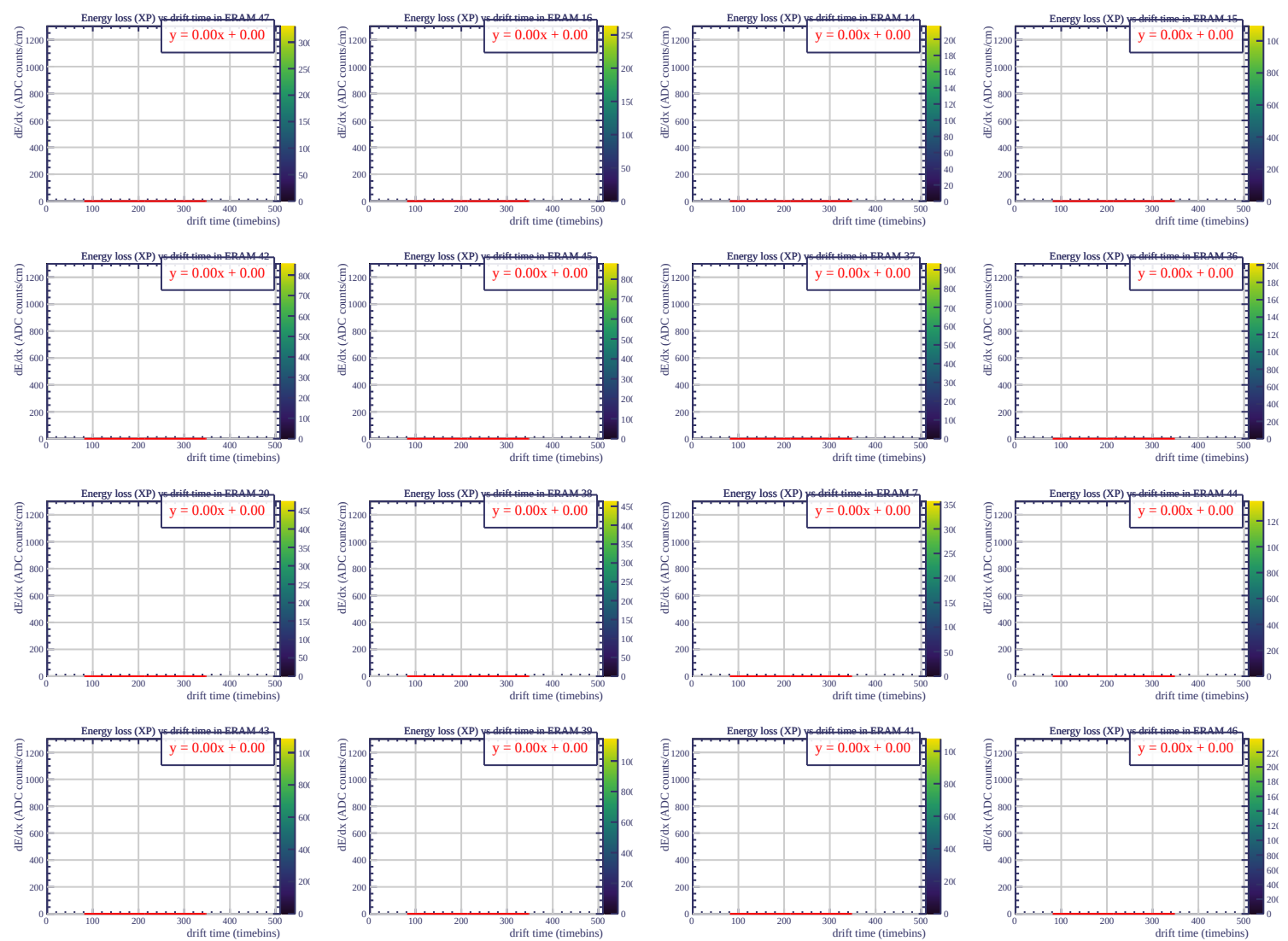


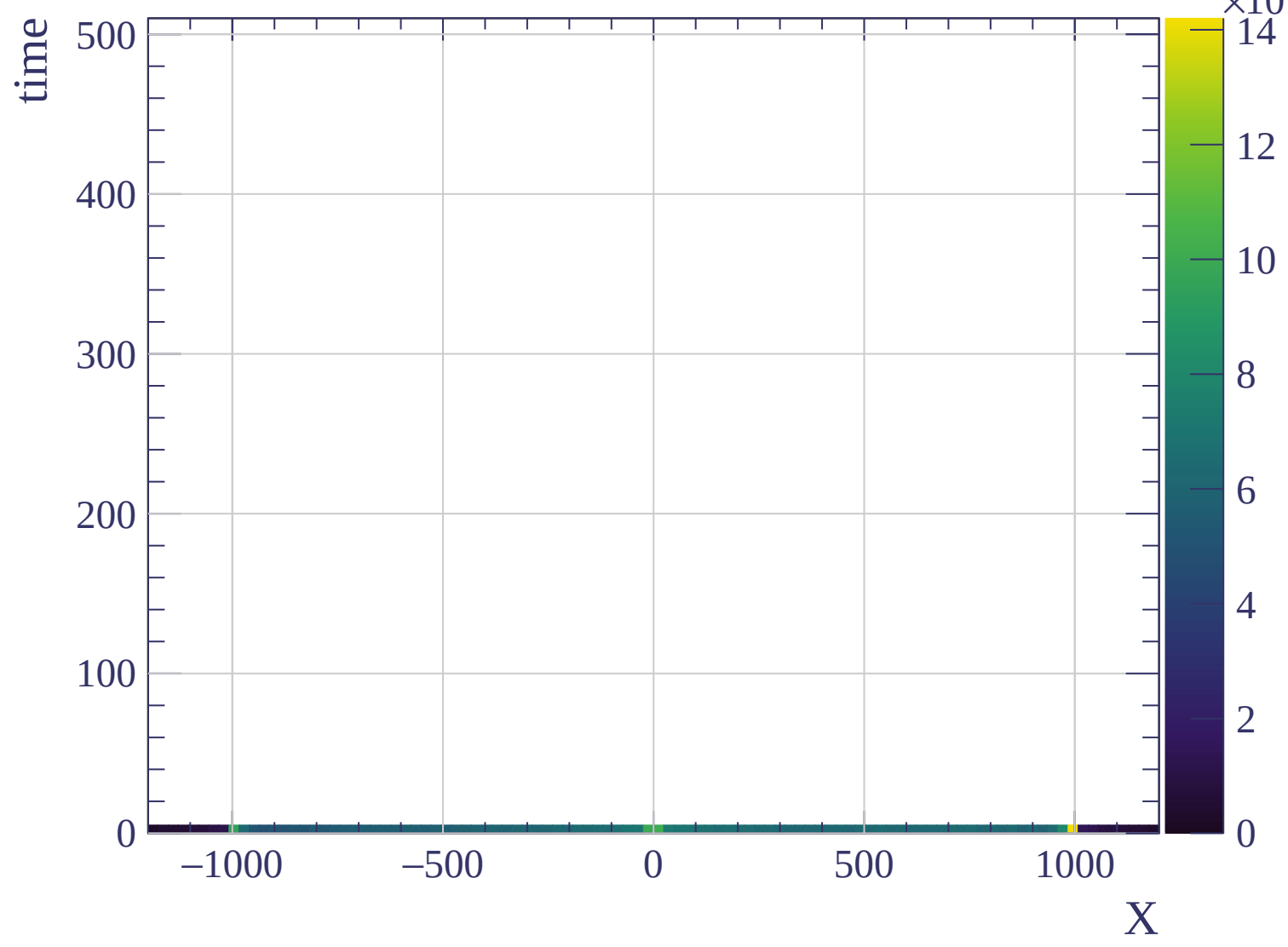


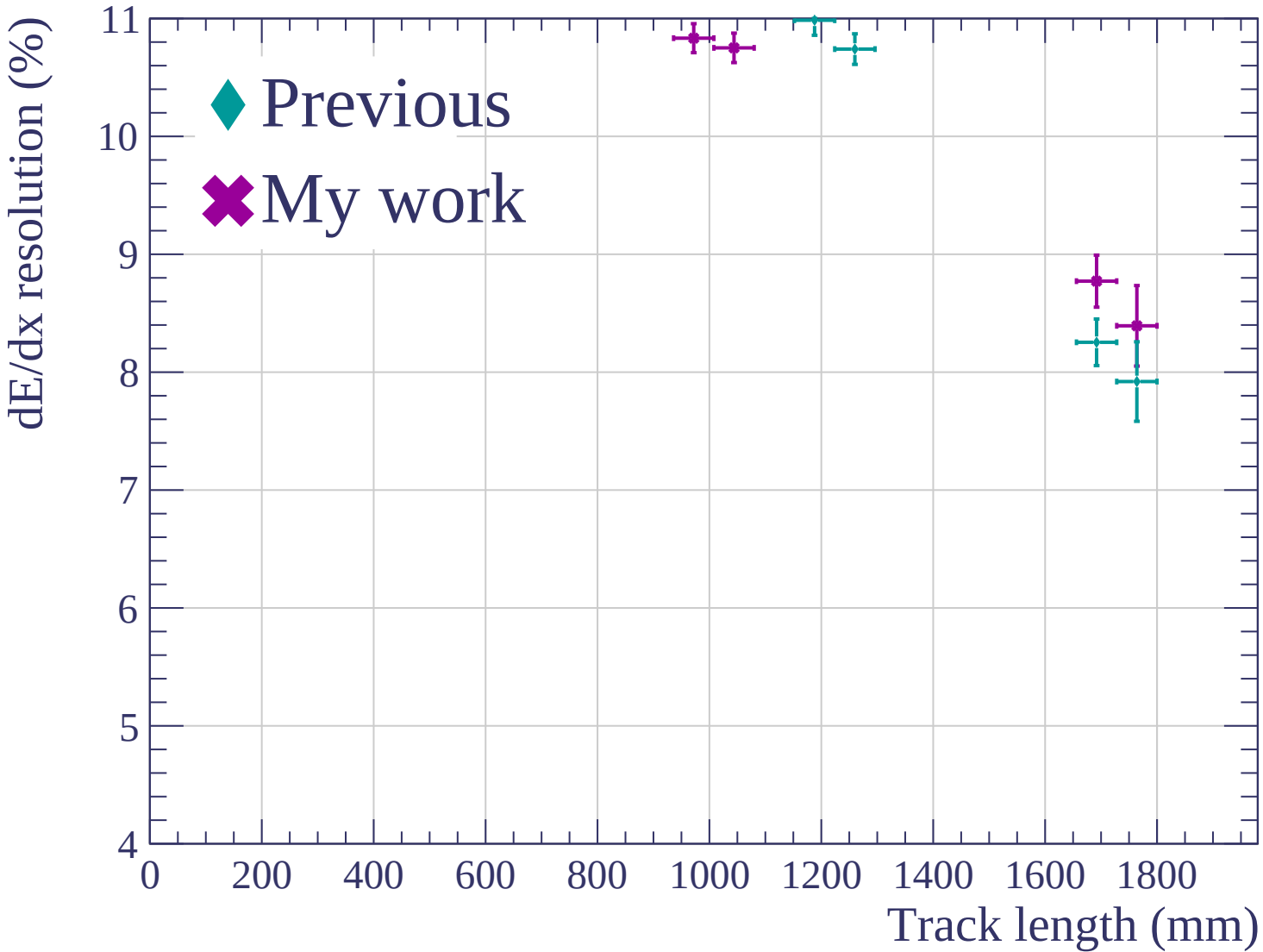


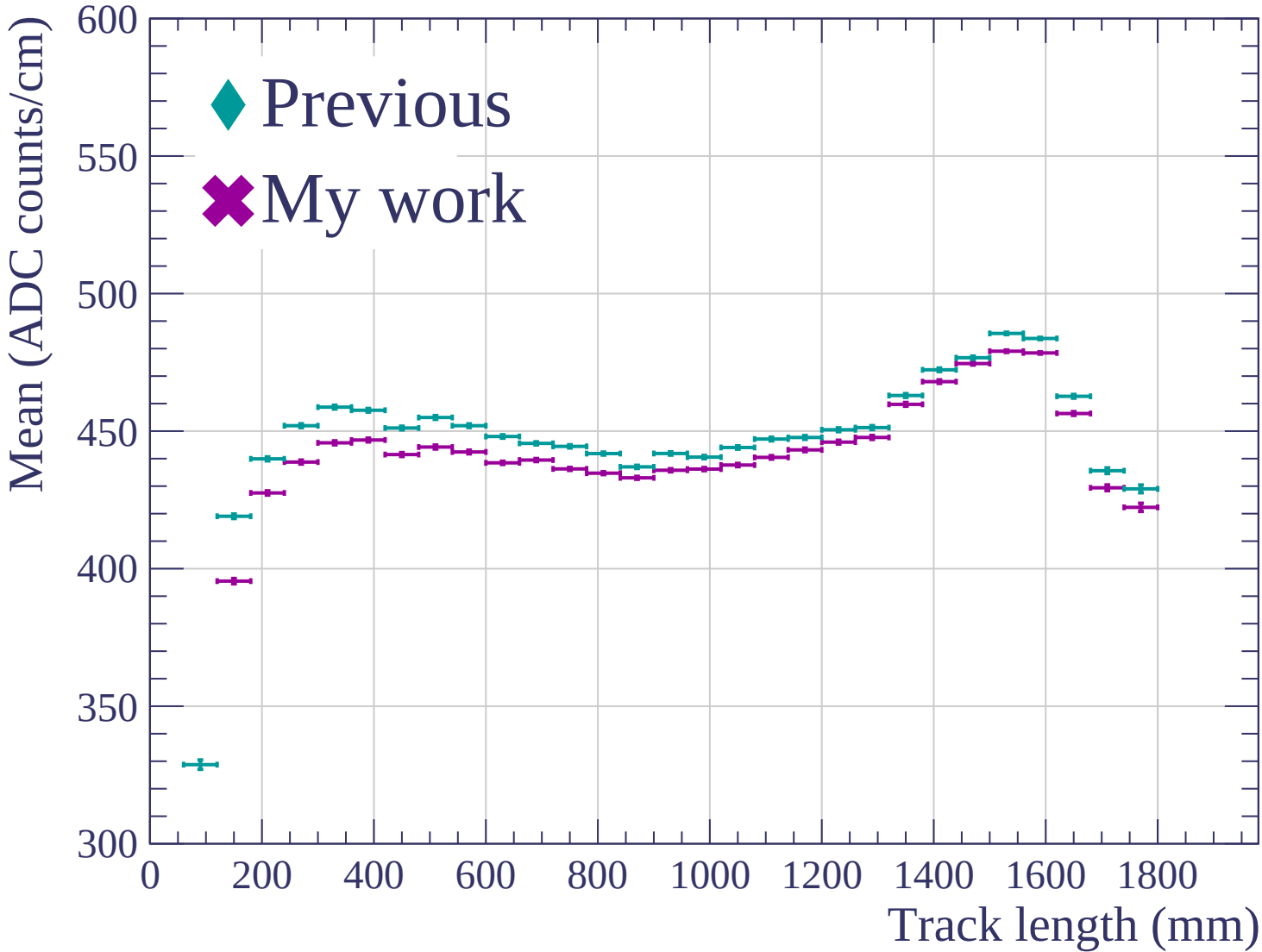


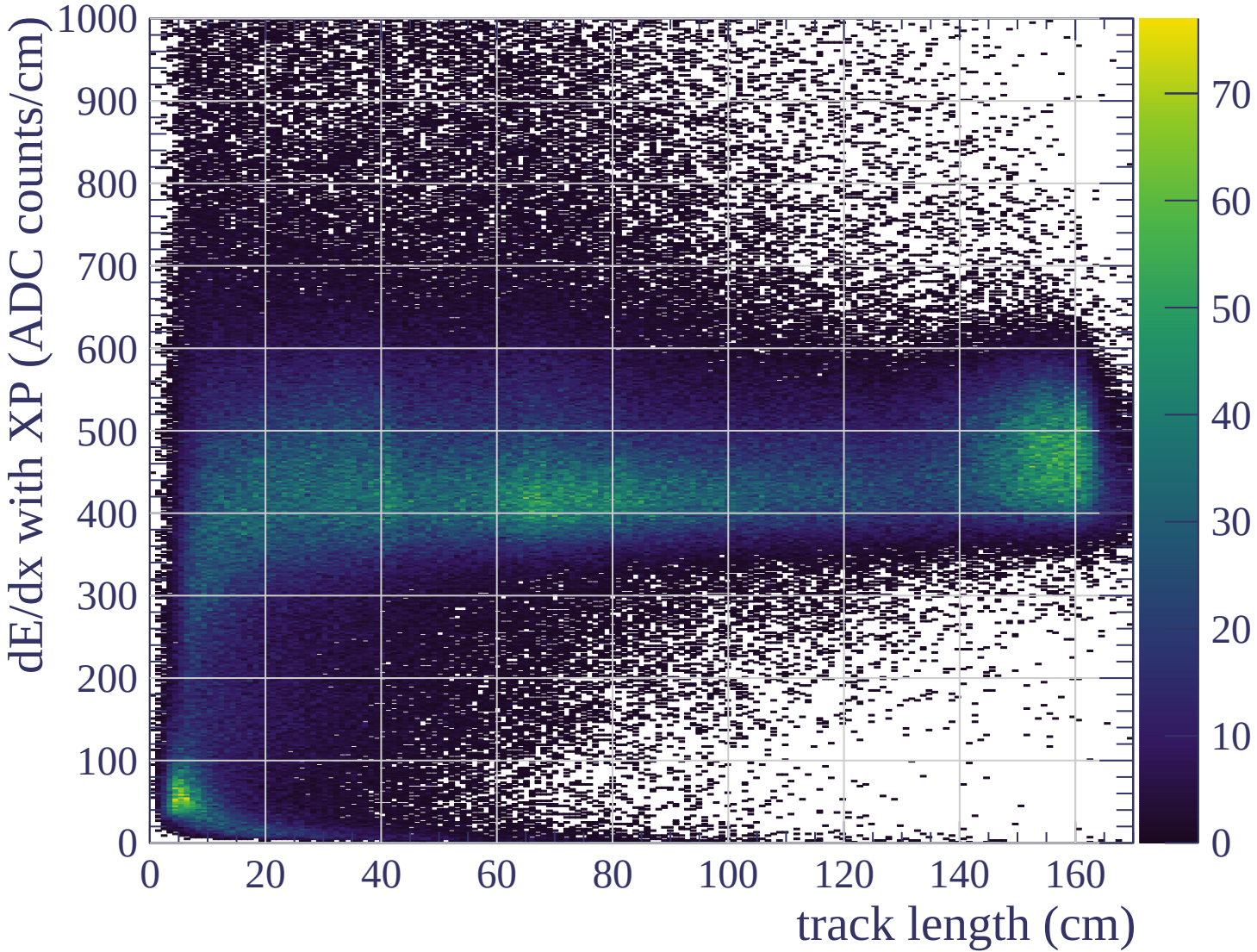


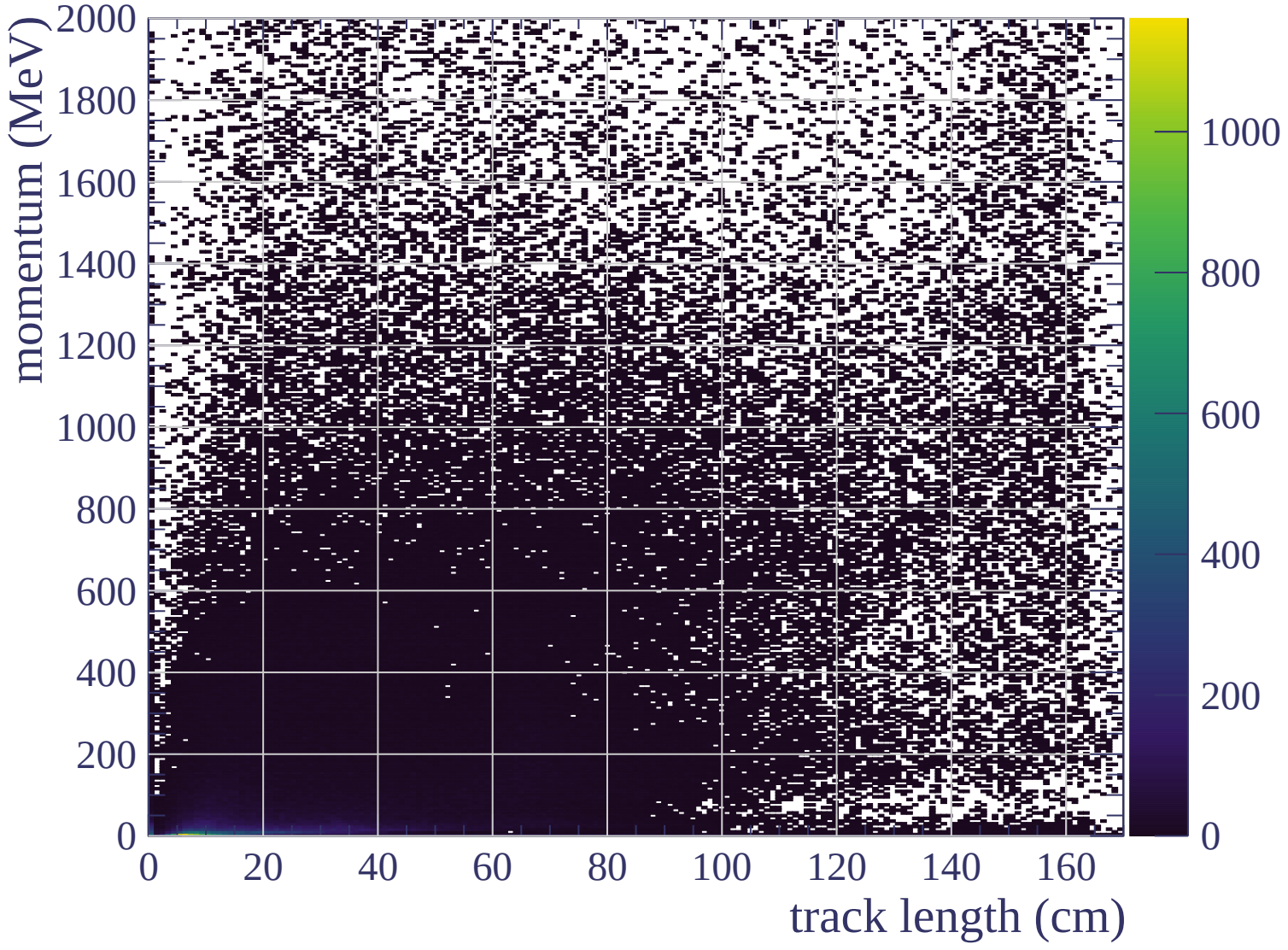


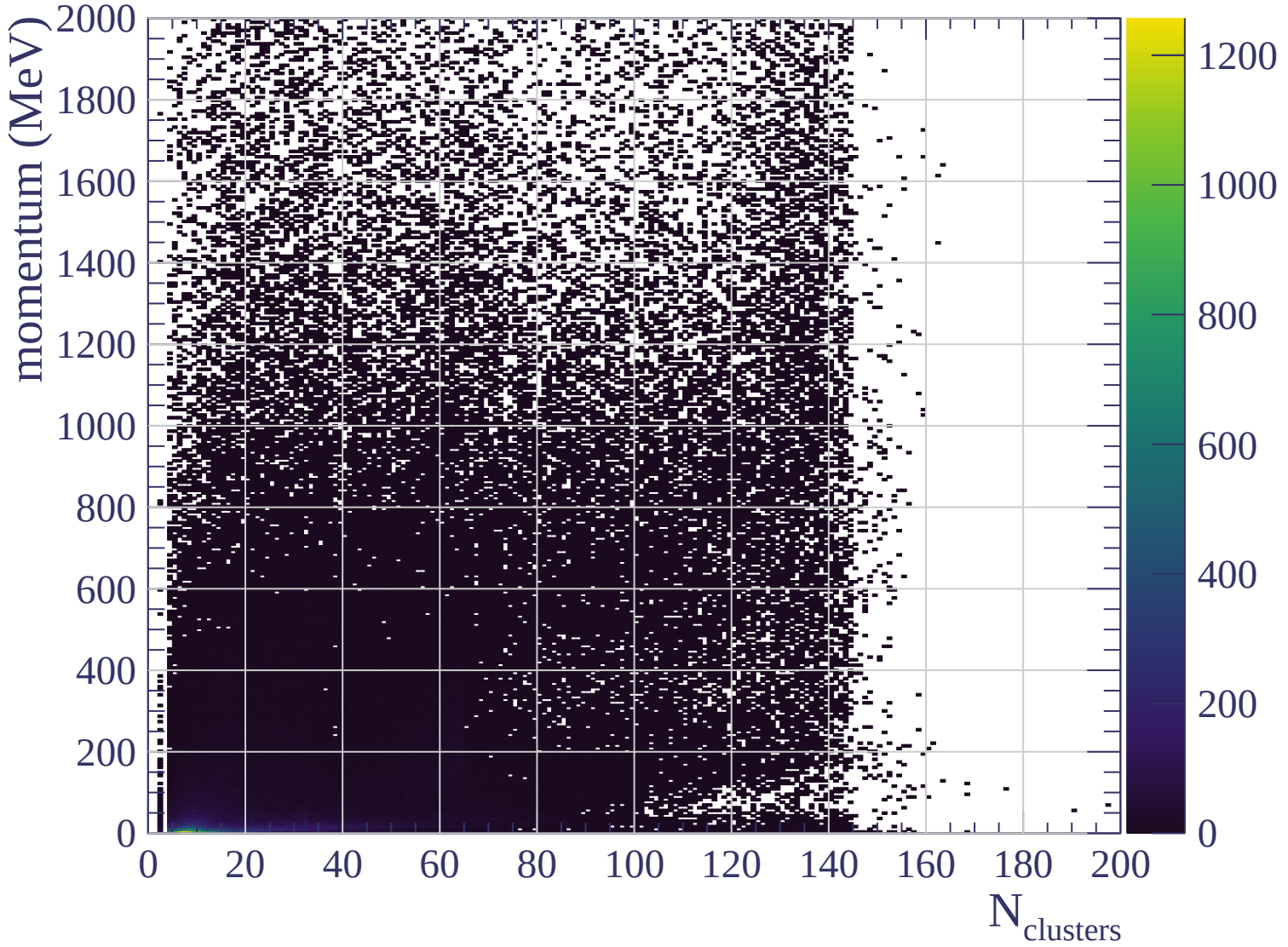


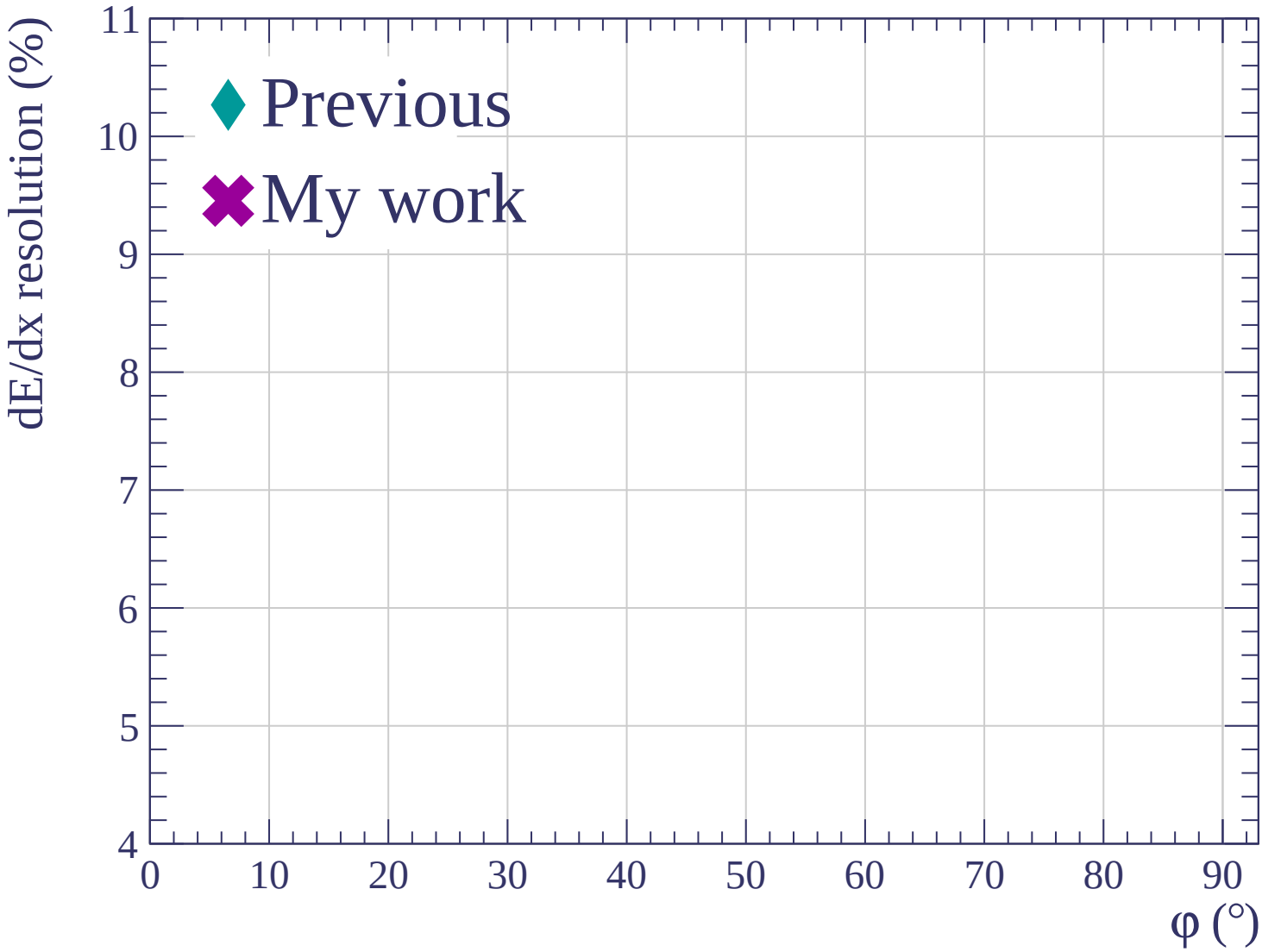


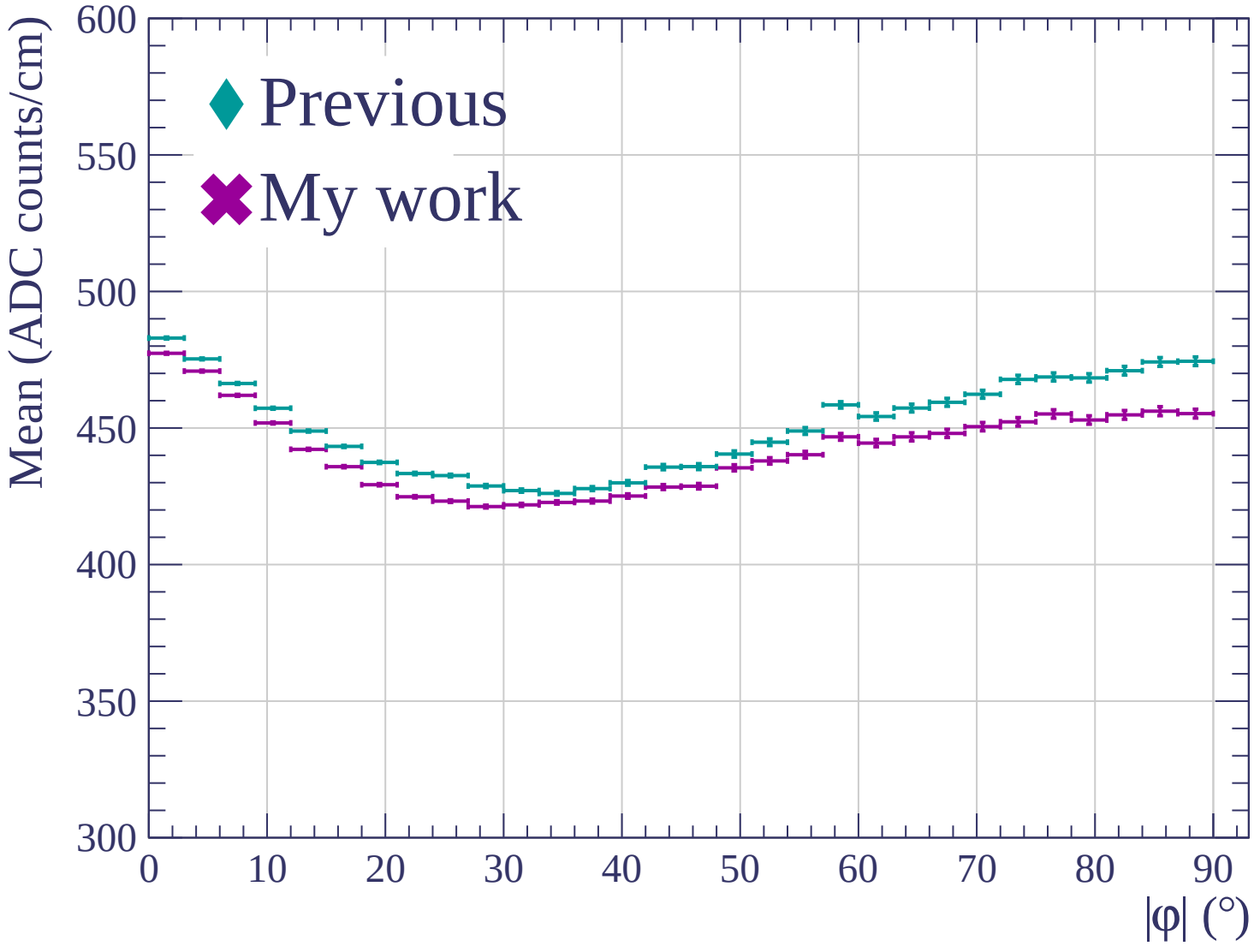


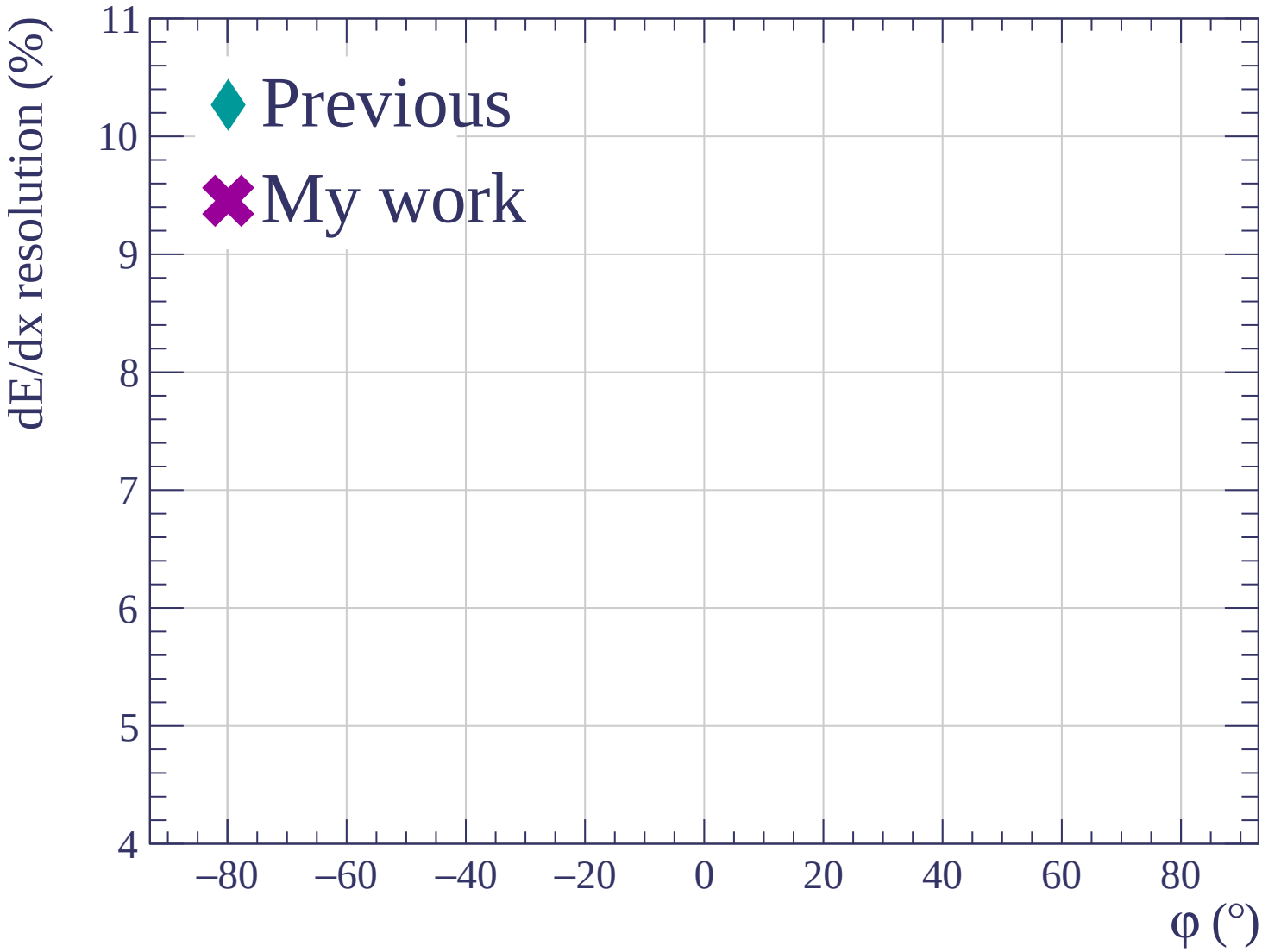


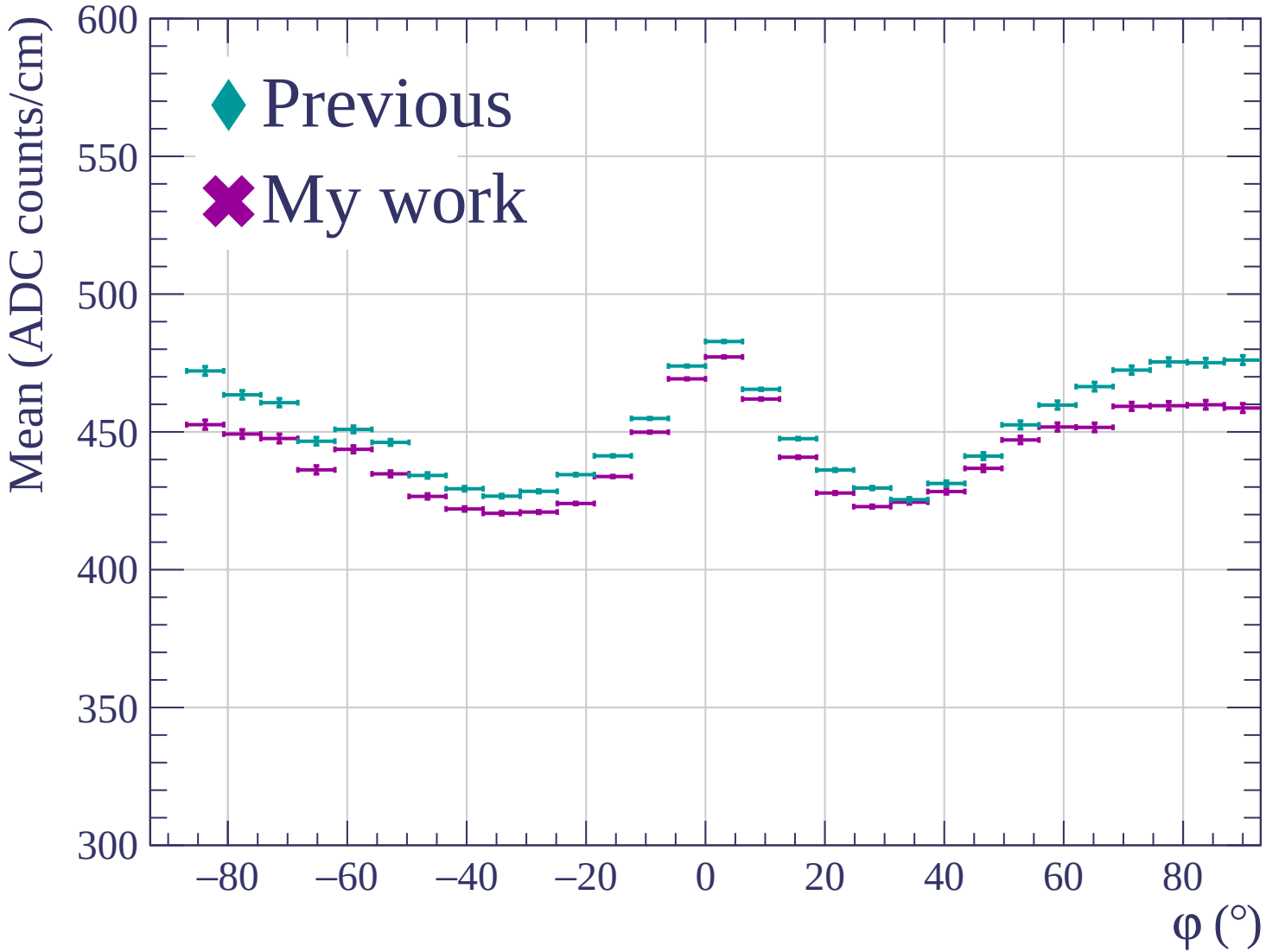


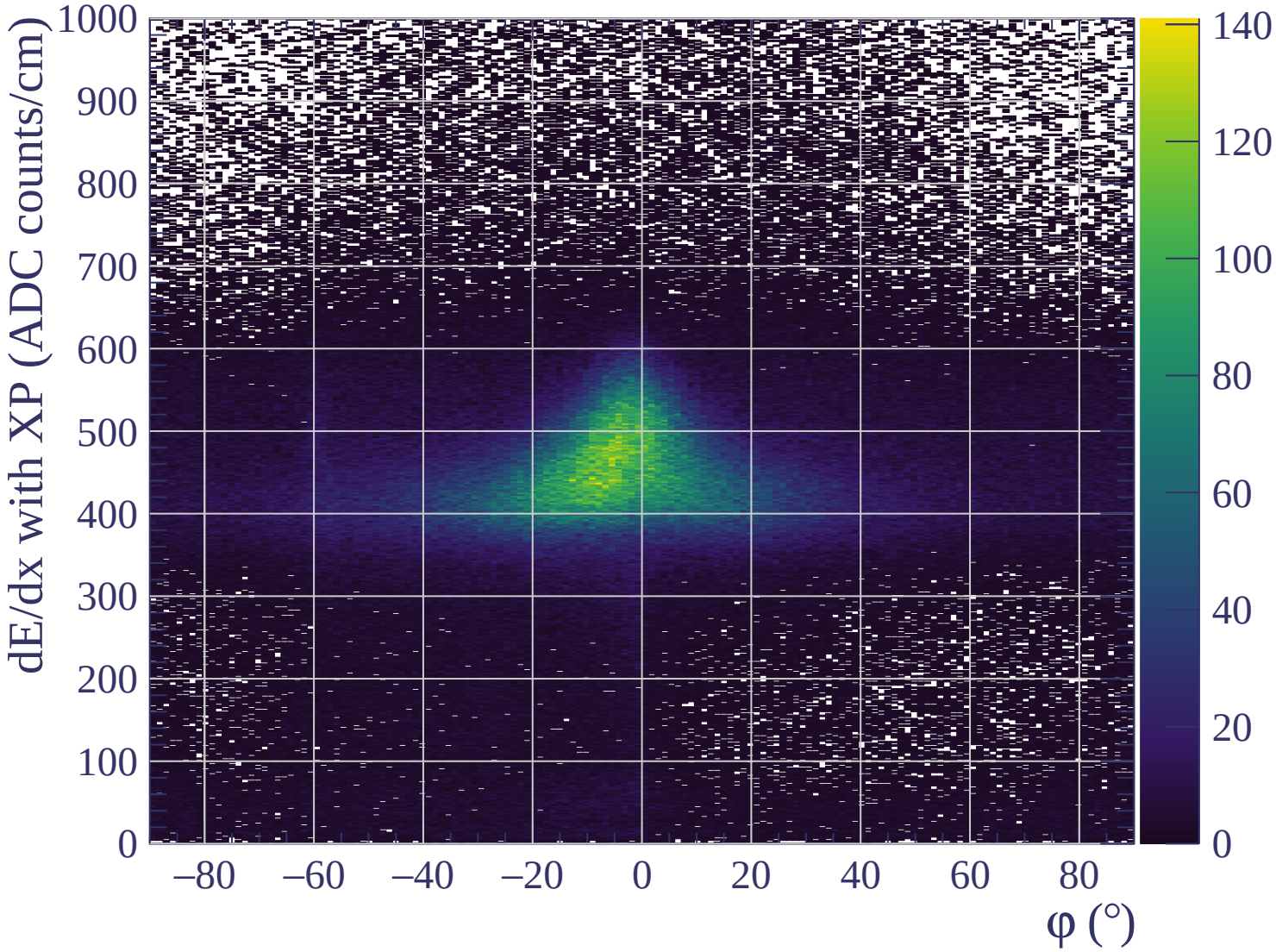


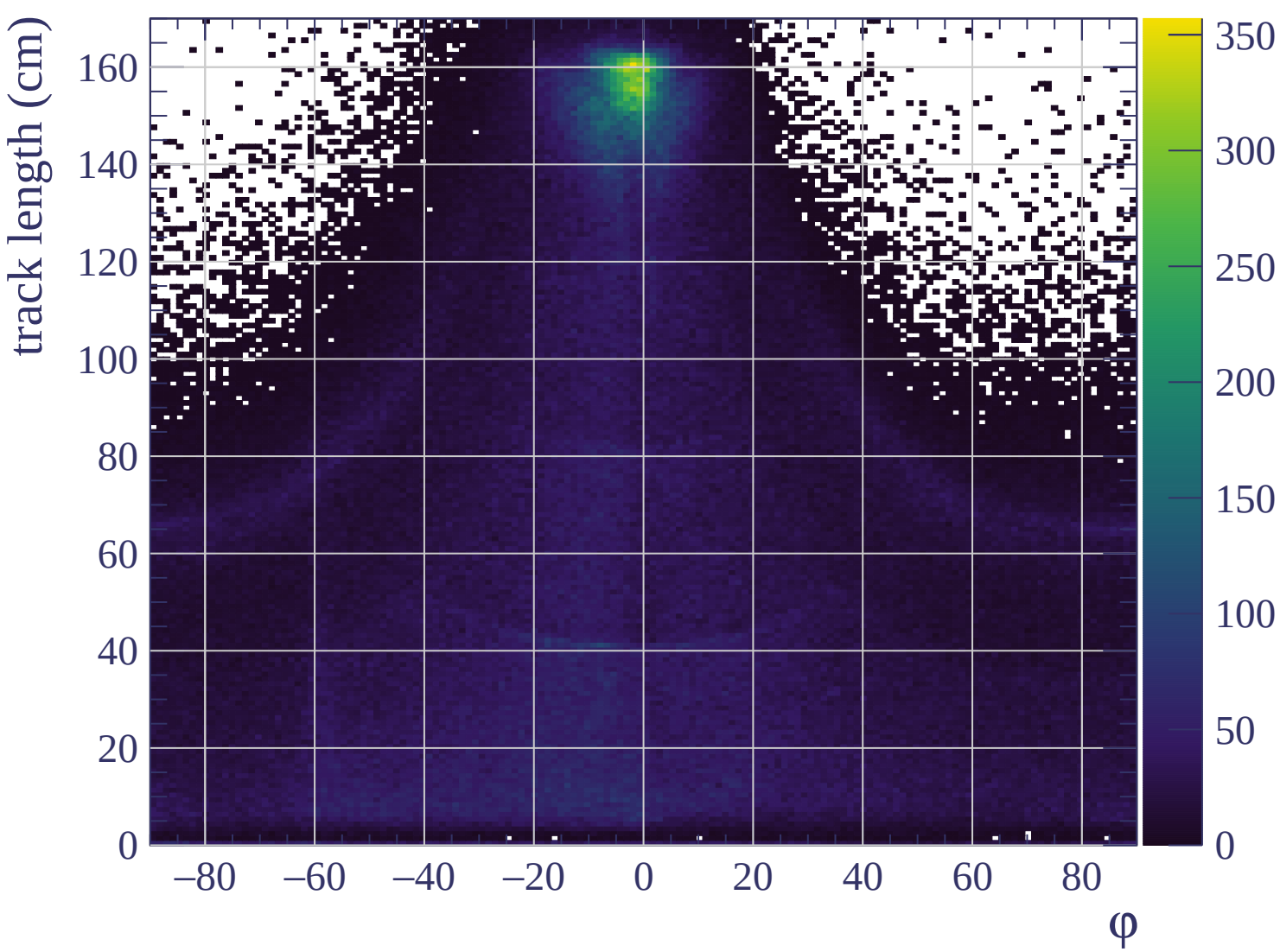


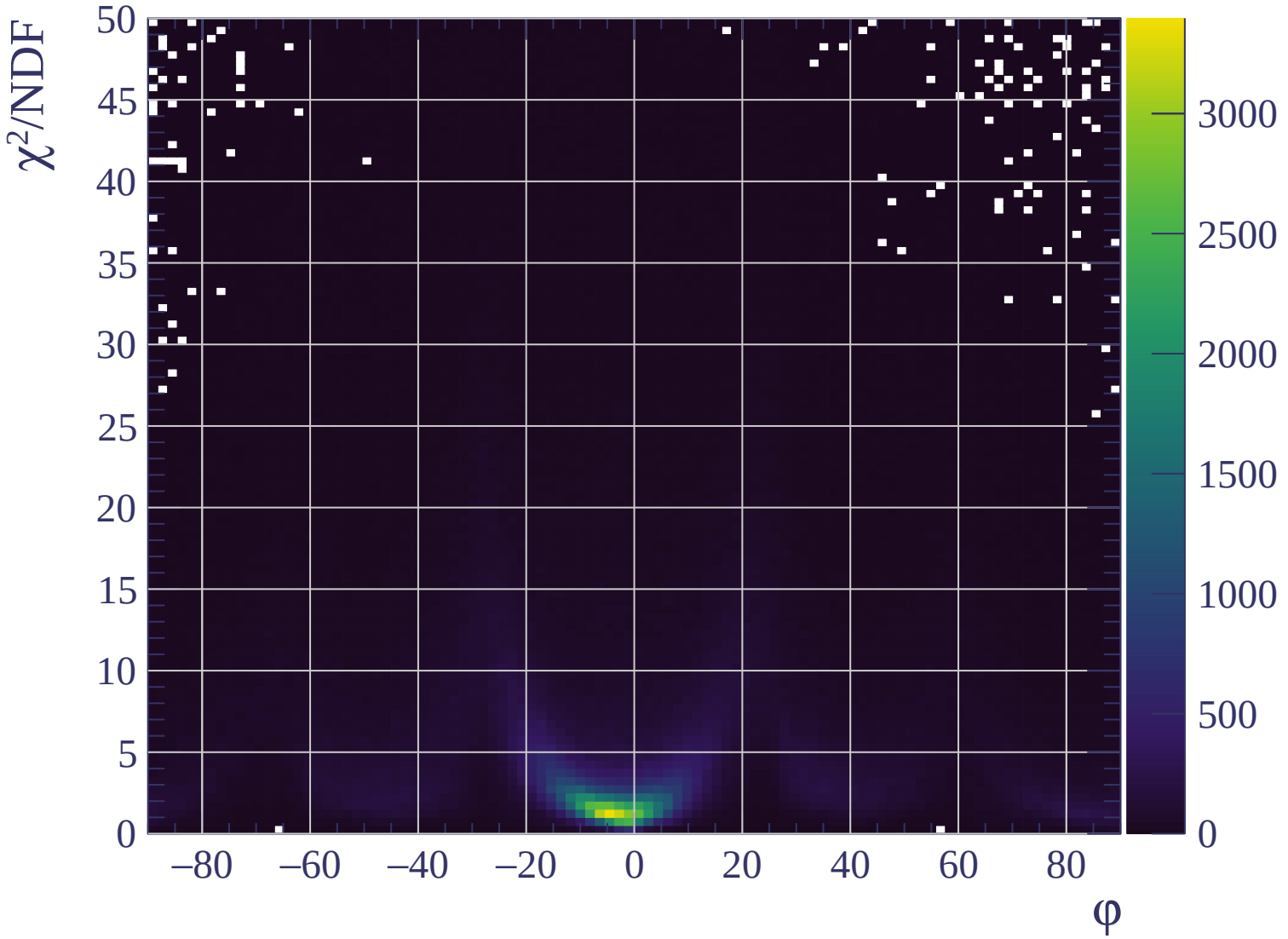


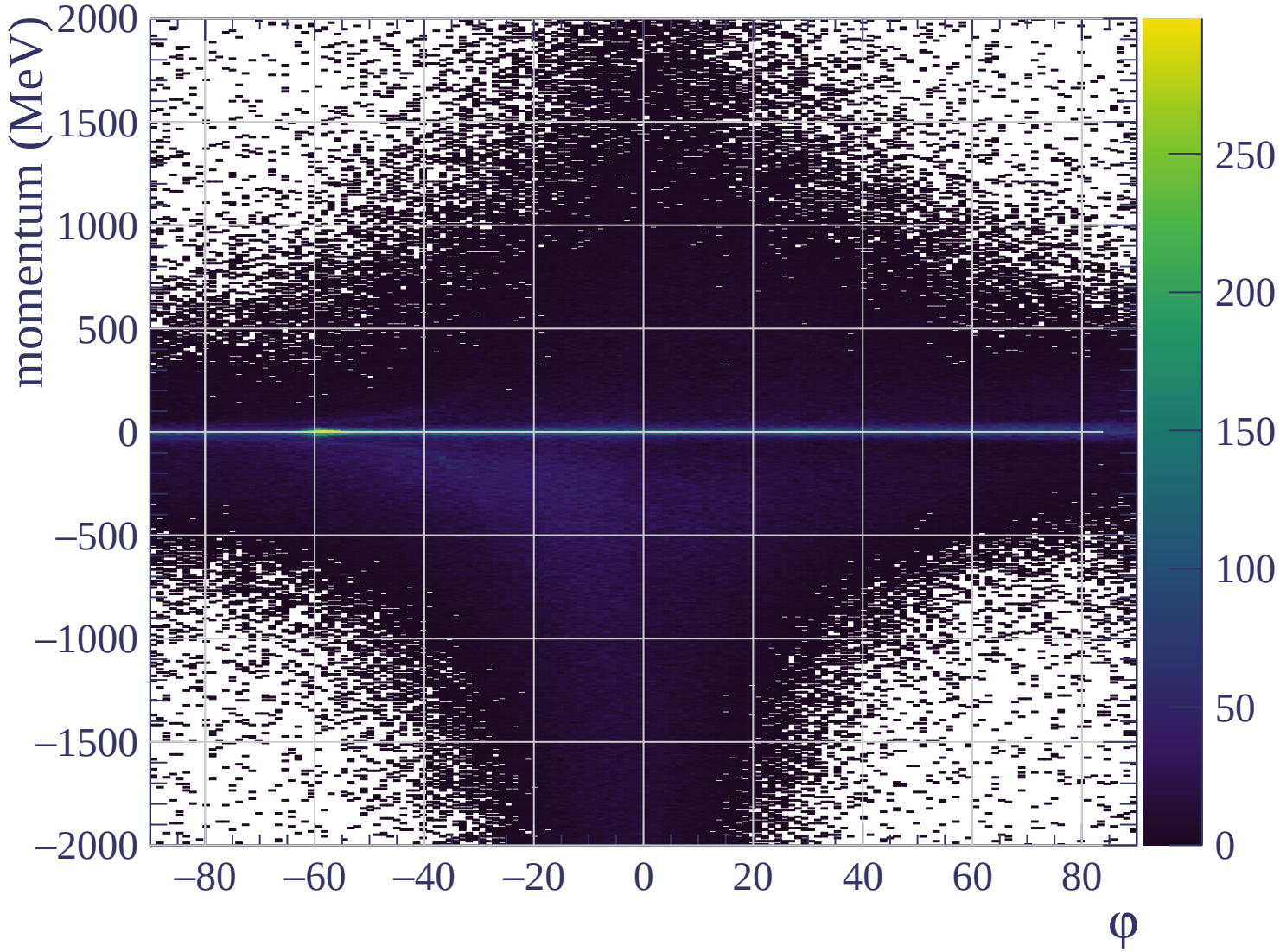


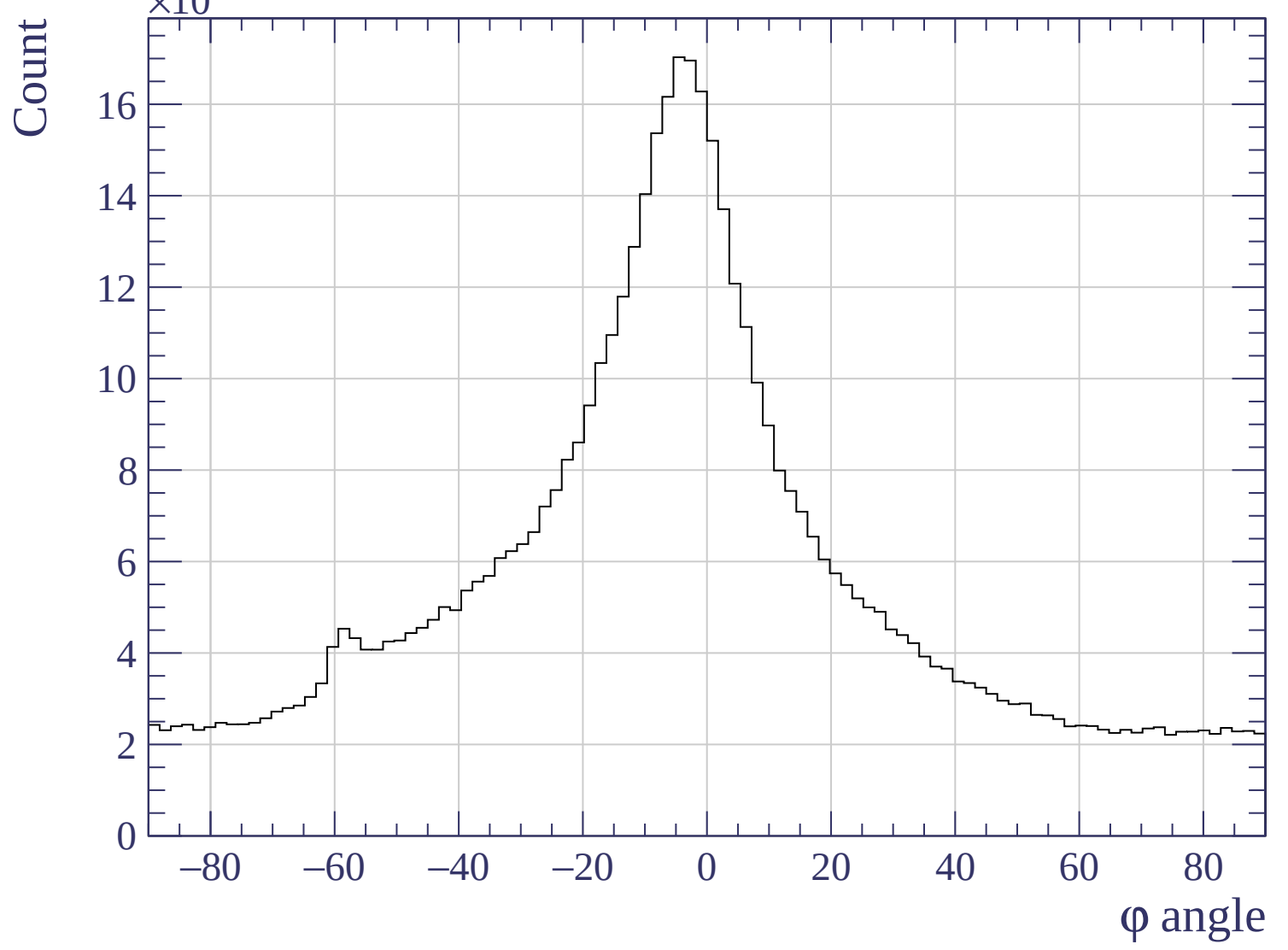


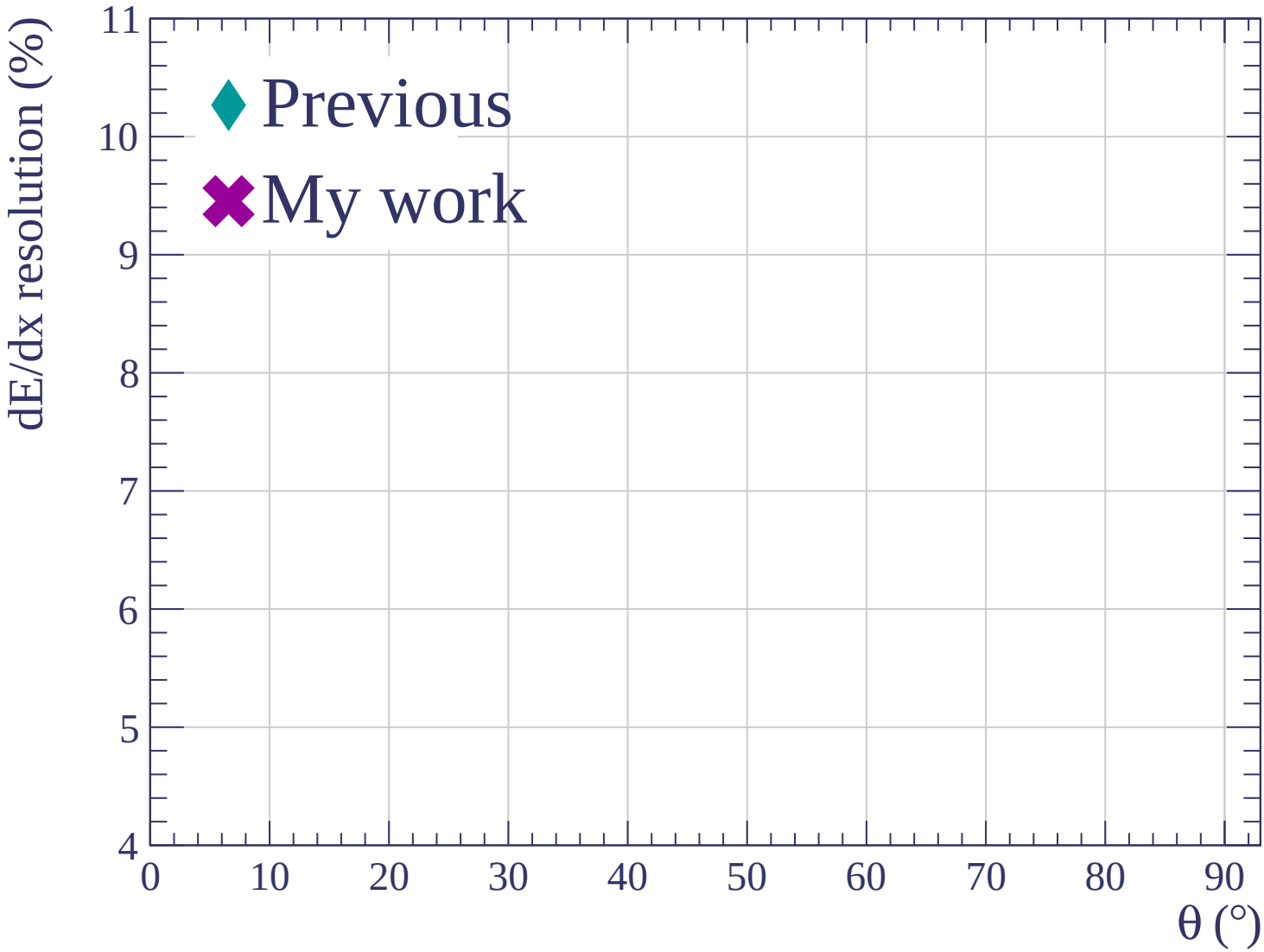


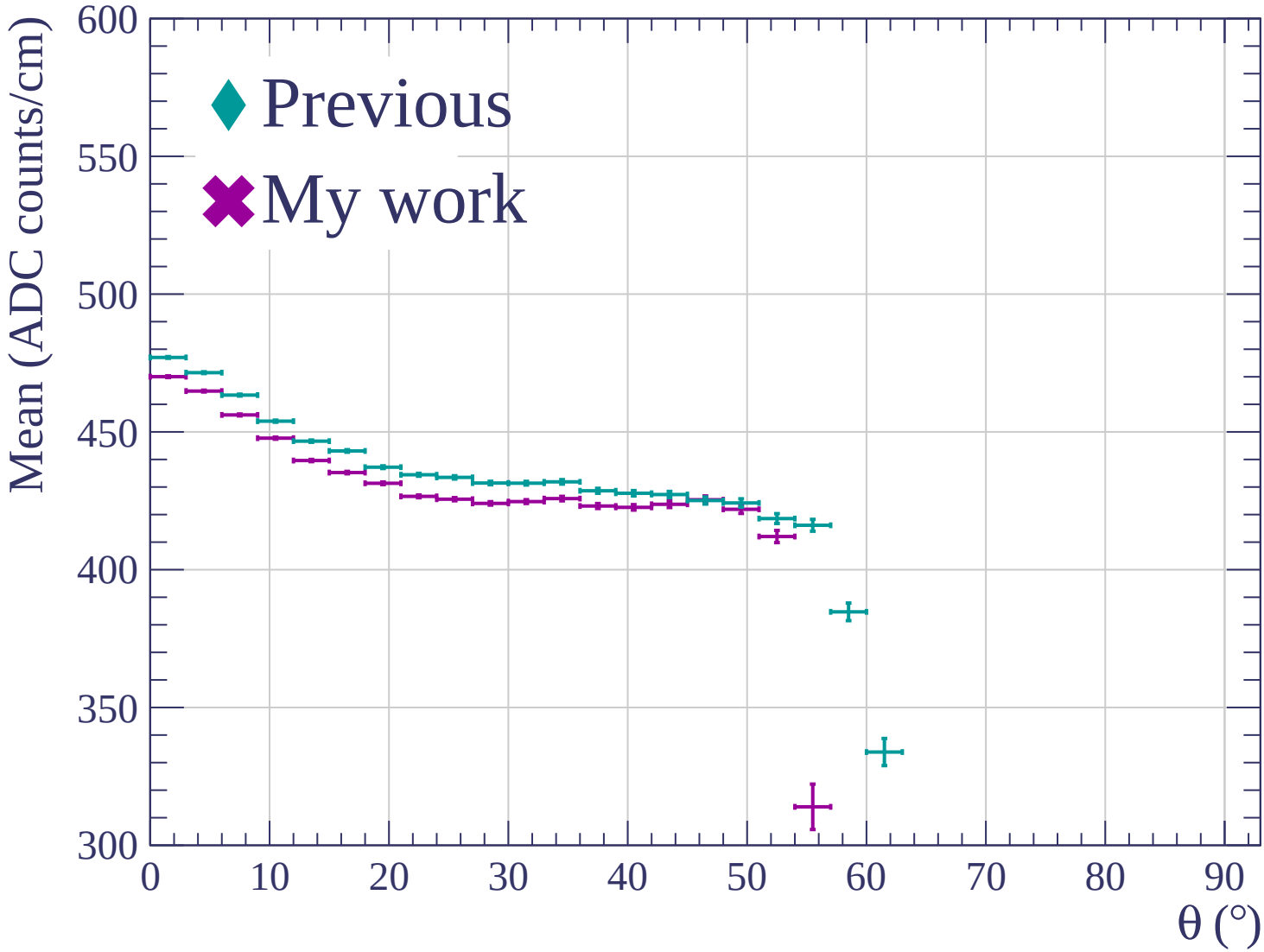


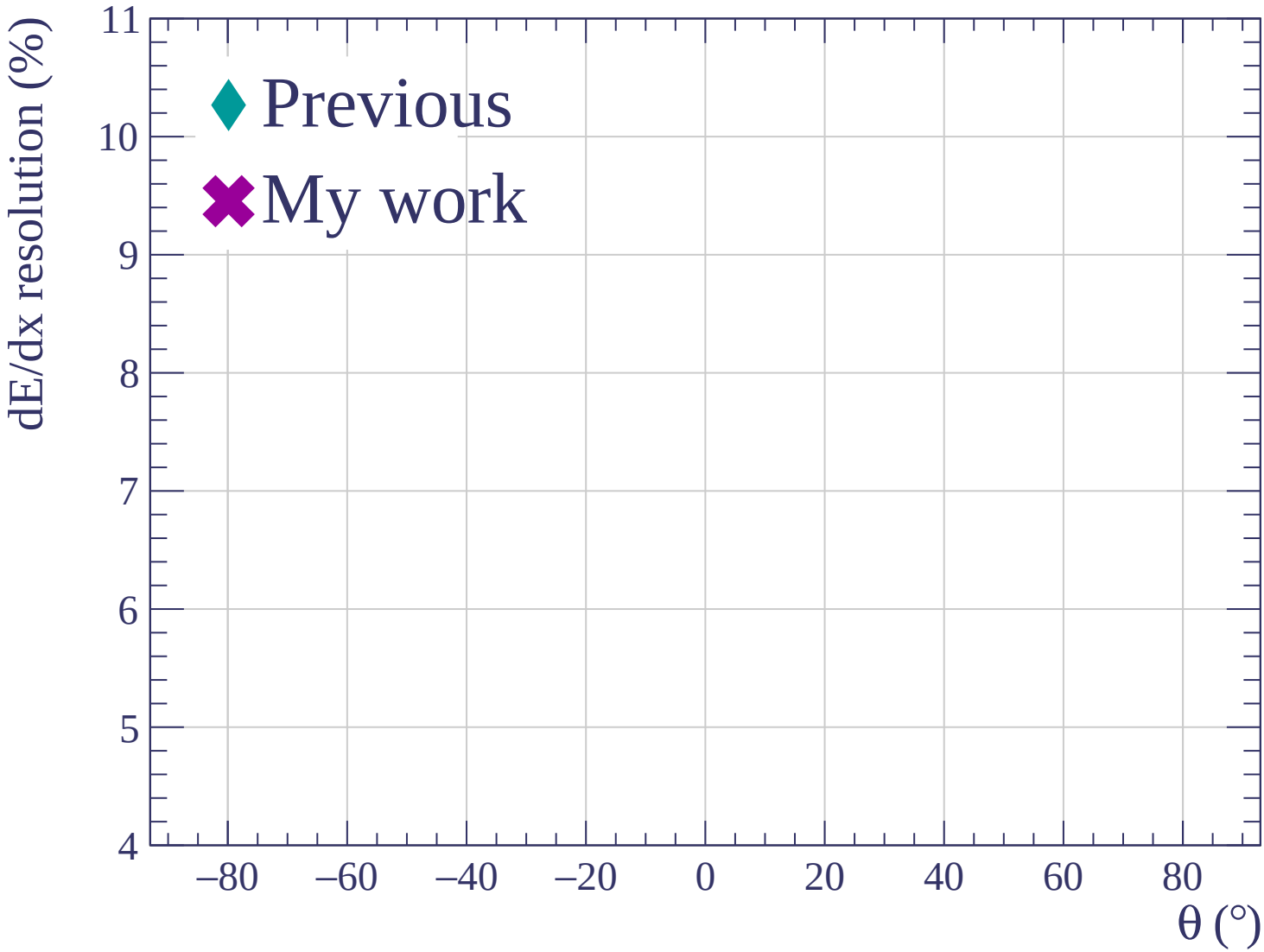


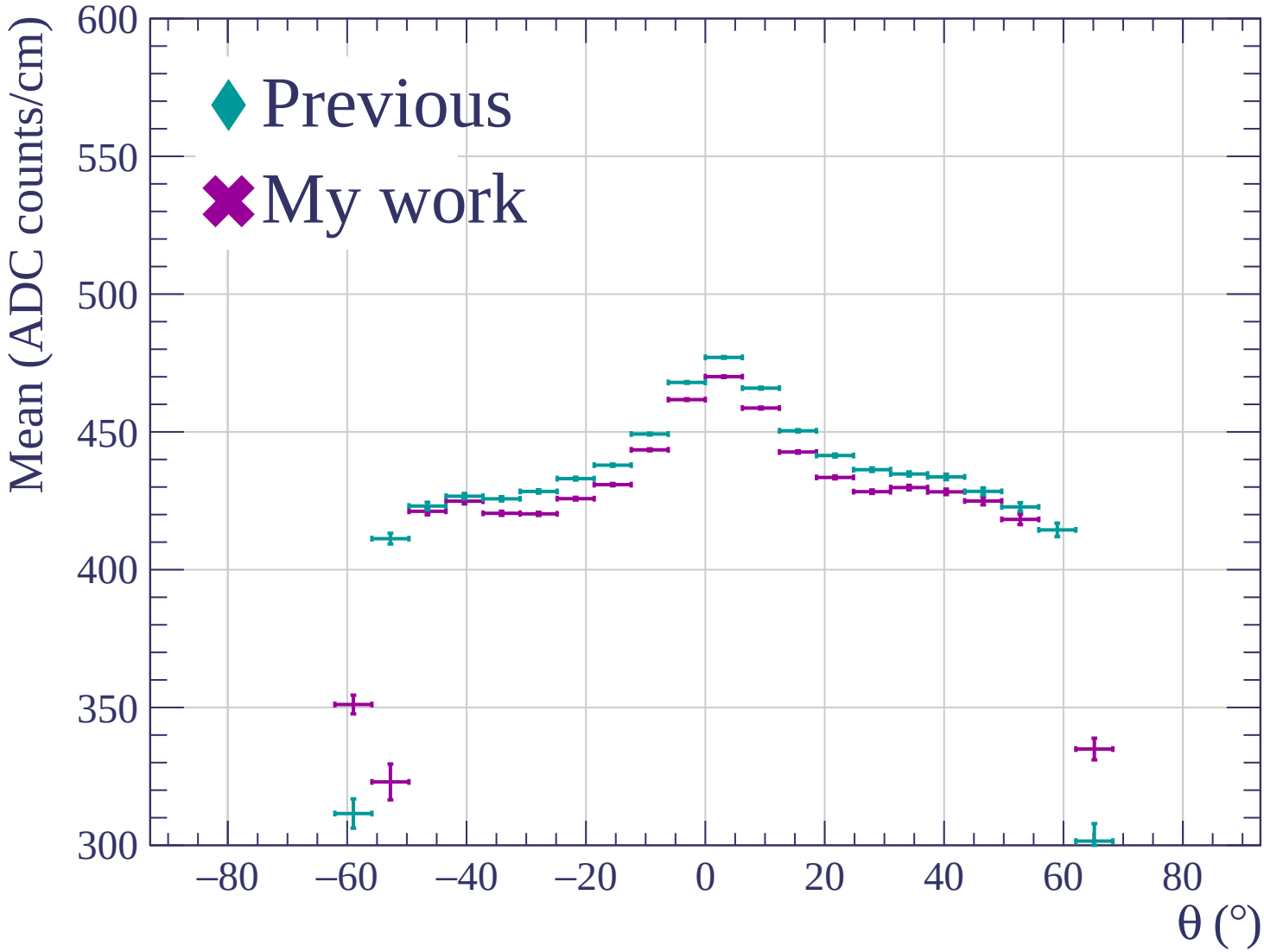


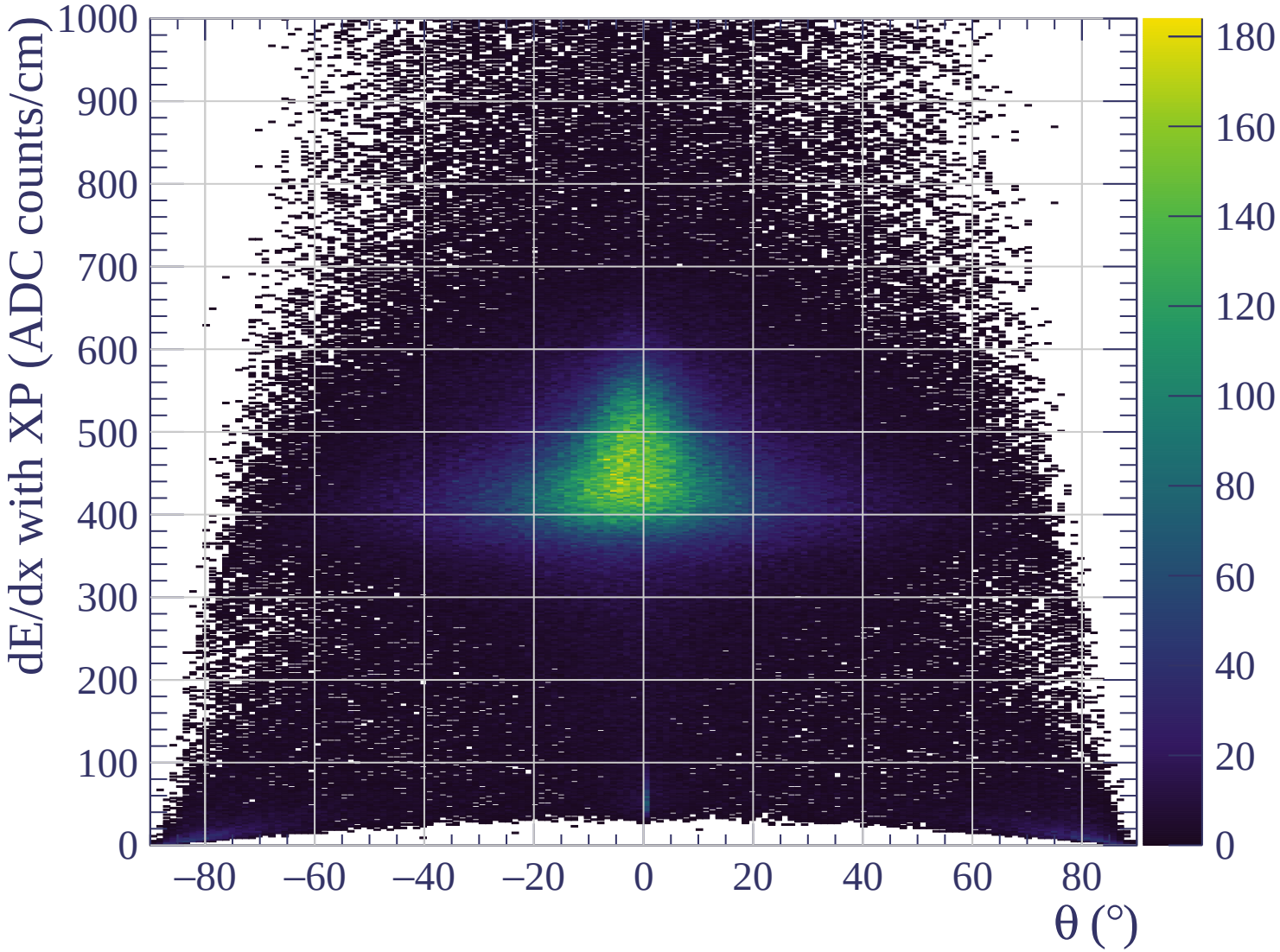


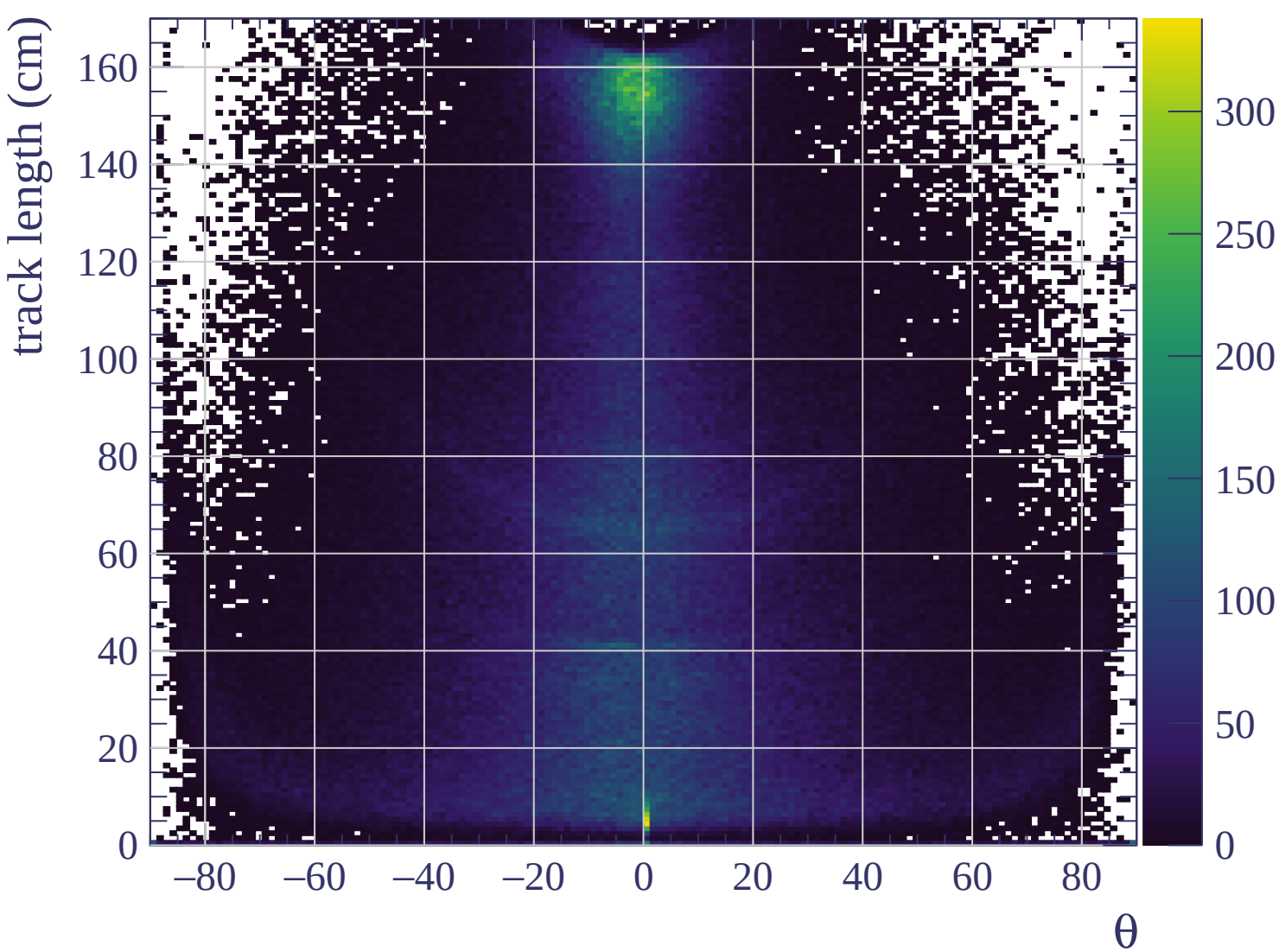


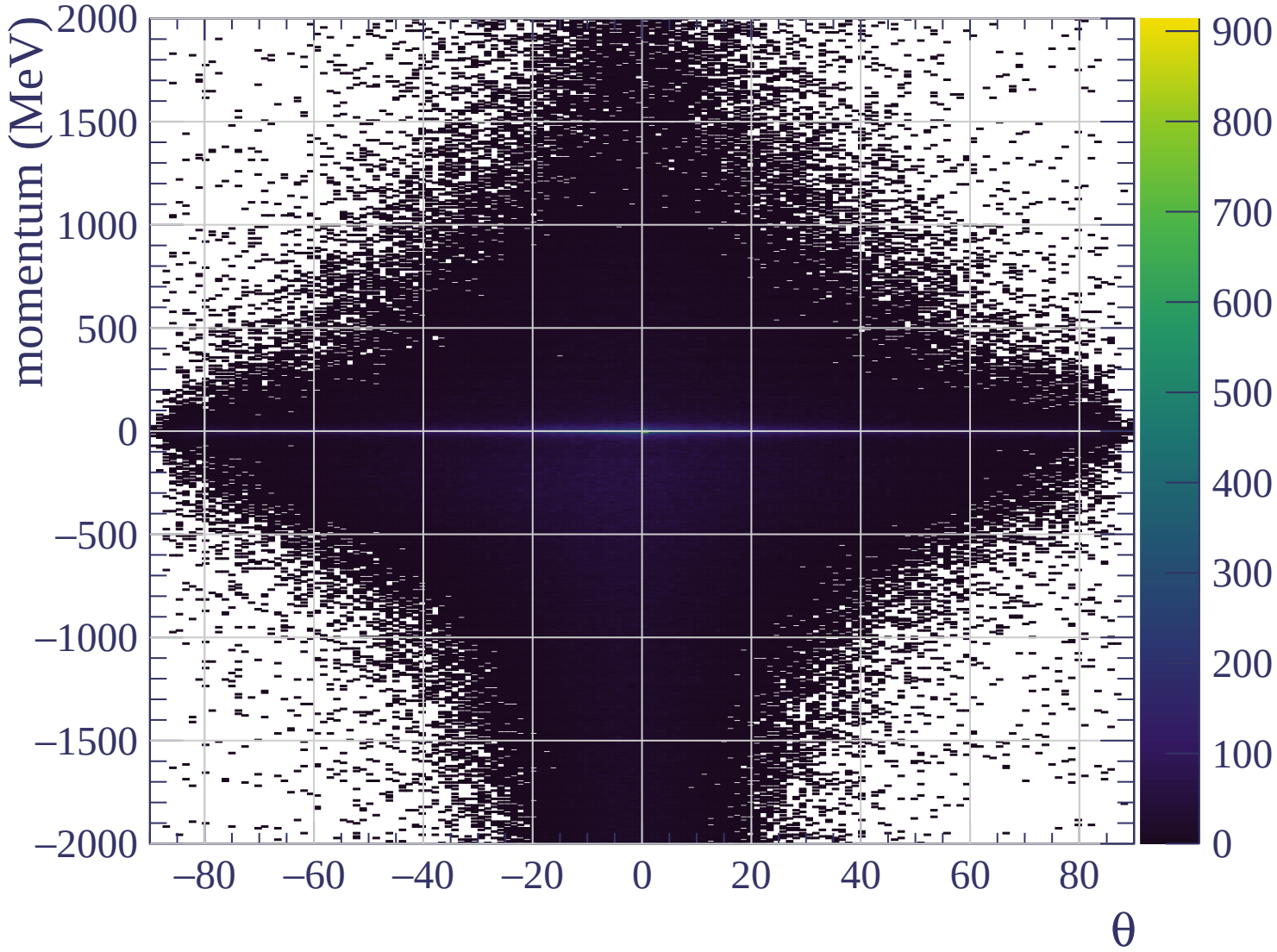


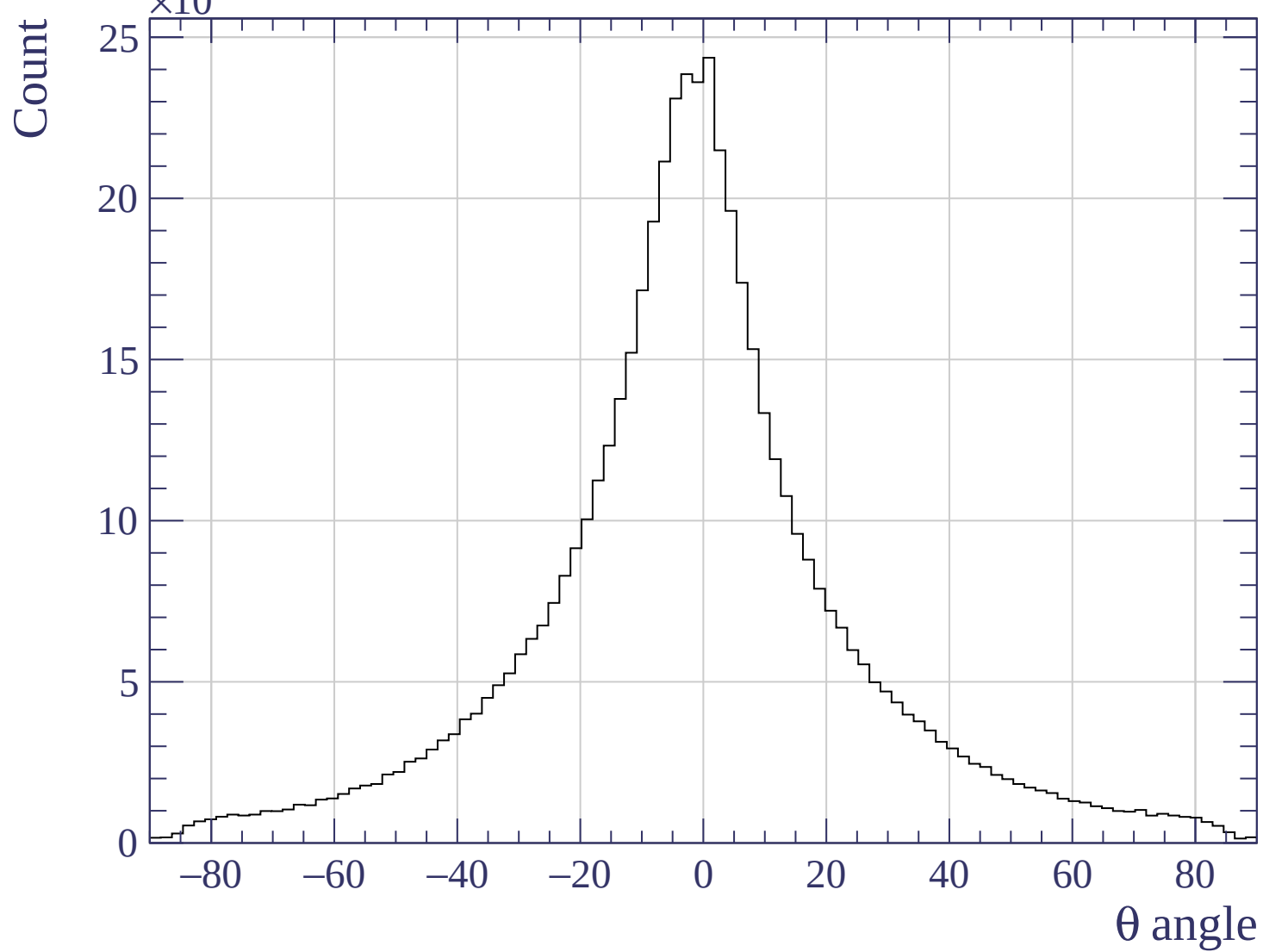


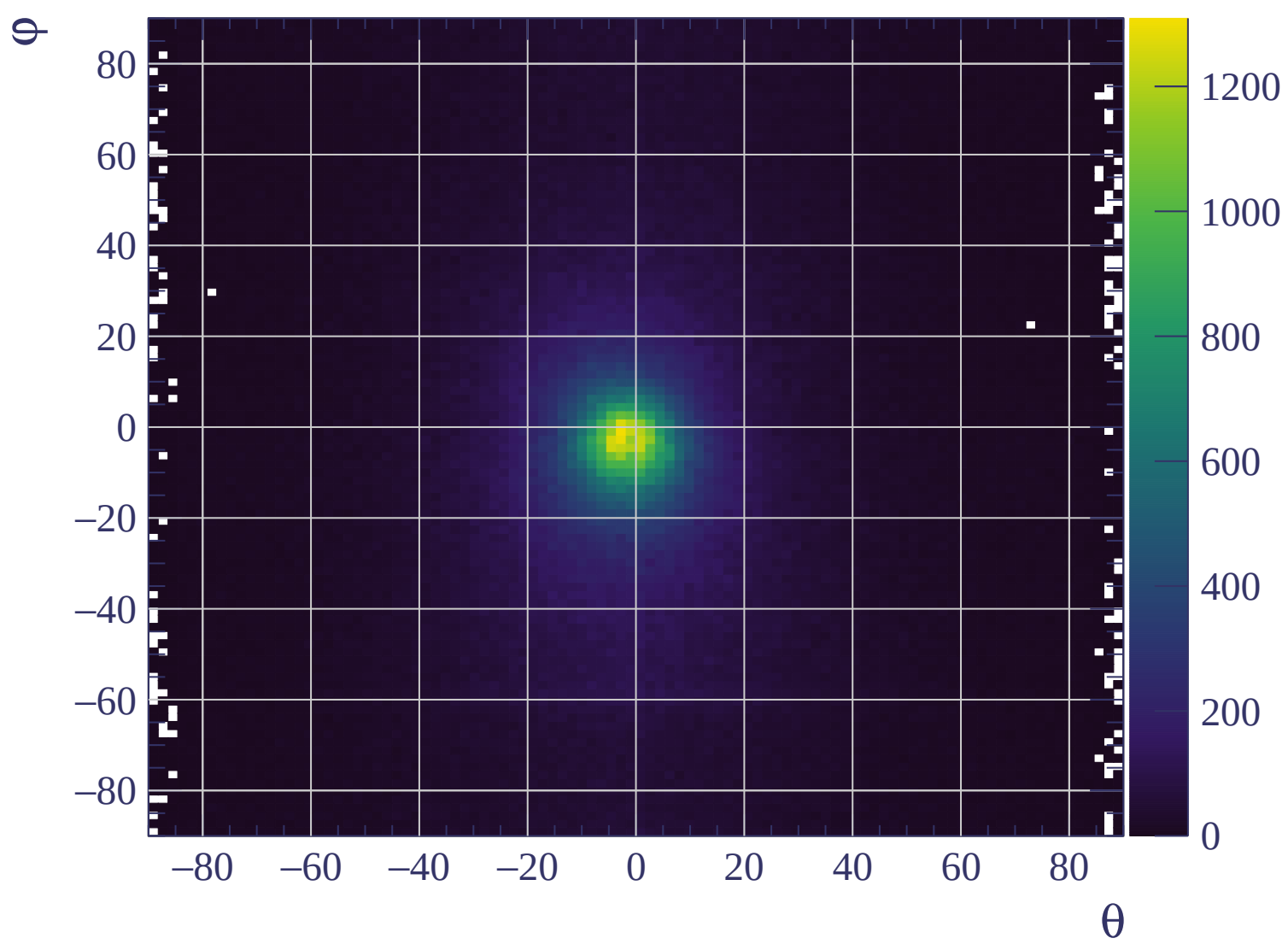


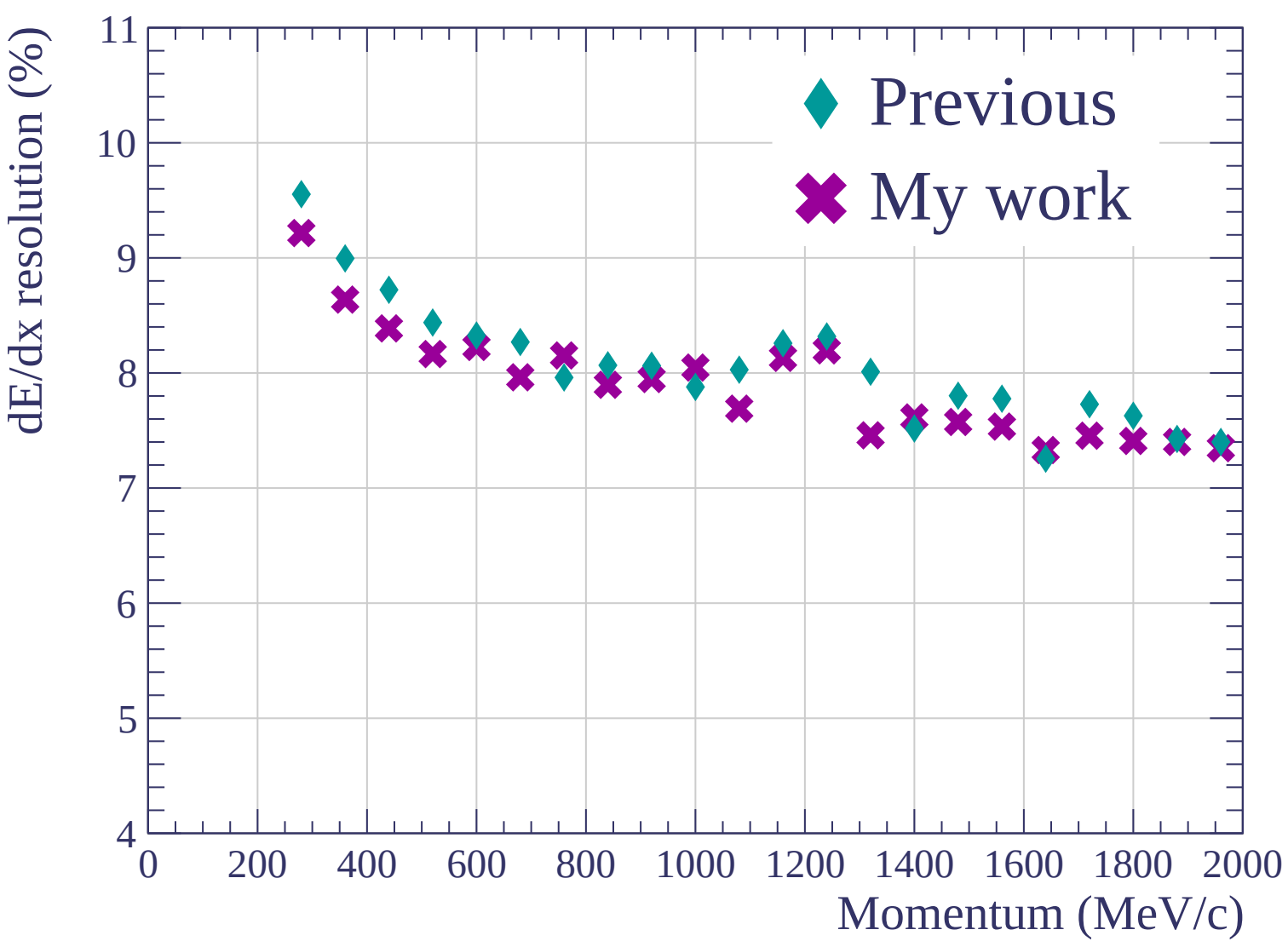


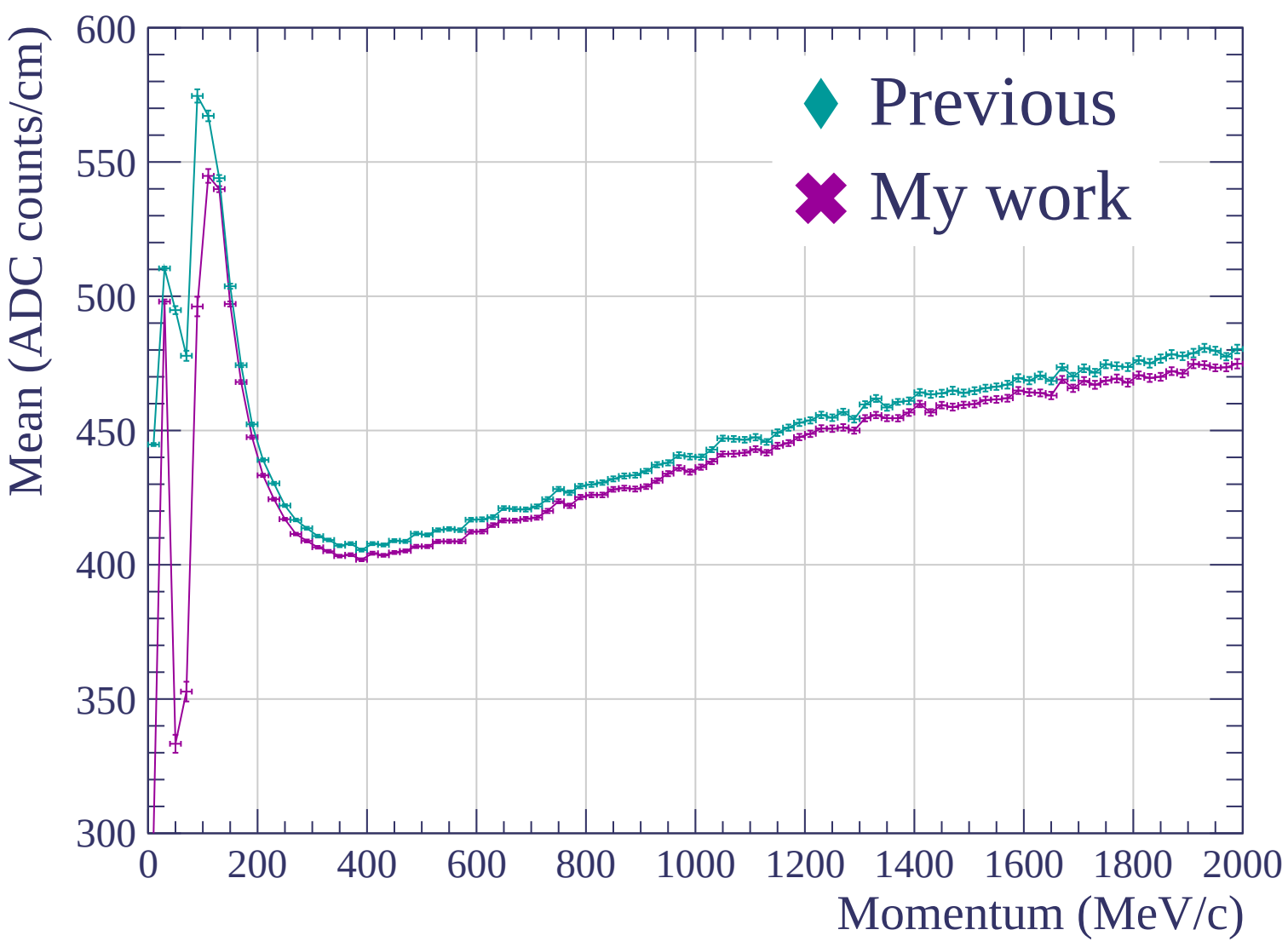


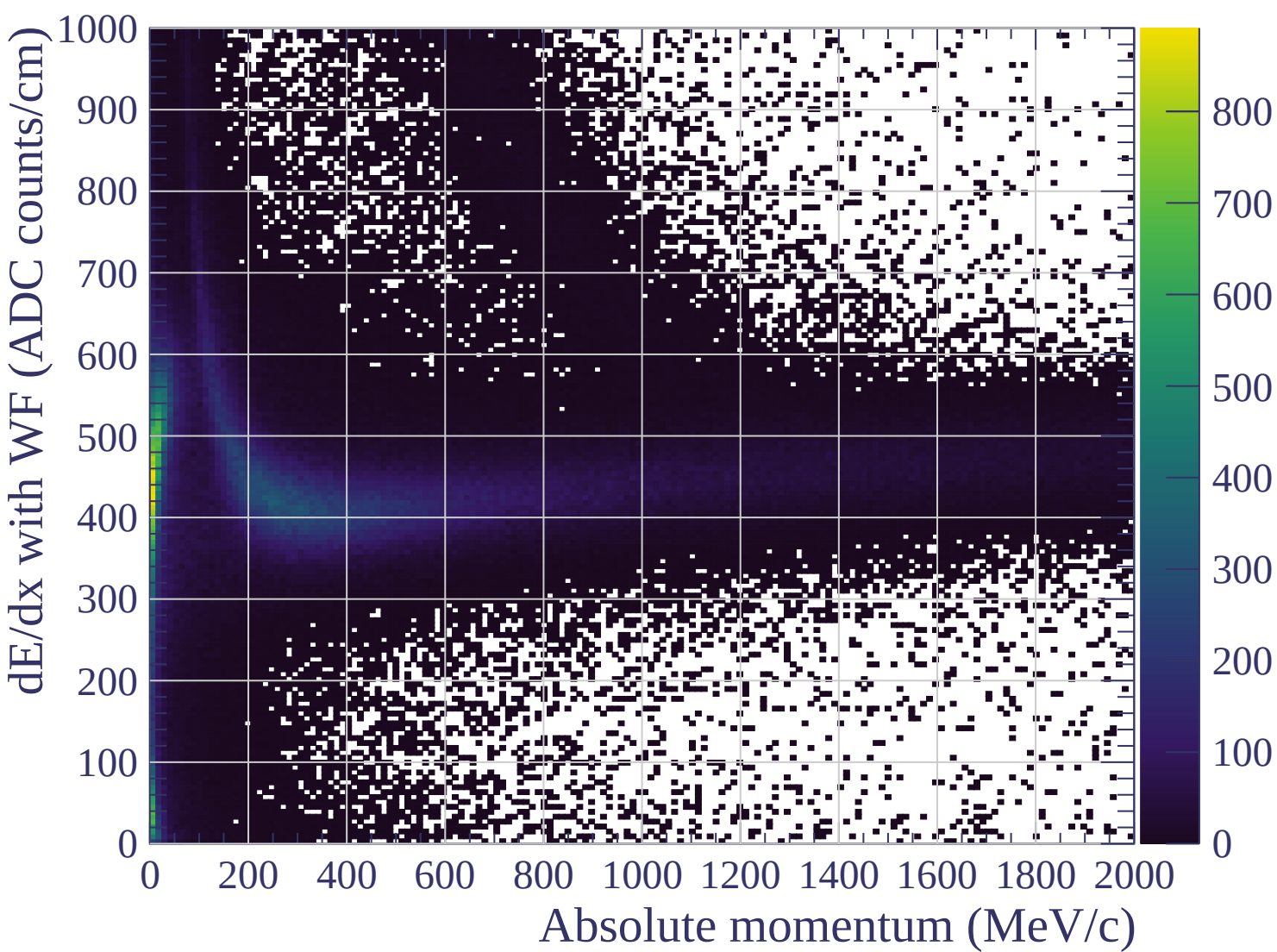


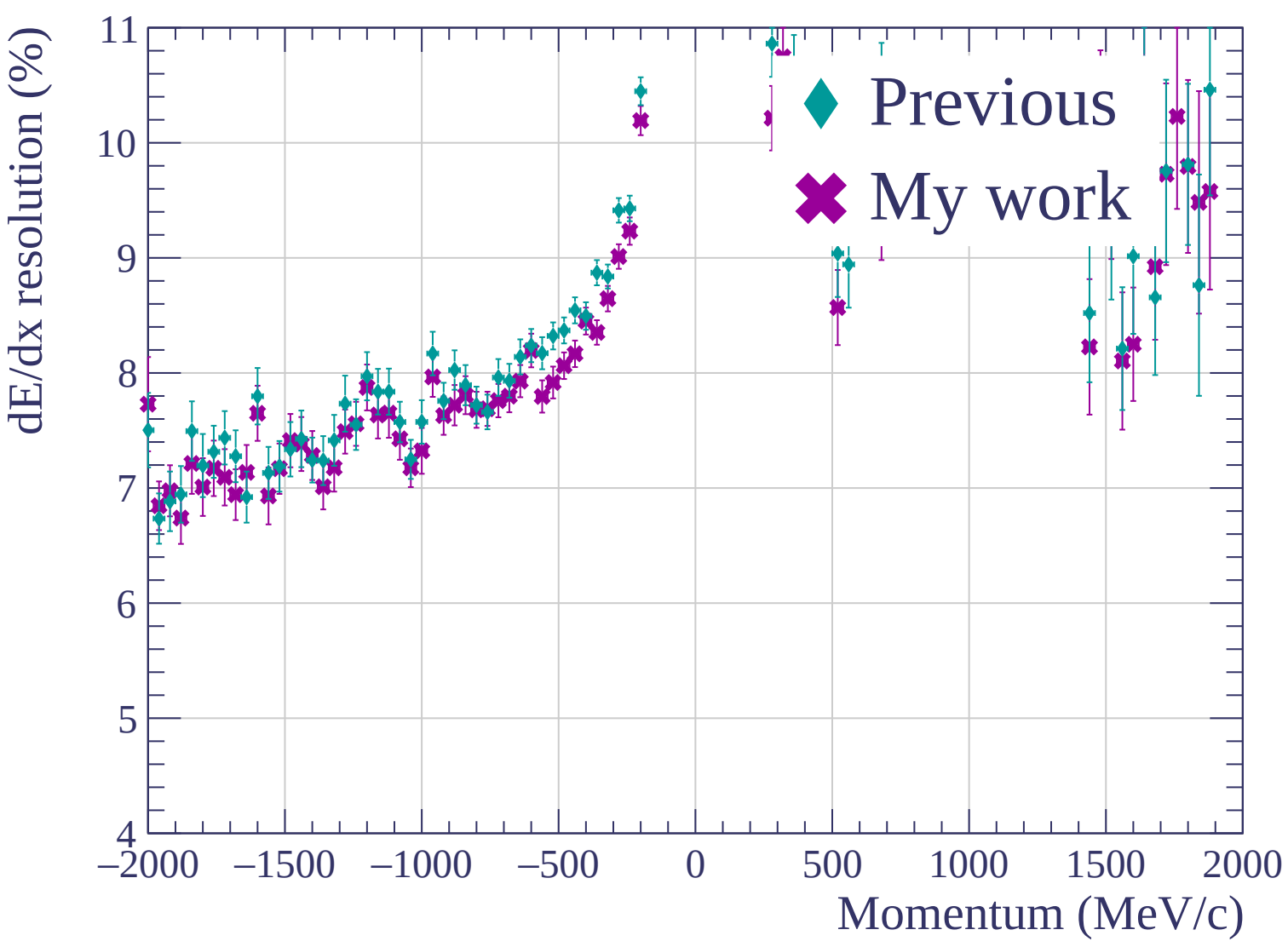


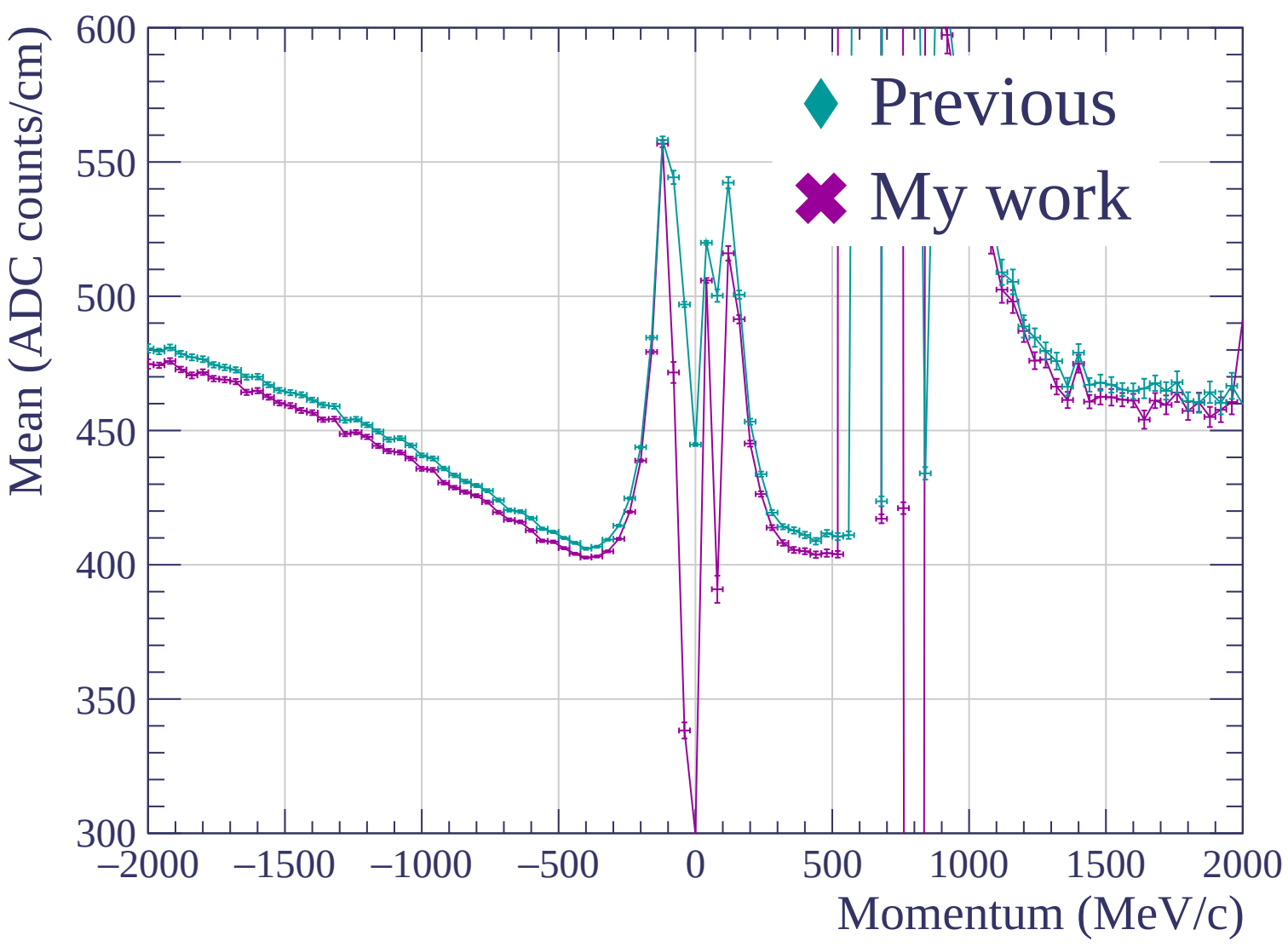


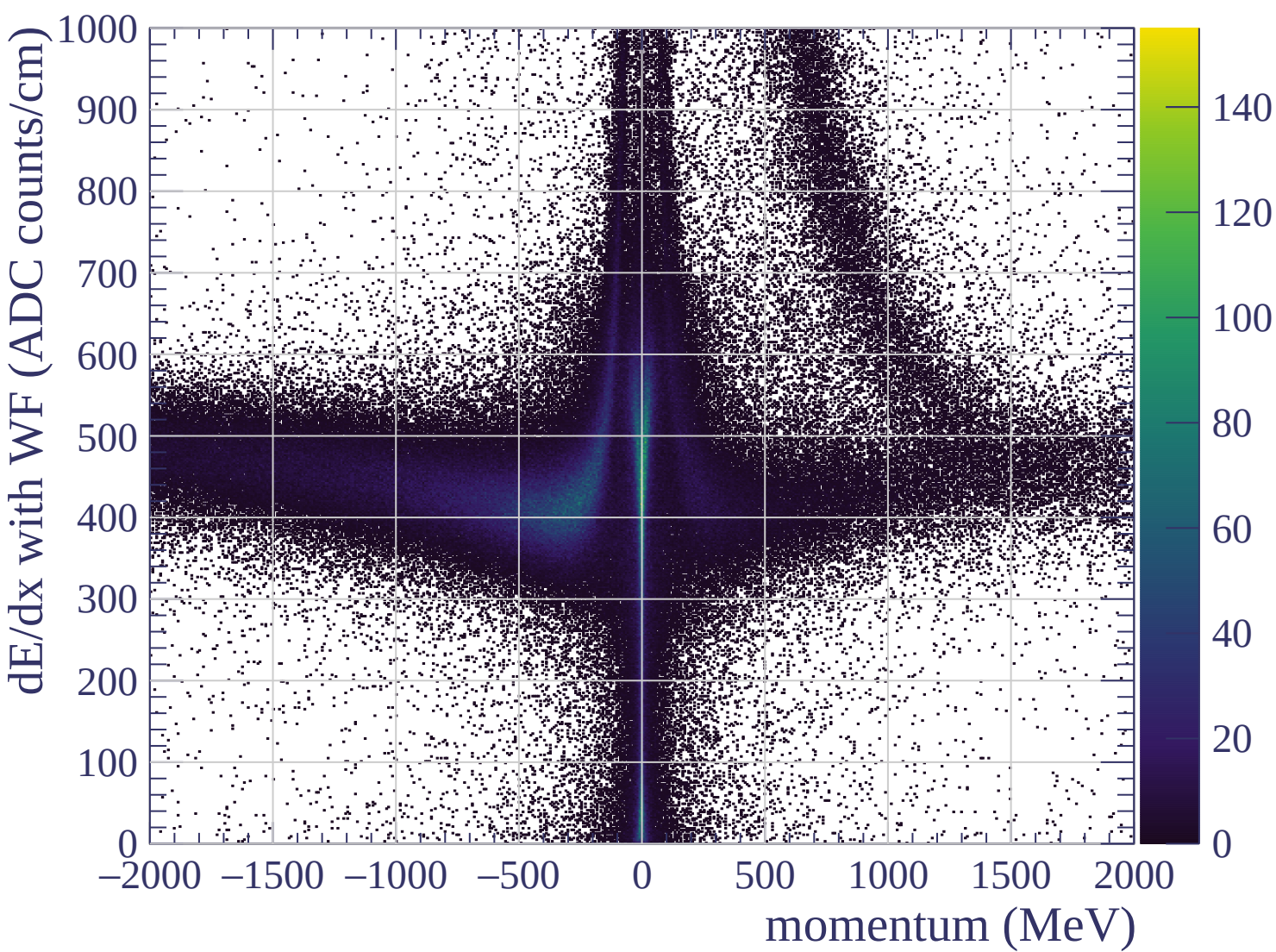


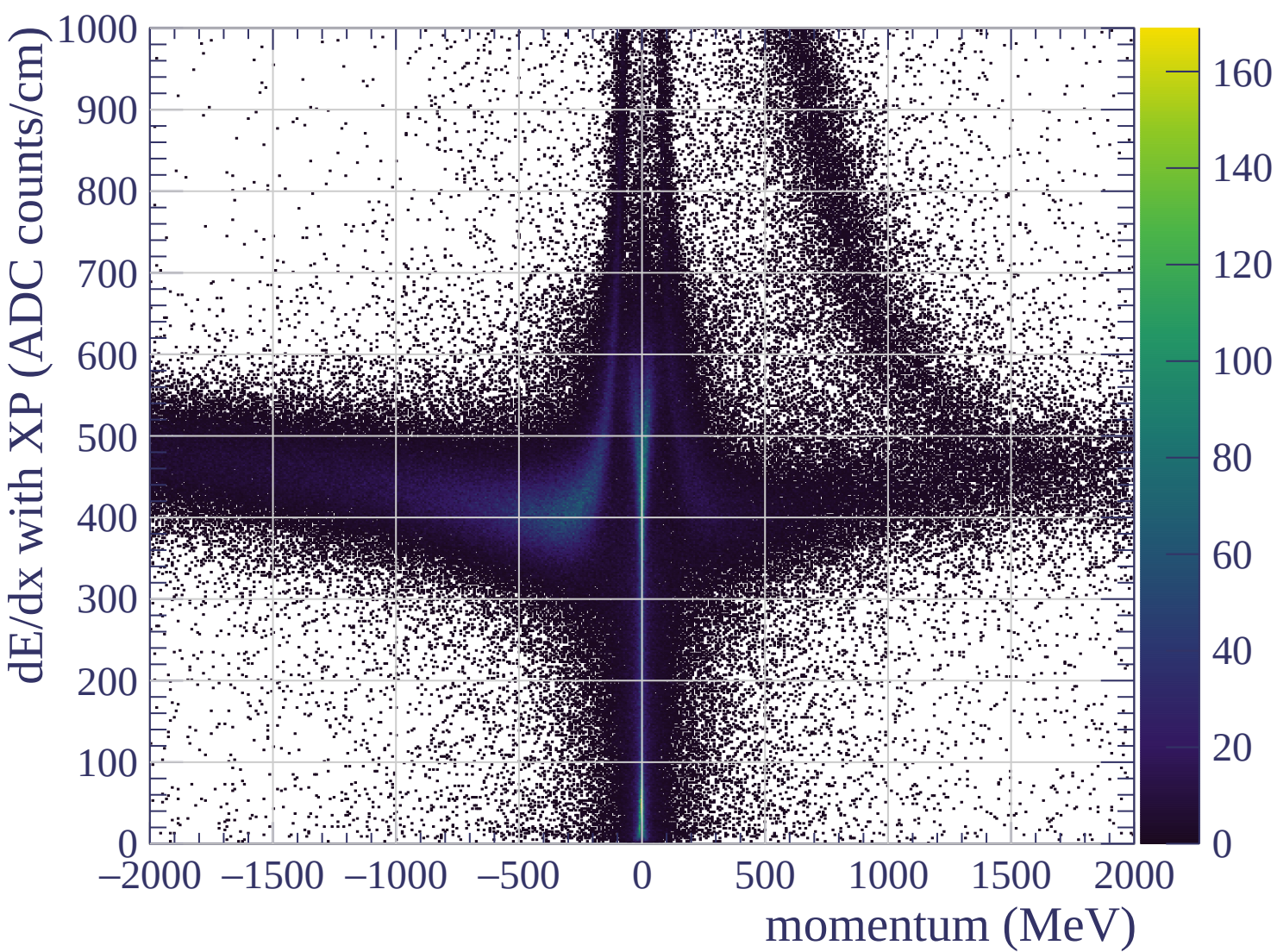


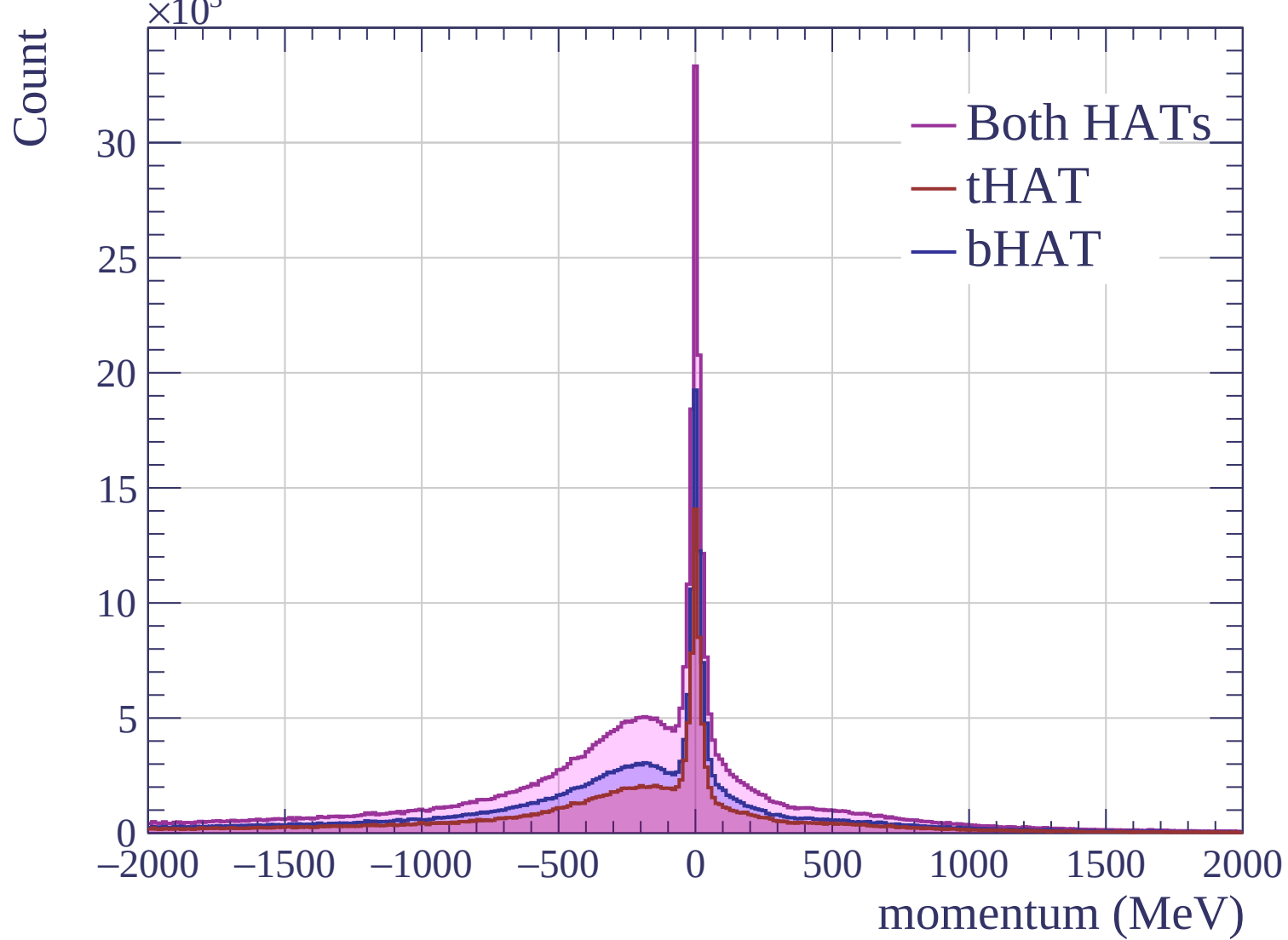


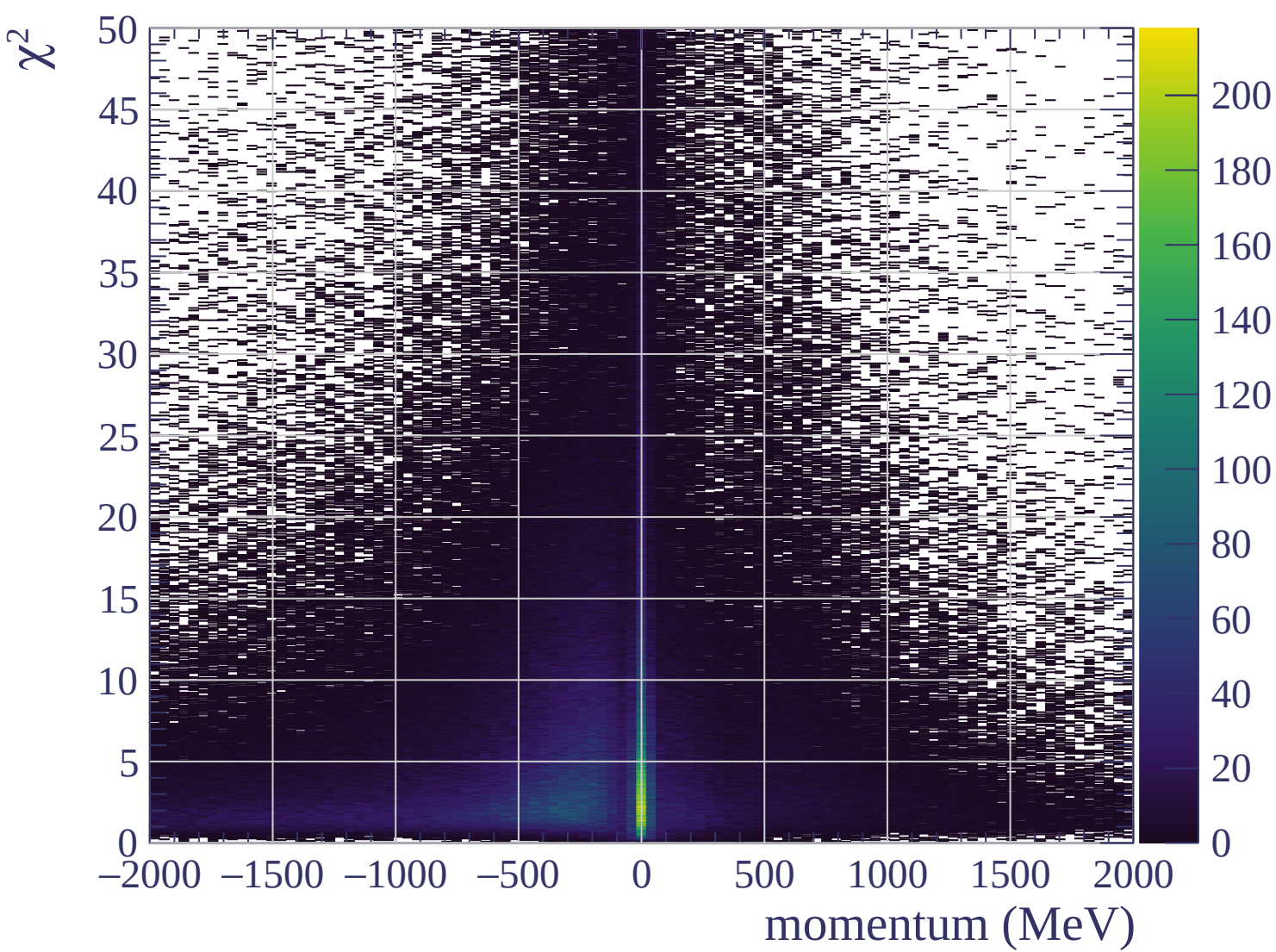




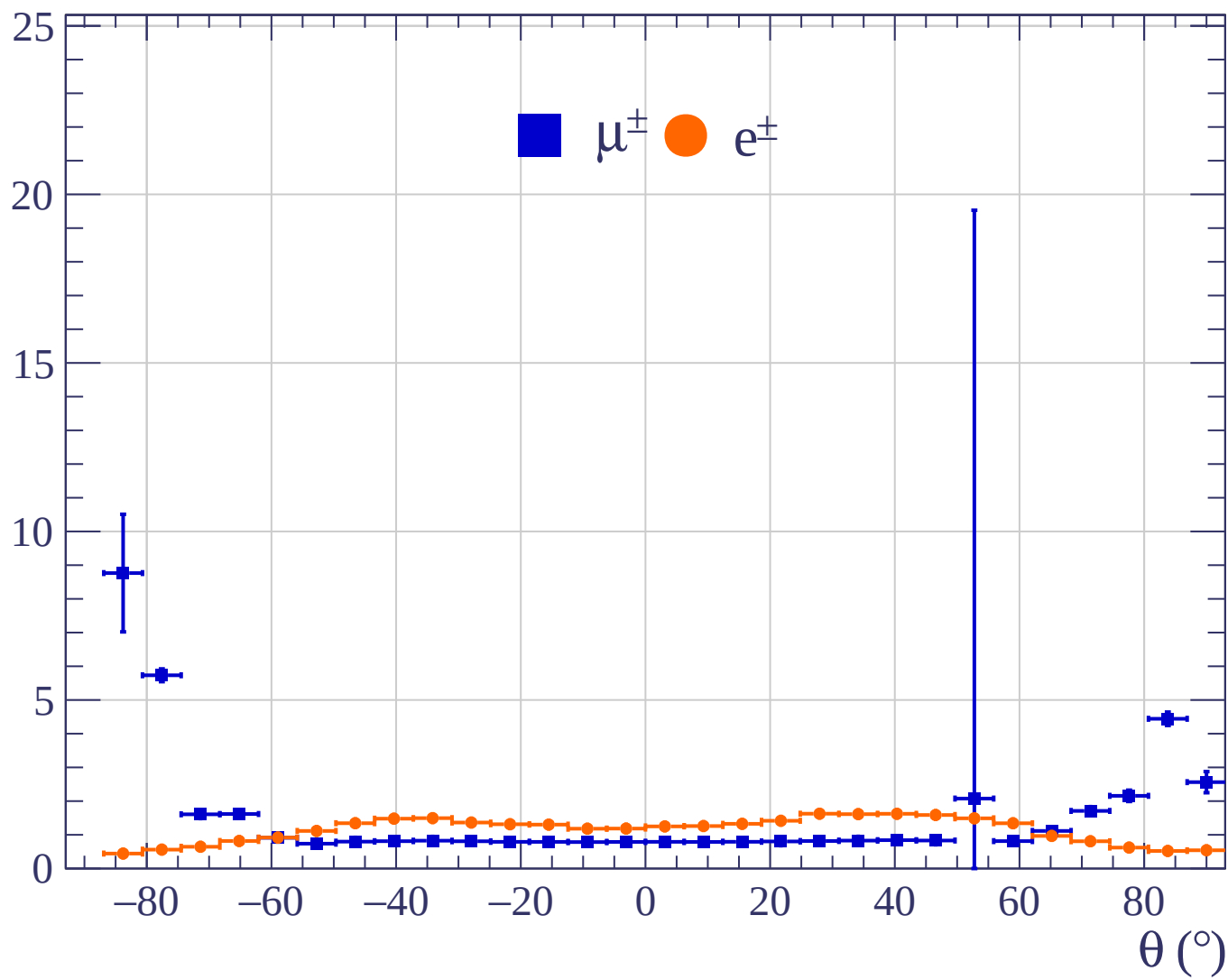




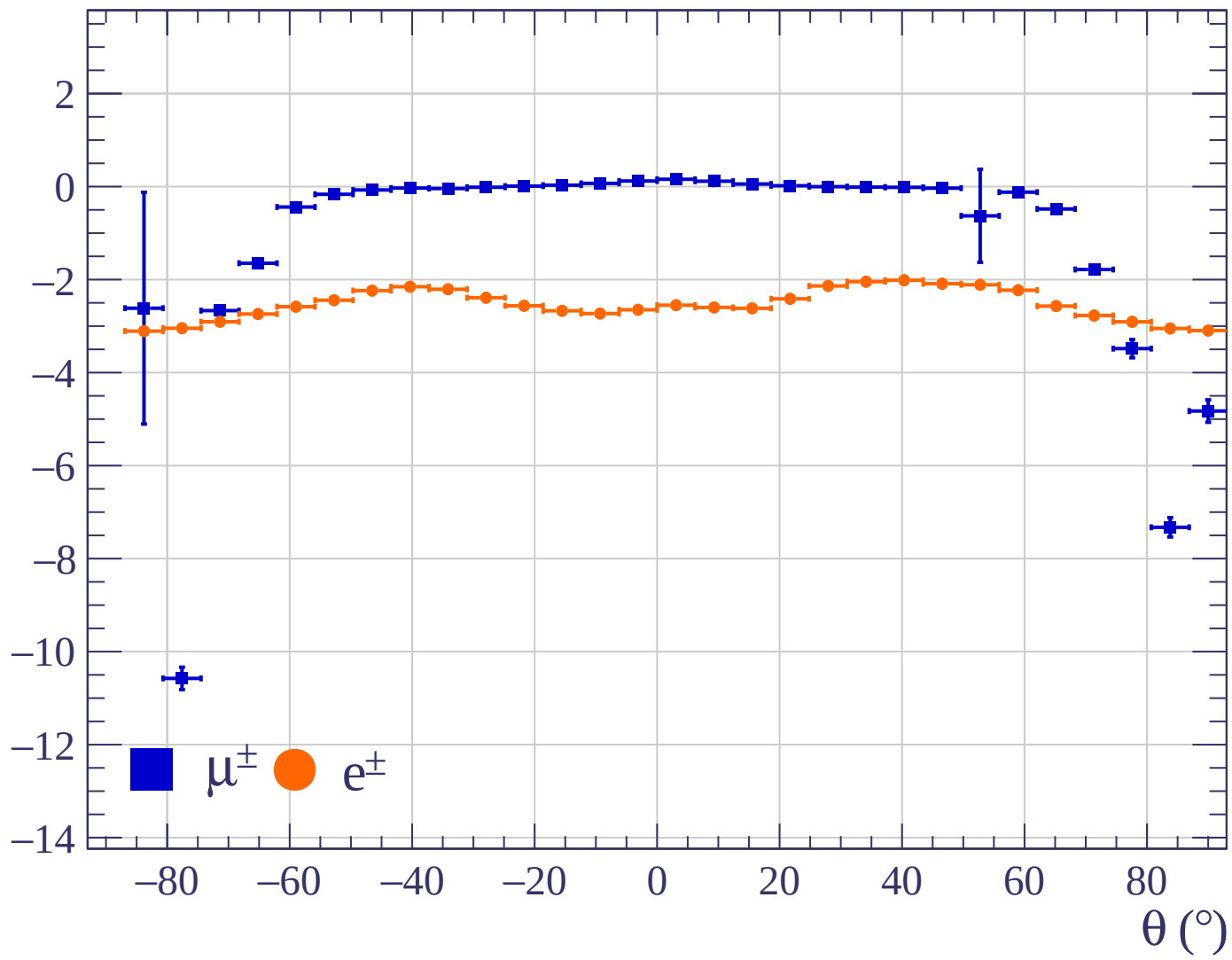




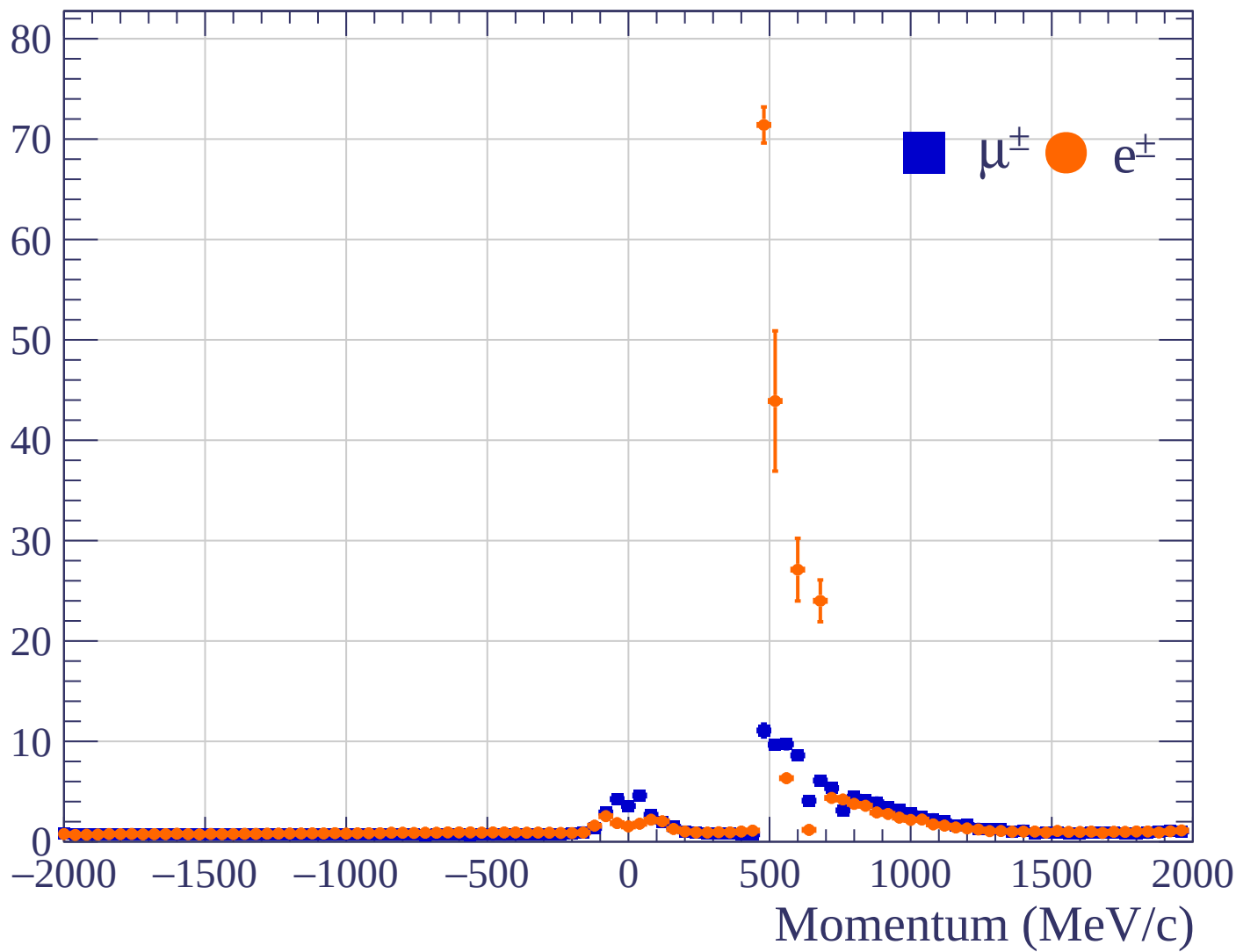
Pulls standard deviation

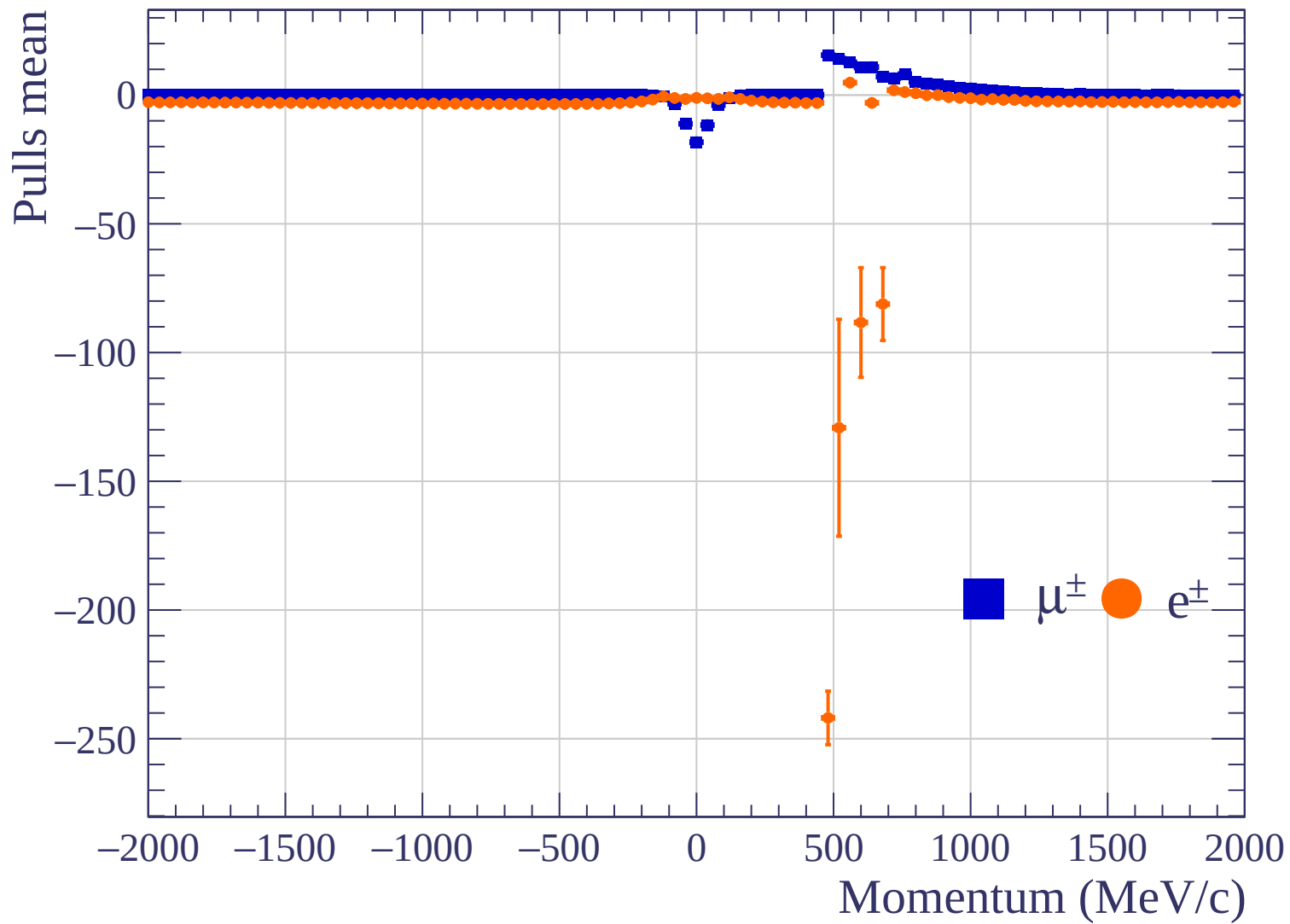


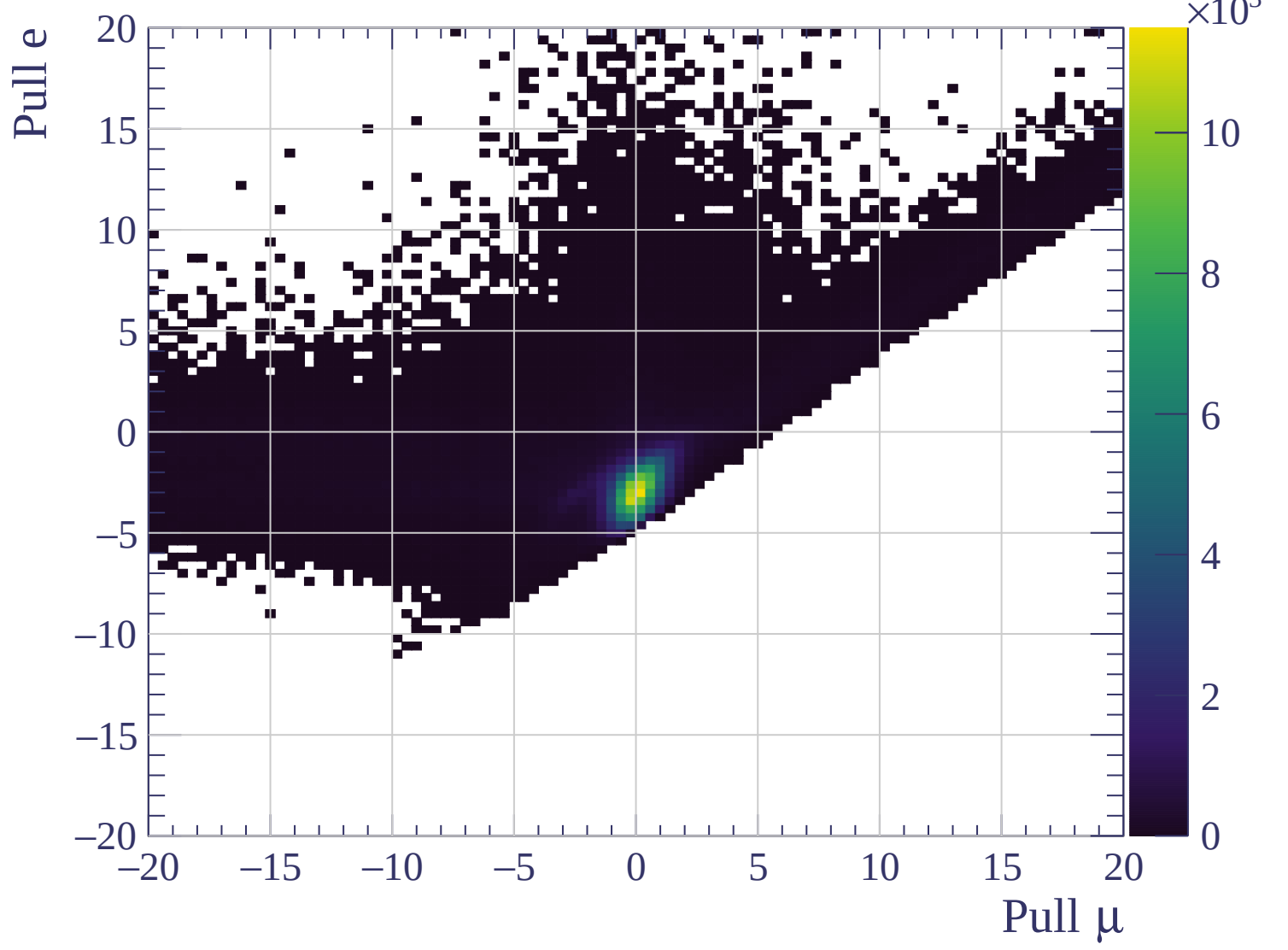
Pulls mean

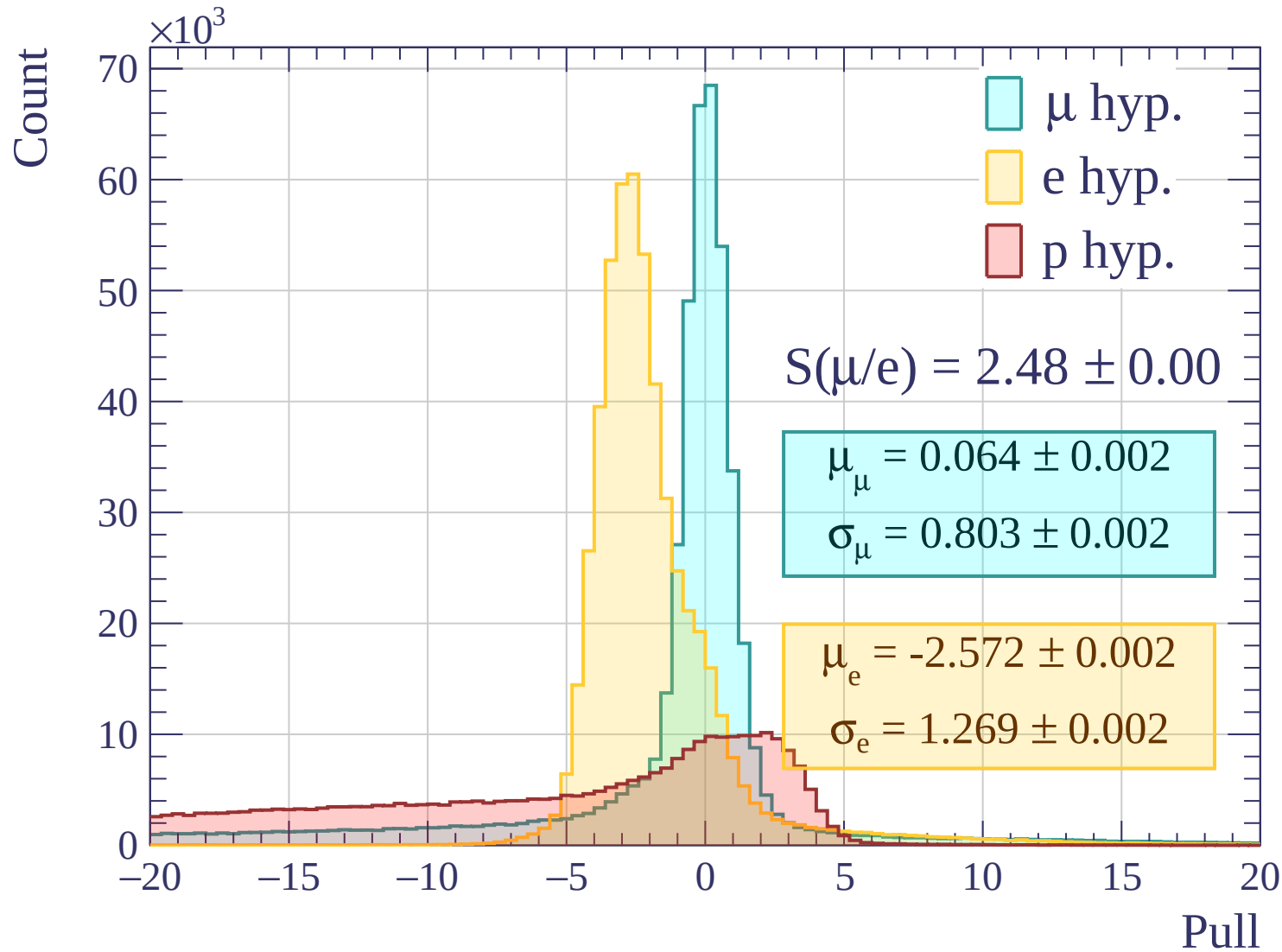


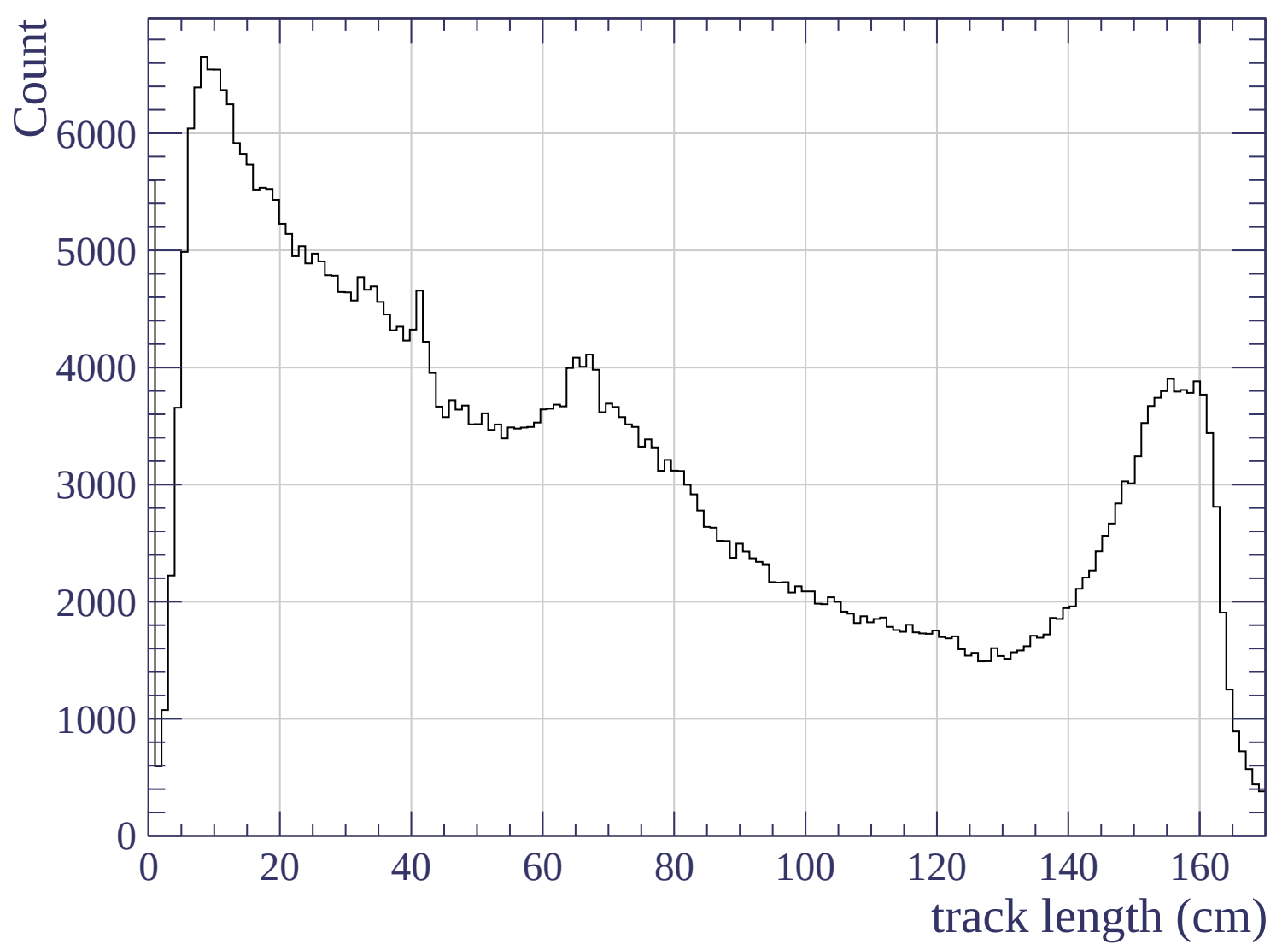
Pulls standard deviation

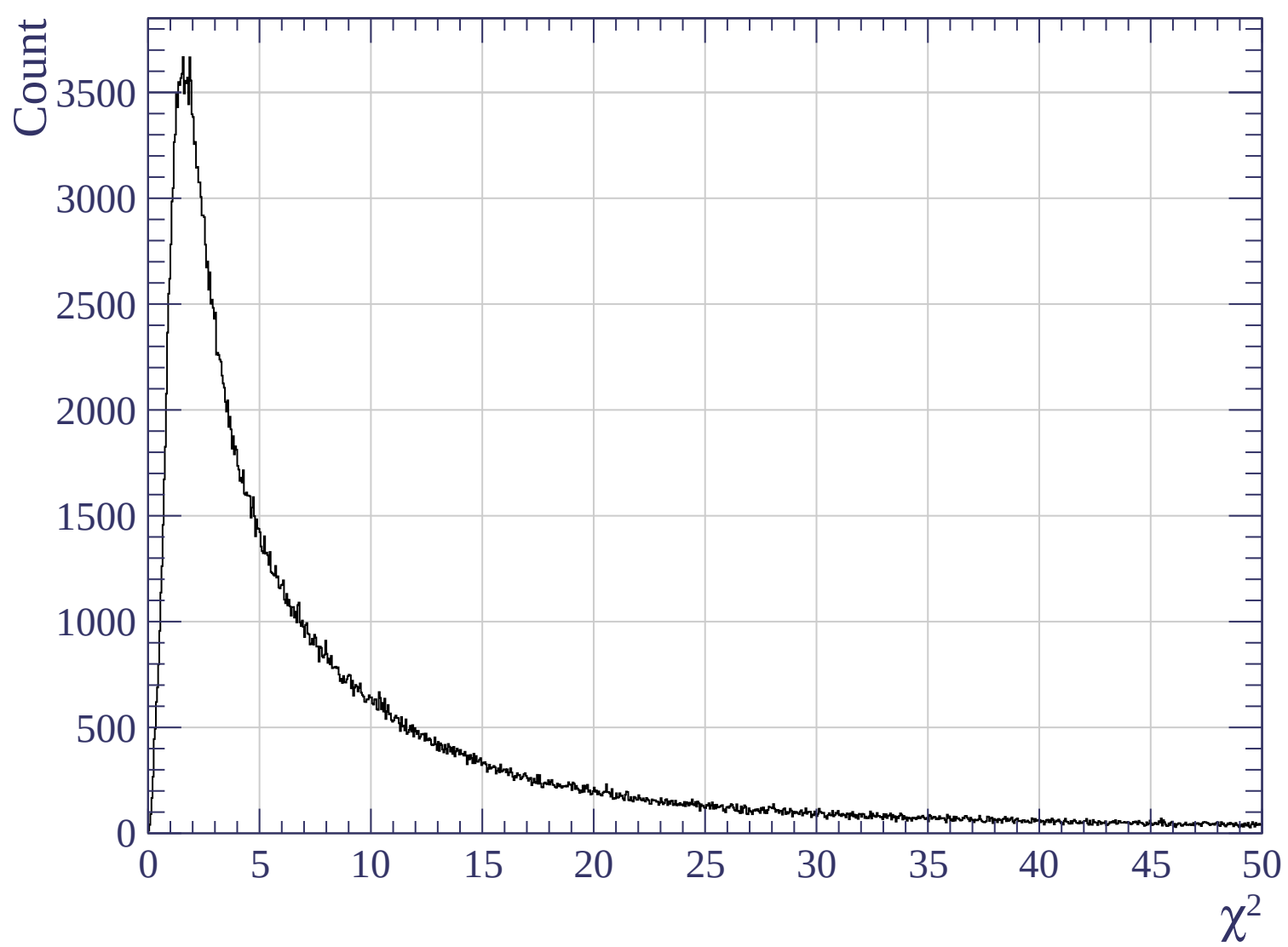


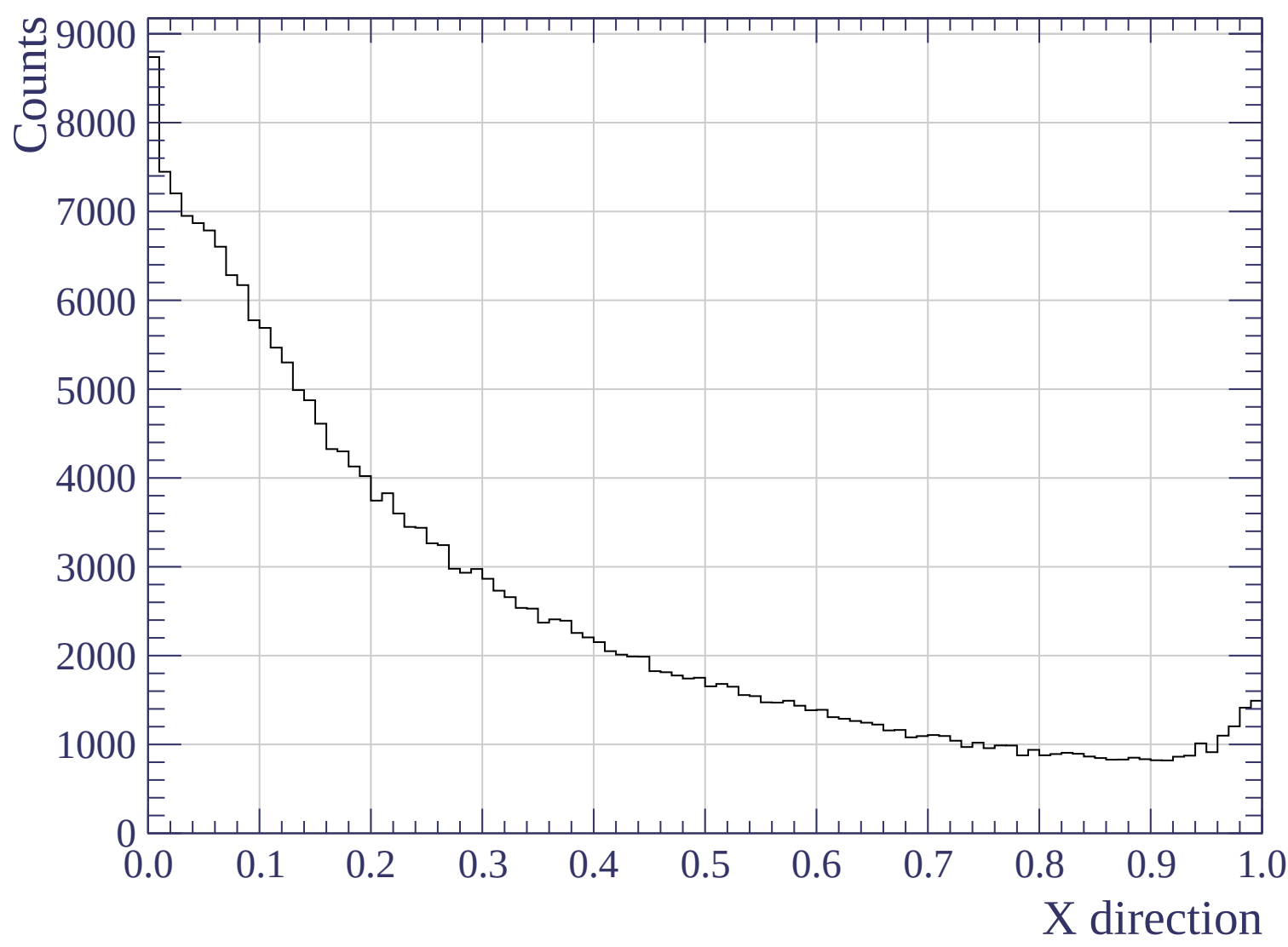


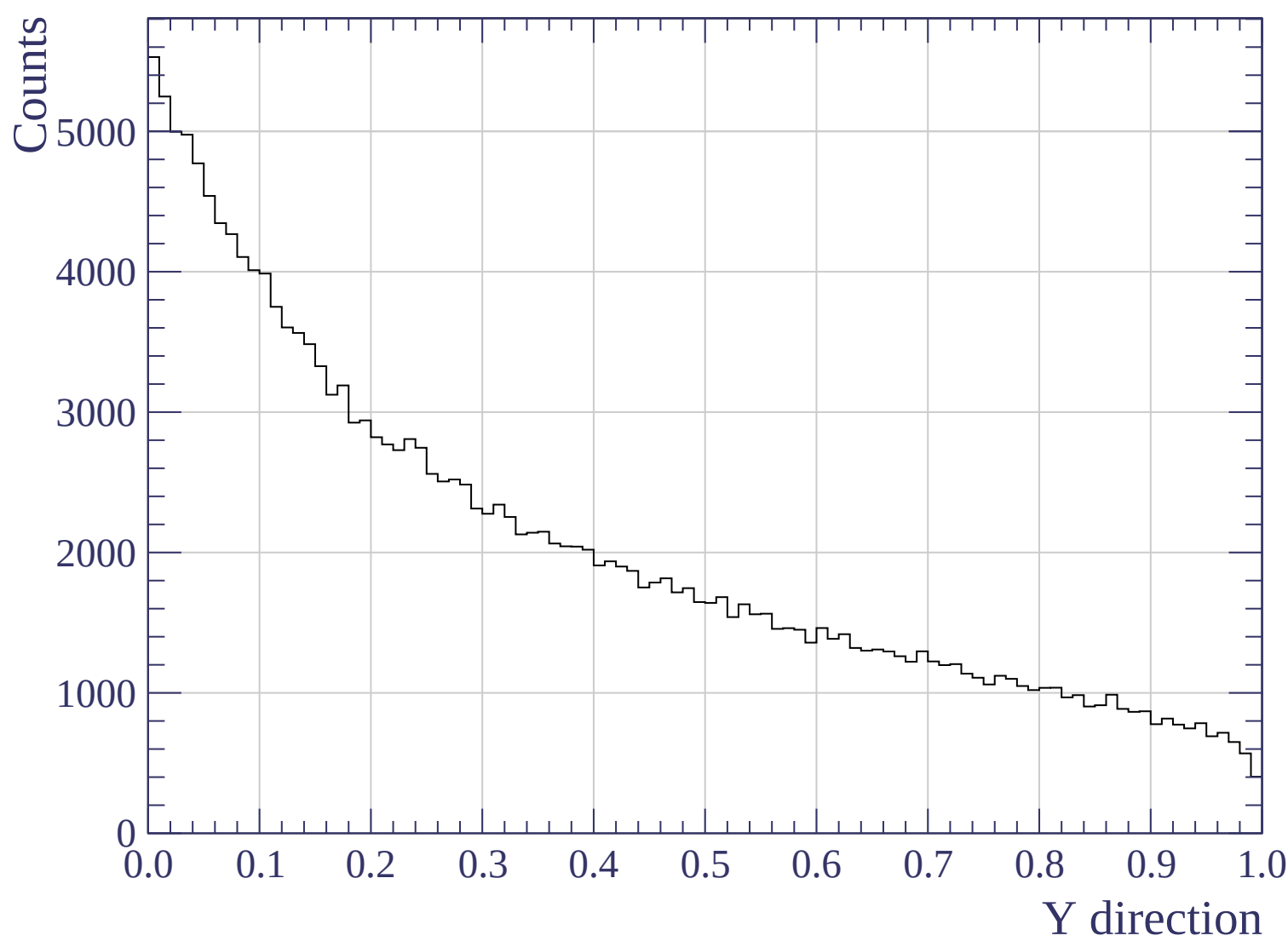


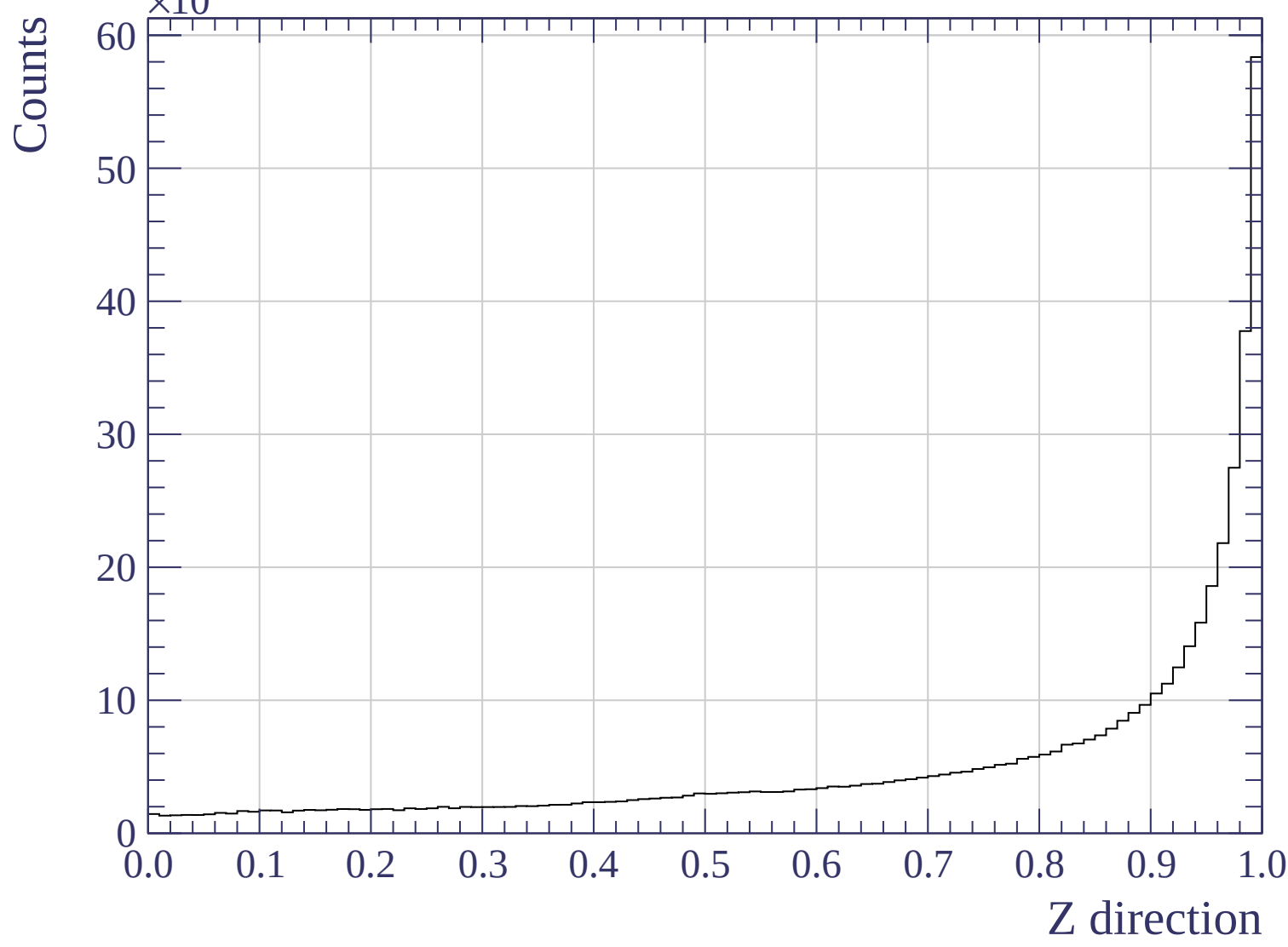


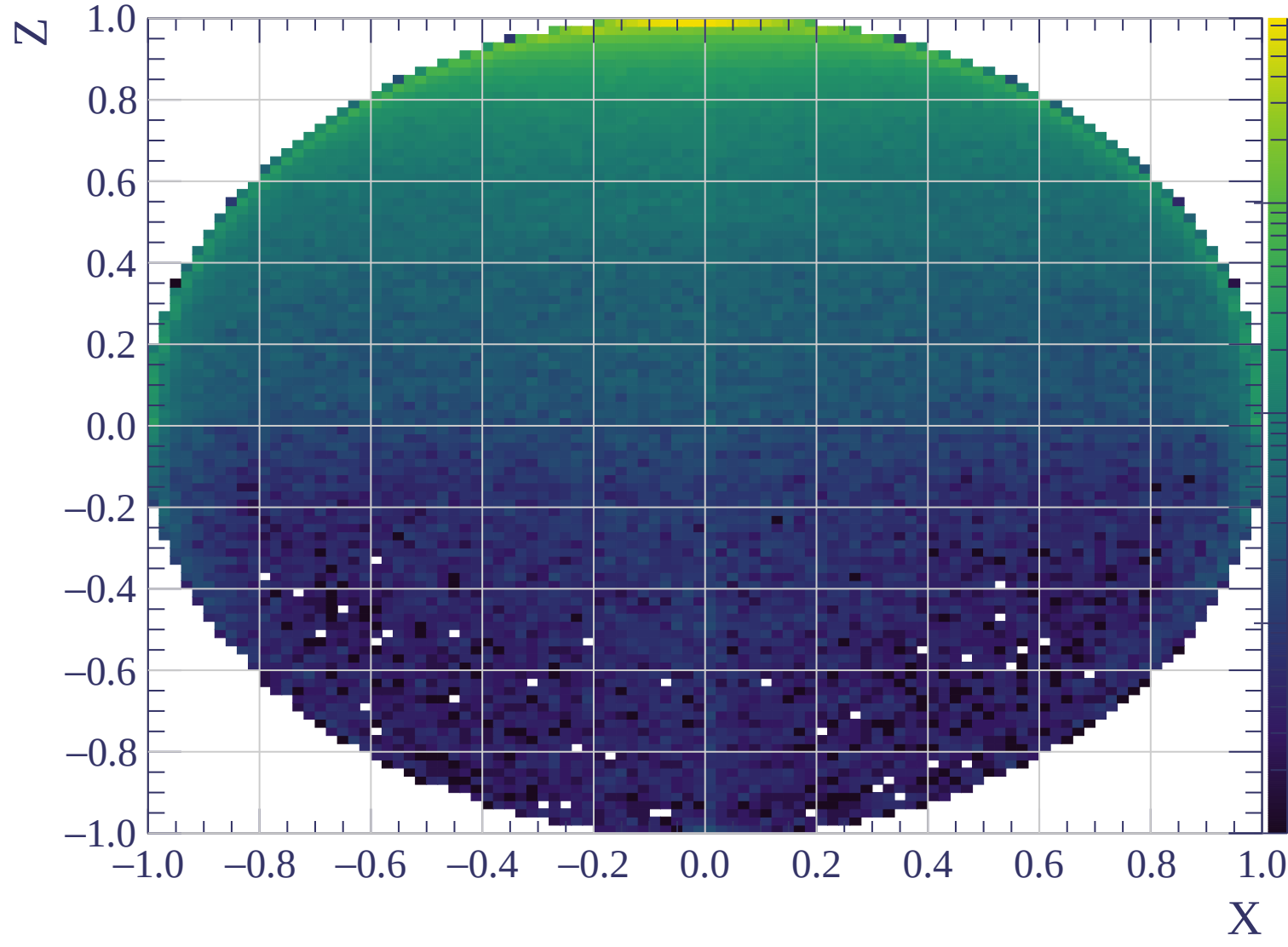


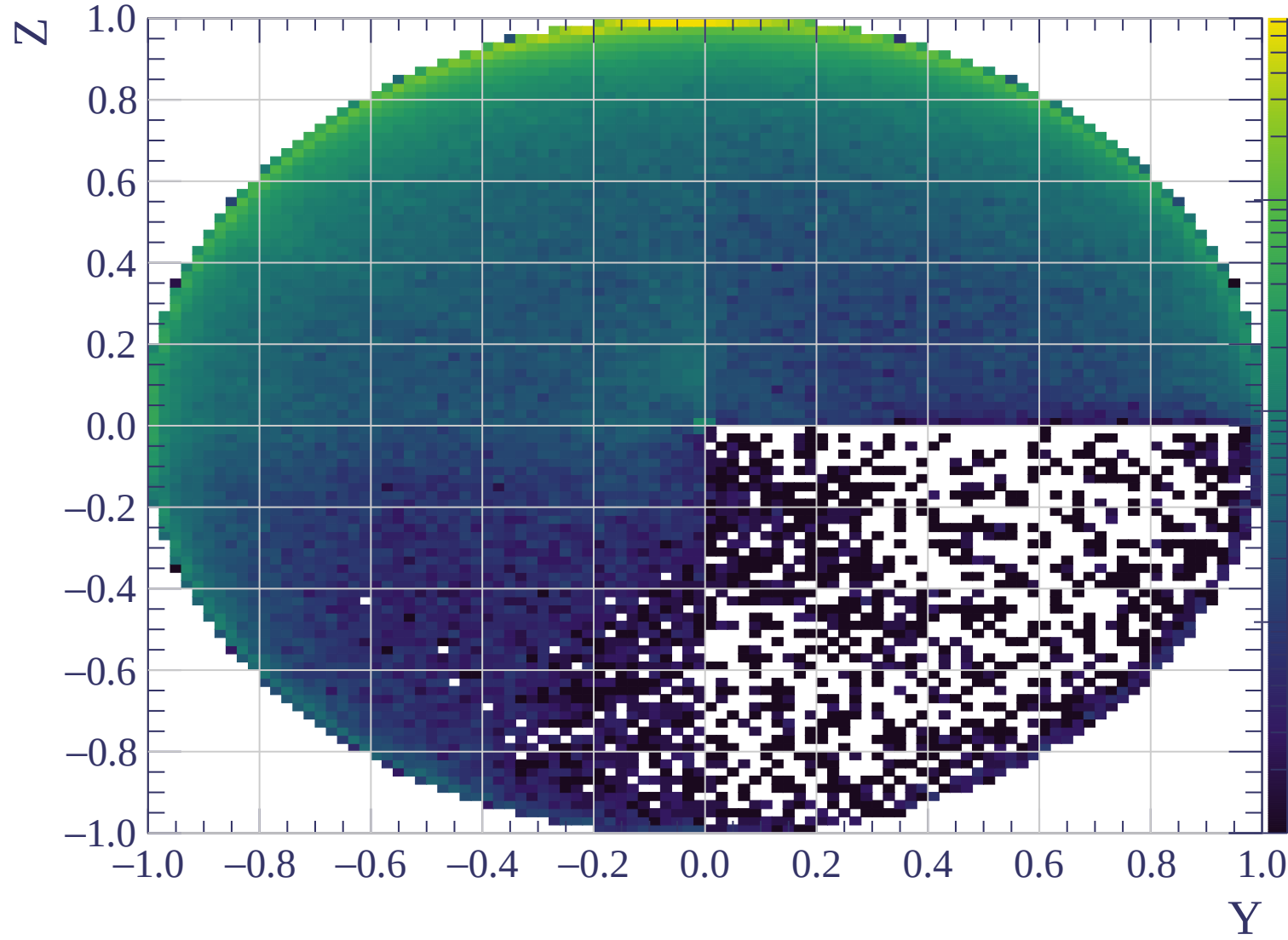


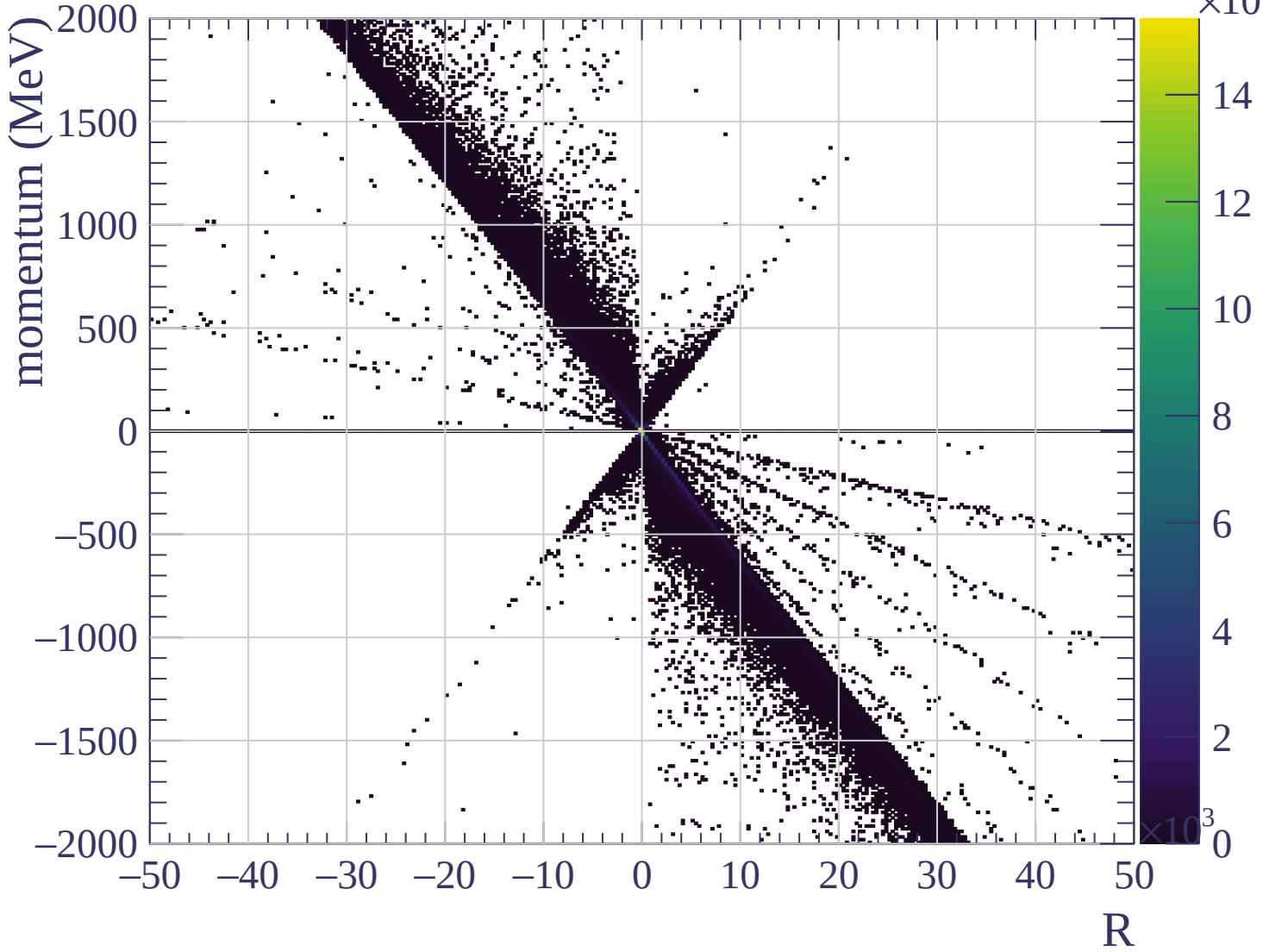


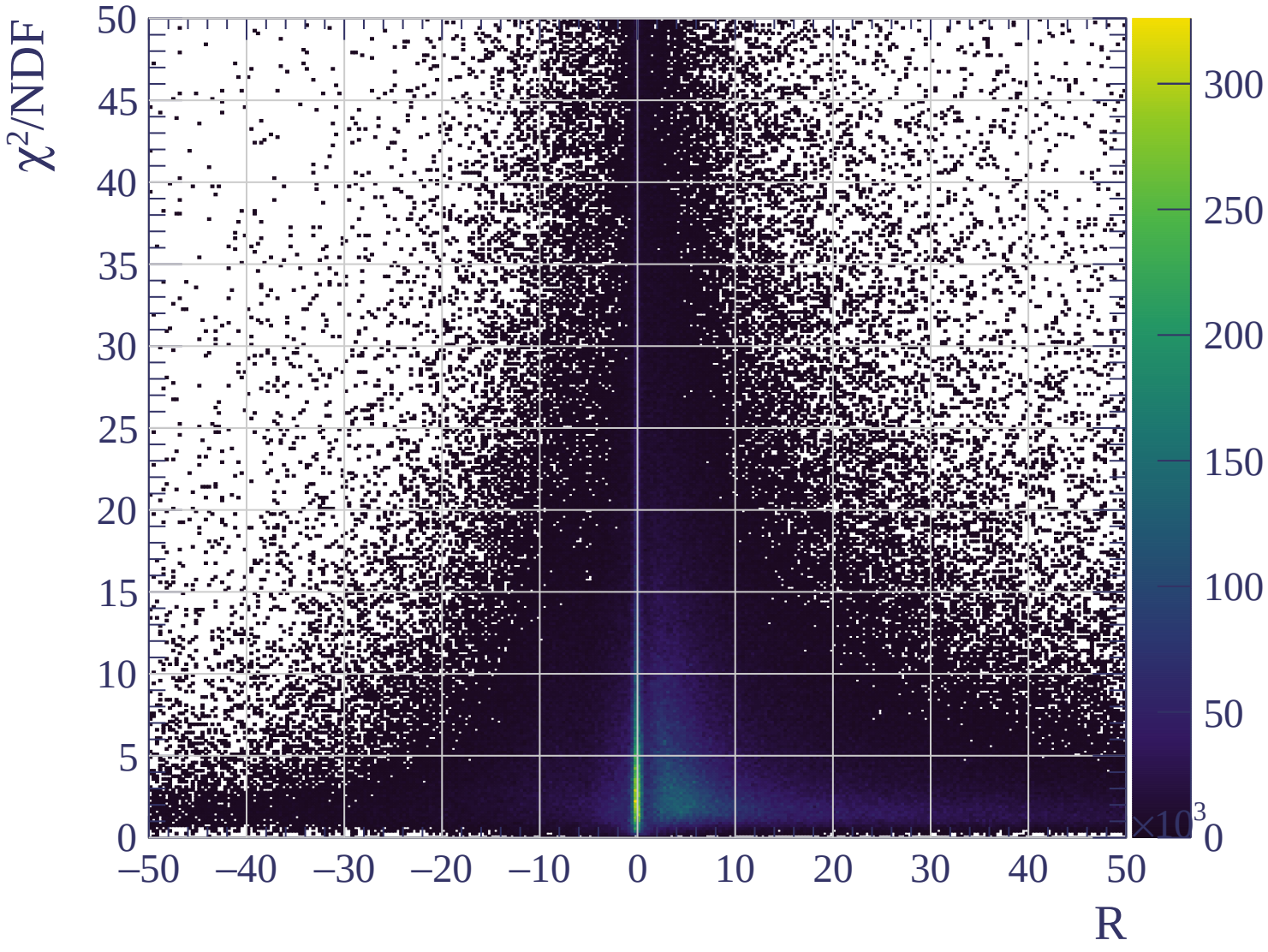






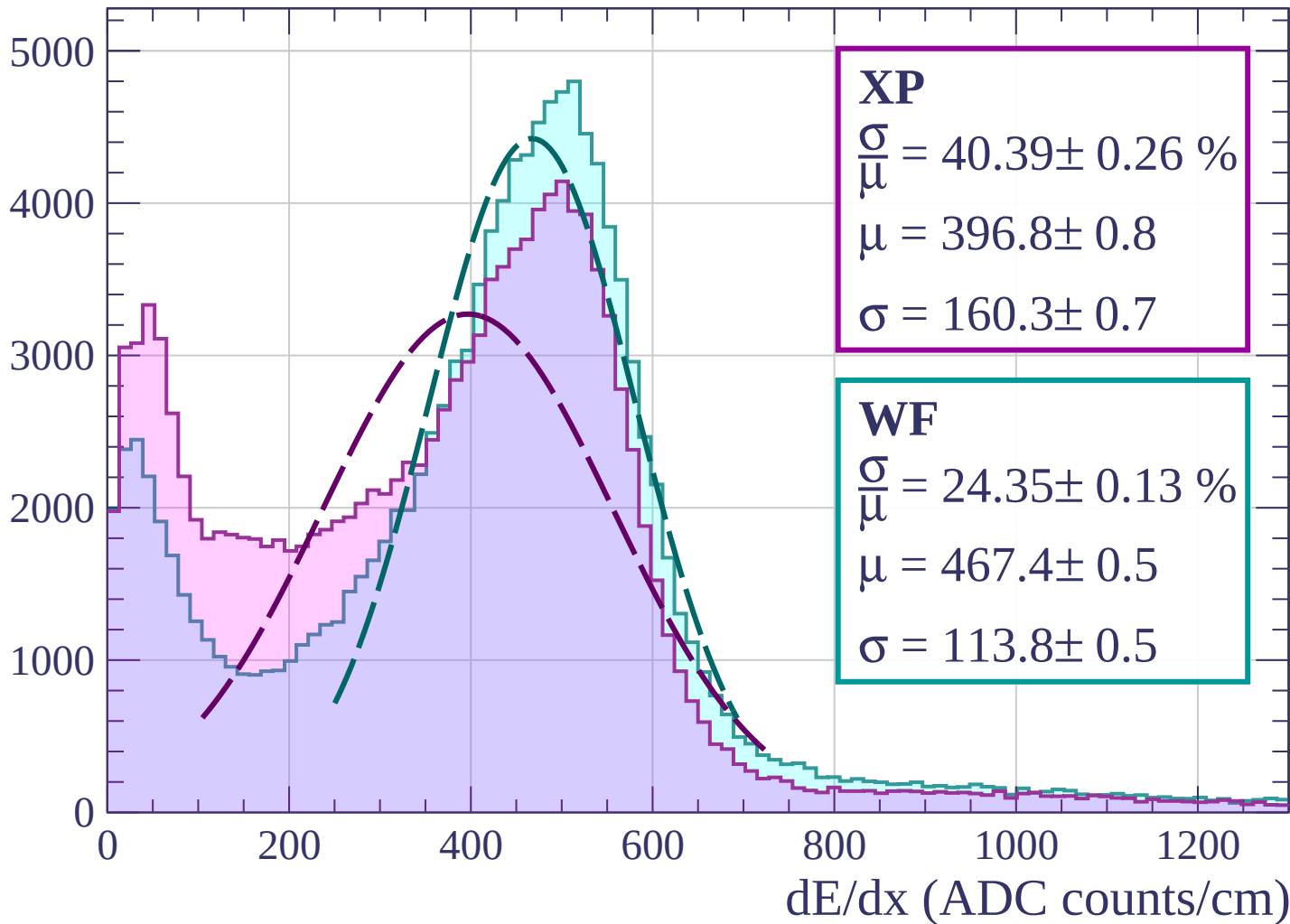




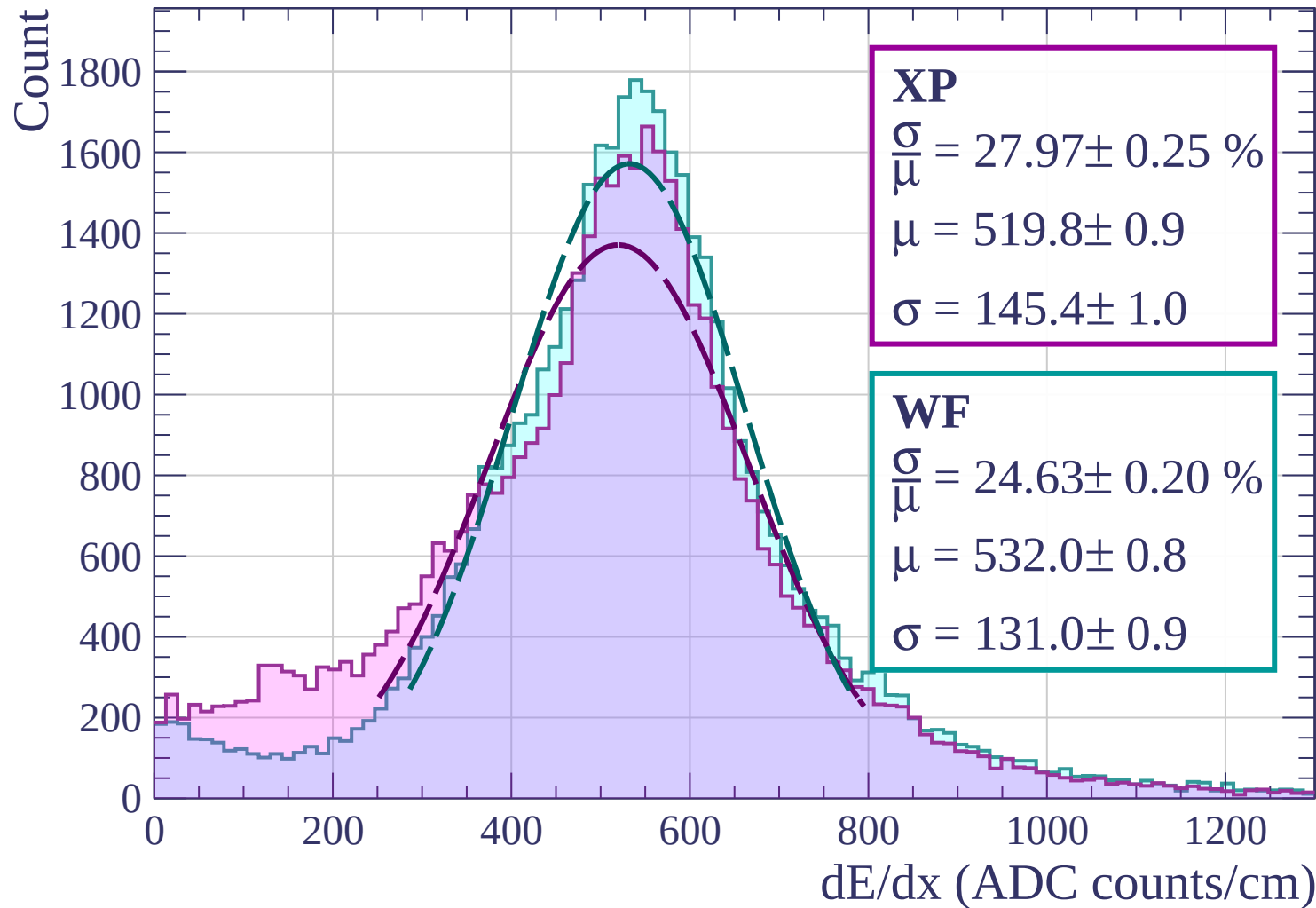


Energy loss | $0 < p < 80$

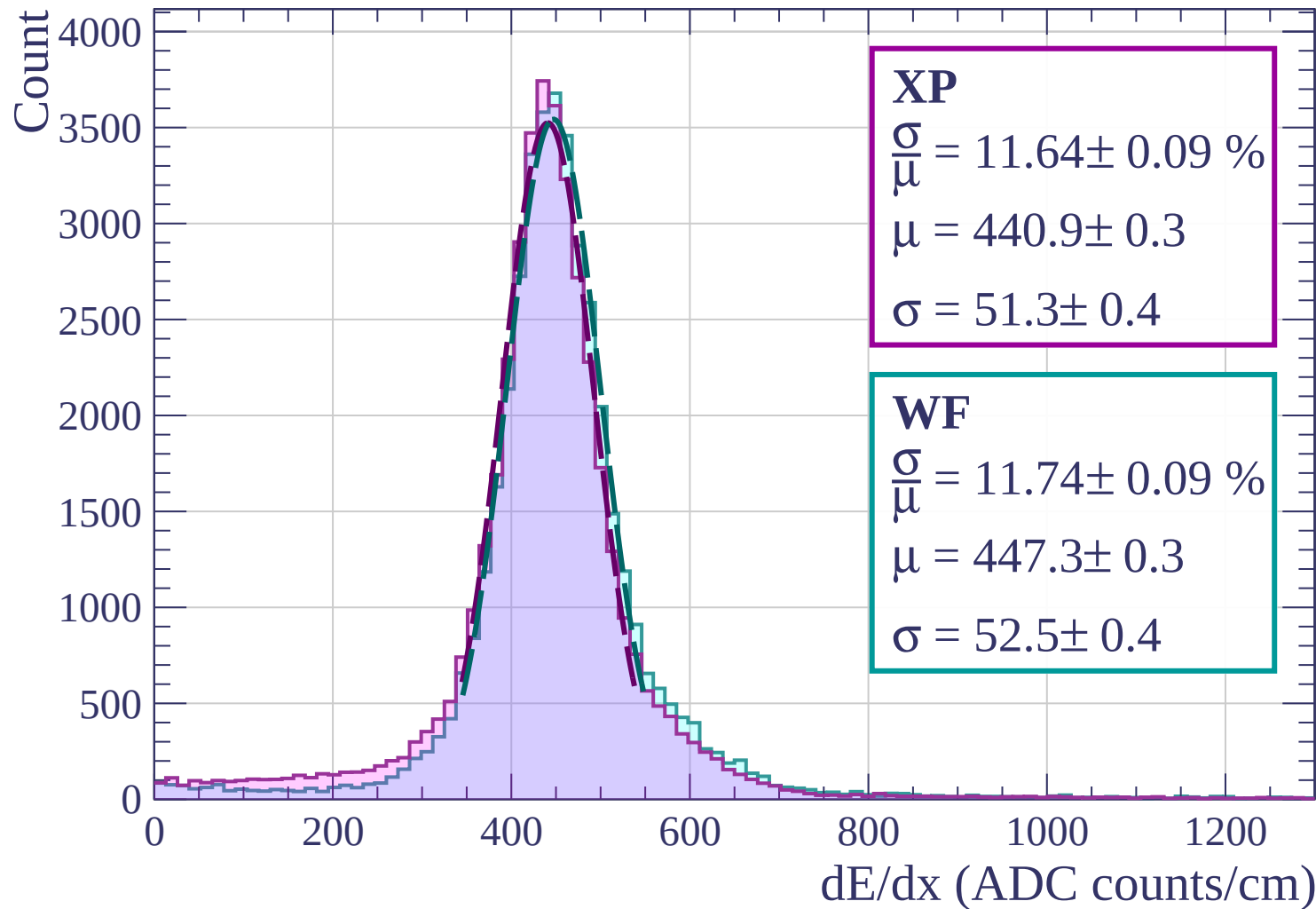
Count



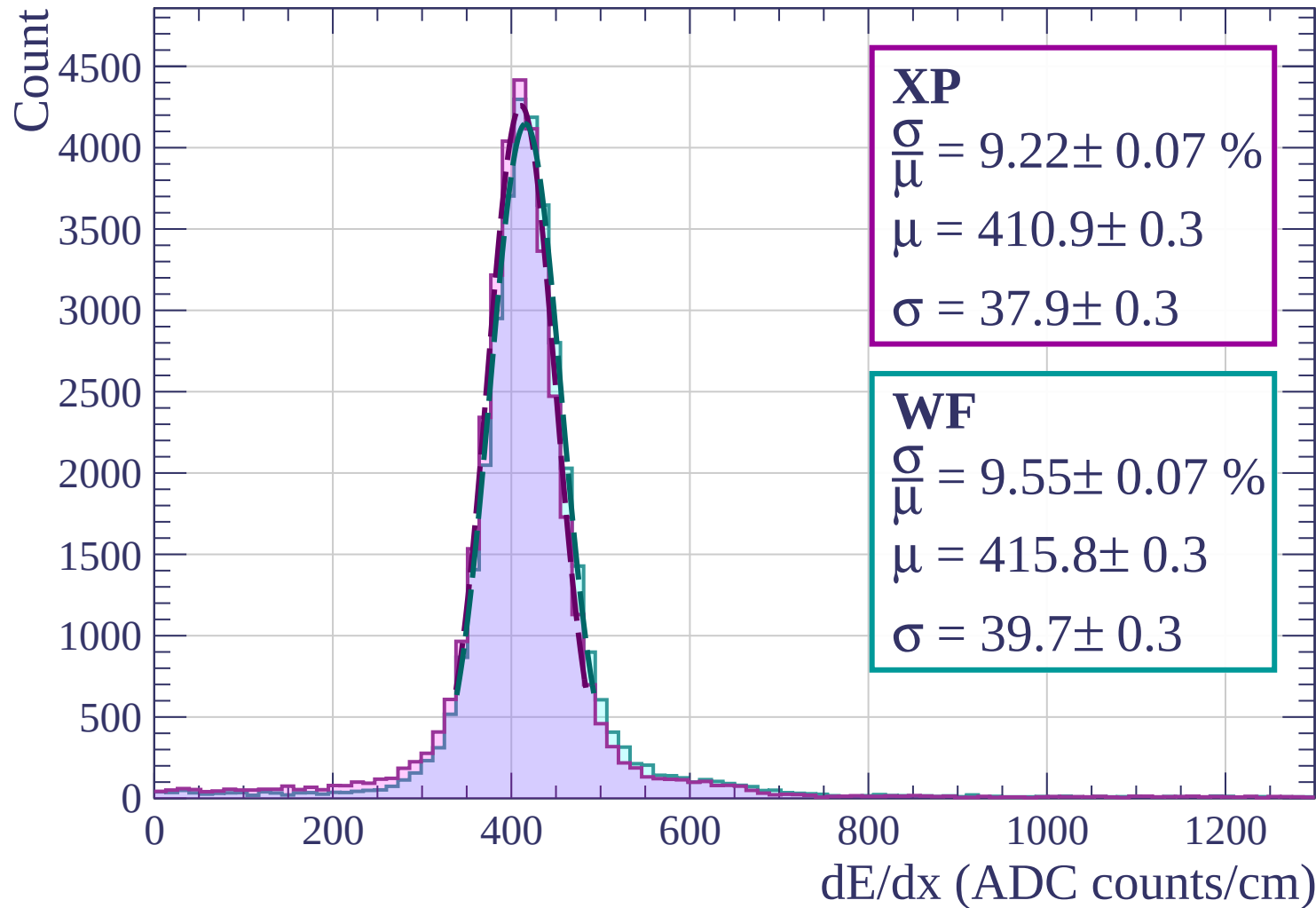
Energy loss | $80 < p < 160$



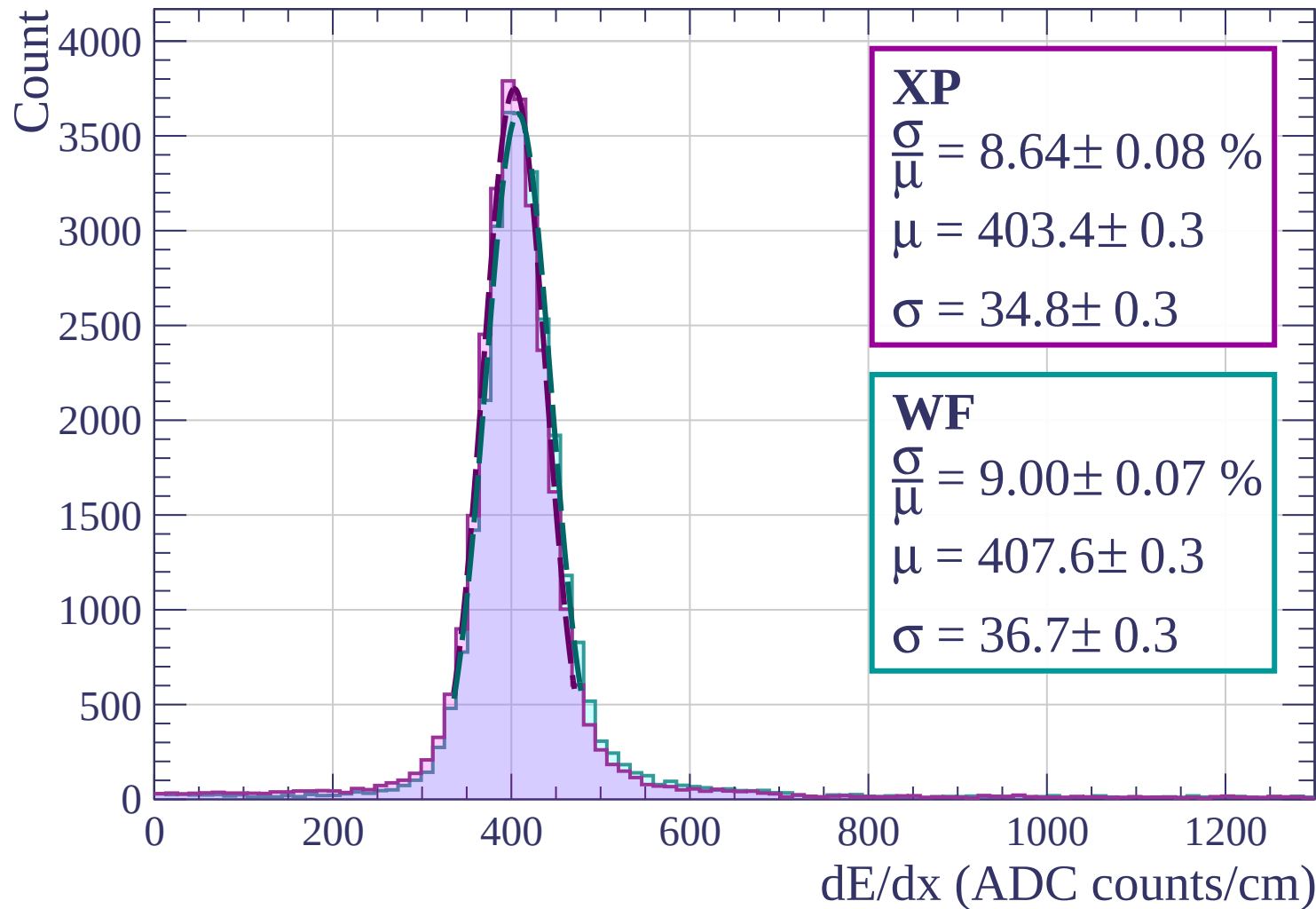
Energy loss | $160 < p < 240$



Energy loss | $240 < p < 320$

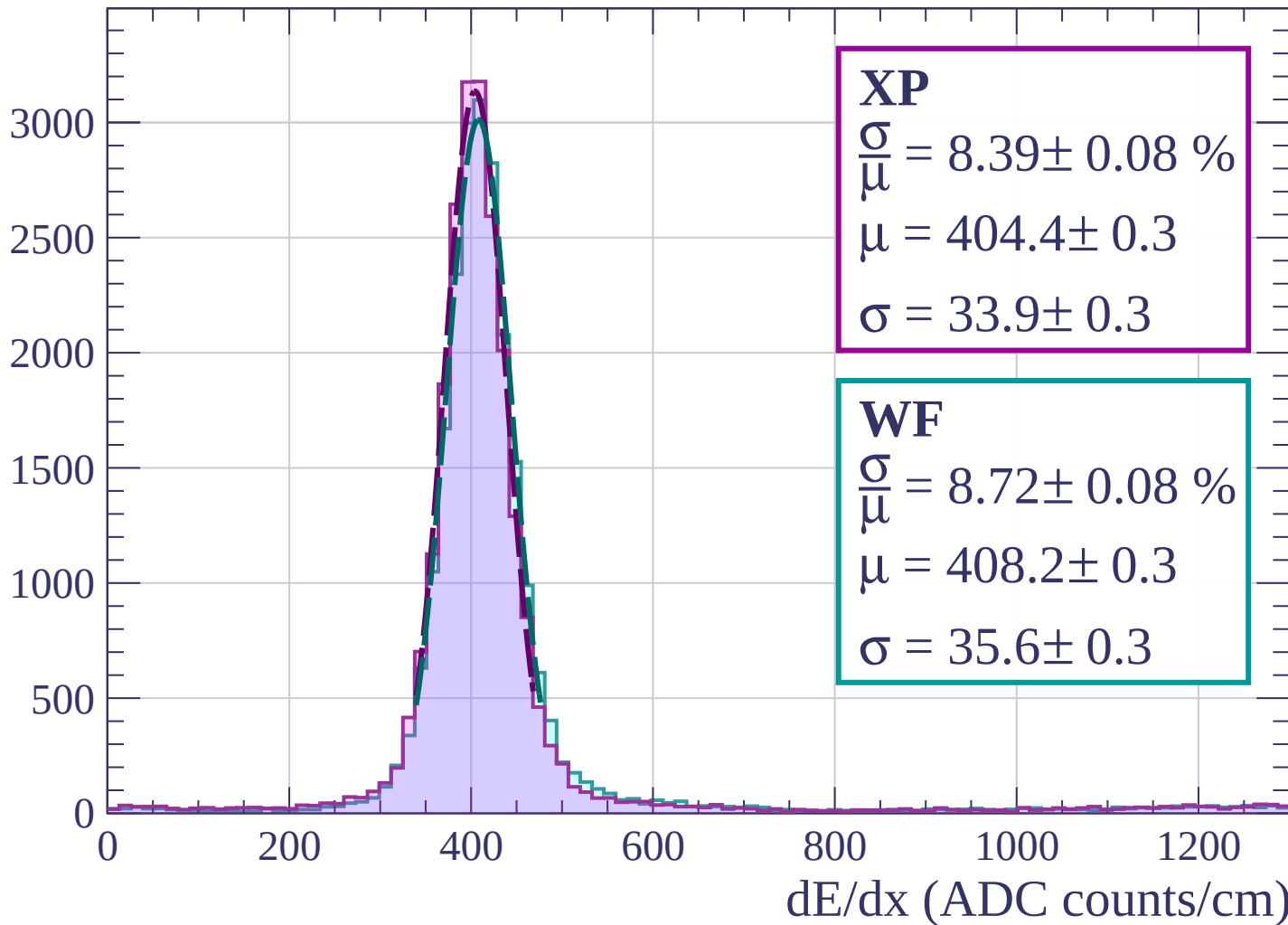


Energy loss | $320 < p < 400$

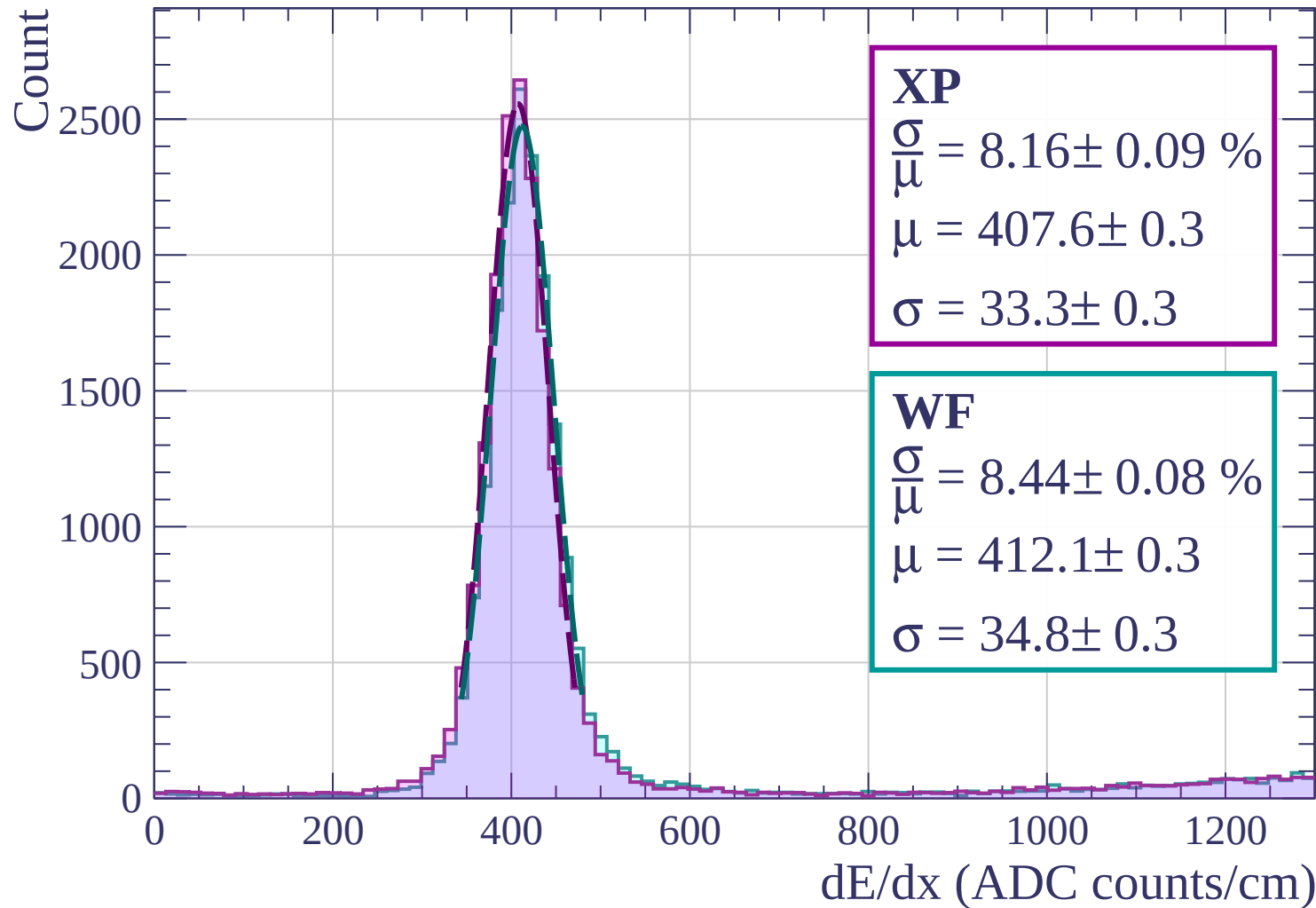


Energy loss | $400 < p < 480$

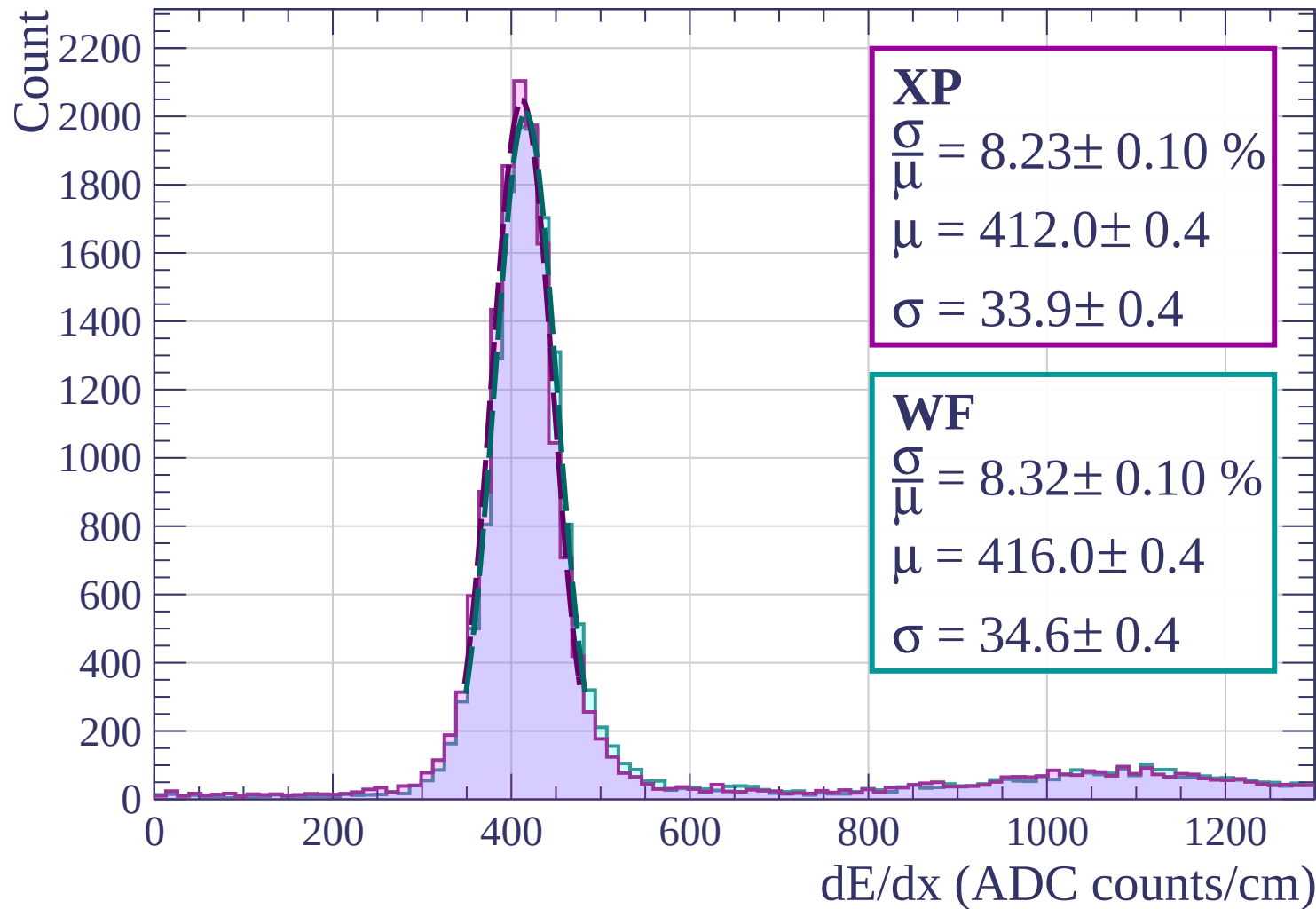
Count



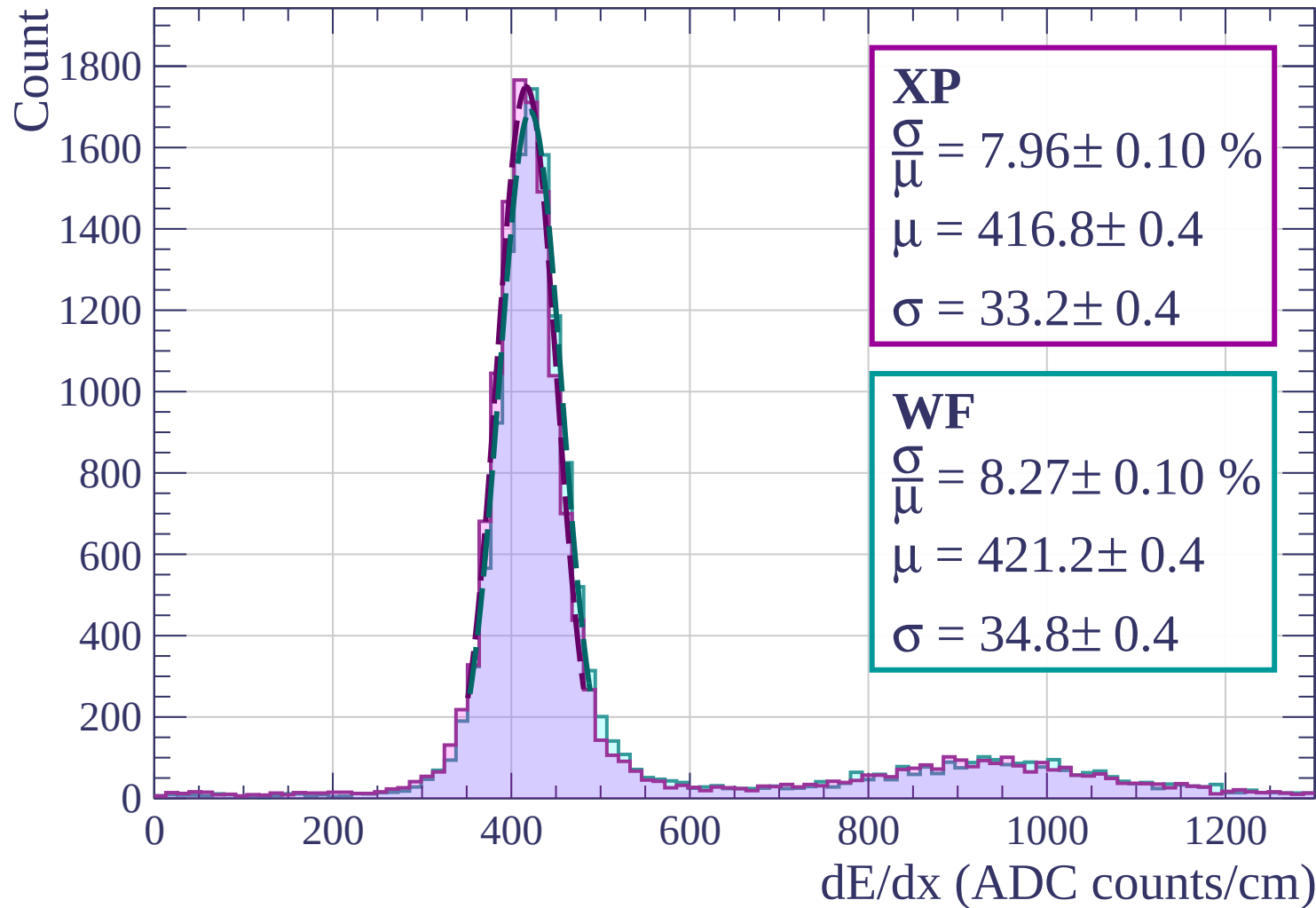
Energy loss | $480 < p < 560$



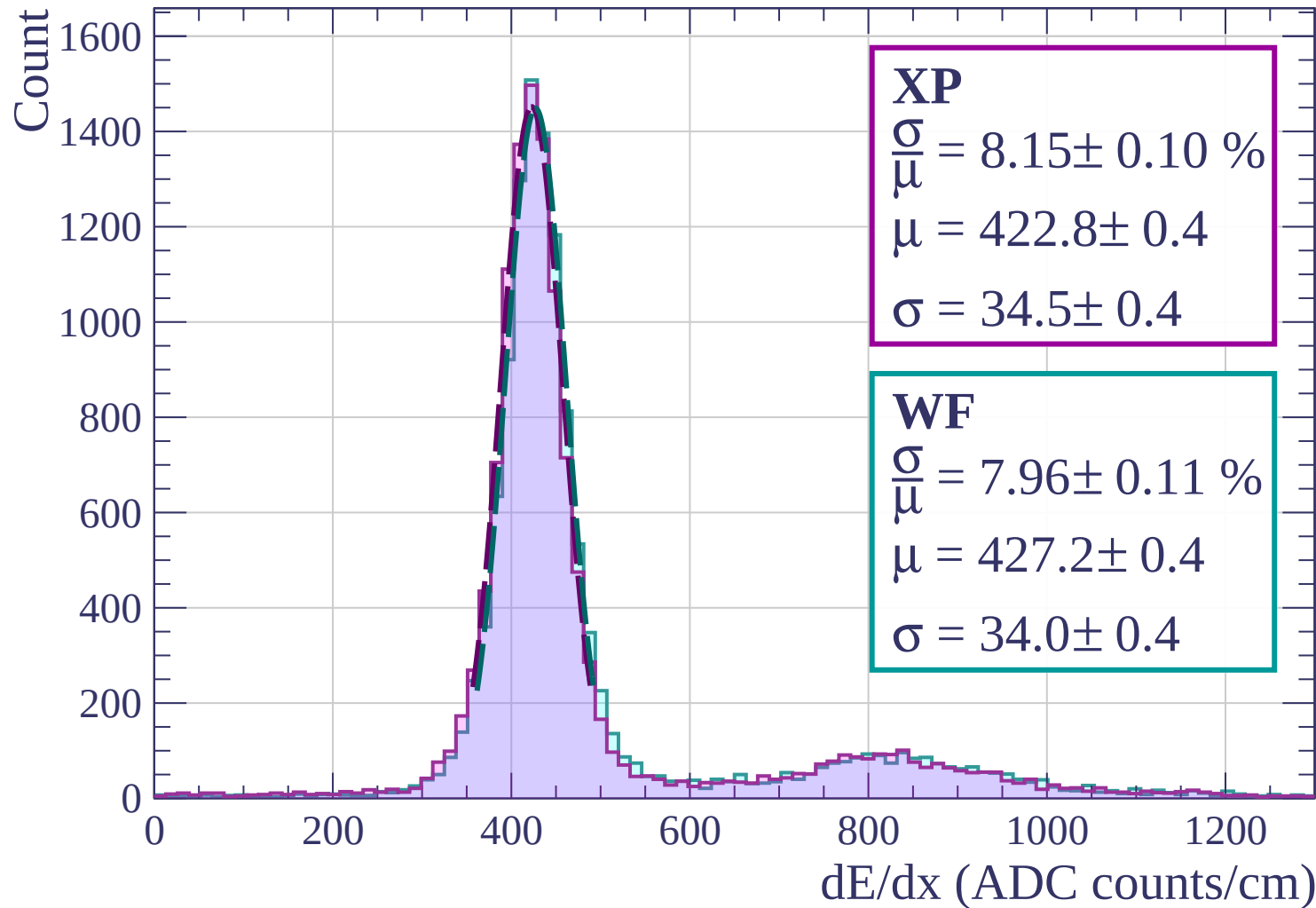
Energy loss | $560 < p < 640$



Energy loss | $640 < p < 720$

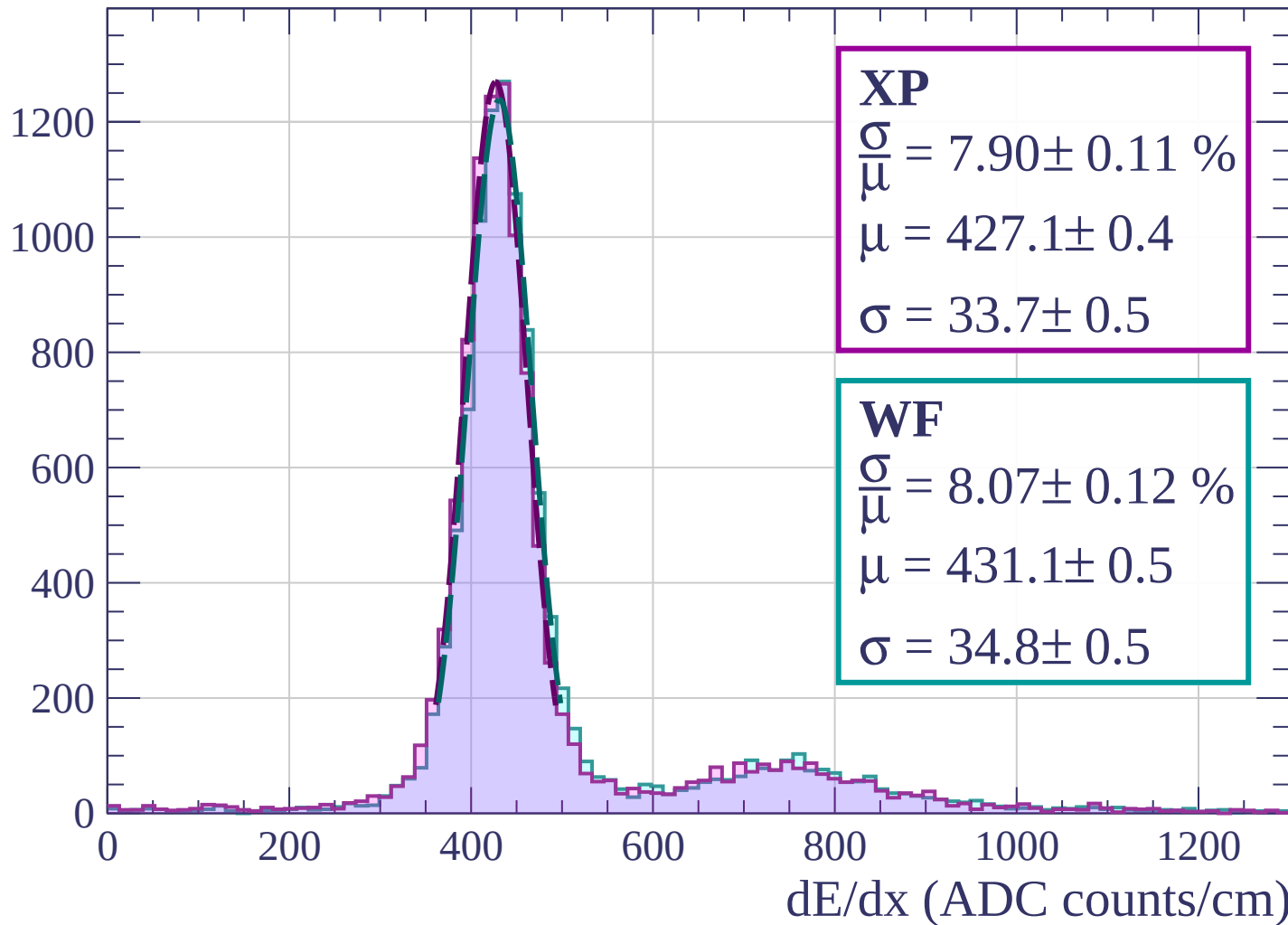


Energy loss | $720 < p < 800$

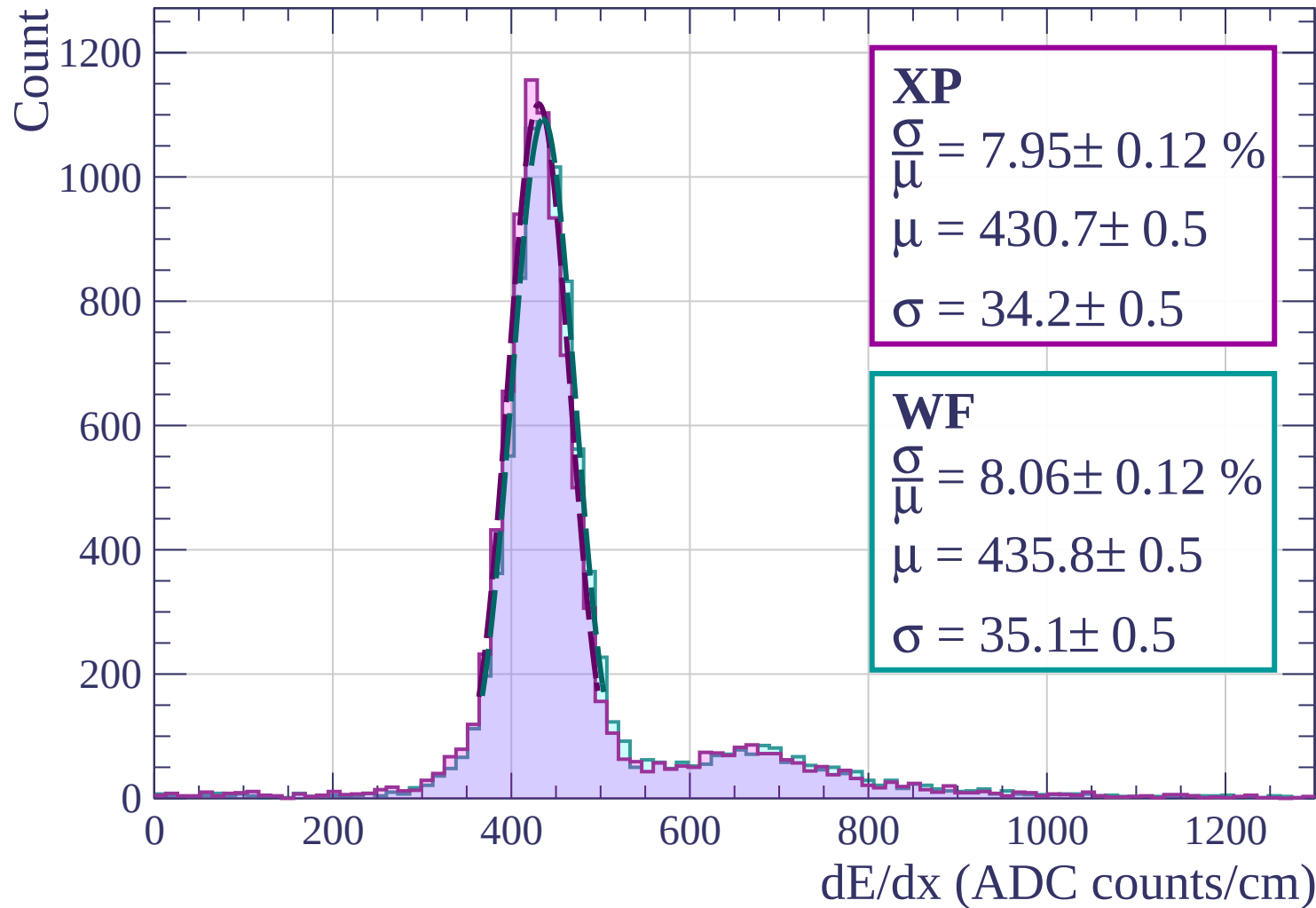


Energy loss | $800 < p < 880$

Count



Energy loss | $880 < p < 960$



Energy loss | $960 < p < 1040$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.12 \%$$

$$\mu = 436.3 \pm 0.5$$

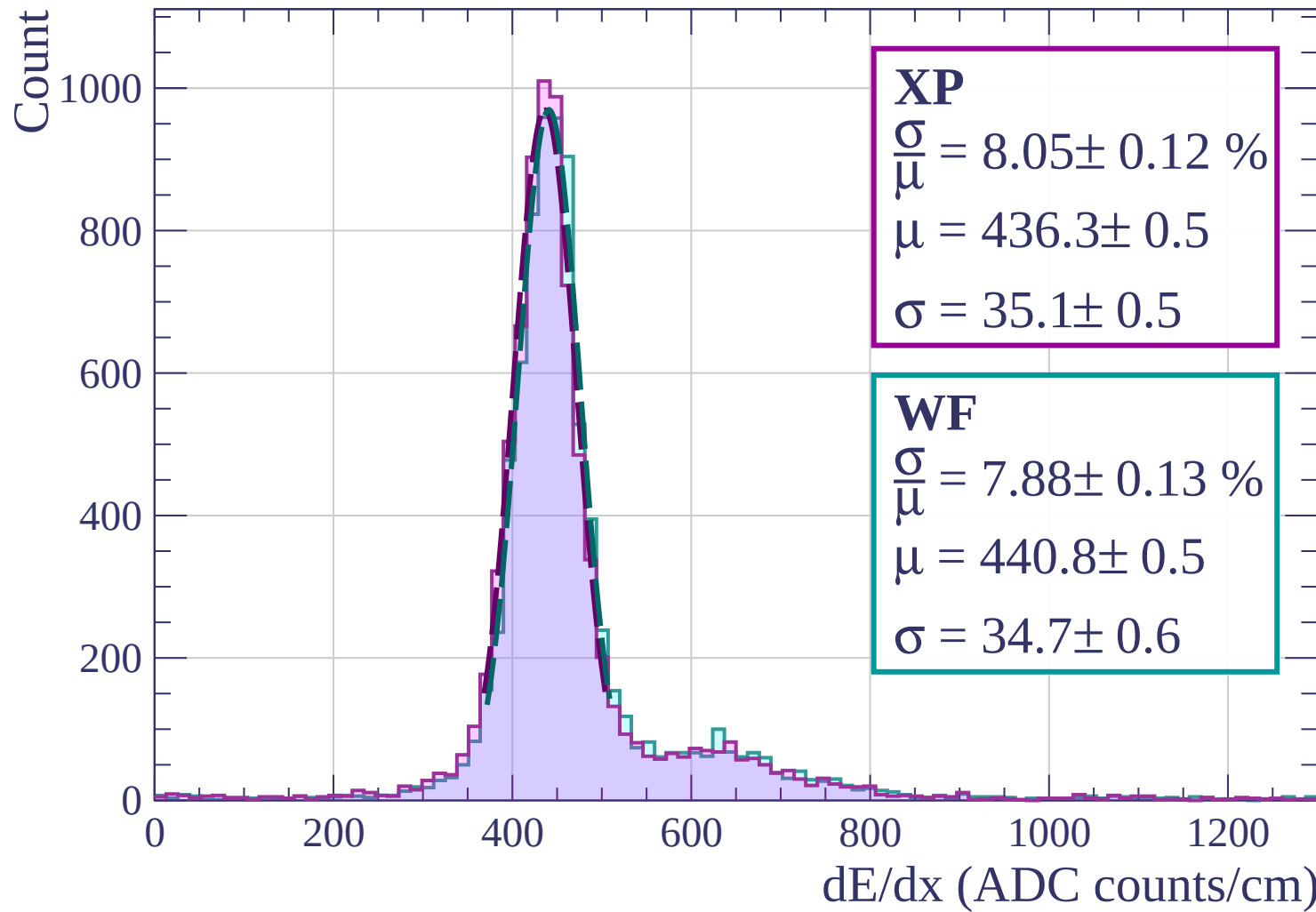
$$\sigma = 35.1 \pm 0.5$$

WF

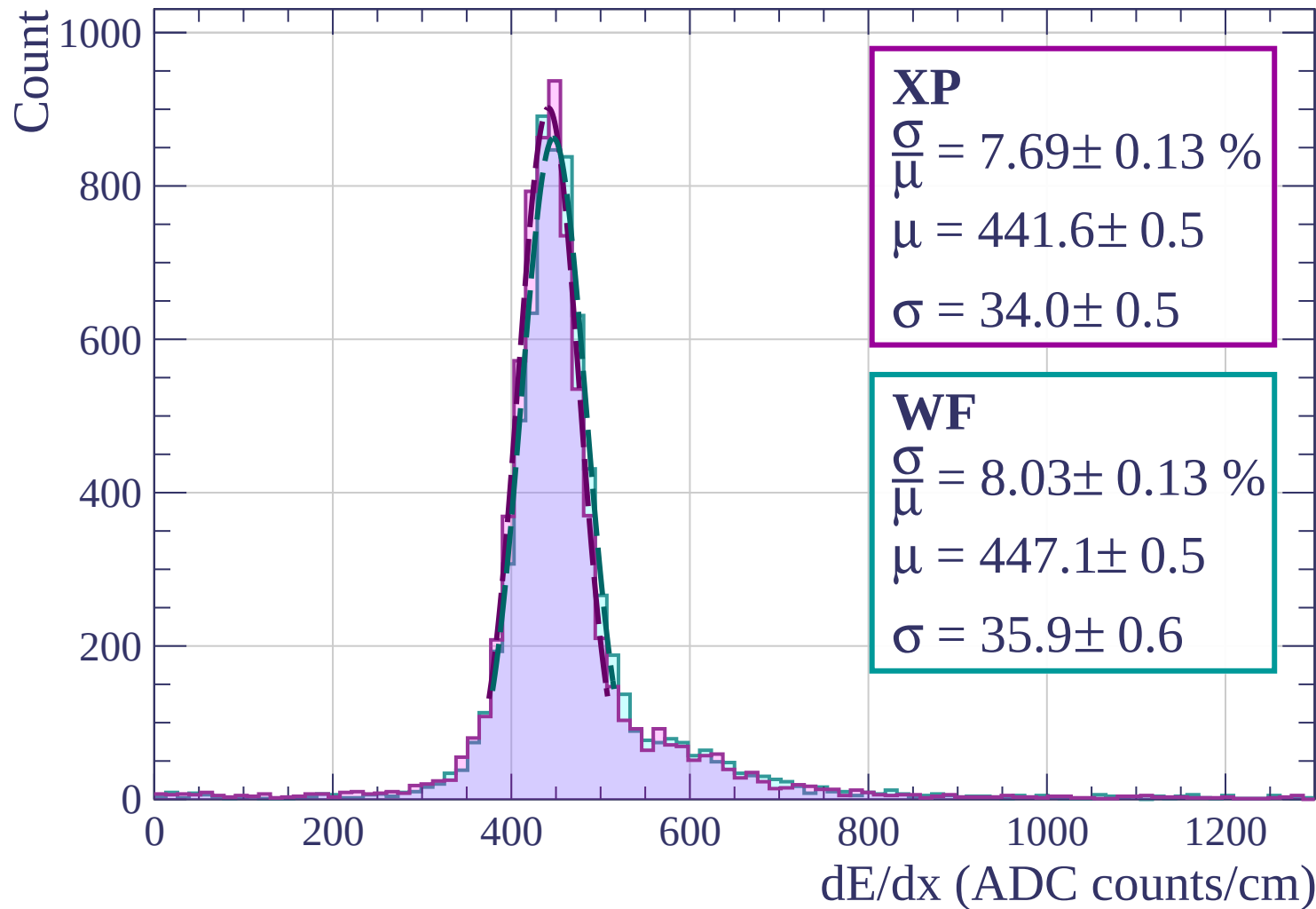
$$\frac{\sigma}{\mu} = 7.88 \pm 0.13 \%$$

$$\mu = 440.8 \pm 0.5$$

$$\sigma = 34.7 \pm 0.6$$

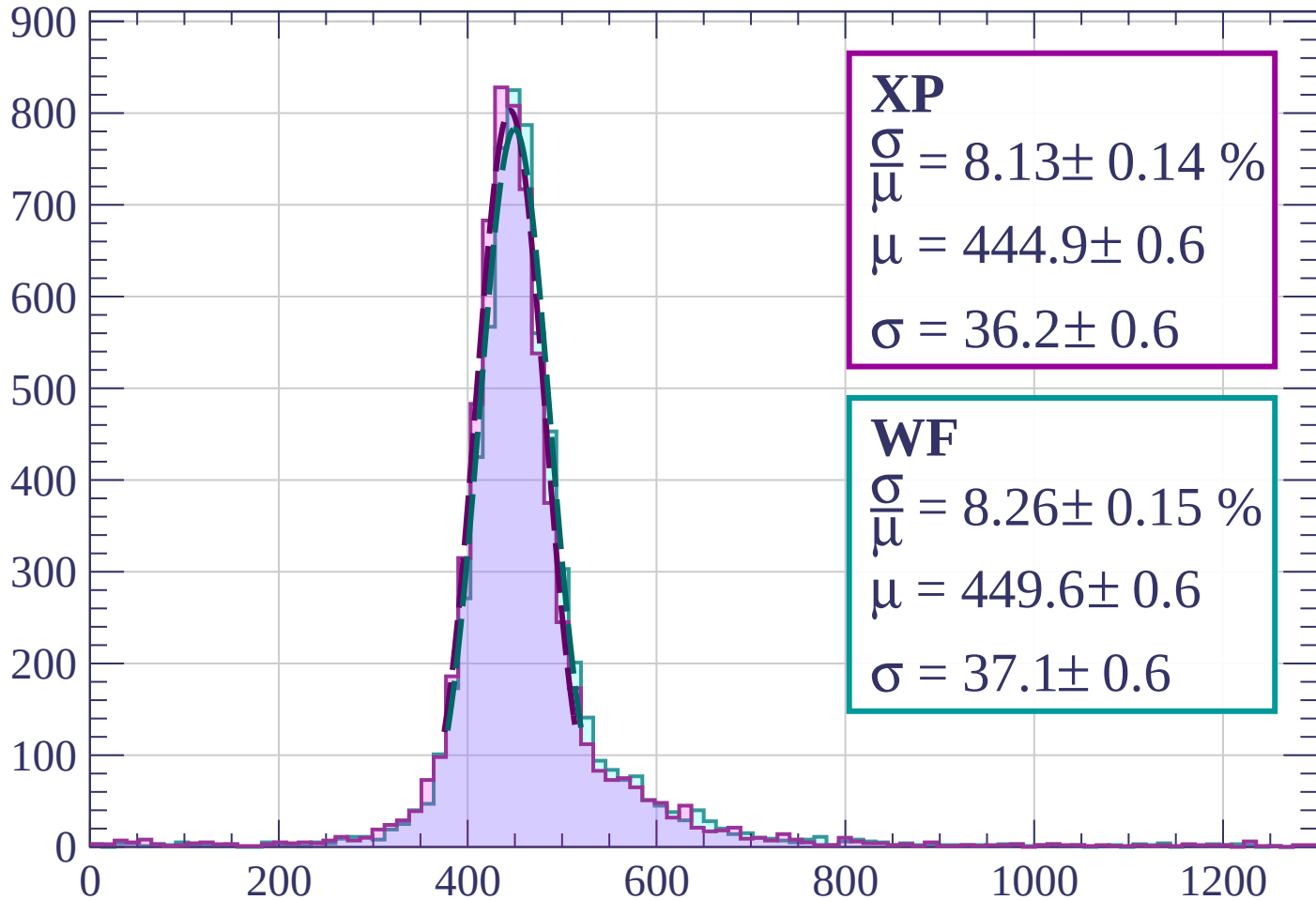


Energy loss | $1040 < p < 1120$



Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.13 \pm 0.14 \%$$

$$\mu = 444.9 \pm 0.6$$

$$\sigma = 36.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.26 \pm 0.15 \%$$

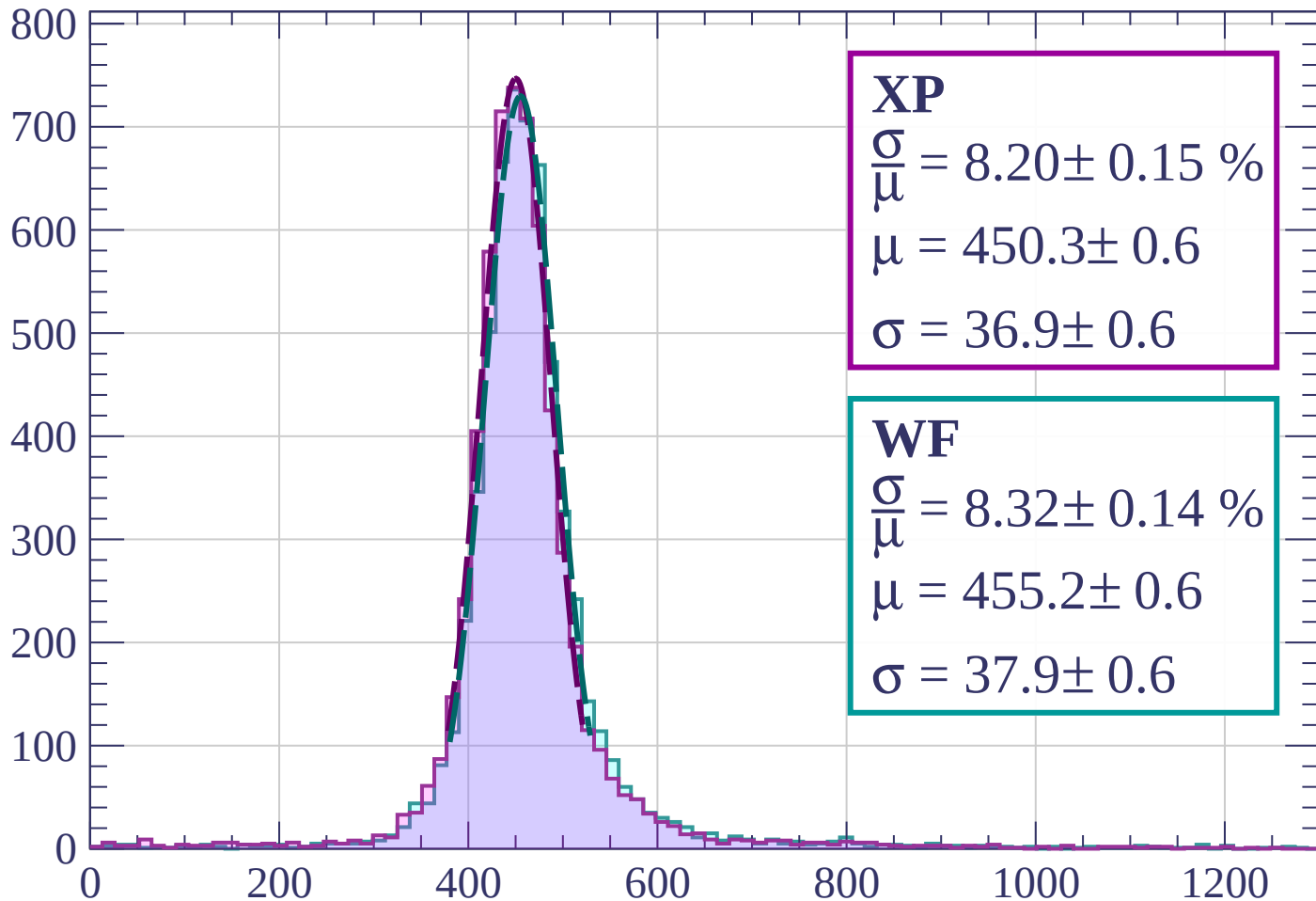
$$\mu = 449.6 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.20 \pm 0.15 \%$$

$$\mu = 450.3 \pm 0.6$$

$$\sigma = 36.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.32 \pm 0.14 \%$$

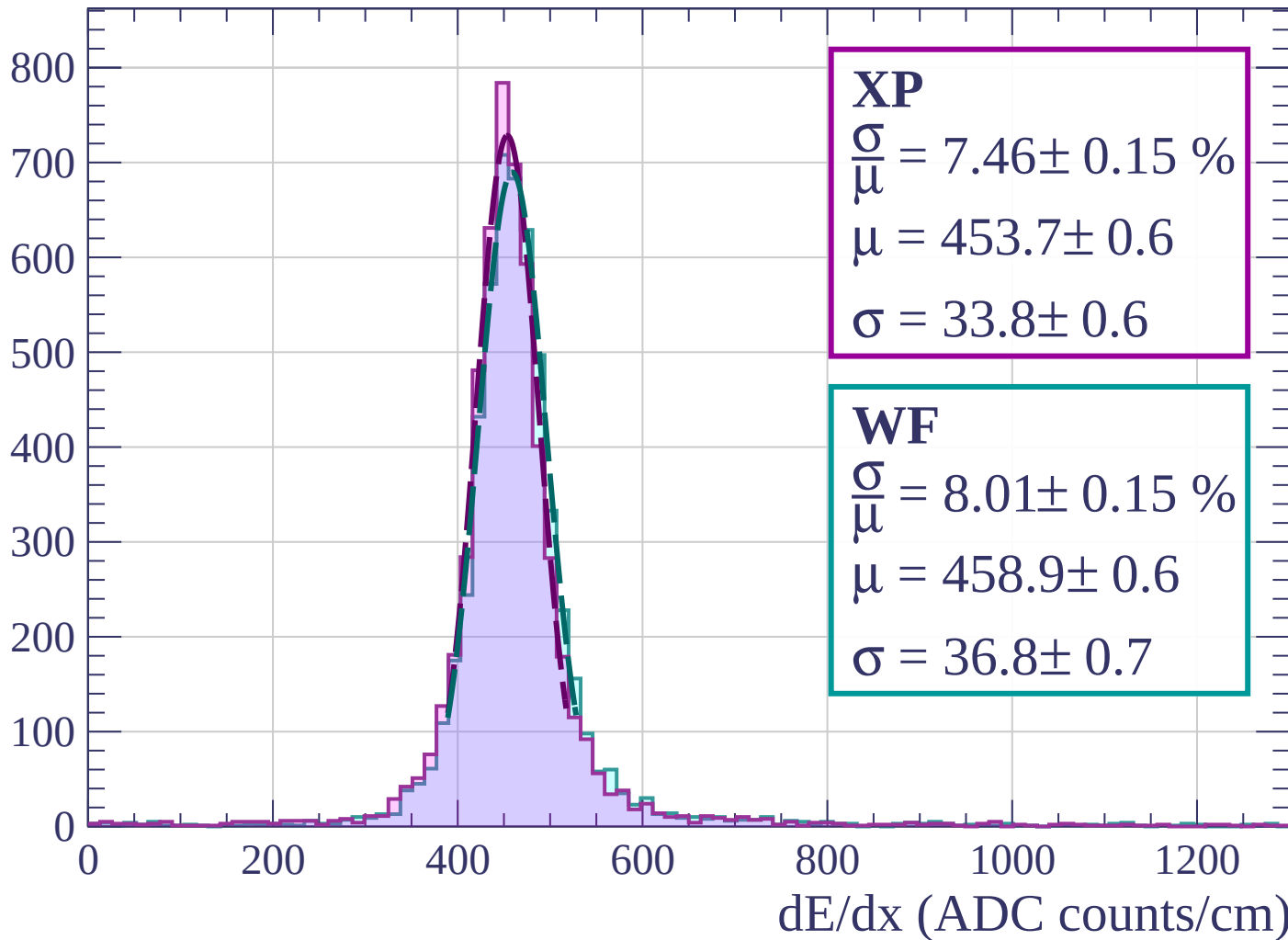
$$\mu = 455.2 \pm 0.6$$

$$\sigma = 37.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1360$

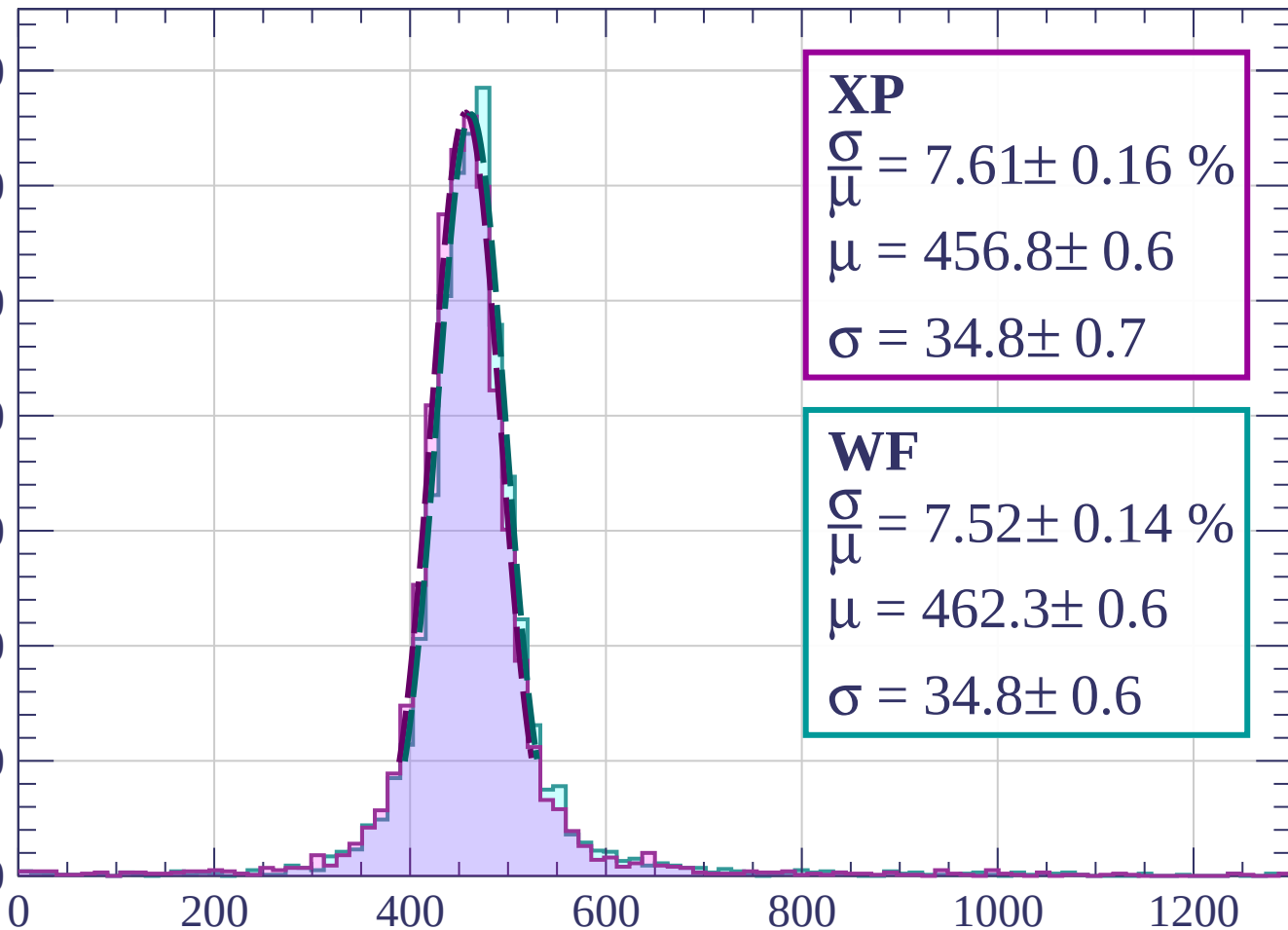
Count



Energy loss | $1360 < p < 1440$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 7.61 \pm 0.16 \%$$

$$\mu = 456.8 \pm 0.6$$

$$\sigma = 34.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.52 \pm 0.14 \%$$

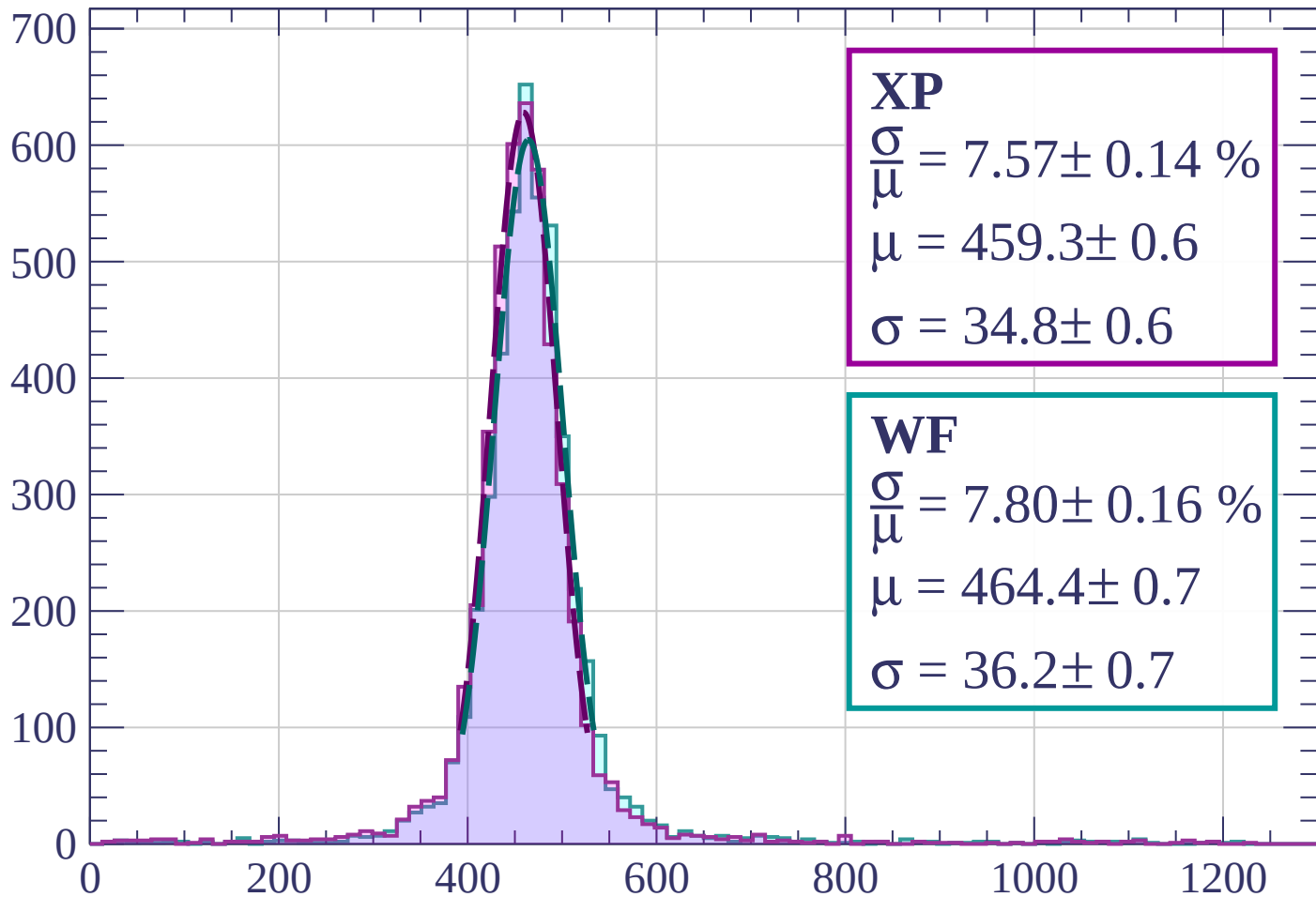
$$\mu = 462.3 \pm 0.6$$

$$\sigma = 34.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 7.57 \pm 0.14 \%$$

$$\mu = 459.3 \pm 0.6$$

$$\sigma = 34.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.80 \pm 0.16 \%$$

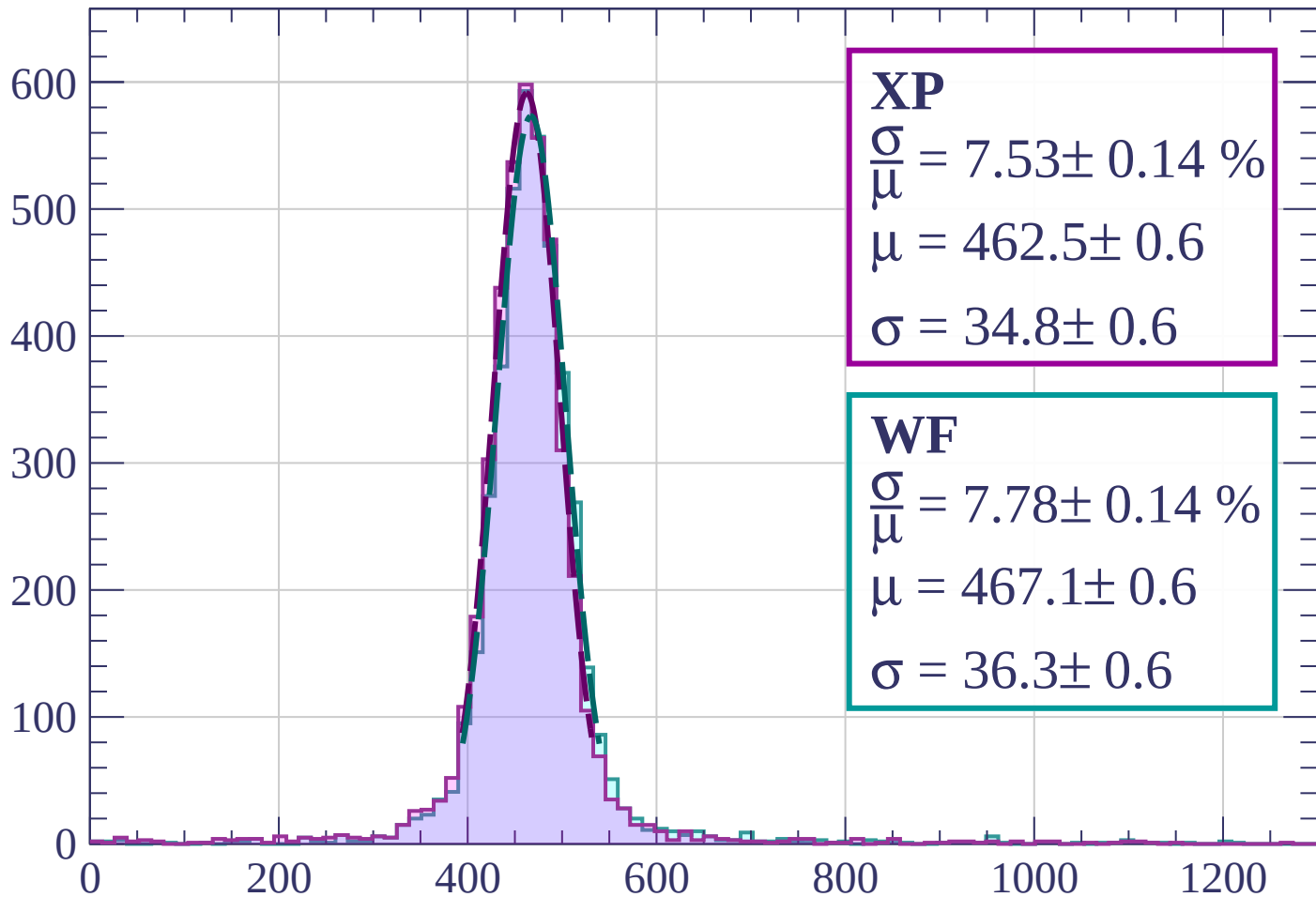
$$\mu = 464.4 \pm 0.7$$

$$\sigma = 36.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.53 \pm 0.14 \%$$

$$\mu = 462.5 \pm 0.6$$

$$\sigma = 34.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.78 \pm 0.14 \%$$

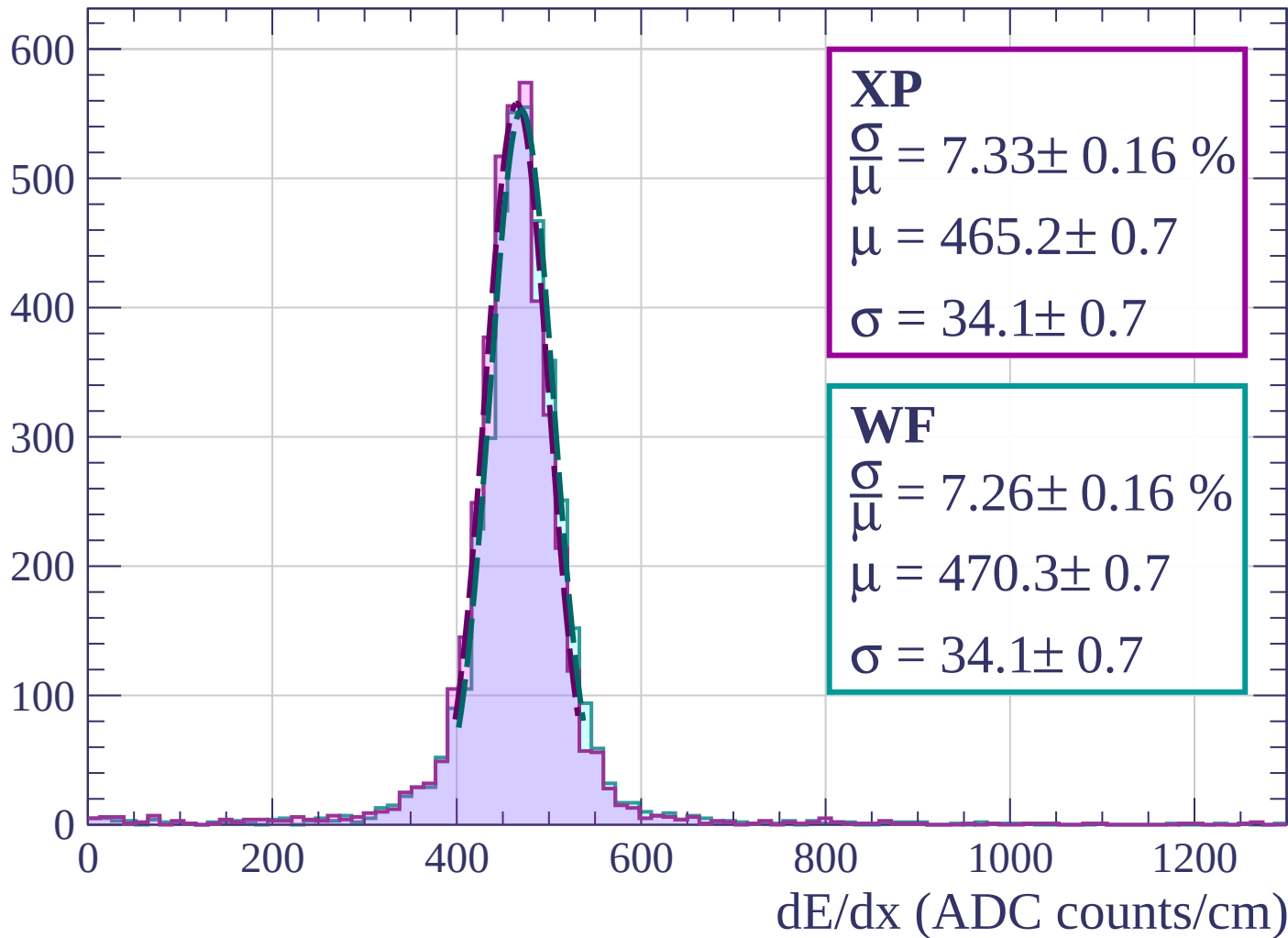
$$\mu = 467.1 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

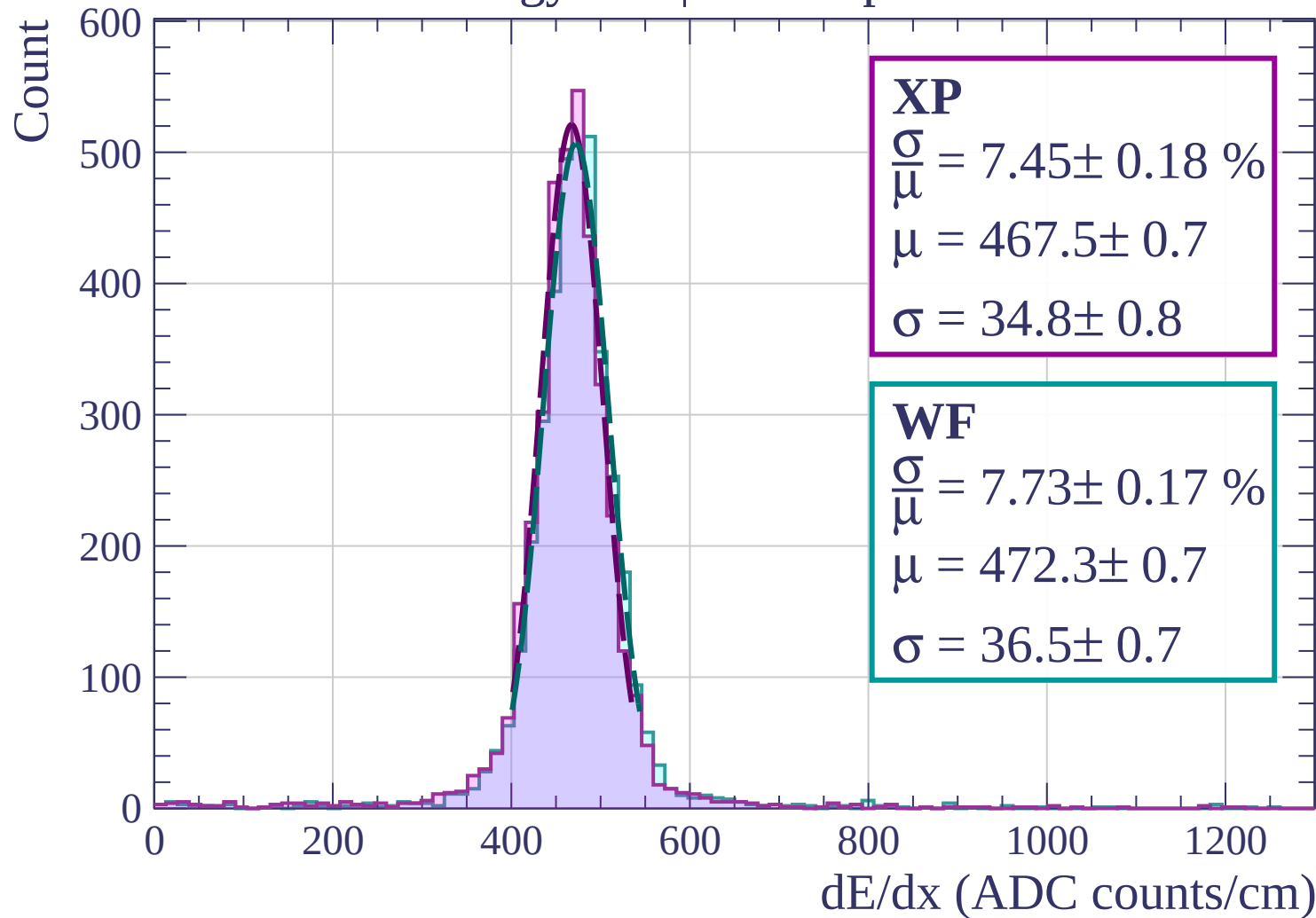
dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count



Energy loss | $1680 < p < 1760$



Energy loss | $1760 < p < 1840$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

XP

$$\frac{\sigma}{\mu} = 7.41 \pm 0.19 \%$$

$$\mu = 469.6 \pm 0.7$$

$$\sigma = 34.8 \pm 0.8$$

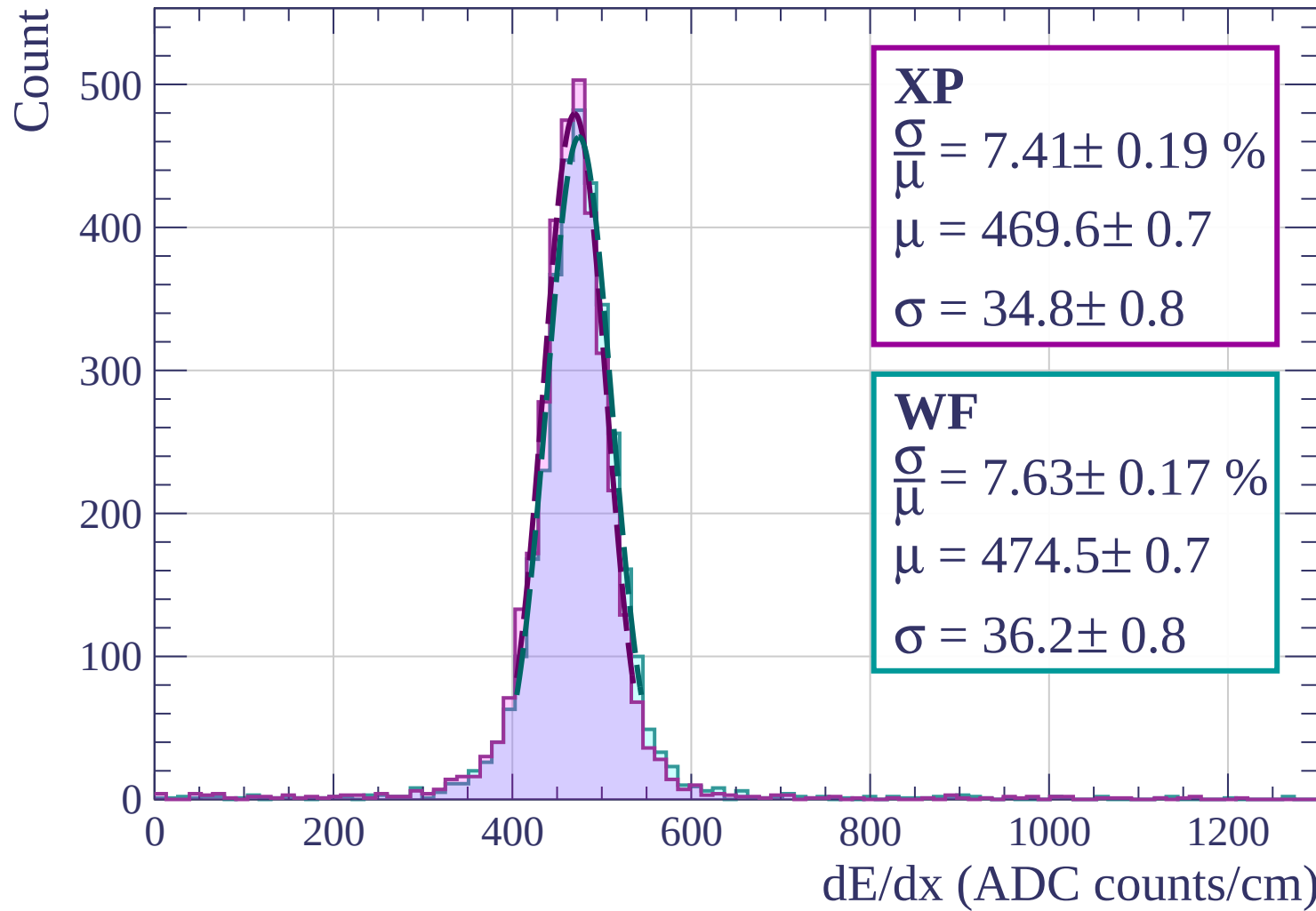
WF

$$\frac{\sigma}{\mu} = 7.63 \pm 0.17 \%$$

$$\mu = 474.5 \pm 0.7$$

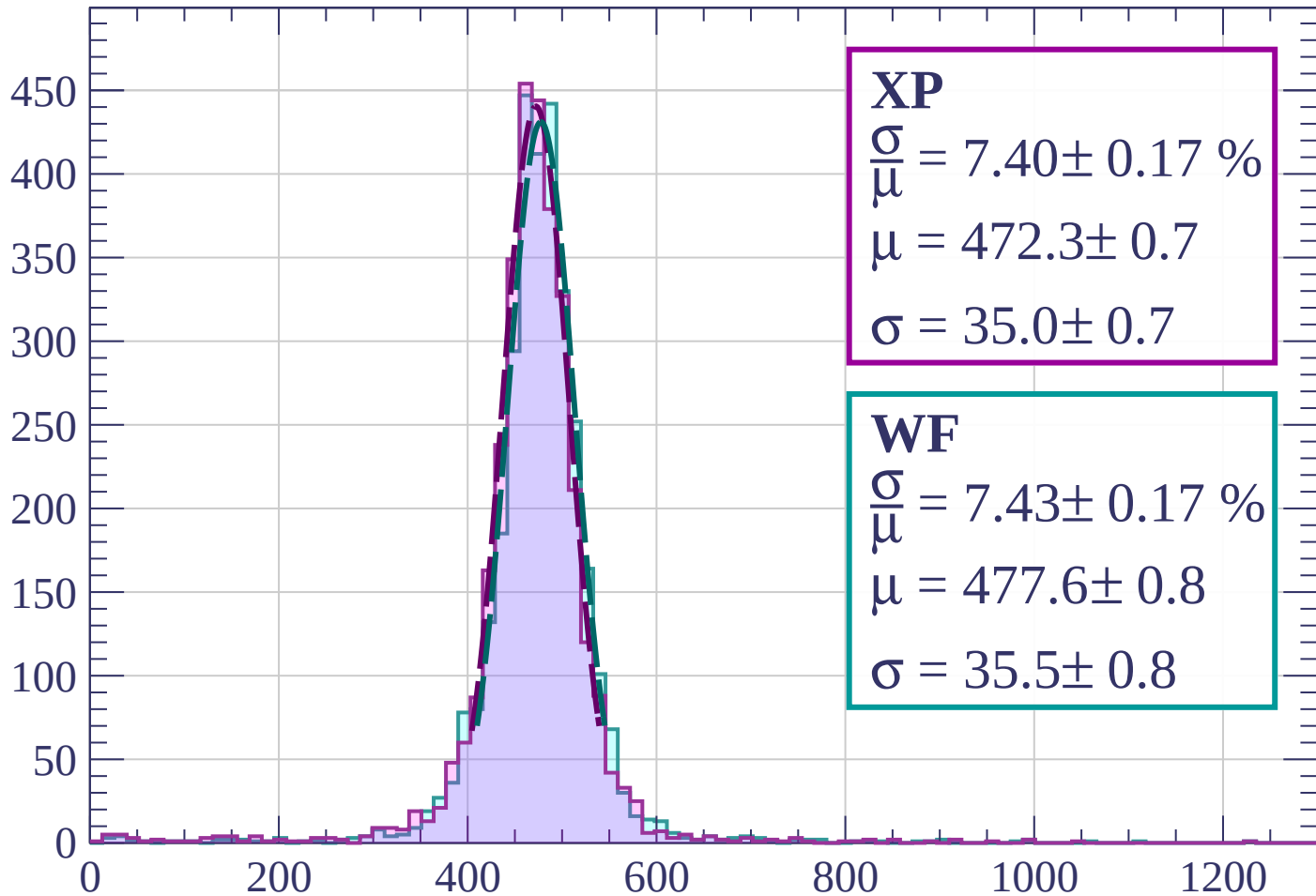
$$\sigma = 36.2 \pm 0.8$$

dE/dx (ADC counts/cm)



Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.17 \%$$

$$\mu = 472.3 \pm 0.7$$

$$\sigma = 35.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.43 \pm 0.17 \%$$

$$\mu = 477.6 \pm 0.8$$

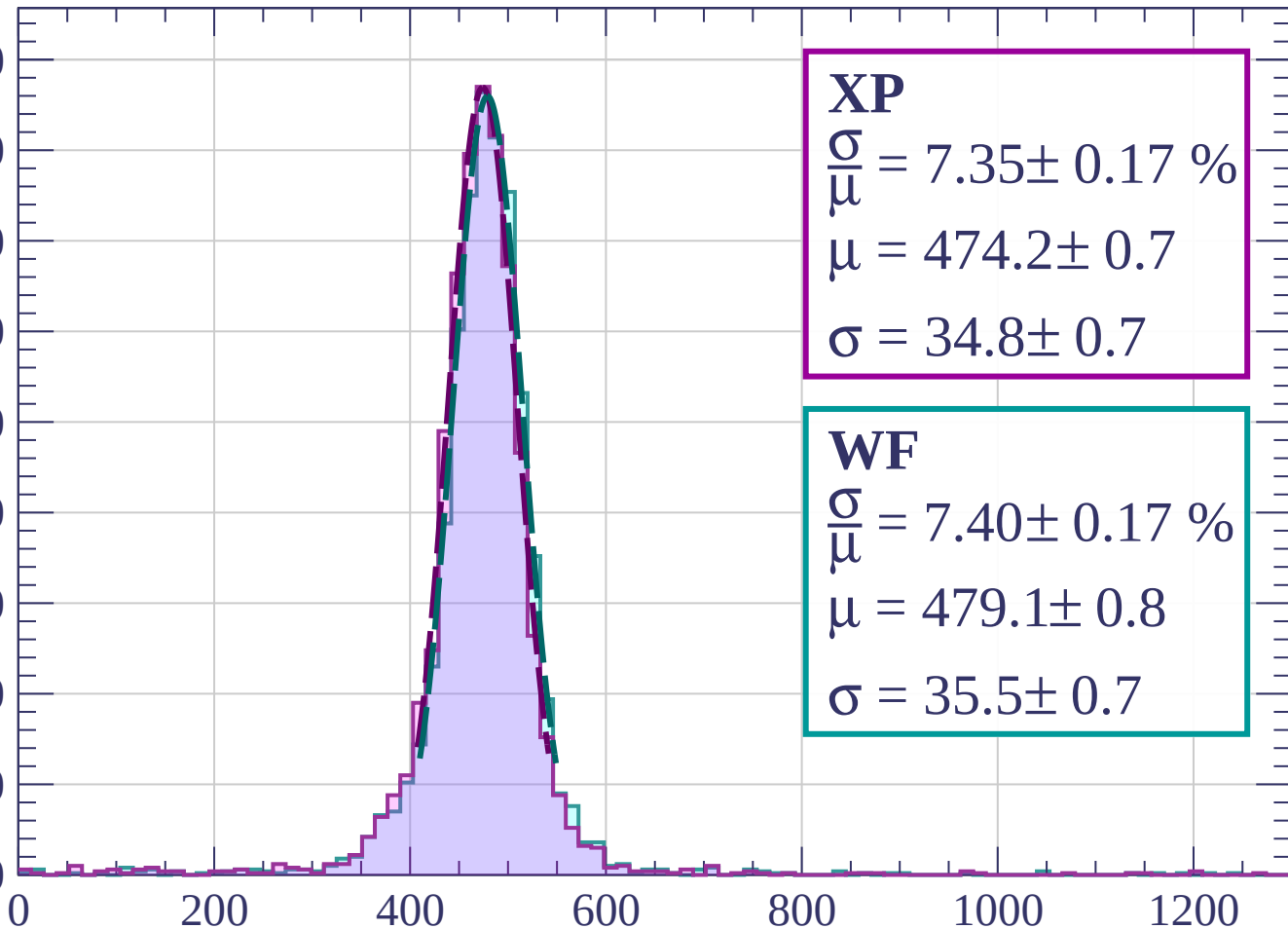
$$\sigma = 35.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 2000$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 7.35 \pm 0.17 \%$$

$$\mu = 474.2 \pm 0.7$$

$$\sigma = 34.8 \pm 0.7$$

WF

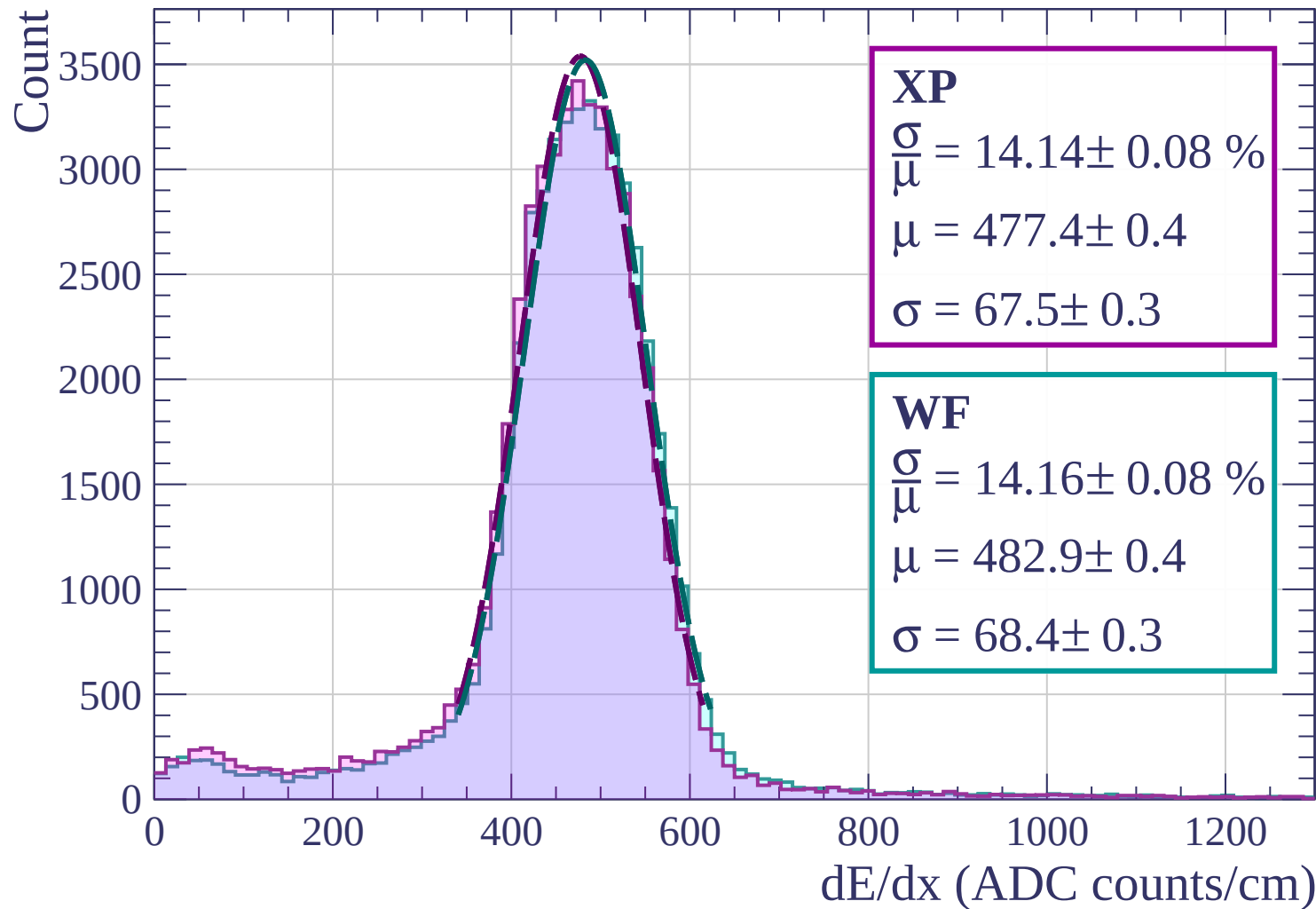
$$\frac{\sigma}{\mu} = 7.40 \pm 0.17 \%$$

$$\mu = 479.1 \pm 0.8$$

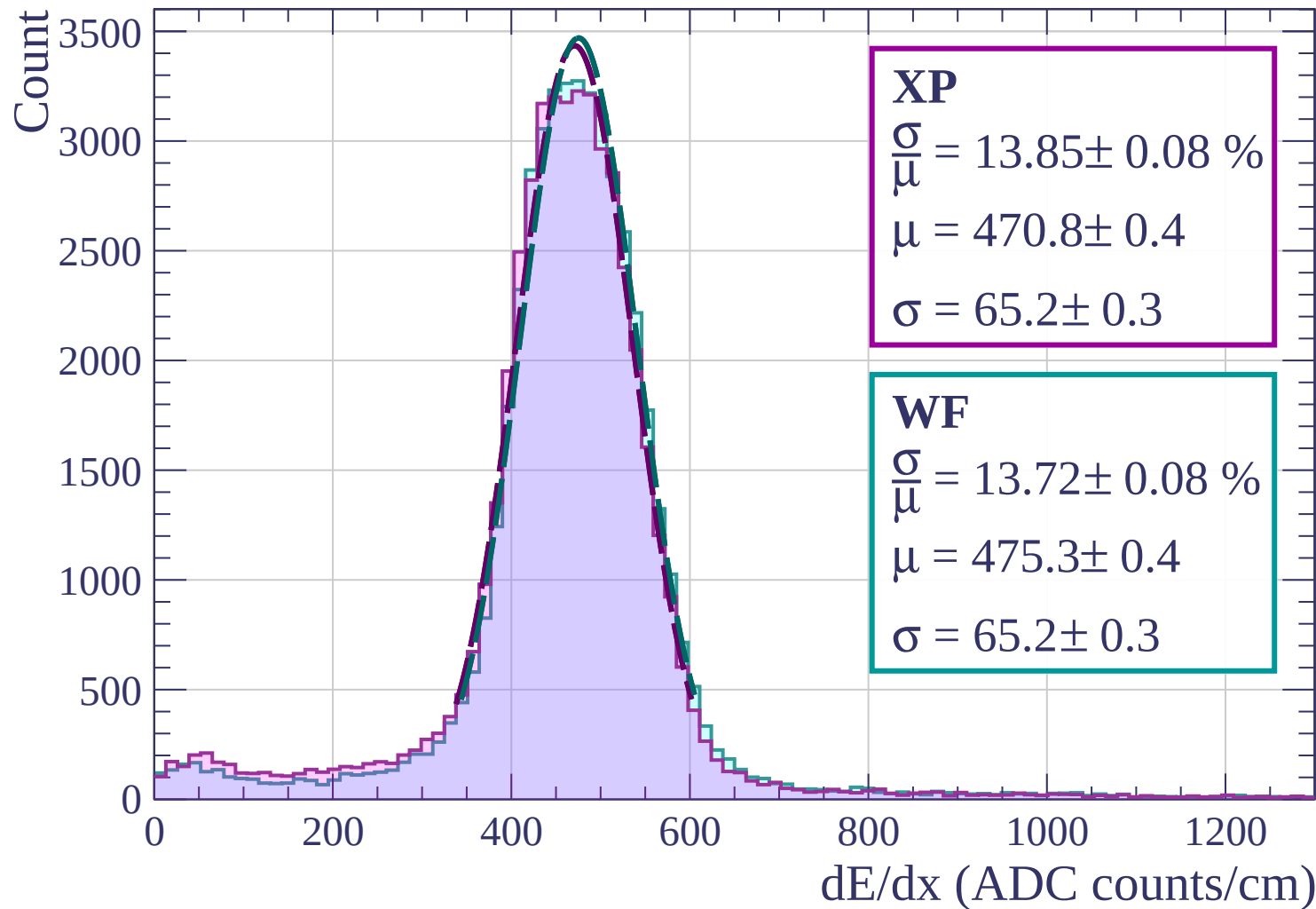
$$\sigma = 35.5 \pm 0.7$$

dE/dx (ADC counts/cm)

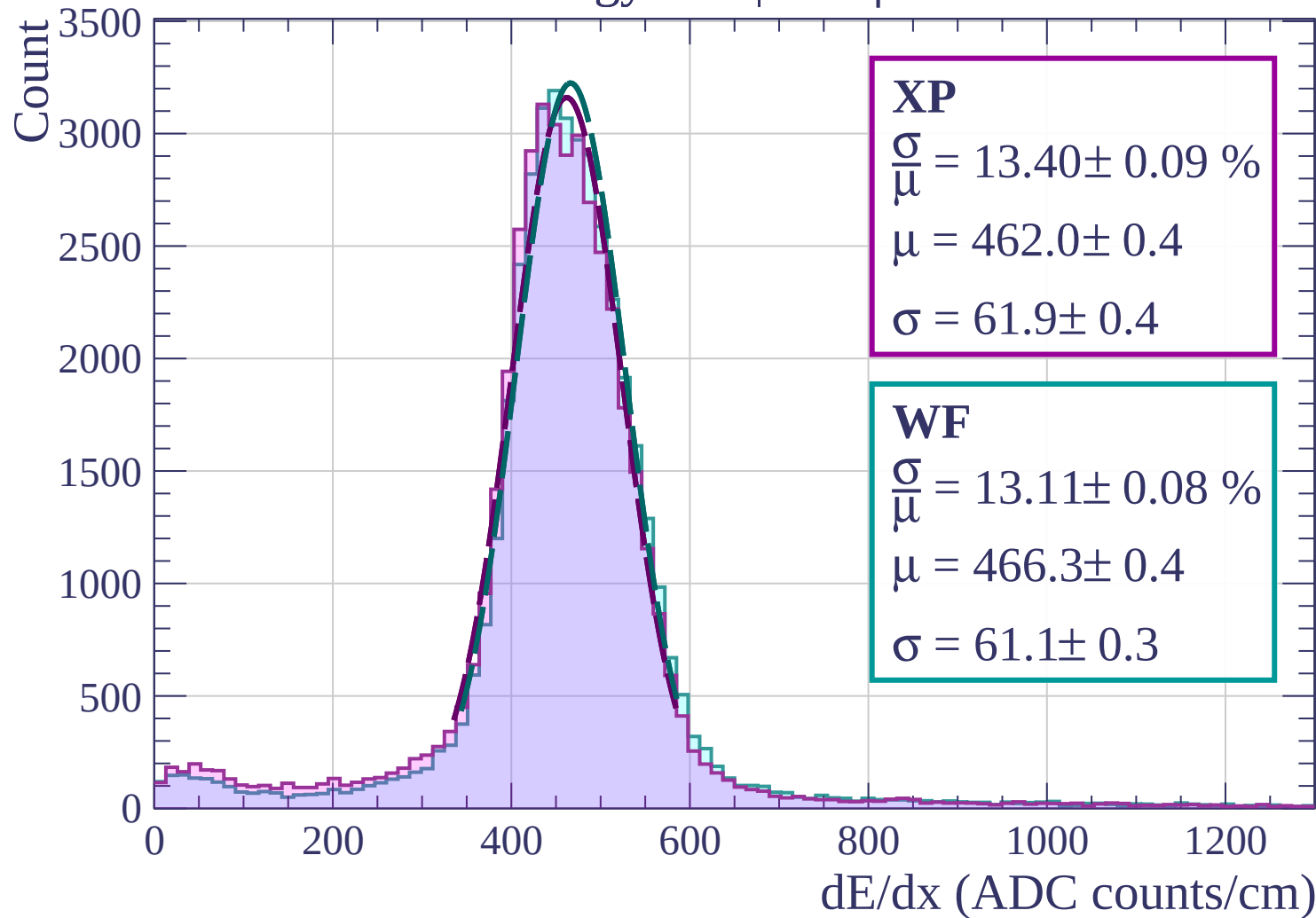
Energy loss | $0 < \phi < 3$



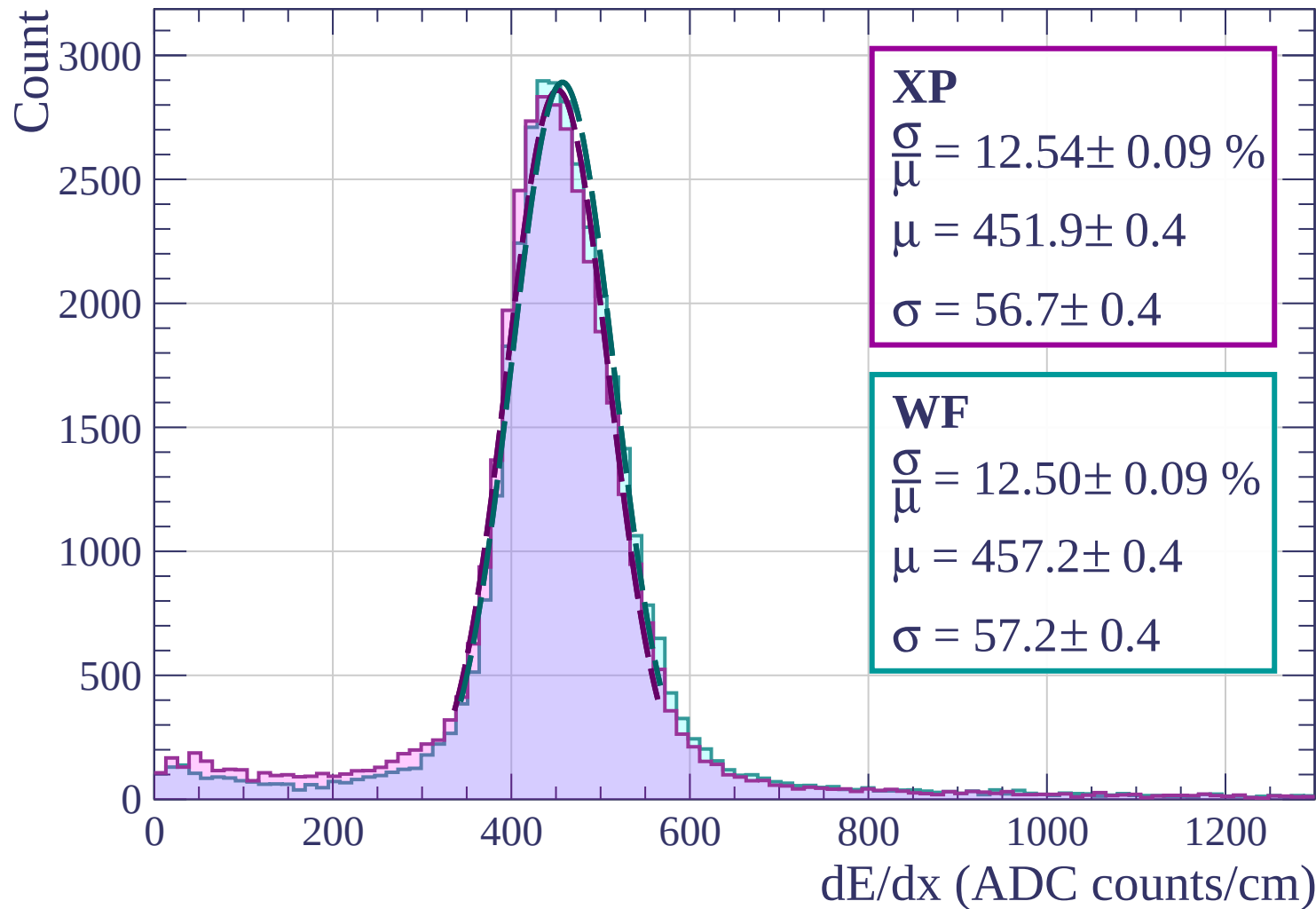
Energy loss | $3 < \phi < 6$



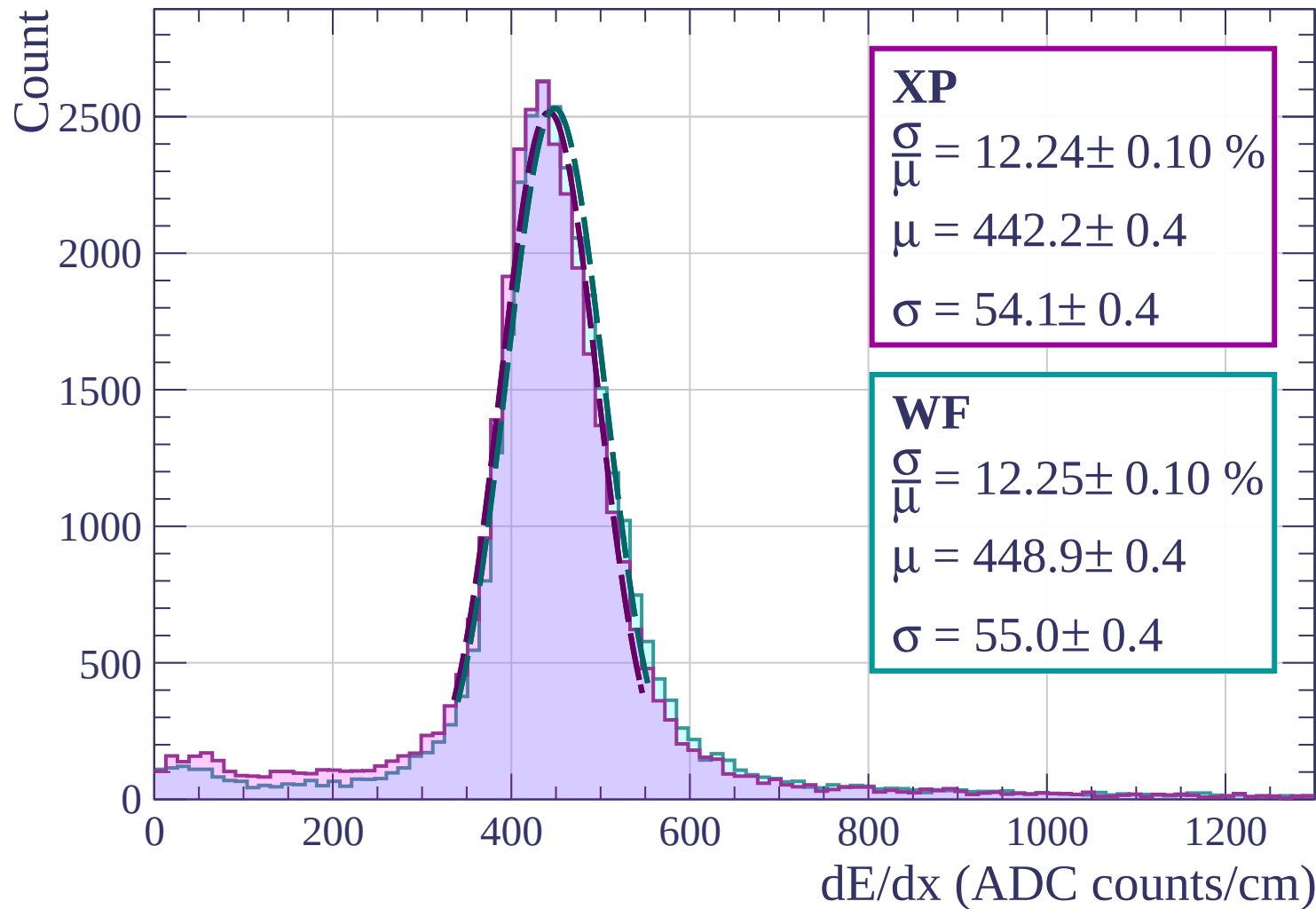
Energy loss | $6 < \phi < 9$



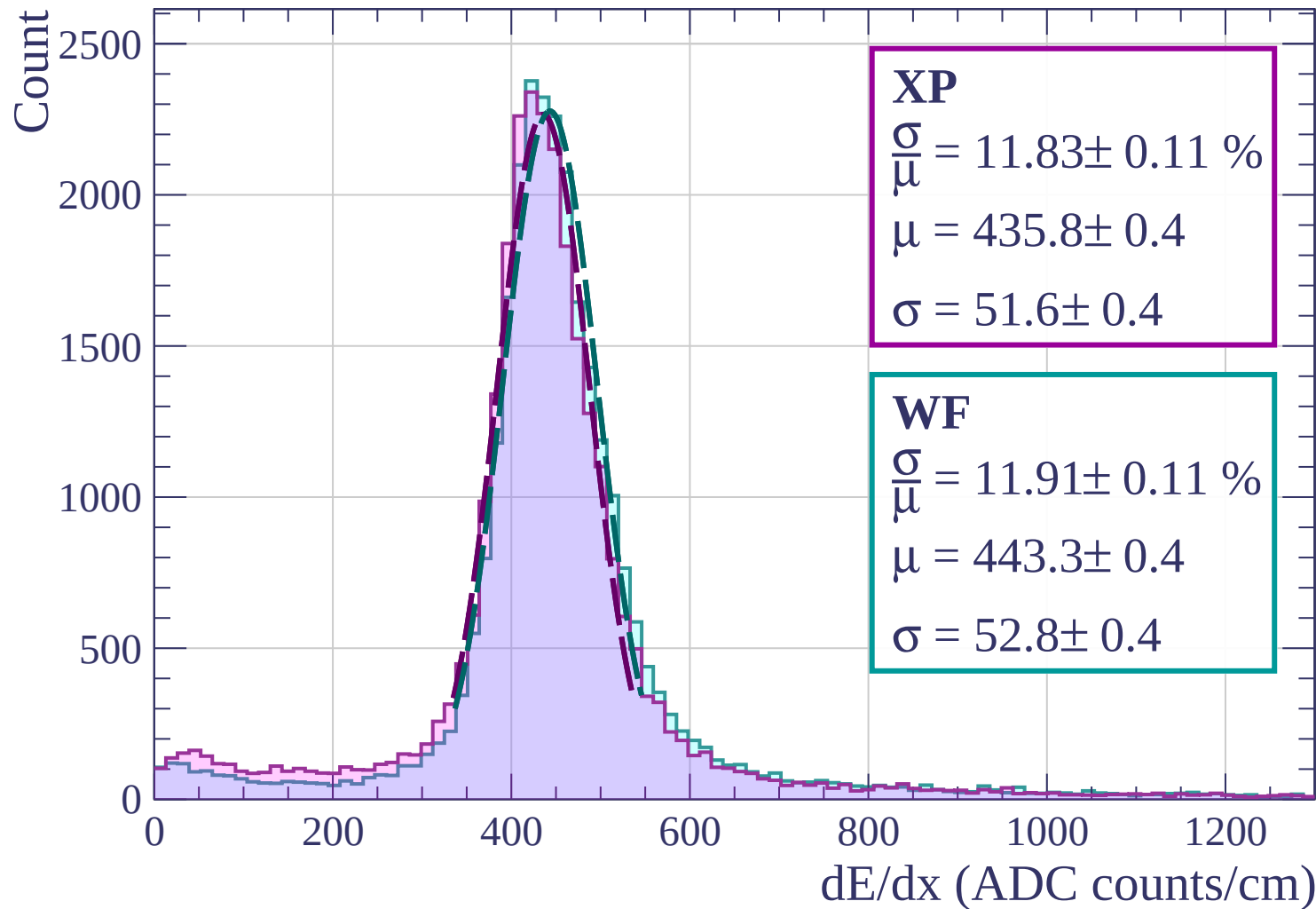
Energy loss | $9 < \phi < 12$



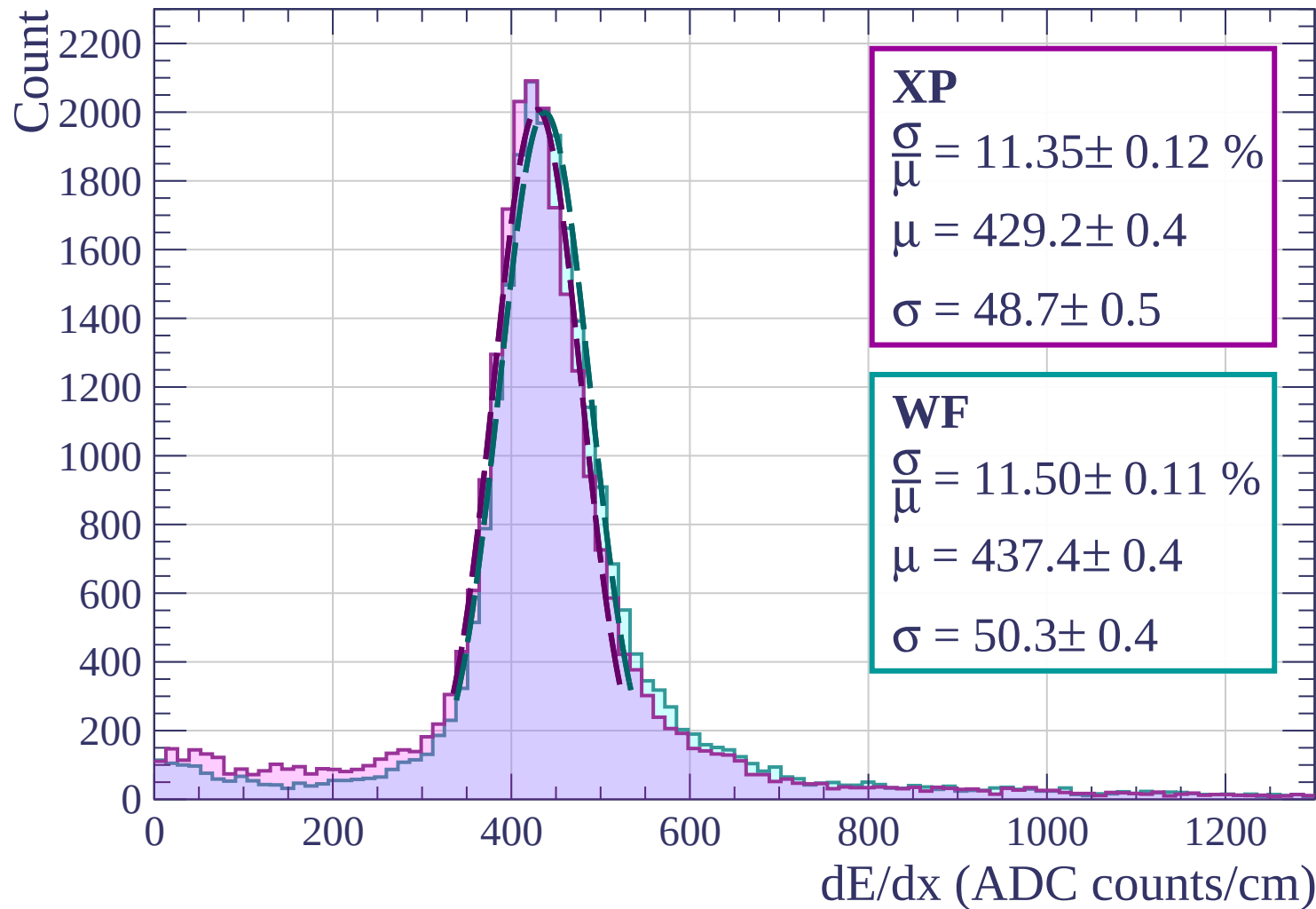
Energy loss | $12 < \phi < 15$



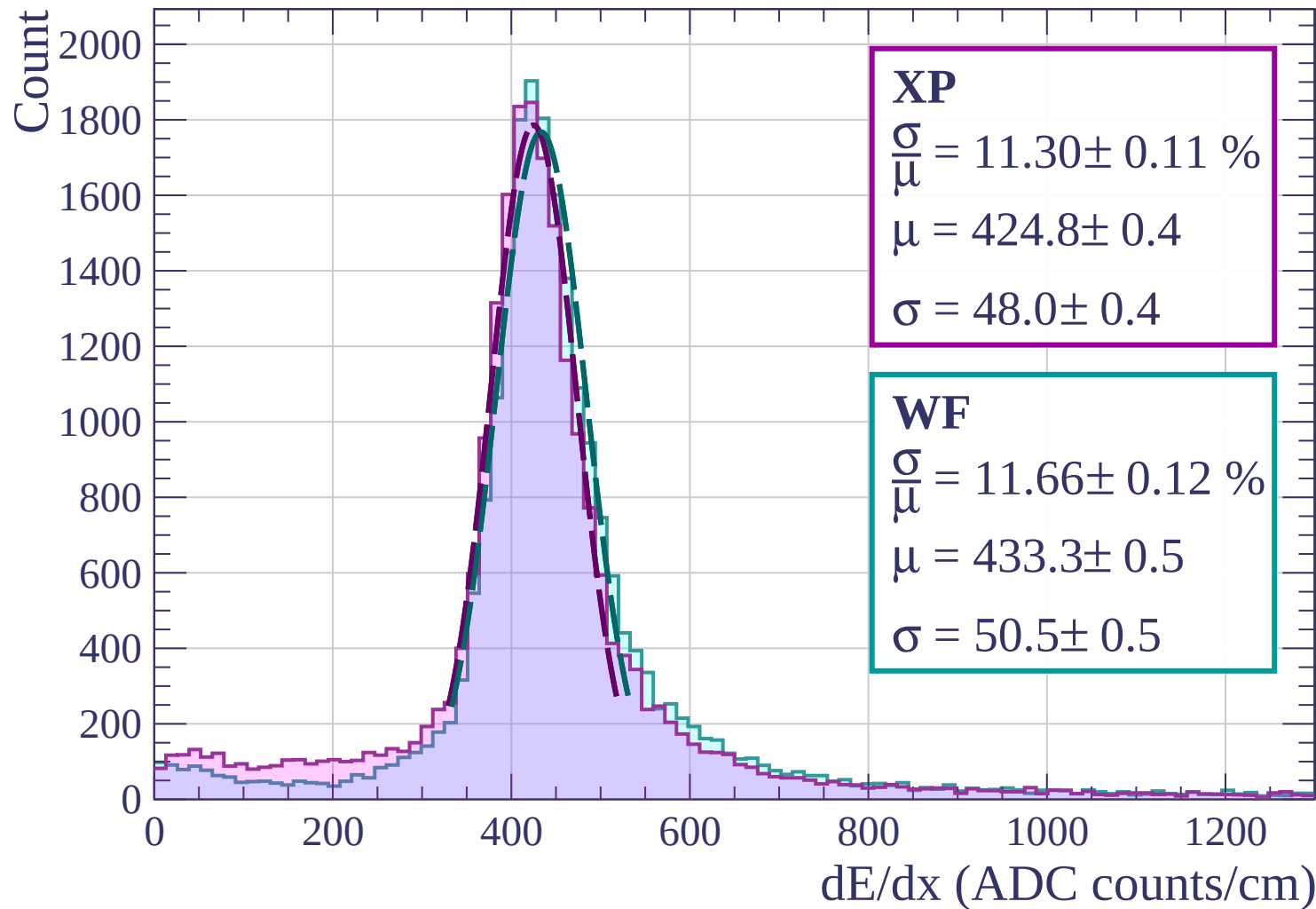
Energy loss | $15 < \phi < 18$



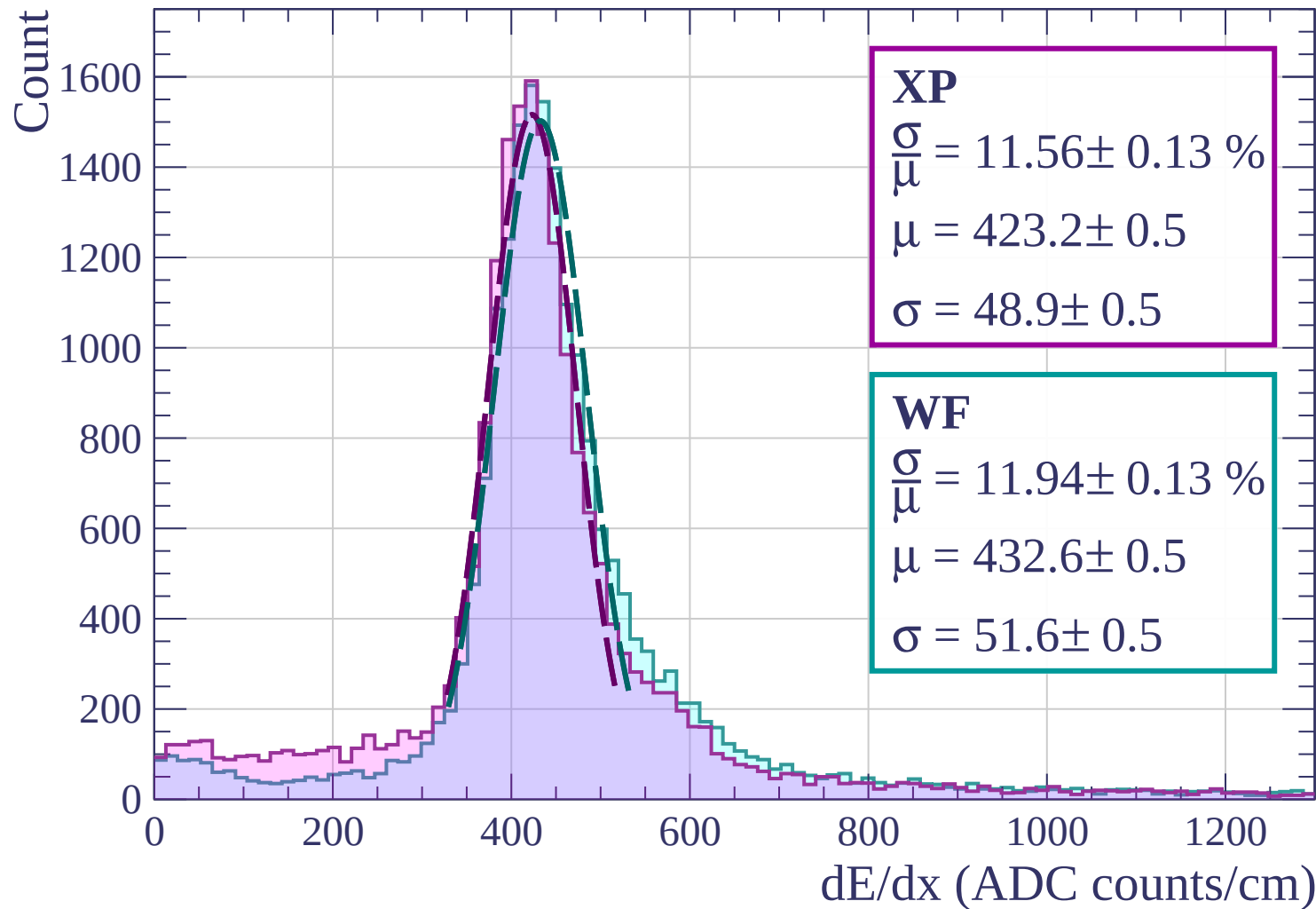
Energy loss | $18 < \phi < 21$



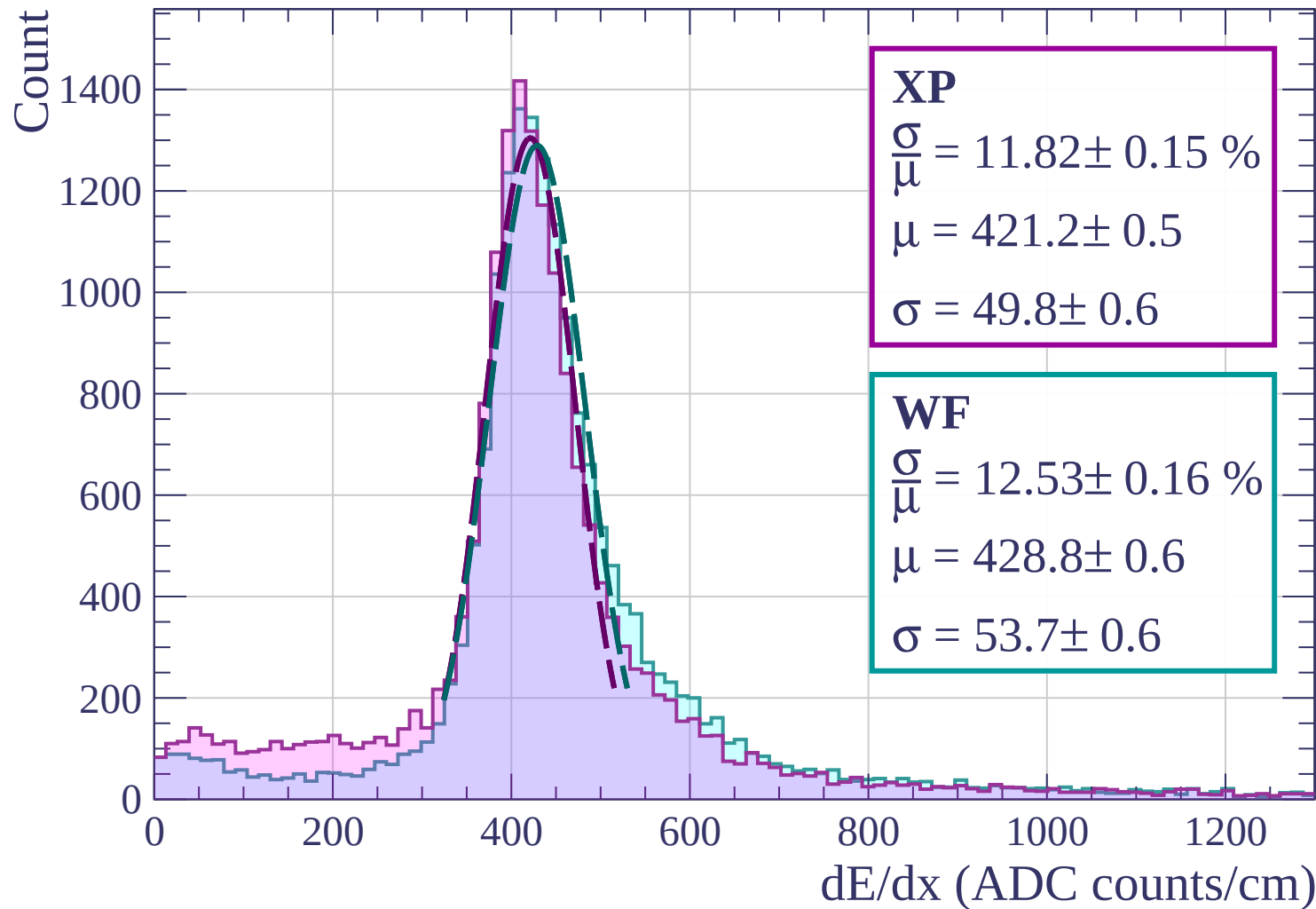
Energy loss | $21 < \phi < 24$



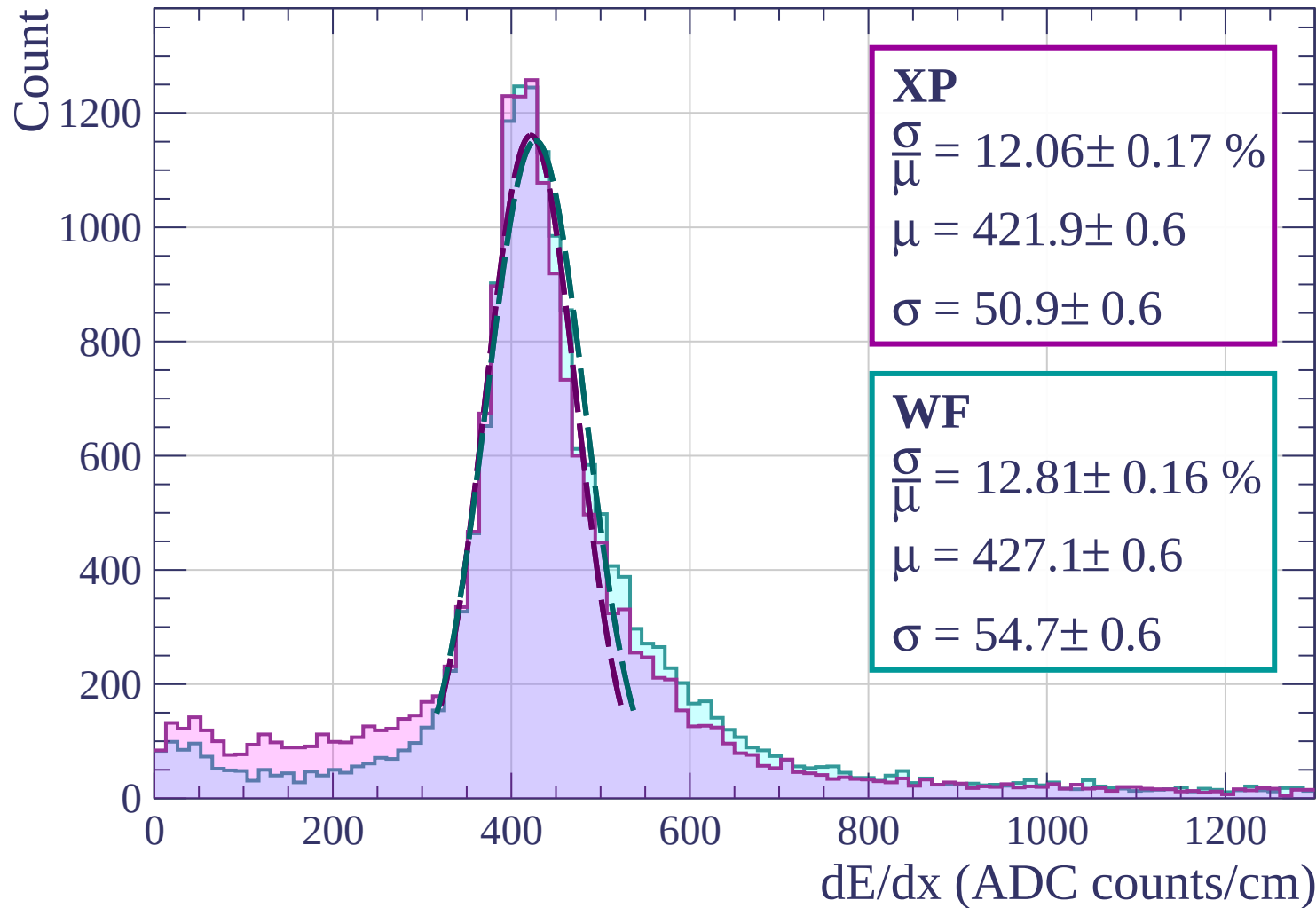
Energy loss | $24 < \phi < 27$



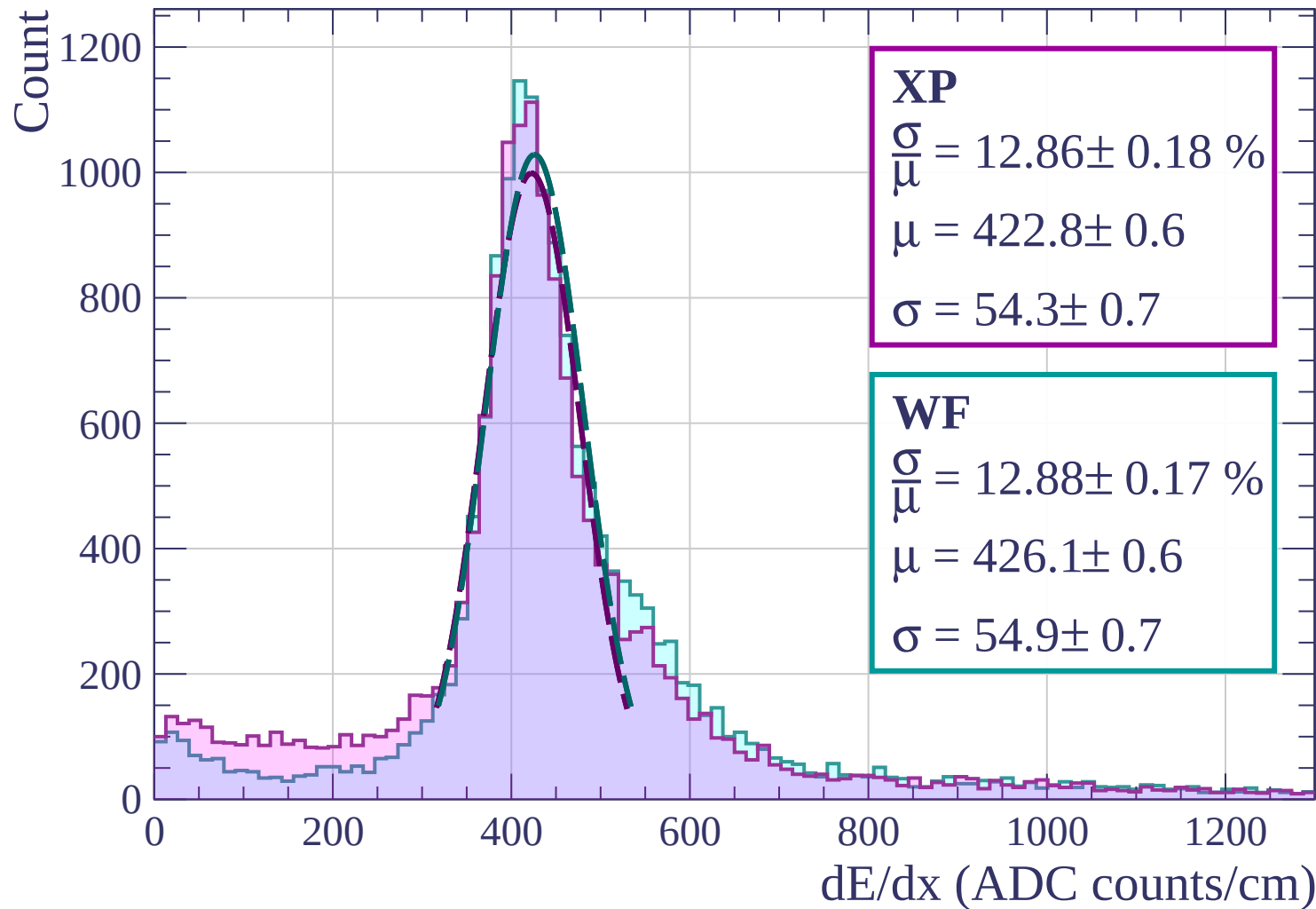
Energy loss | $27 < \phi < 30$



Energy loss | $30 < \phi < 33$



Energy loss | $33 < \phi < 36$



Energy loss | $36 < \phi < 39$

Count

1000

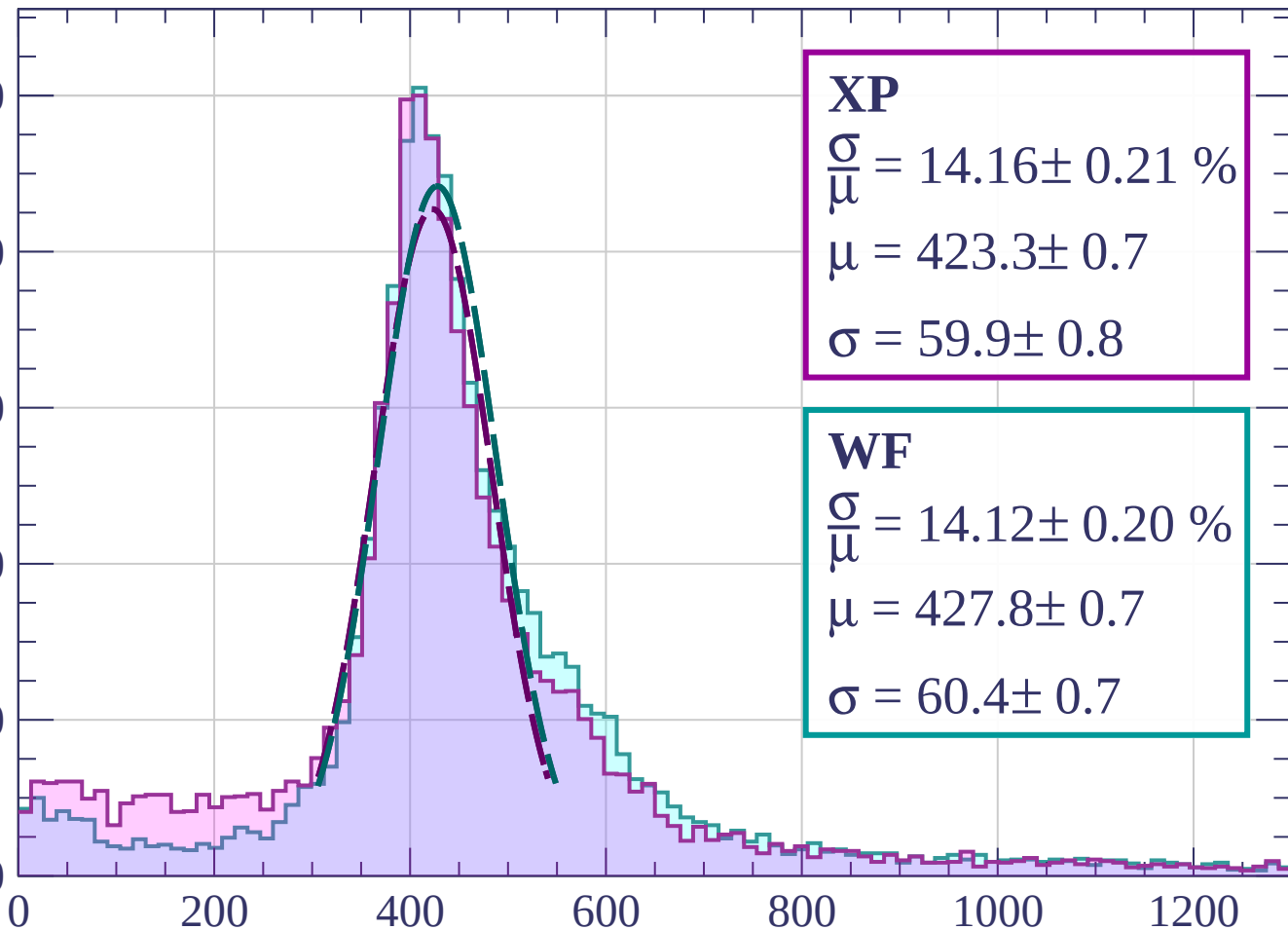
800

600

400

200

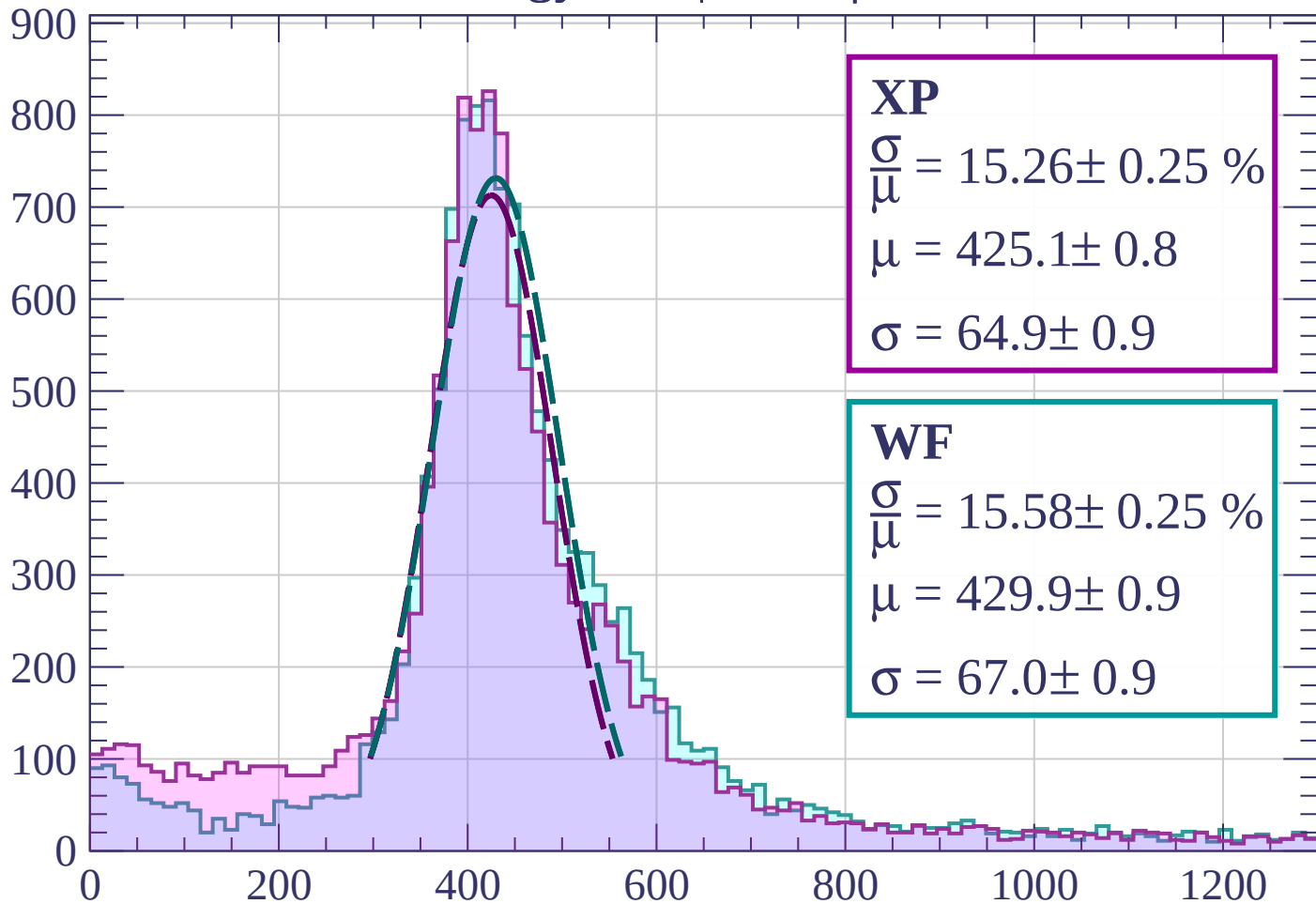
0



dE/dx (ADC counts/cm)

Energy loss | $39 < \phi < 42$

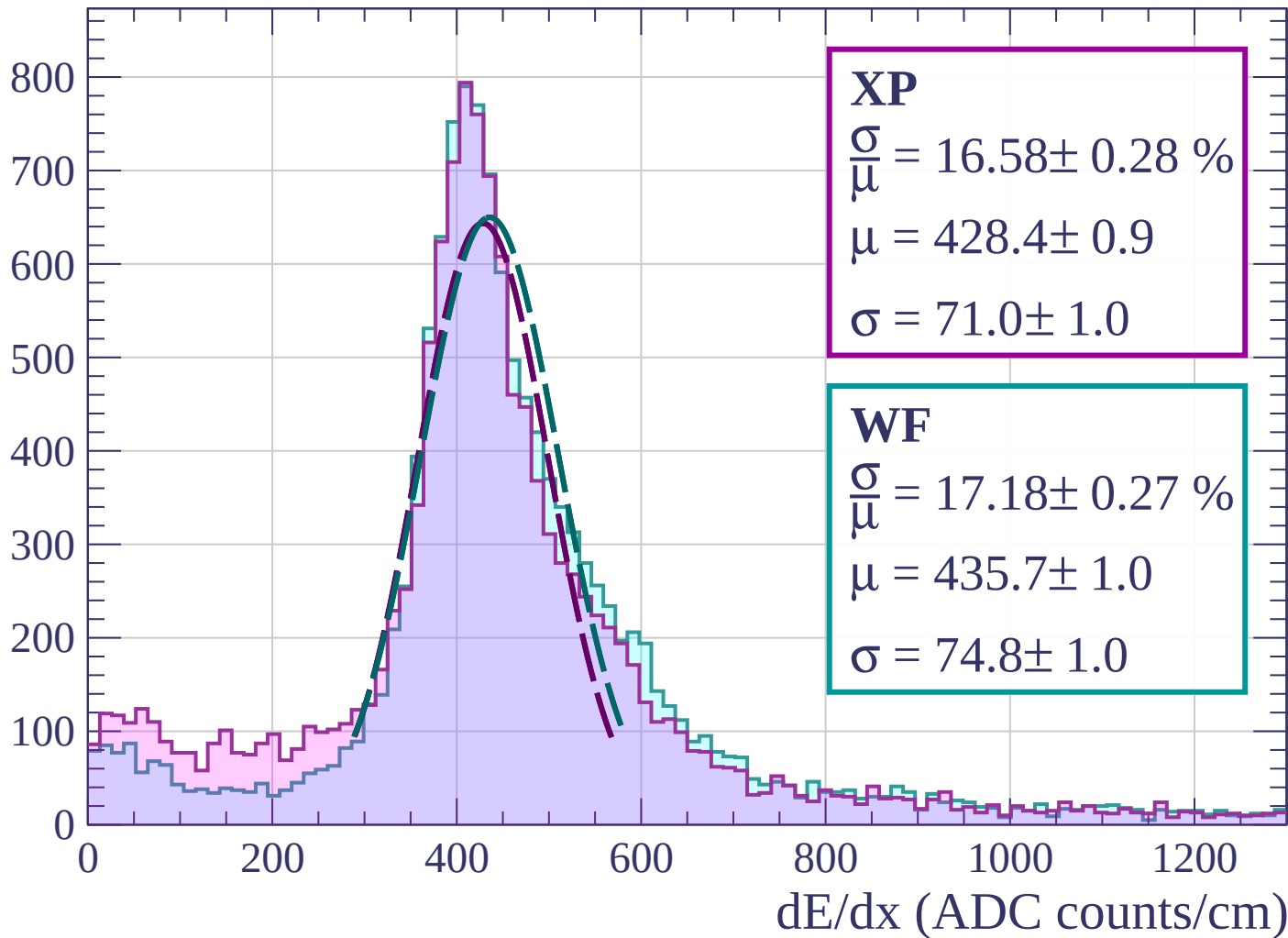
Count



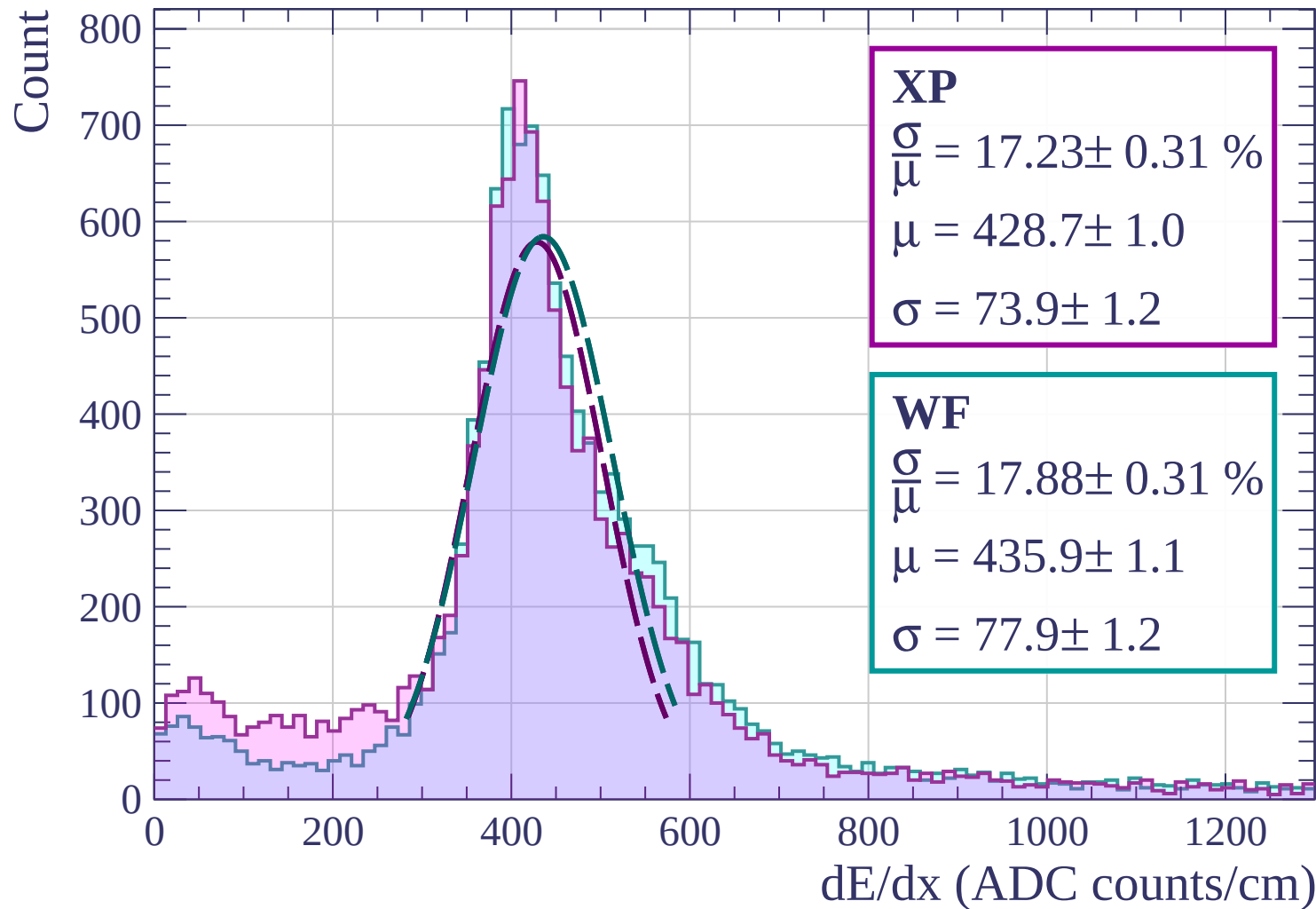
dE/dx (ADC counts/cm)

Energy loss | $42 < \phi < 45$

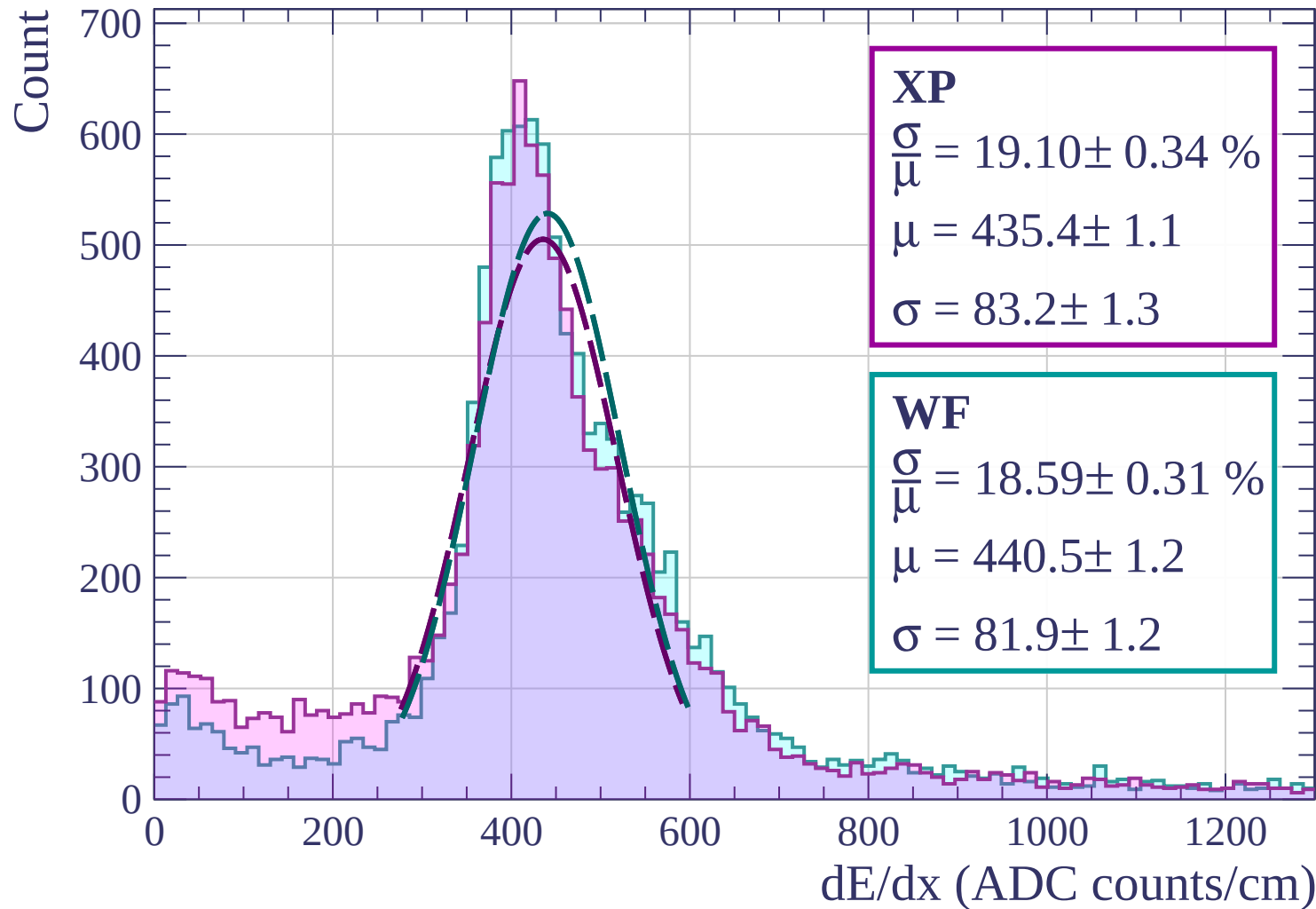
Count



Energy loss | $45 < \phi < 48$

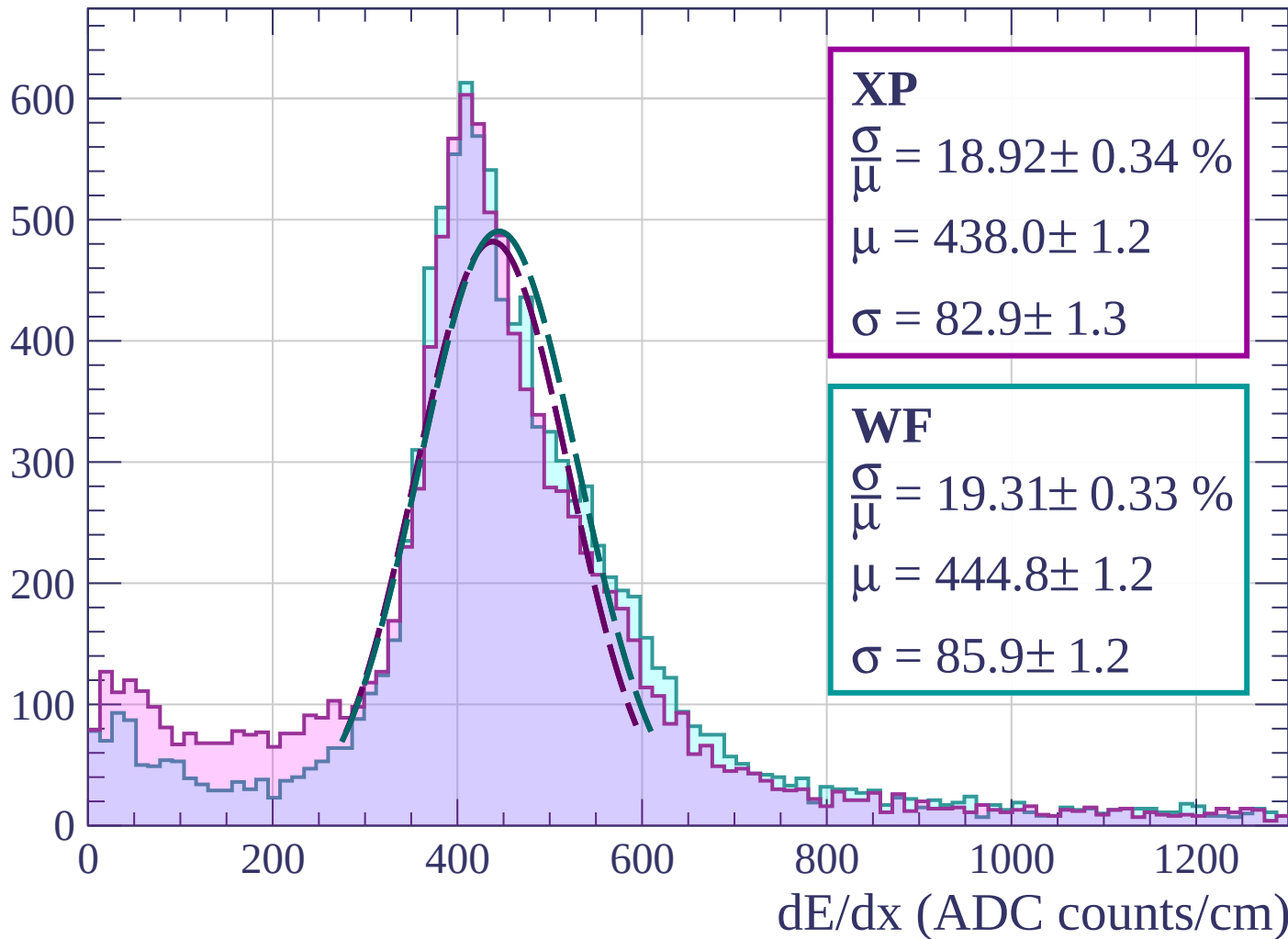


Energy loss | $48 < \phi < 51$



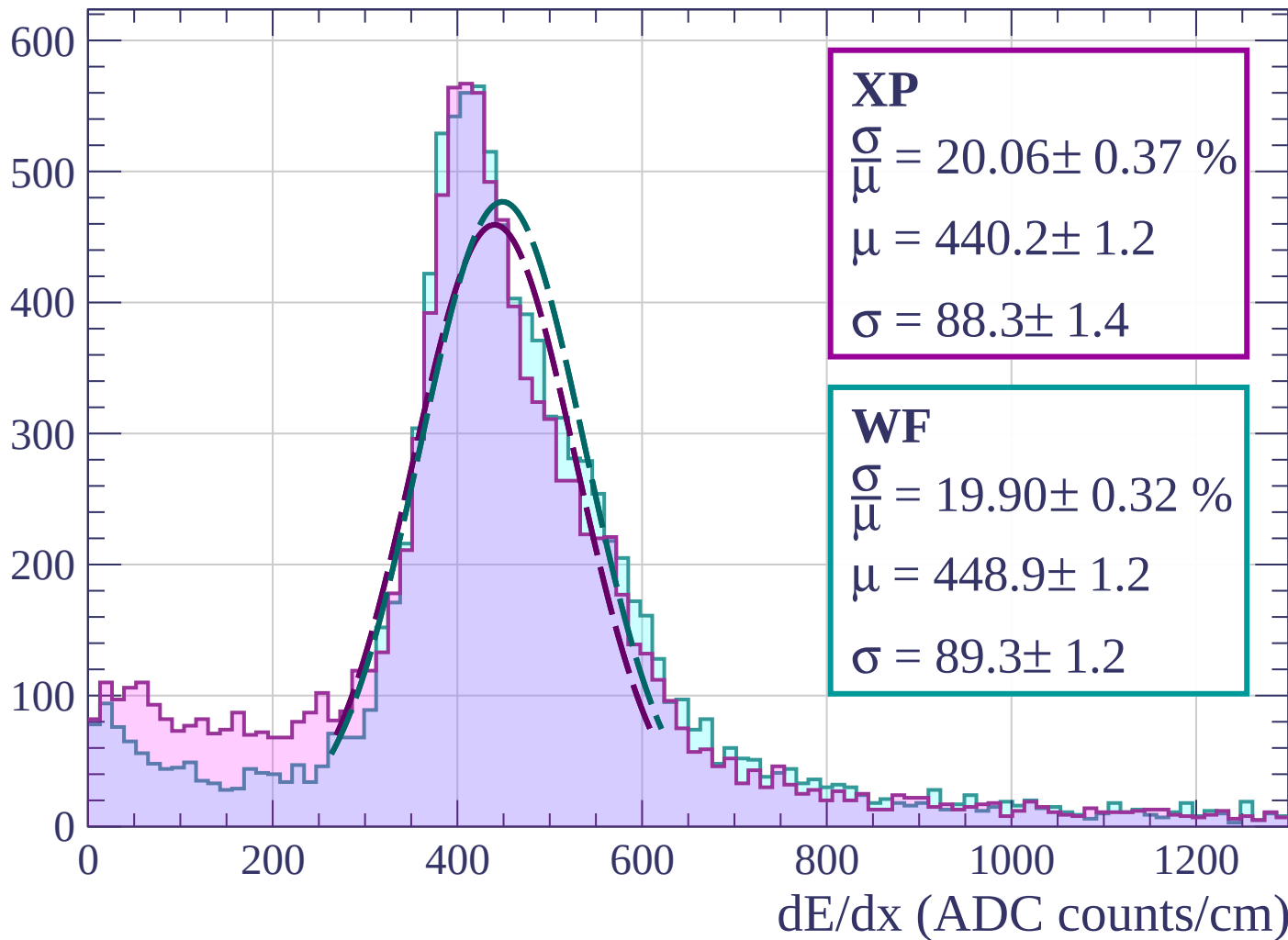
Energy loss | $51 < \phi < 54$

Count



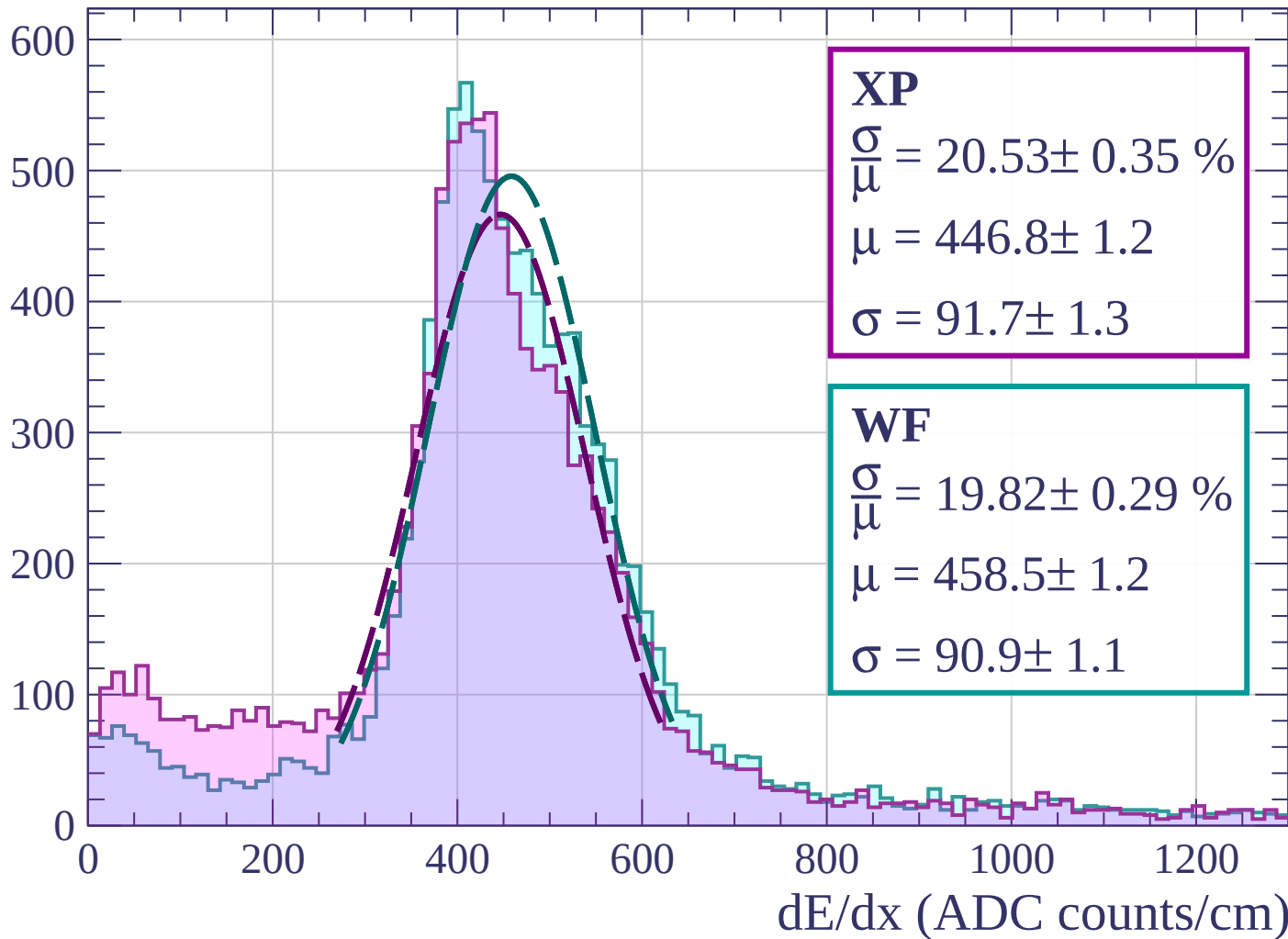
Energy loss | $54 < \phi < 57$

Count



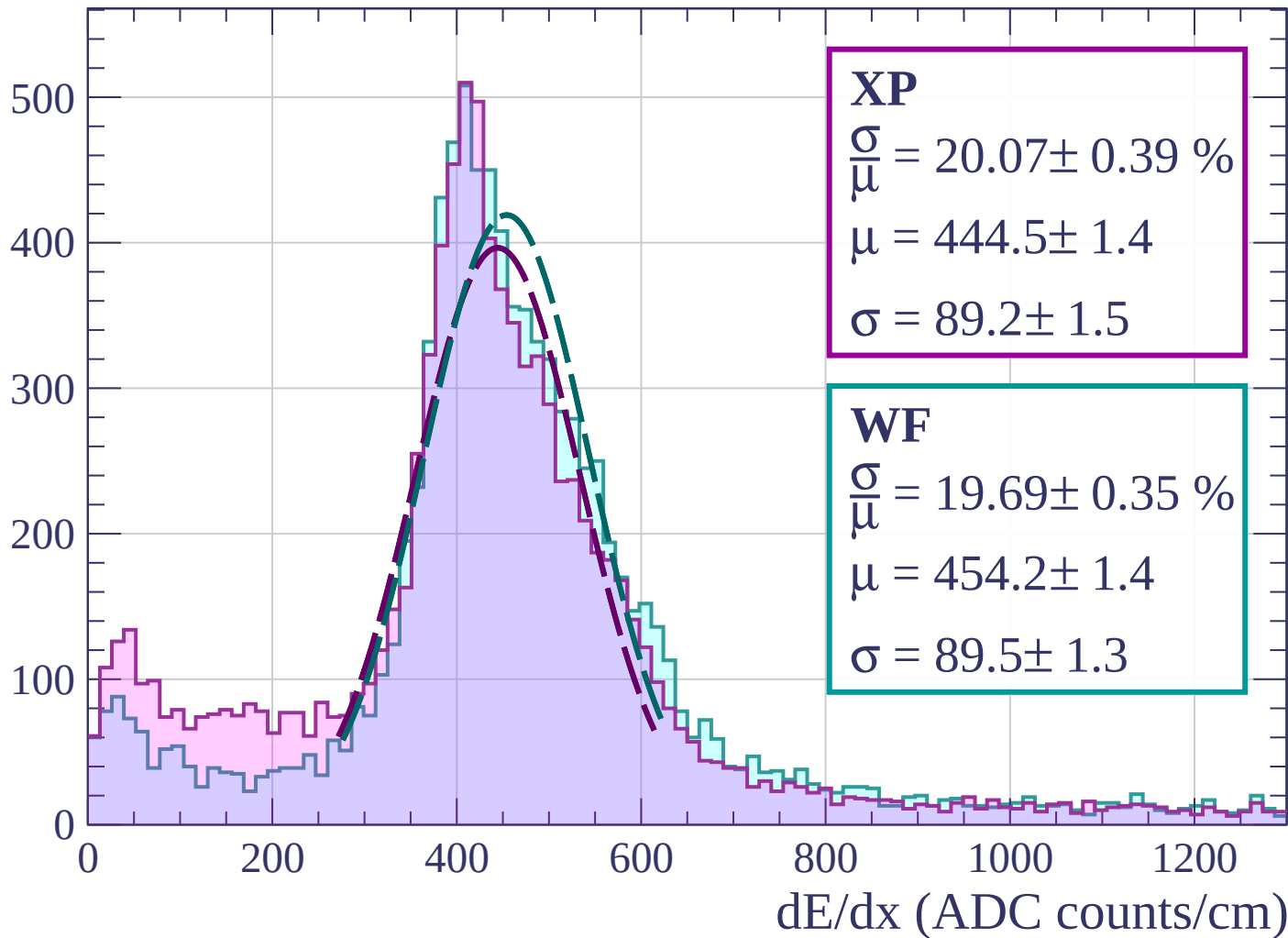
Energy loss | $57 < \phi < 60$

Count

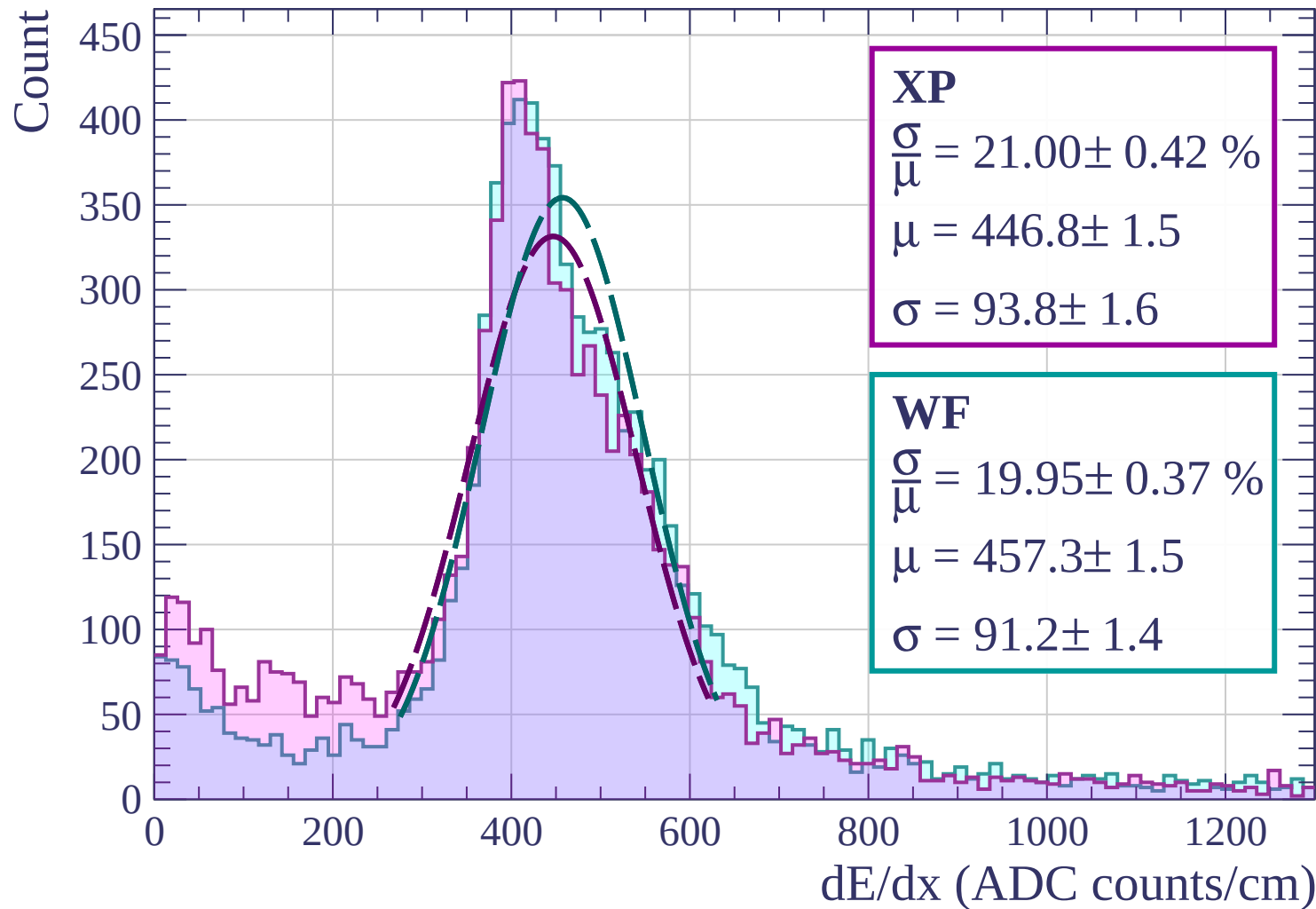


Energy loss | $60 < \phi < 63$

Count

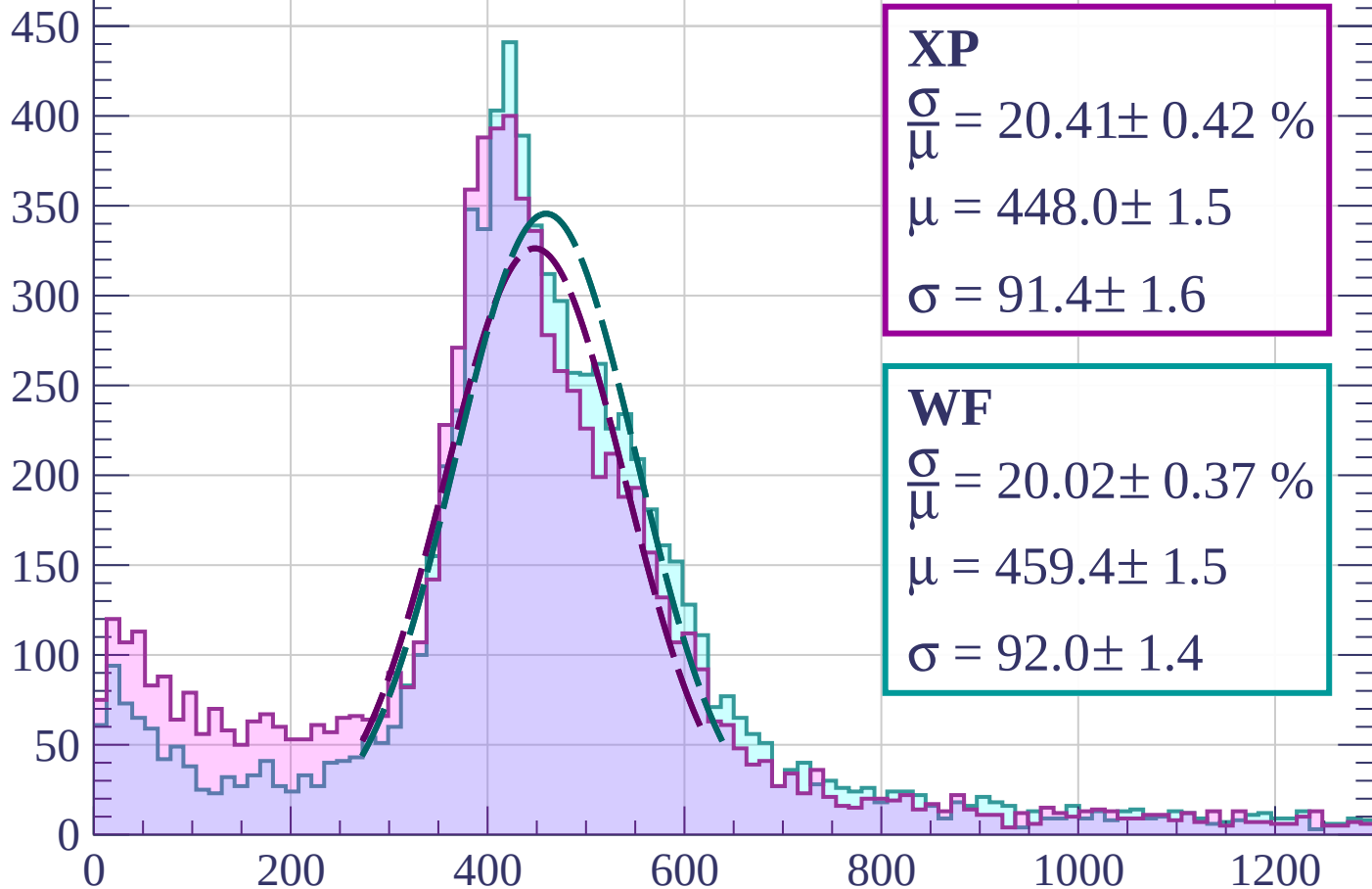


Energy loss | $63 < \phi < 66$



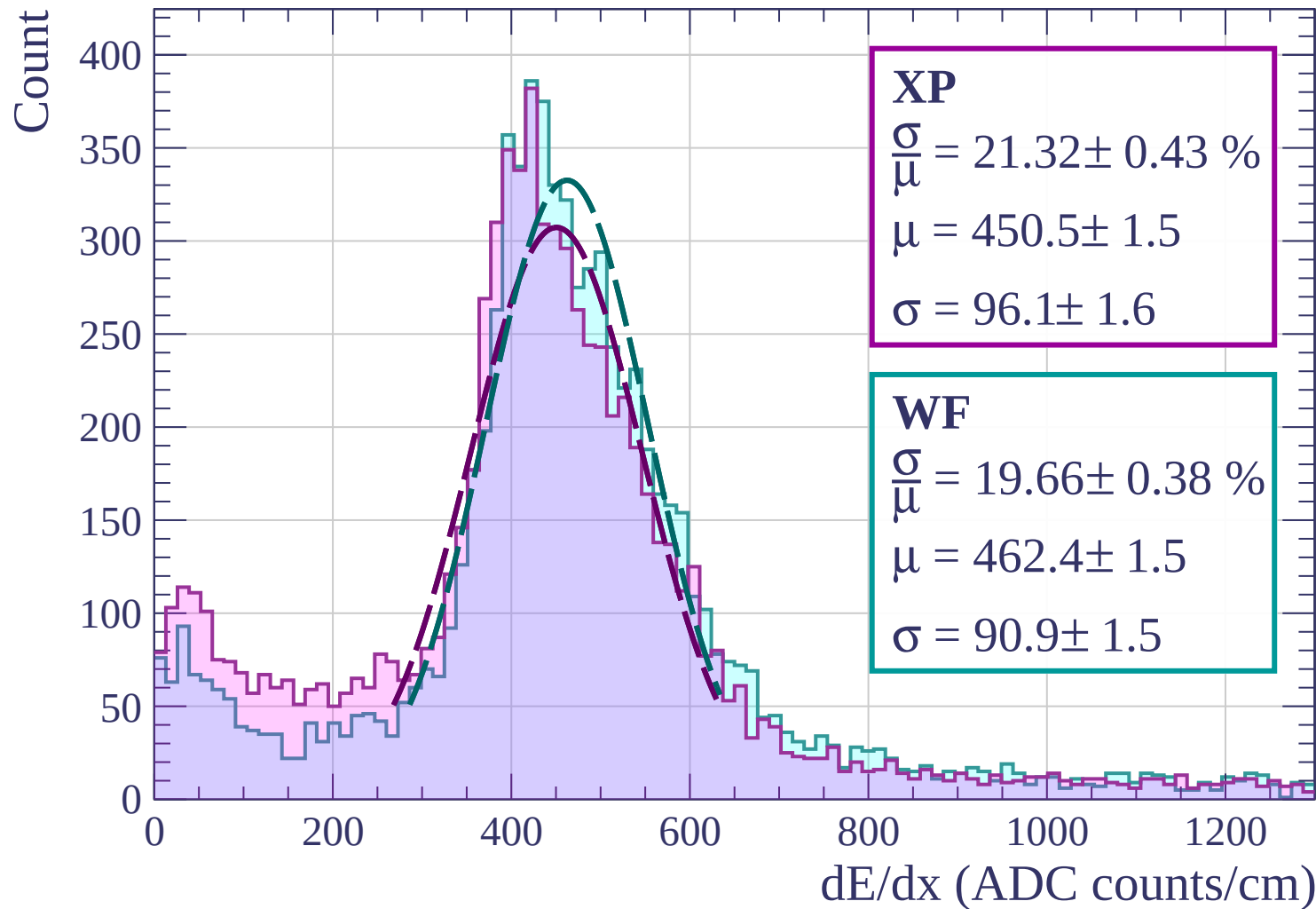
Energy loss | $66 < \phi < 69$

Count



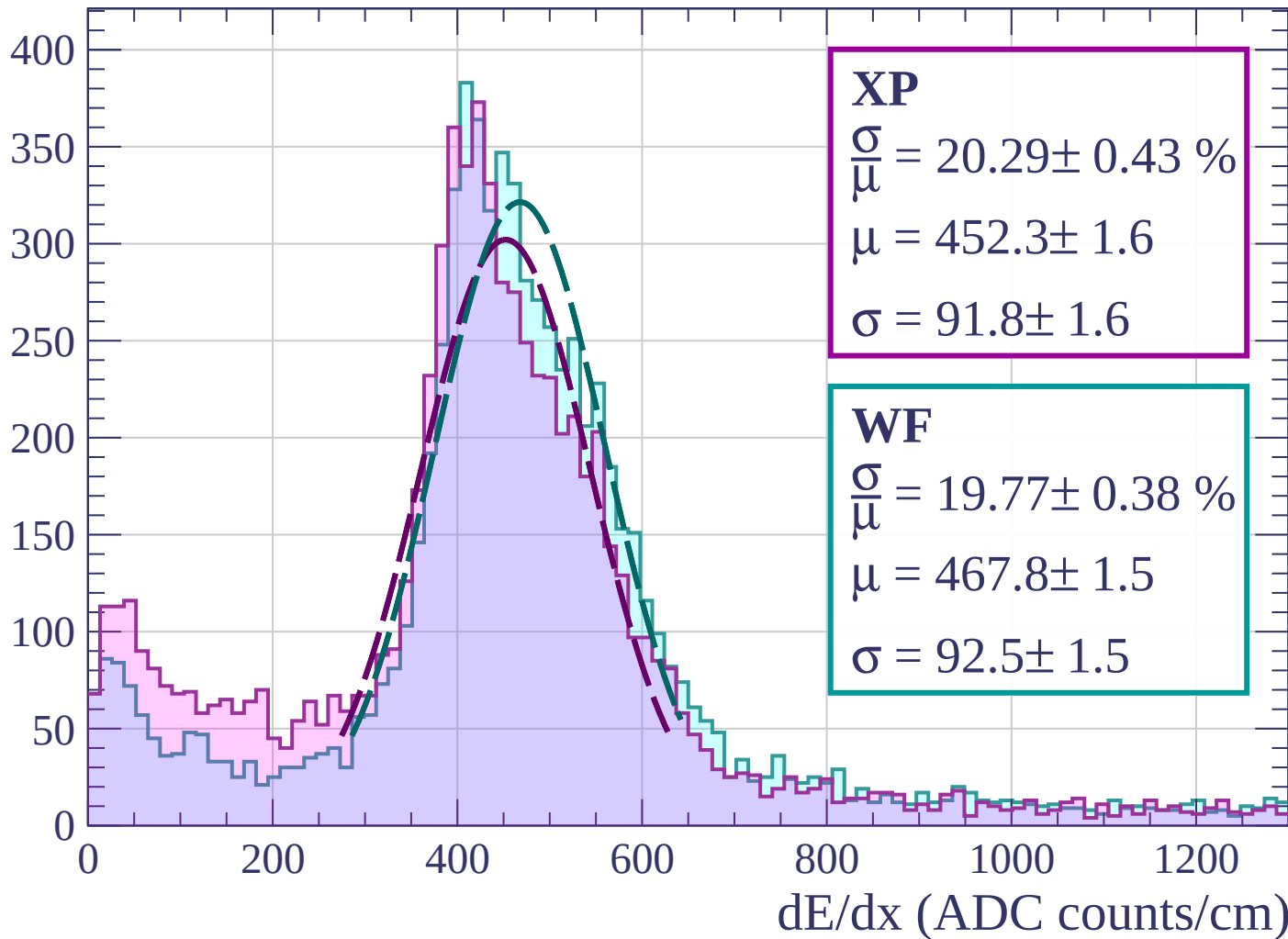
dE/dx (ADC counts/cm)

Energy loss | $69 < \phi < 72$



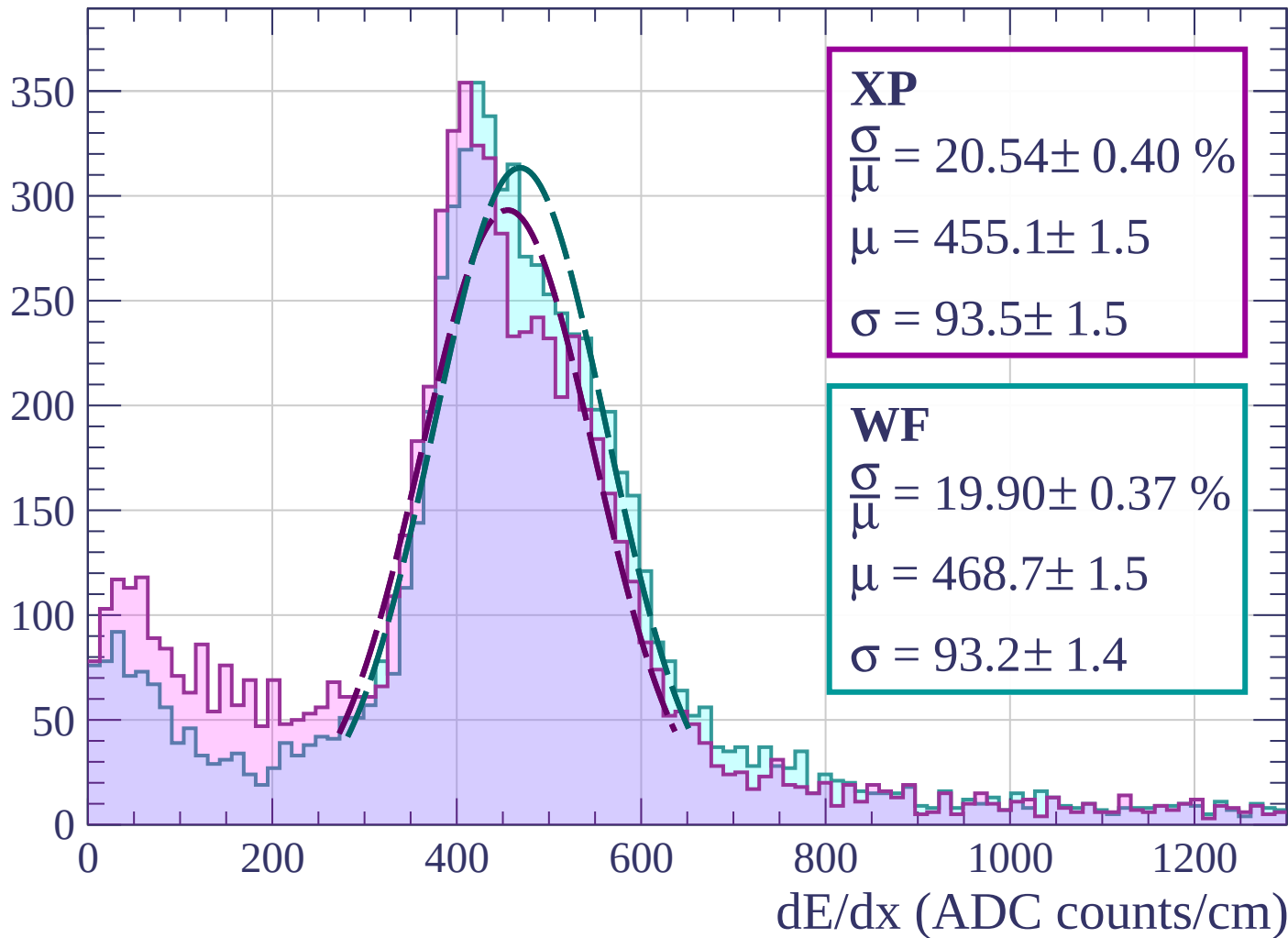
Energy loss | $72 < \phi < 75$

Count



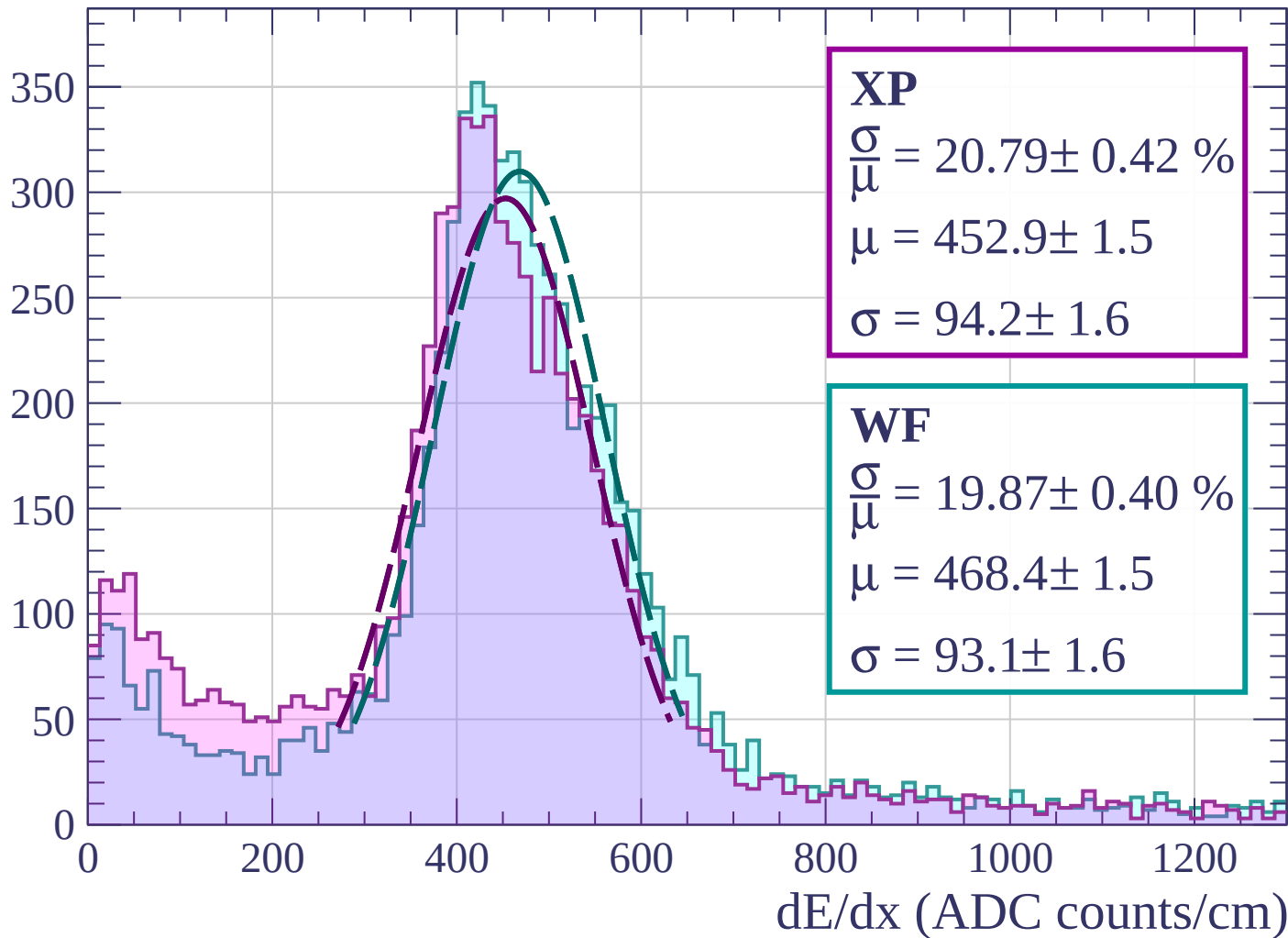
Energy loss | $75 < \phi < 78$

Count



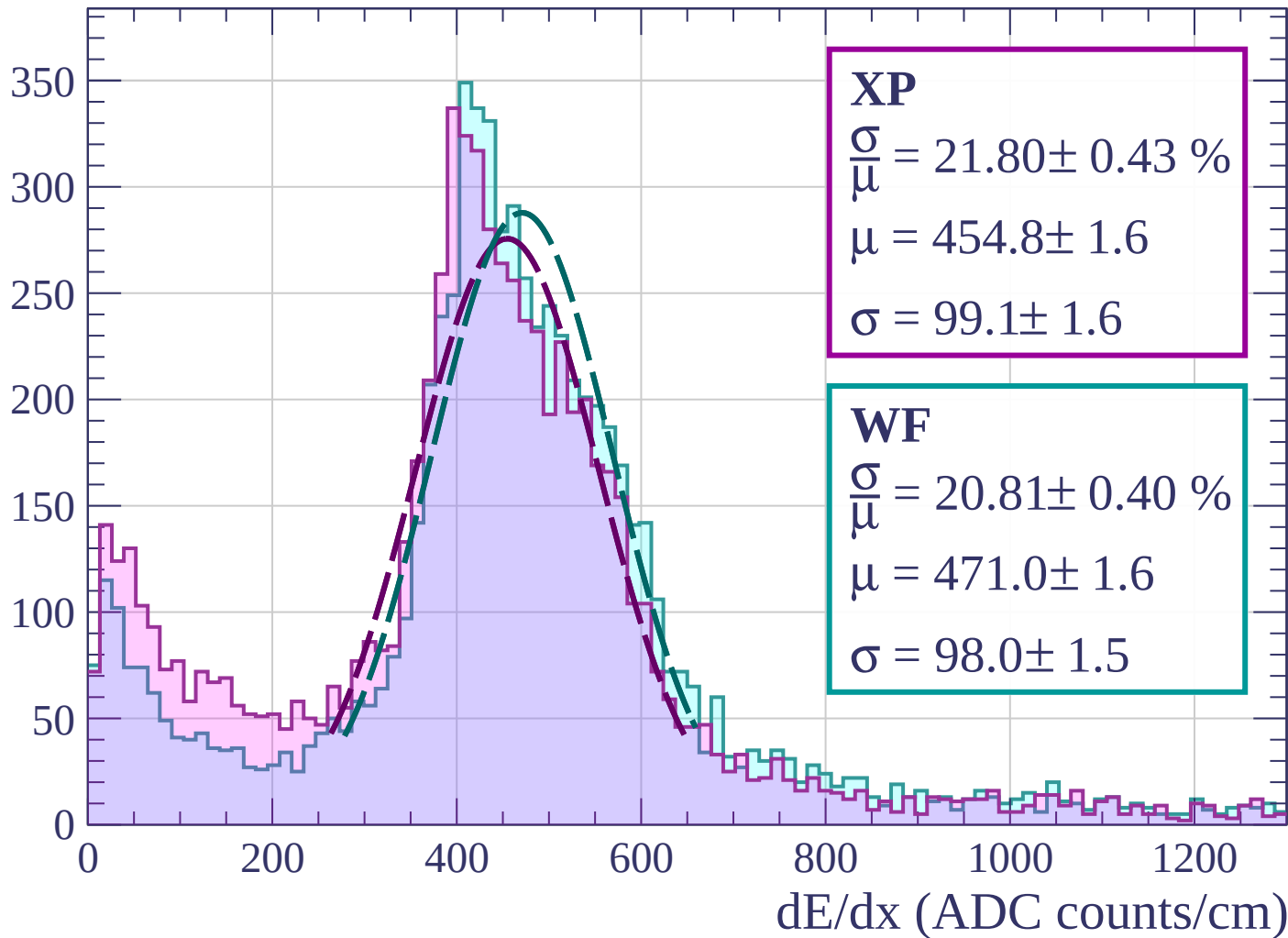
Energy loss | $78 < \phi < 81$

Count



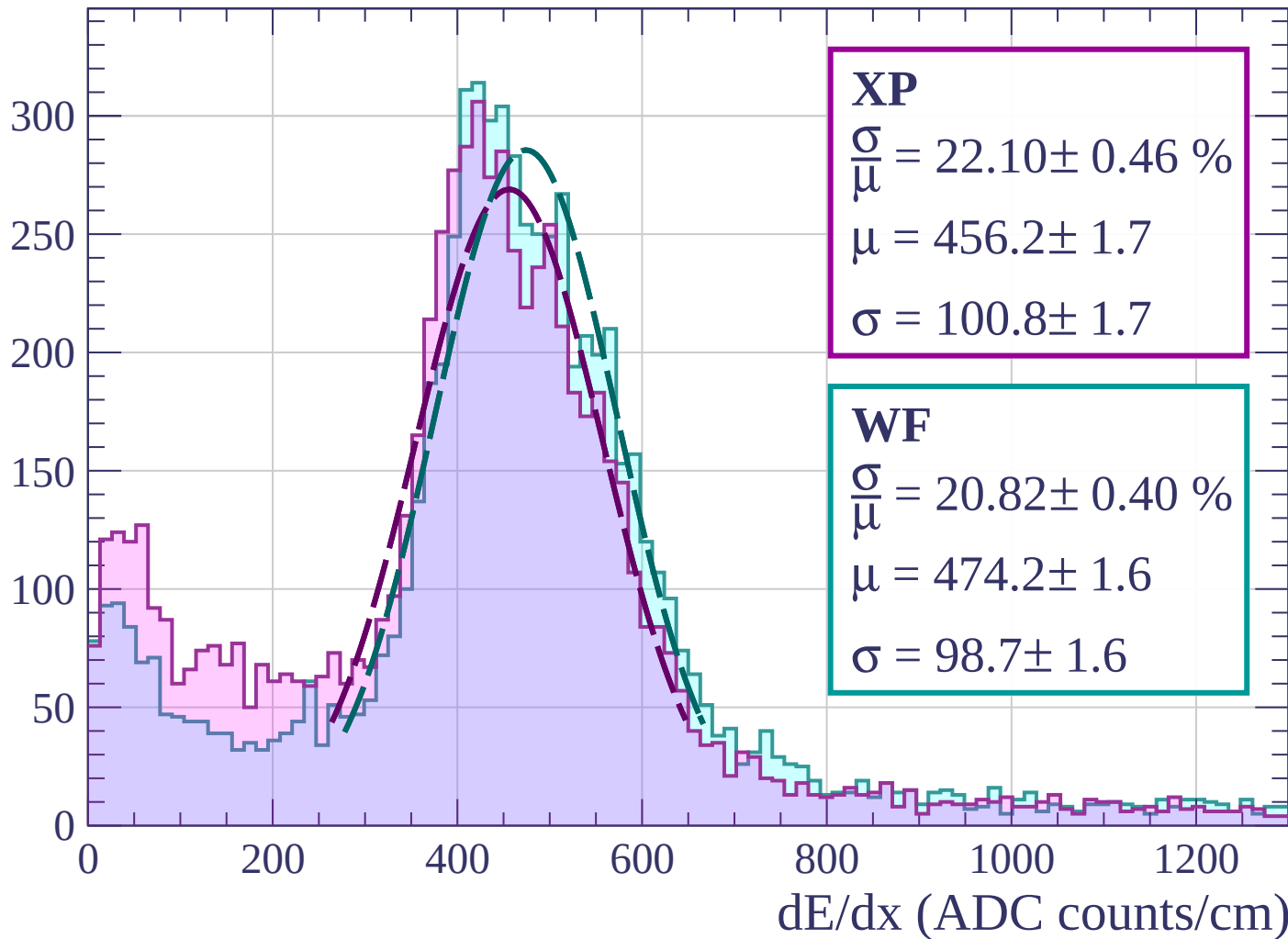
Energy loss | $81 < \phi < 84$

Count



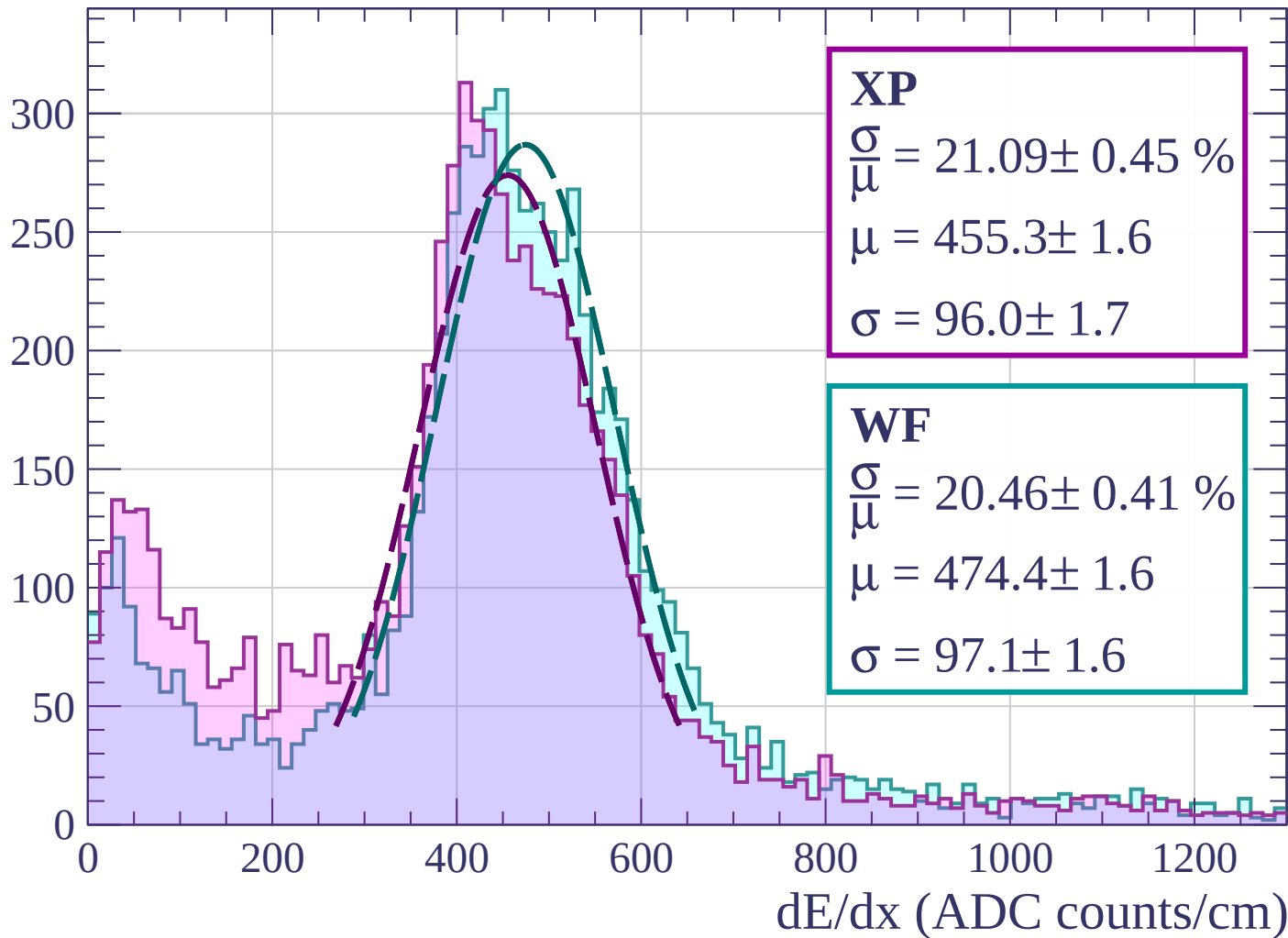
Energy loss | $84 < \phi < 87$

Count

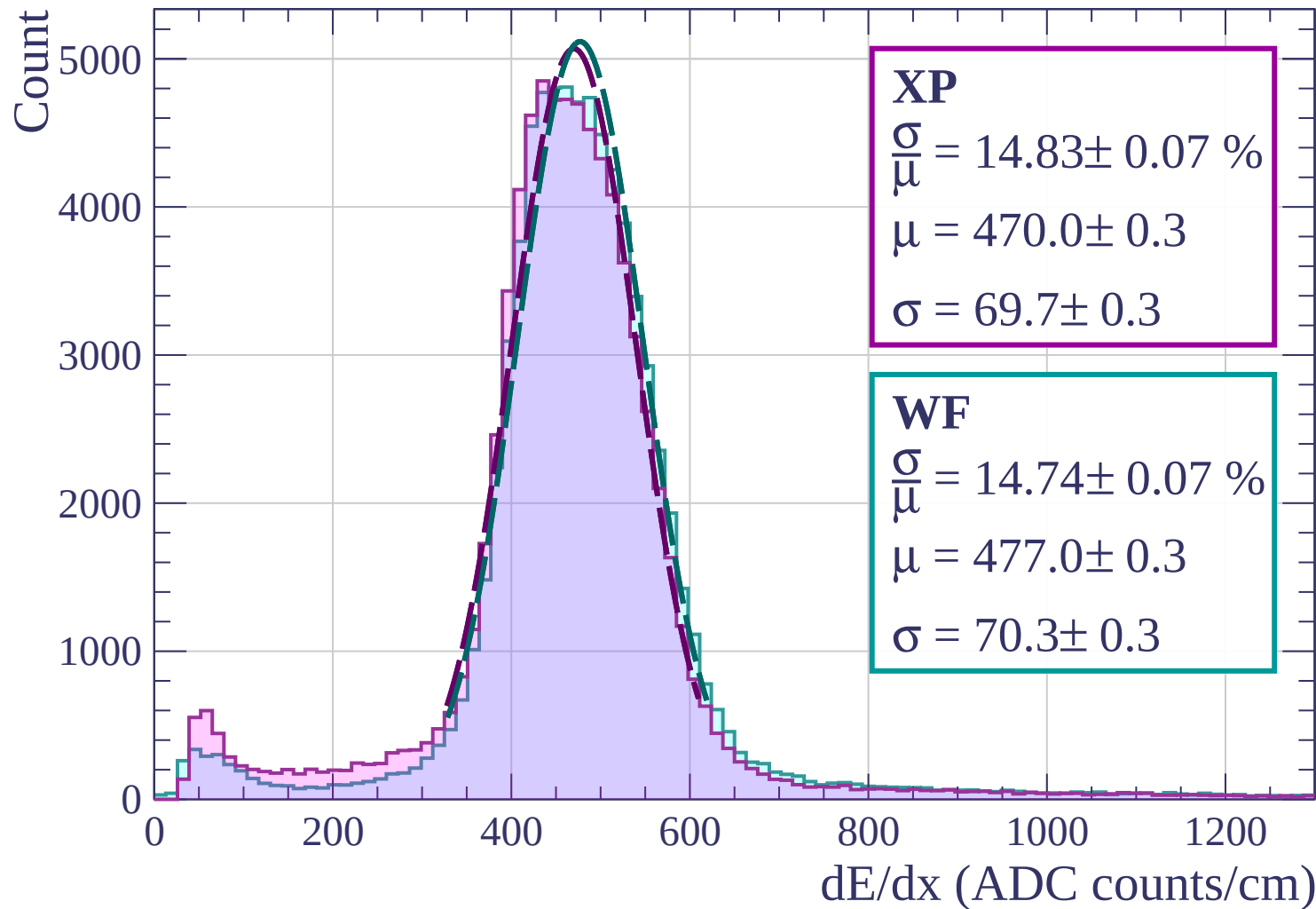


Energy loss | $87 < \phi < 90$

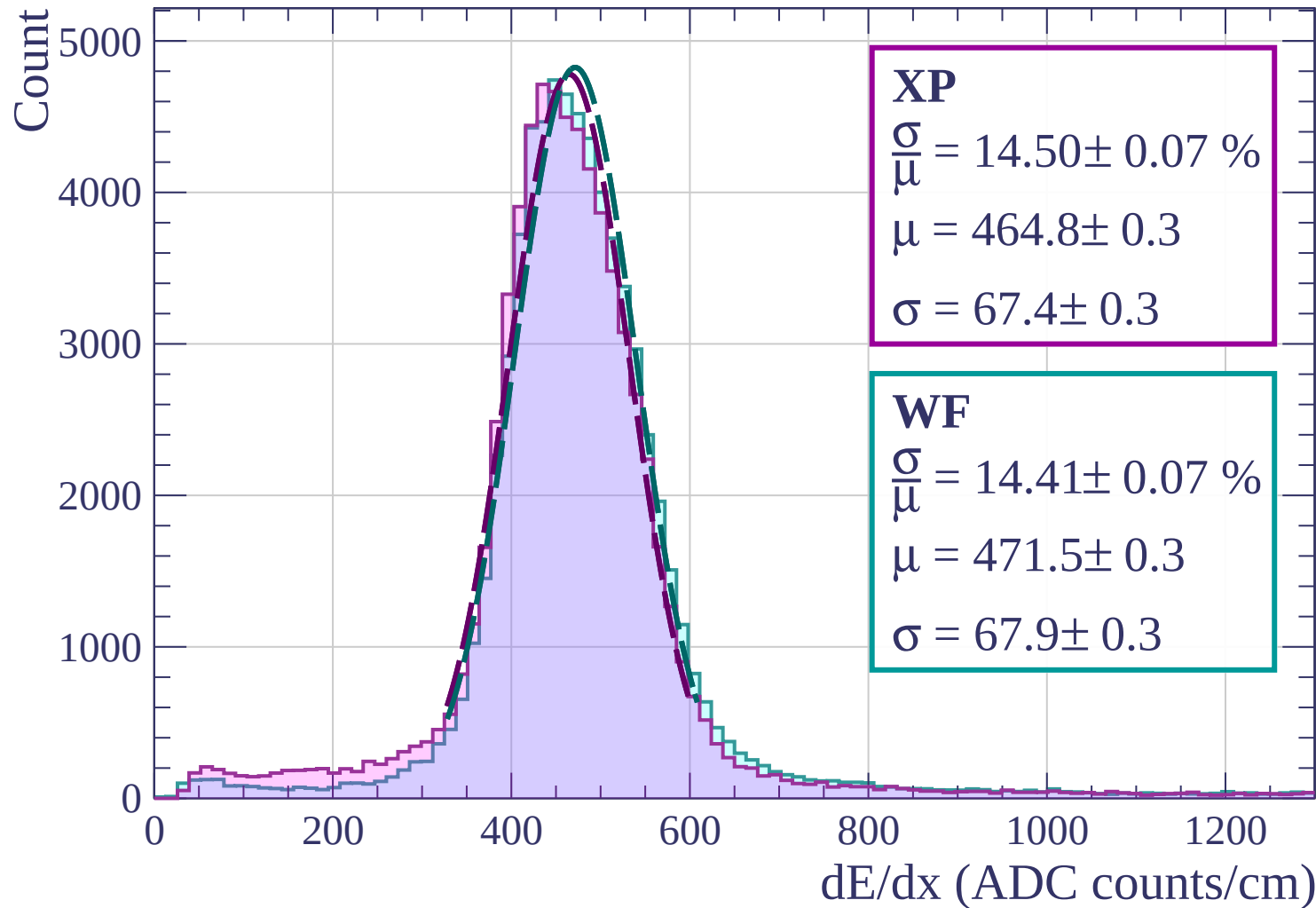
Count



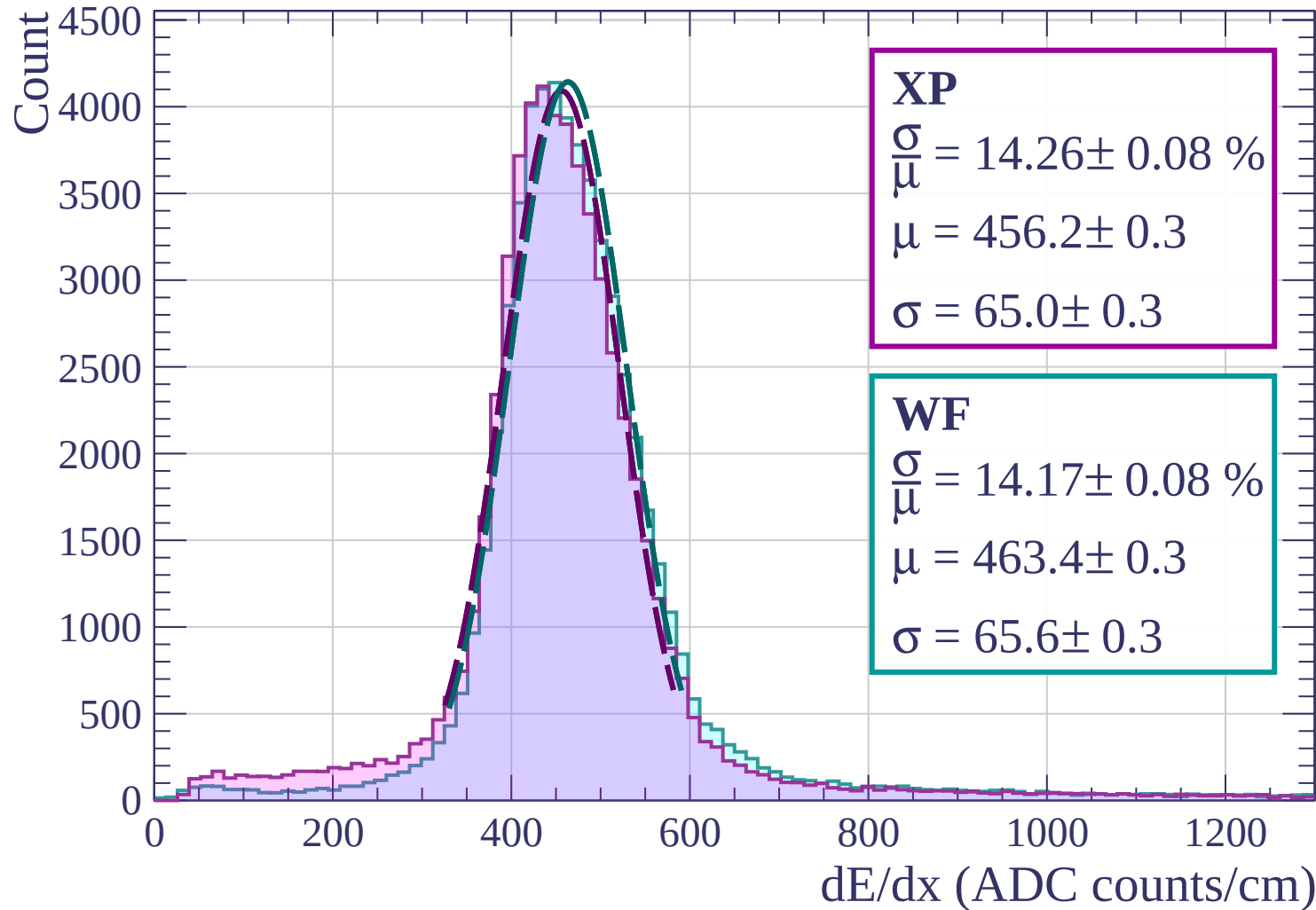
Energy loss | $0 < \theta < 3$



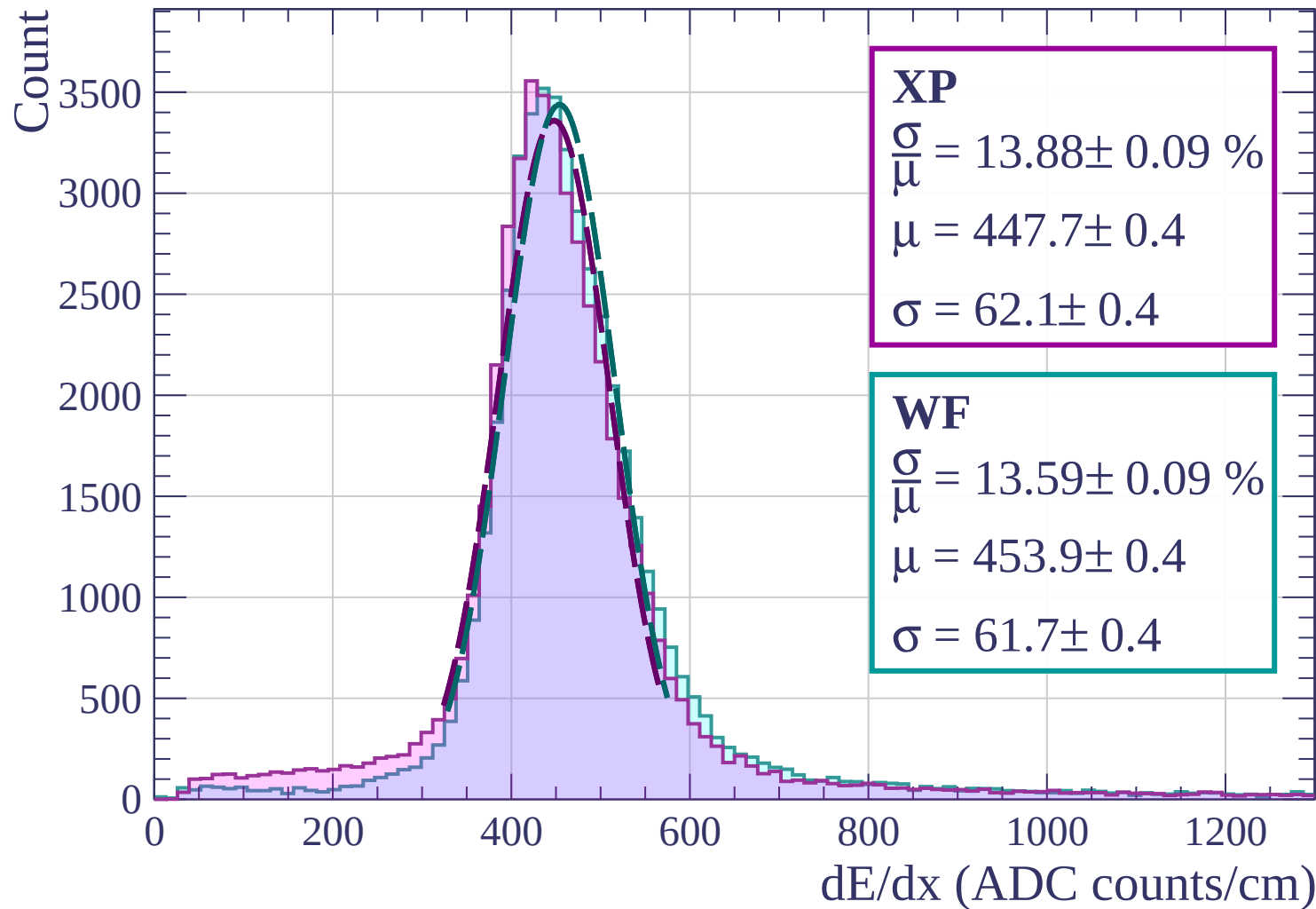
Energy loss | $3 < \theta < 6$



Energy loss | $6 < \theta < 9$



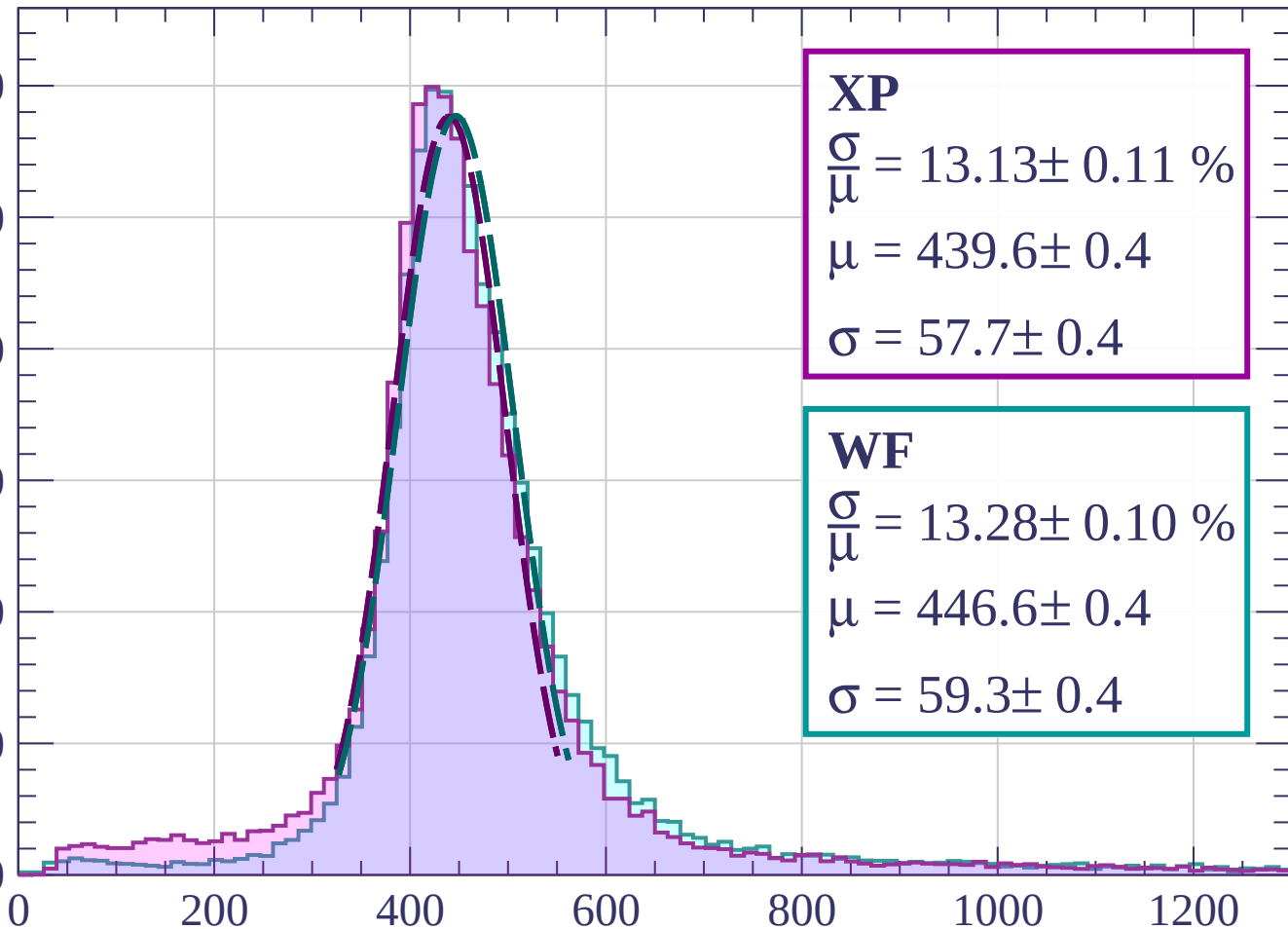
Energy loss | $9 < \theta < 12$



Energy loss | $12 < \theta < 15$

Count

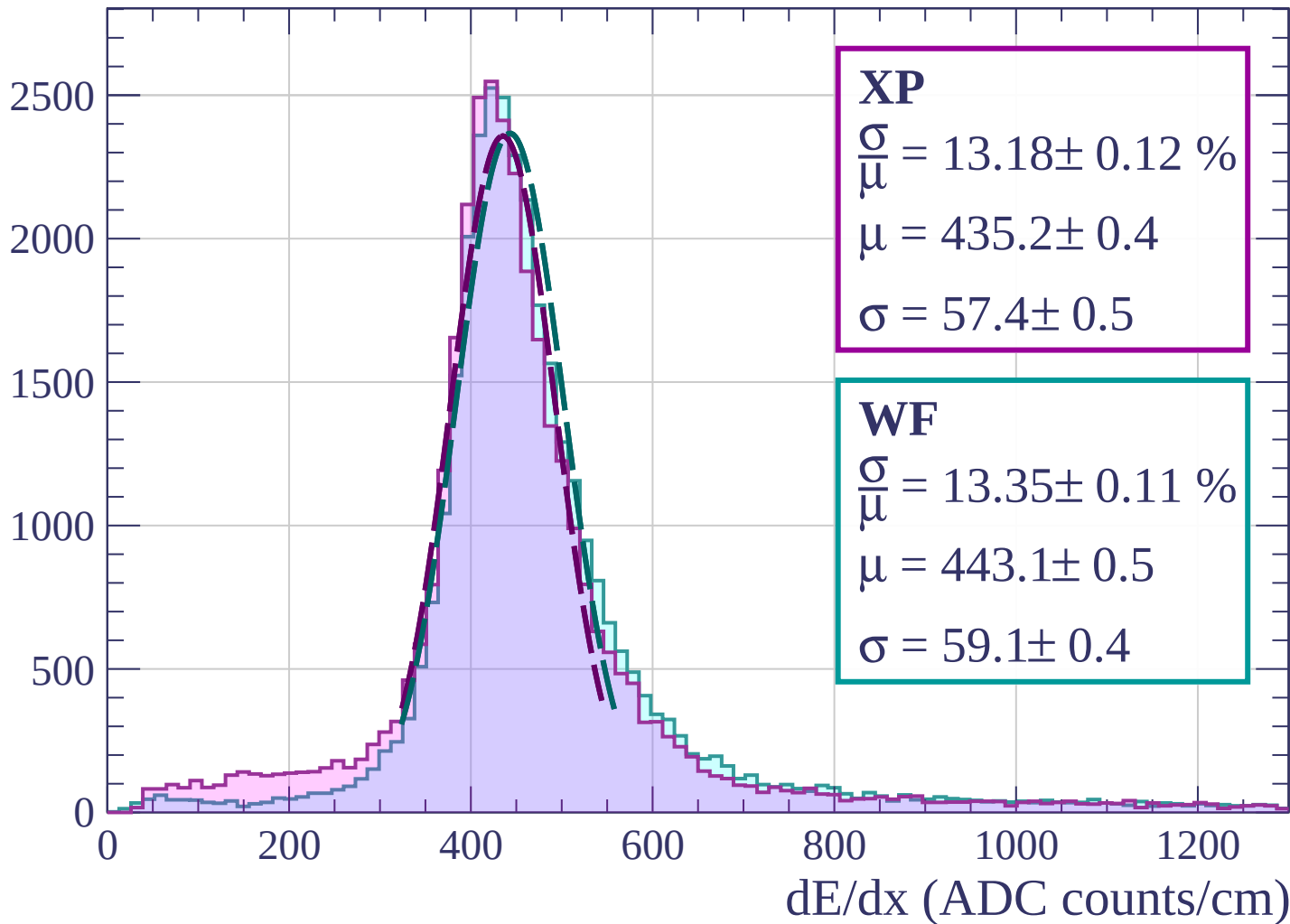
3000
2500
2000
1500
1000
500
0



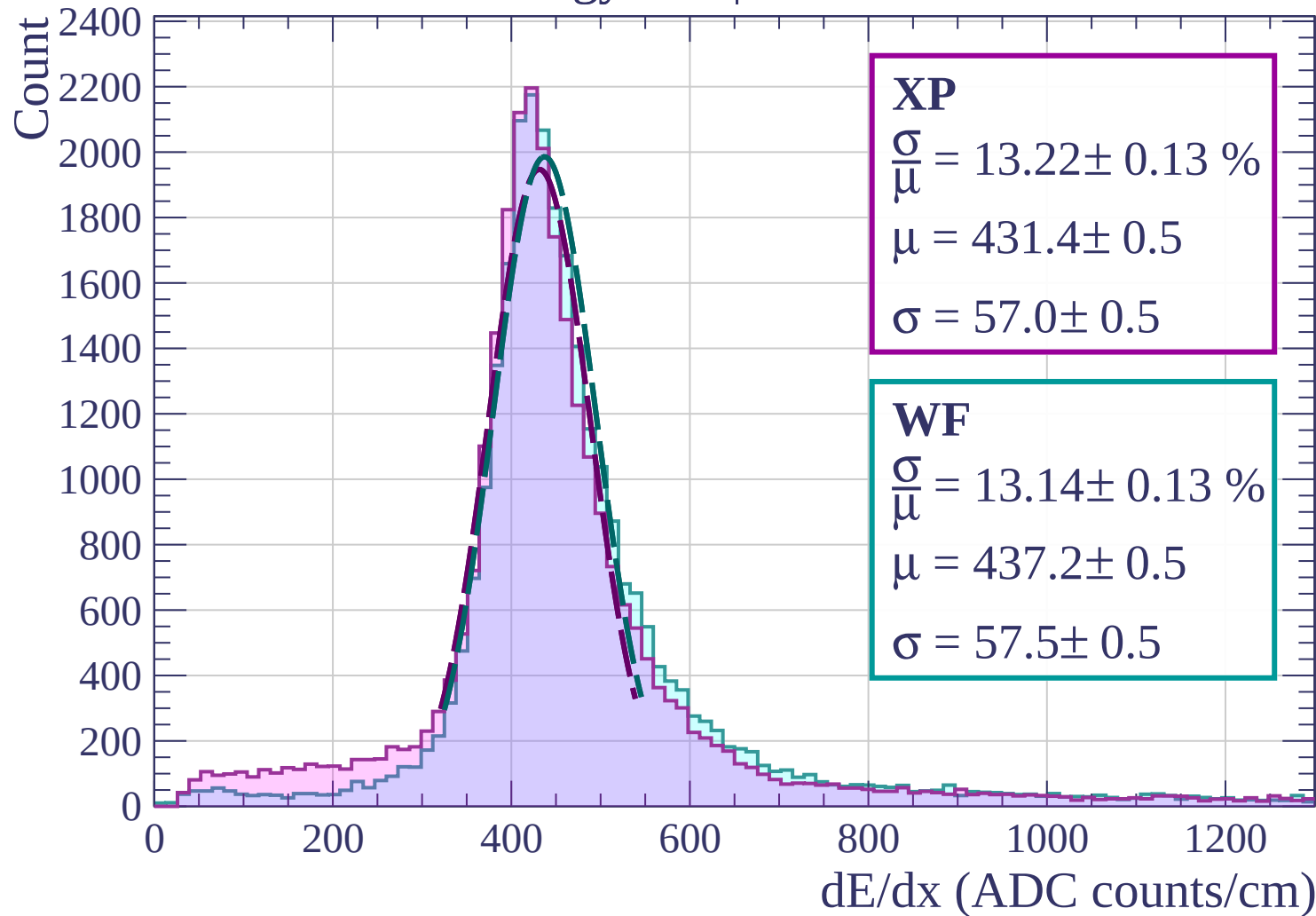
dE/dx (ADC counts/cm)

Energy loss | $15 < \theta < 18$

Count

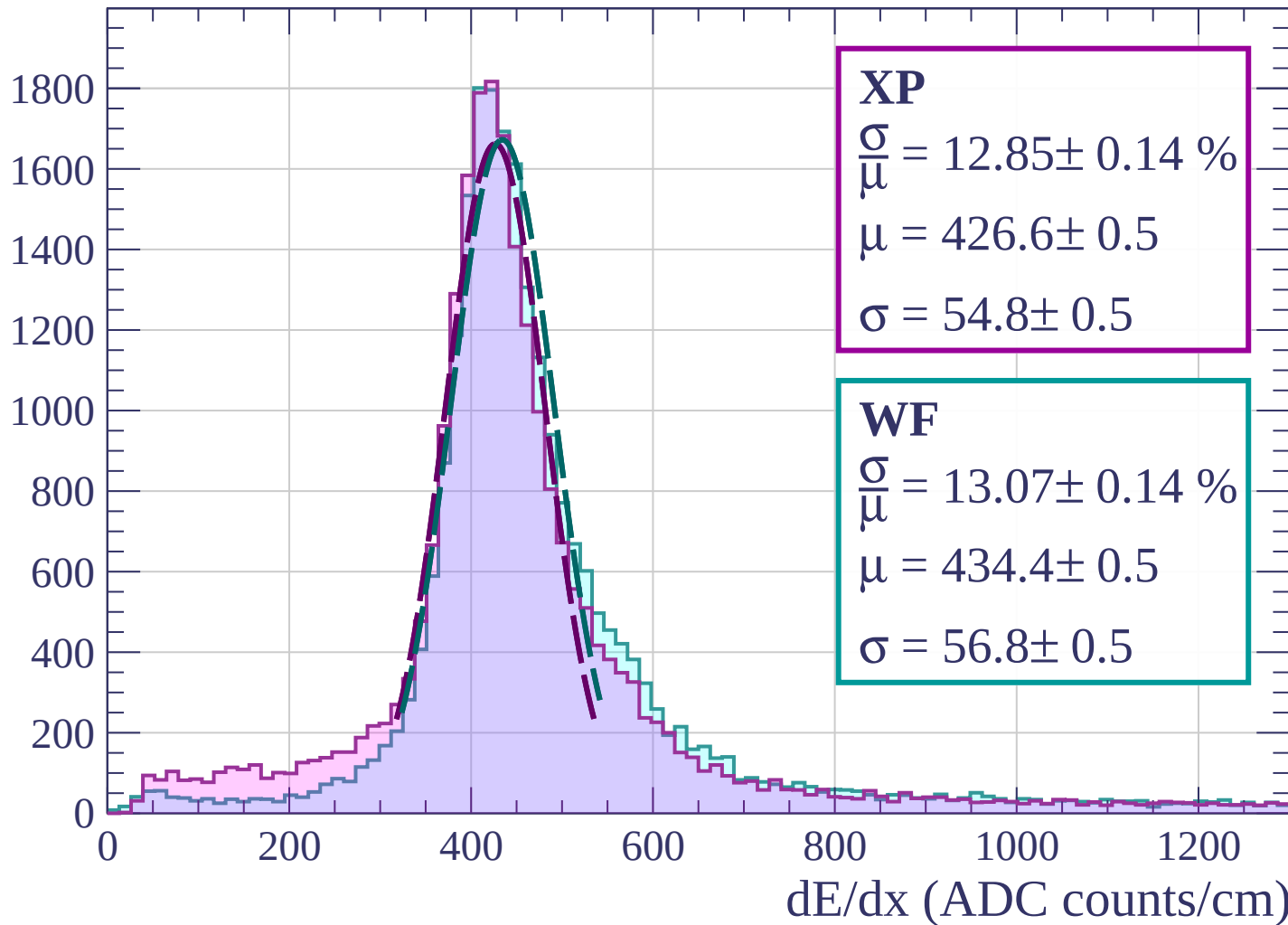


Energy loss | $18 < \theta < 21$

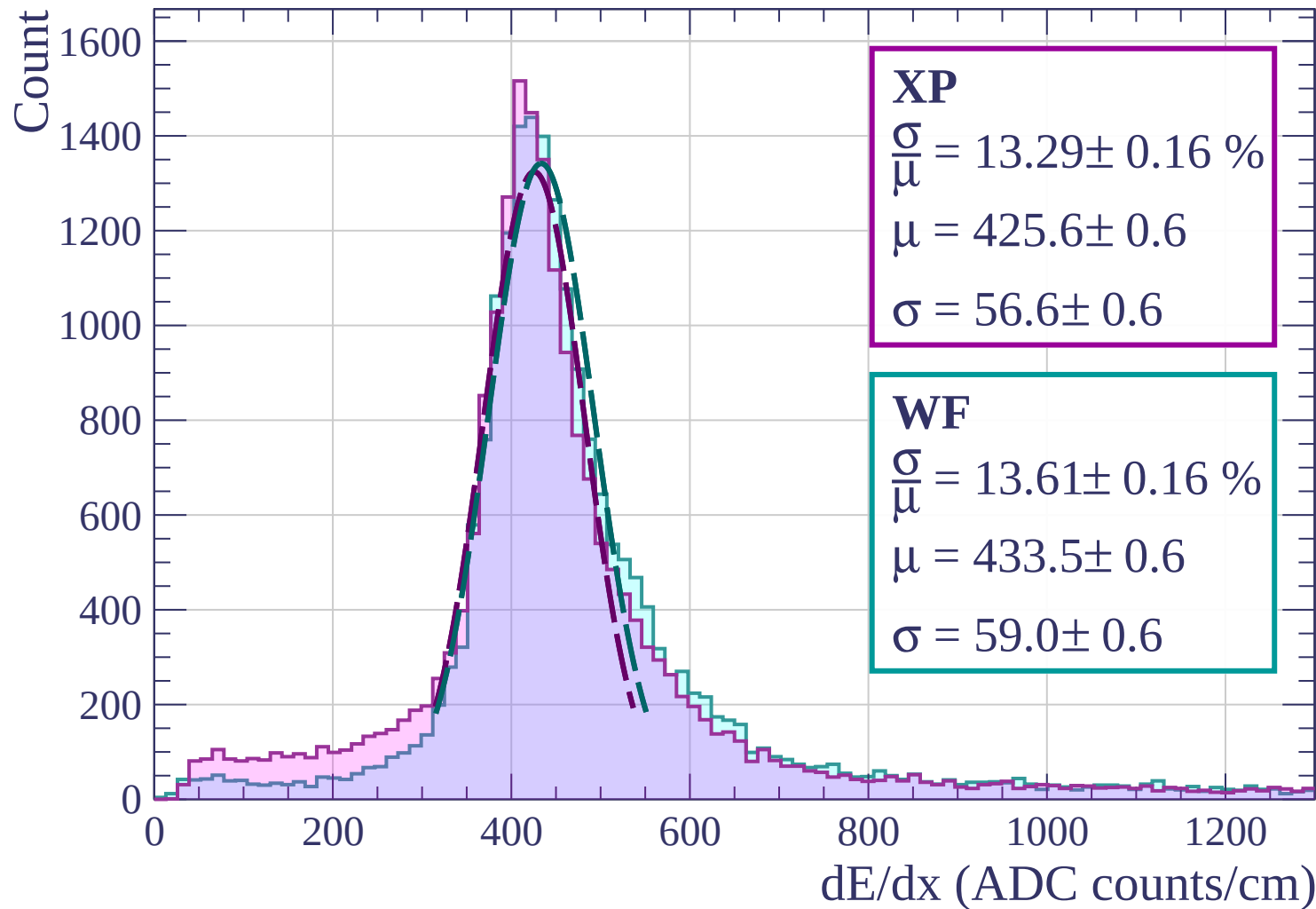


Energy loss | $21 < \theta < 24$

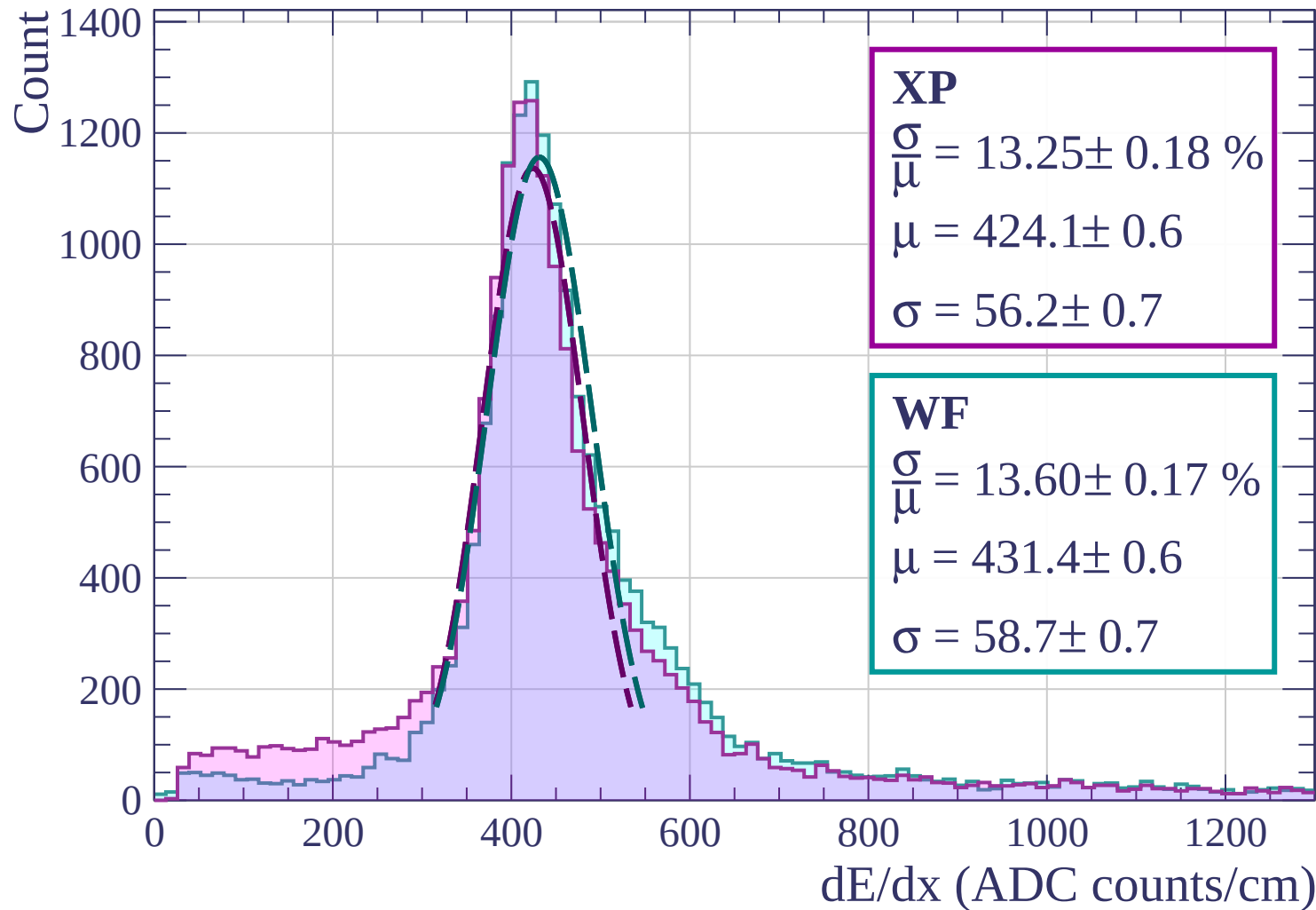
Count



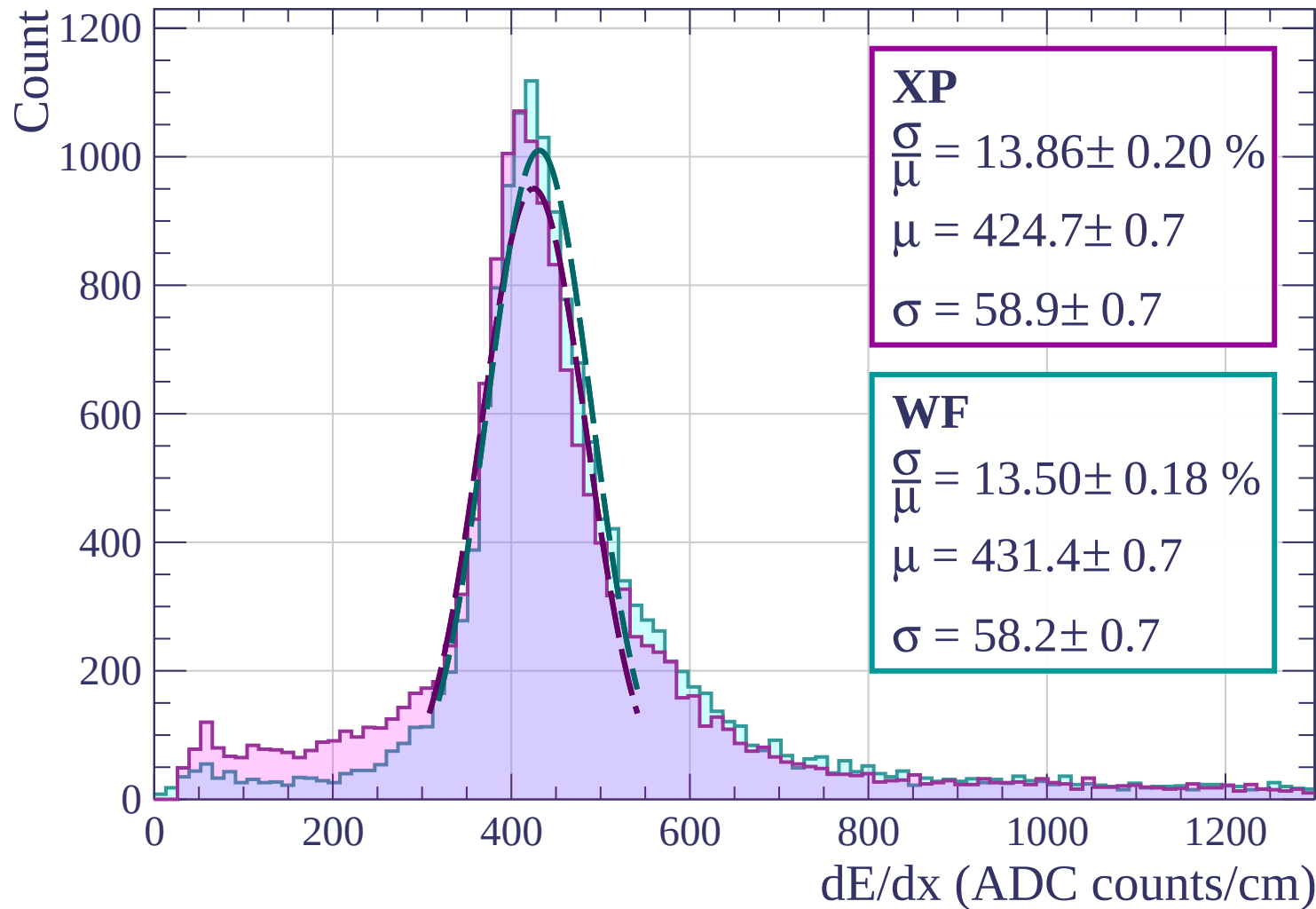
Energy loss | $24 < \theta < 27$



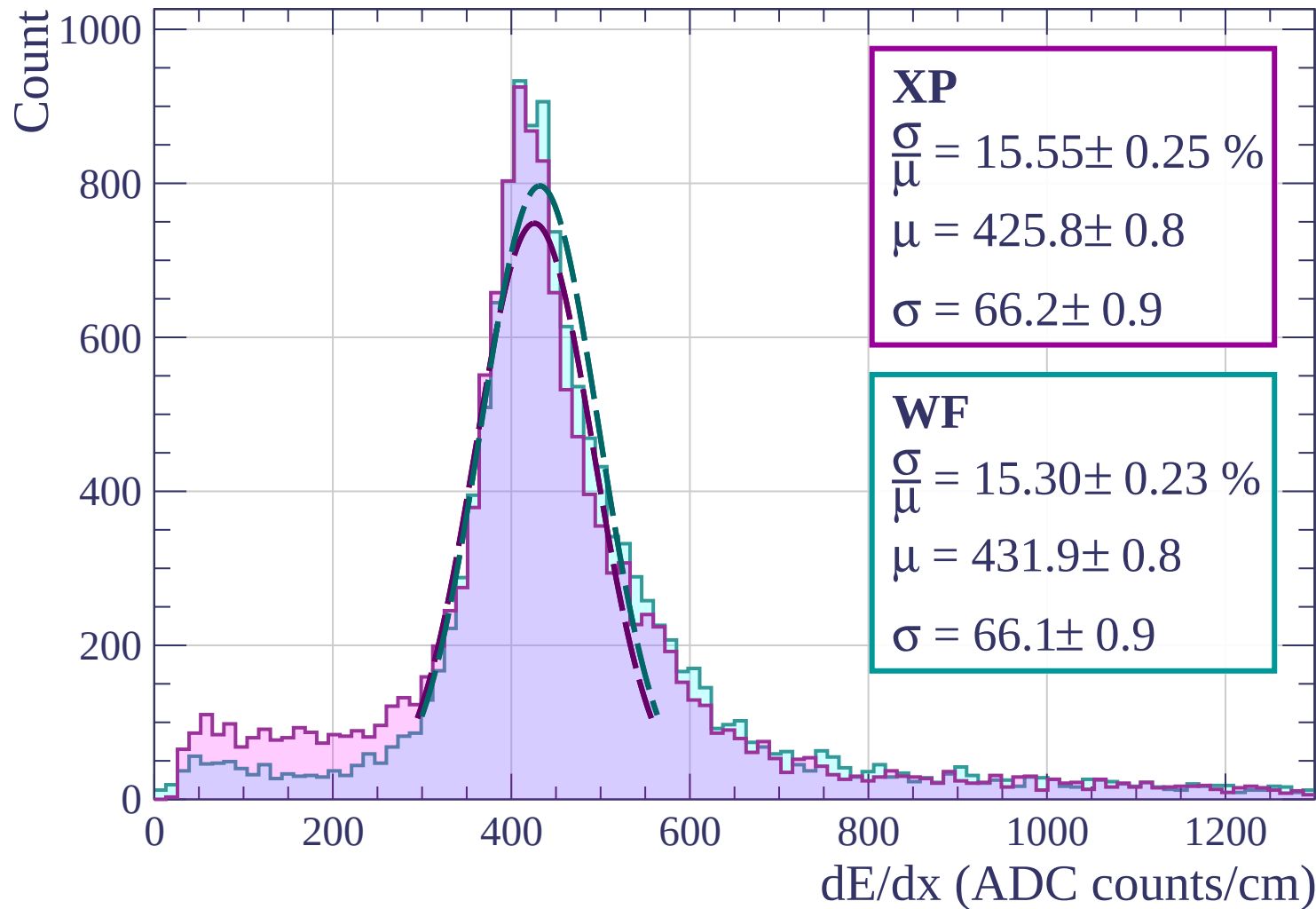
Energy loss | $27 < \theta < 30$



Energy loss | $30 < \theta < 33$

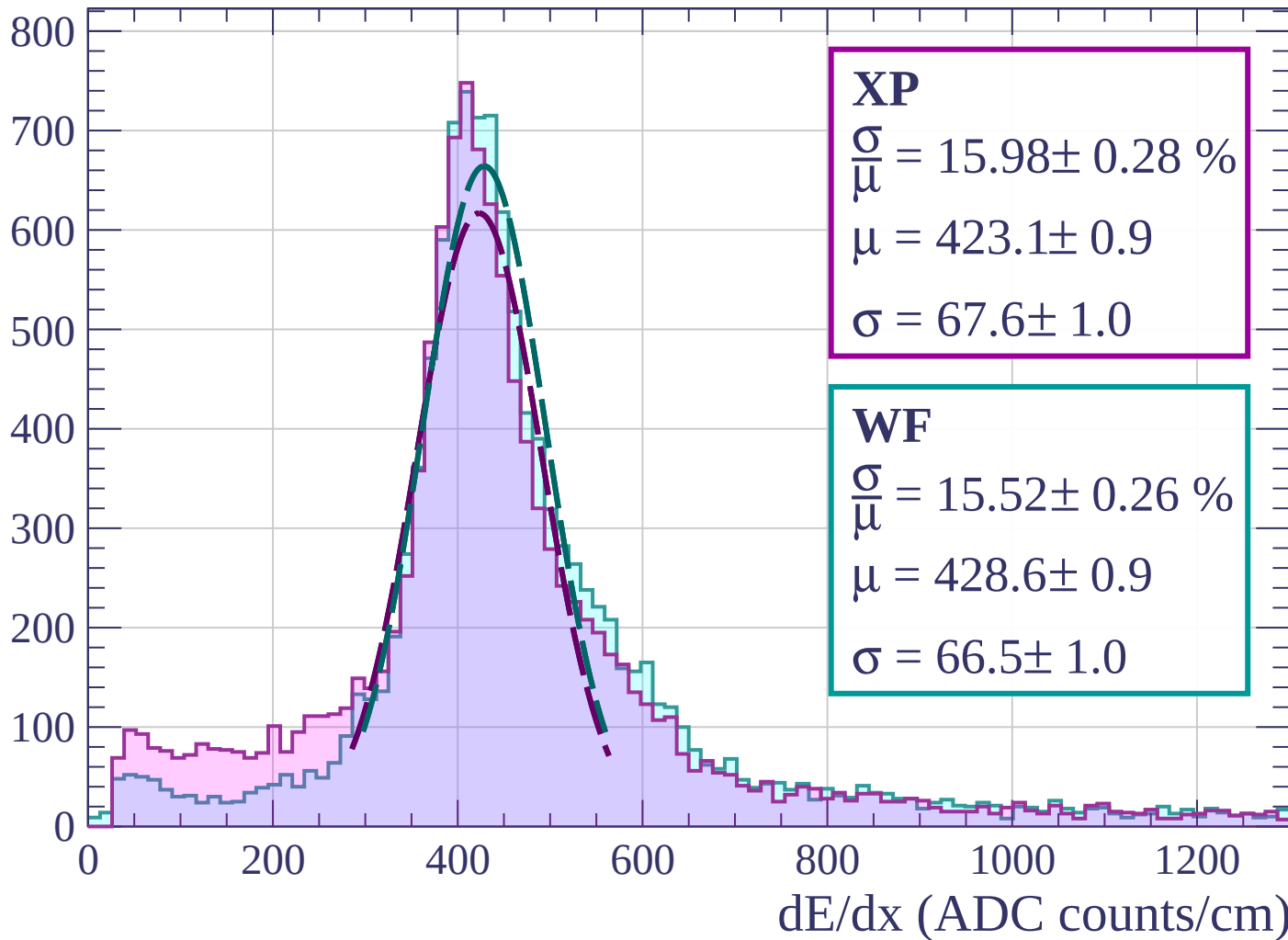


Energy loss | $33 < \theta < 36$



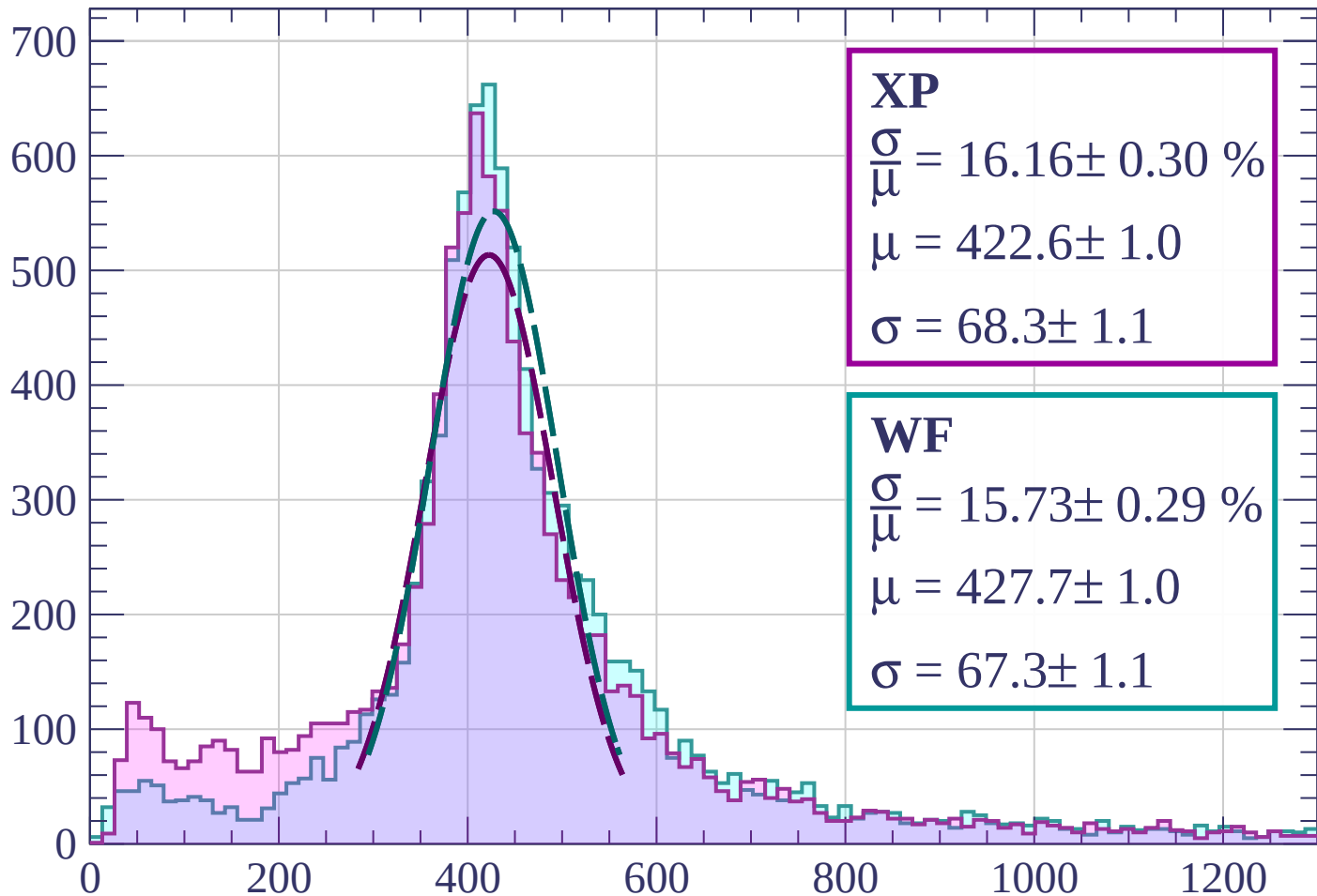
Energy loss | $36 < \theta < 39$

Count



Energy loss | $39 < \theta < 42$

Count



XP

$$\frac{\sigma}{\mu} = 16.16 \pm 0.30 \%$$

$$\mu = 422.6 \pm 1.0$$

$$\sigma = 68.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 15.73 \pm 0.29 \%$$

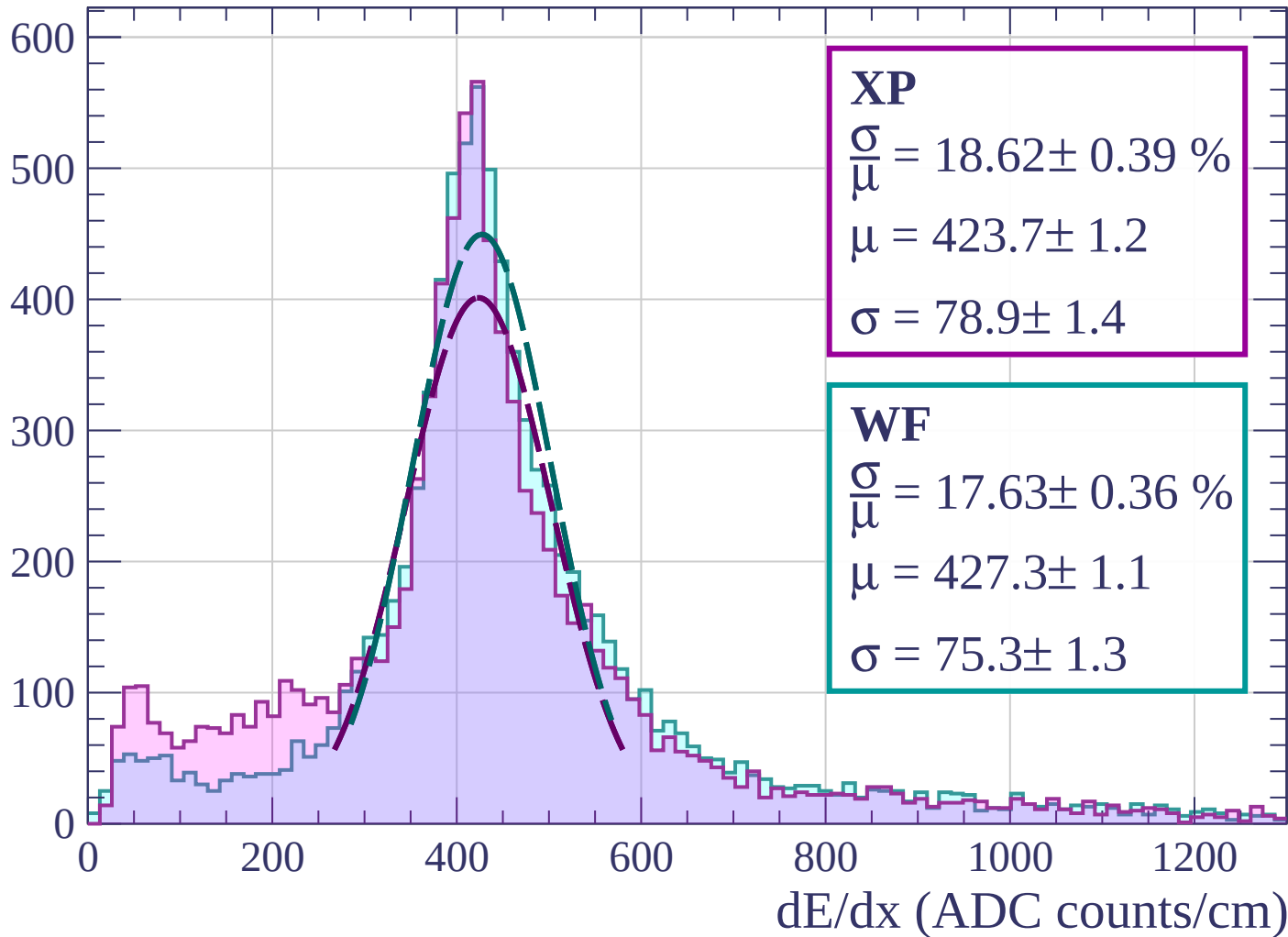
$$\mu = 427.7 \pm 1.0$$

$$\sigma = 67.3 \pm 1.1$$

dE/dx (ADC counts/cm)

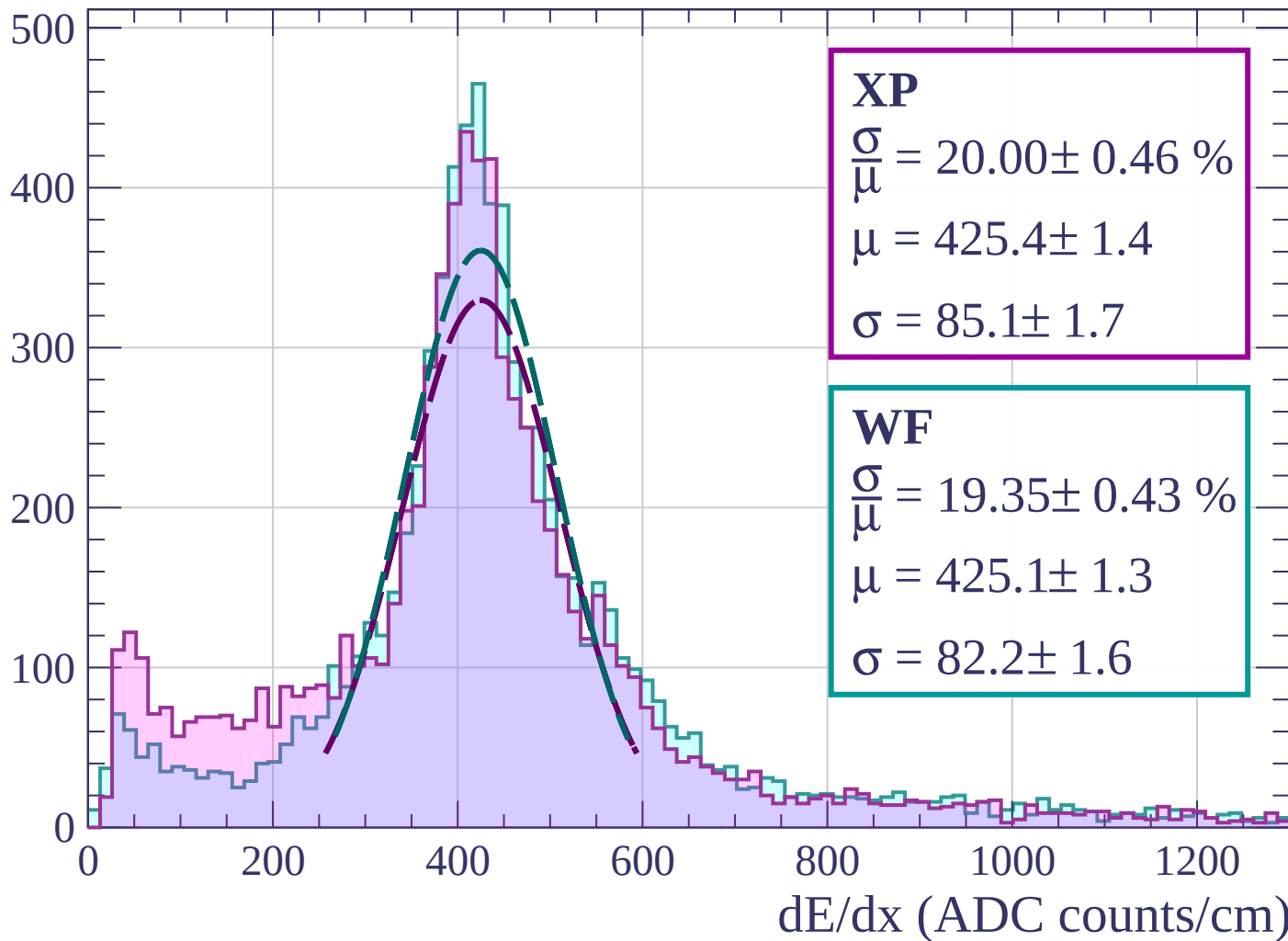
Energy loss | $42 < \theta < 45$

Count



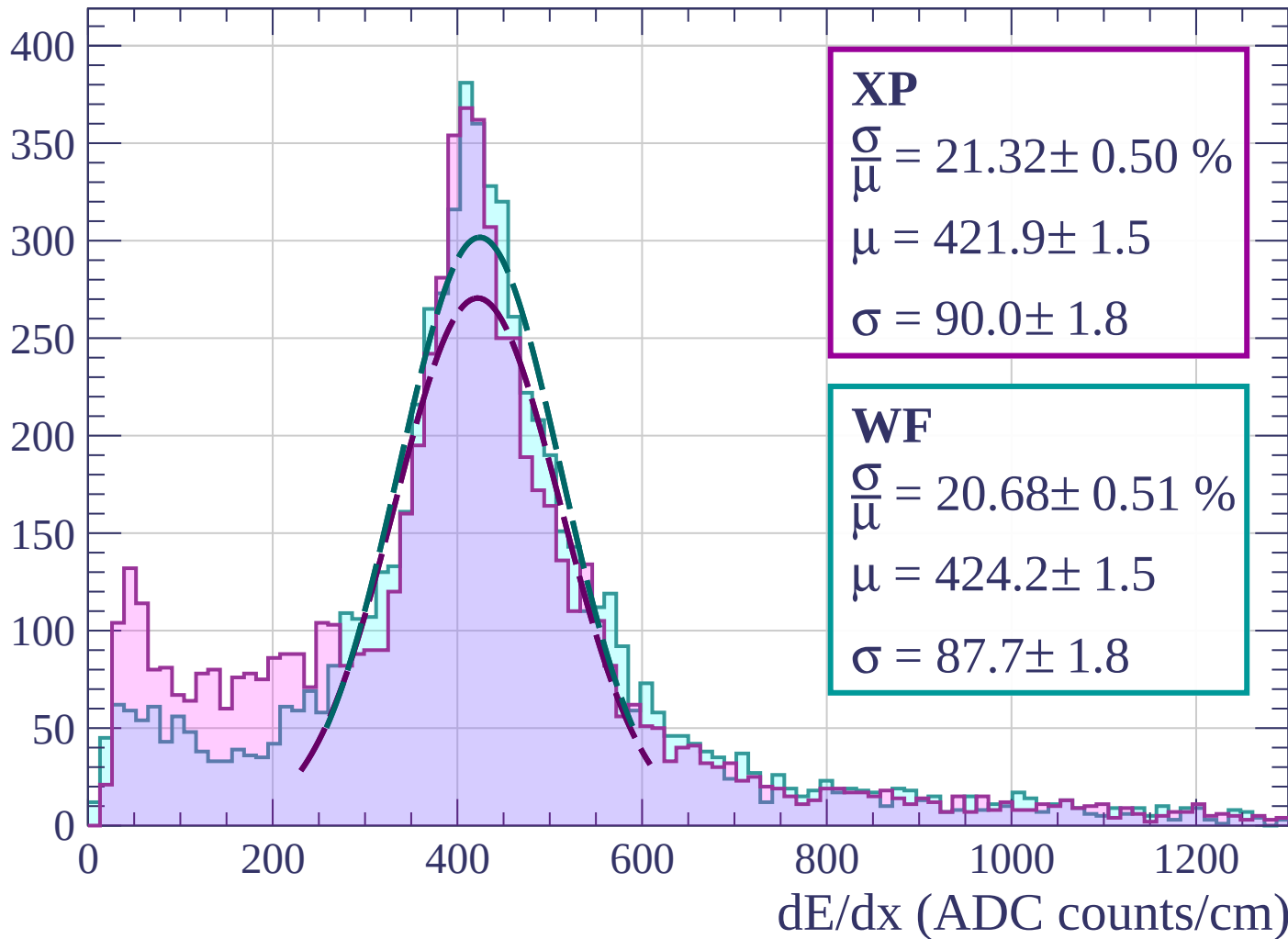
Energy loss | $45 < \theta < 48$

Count



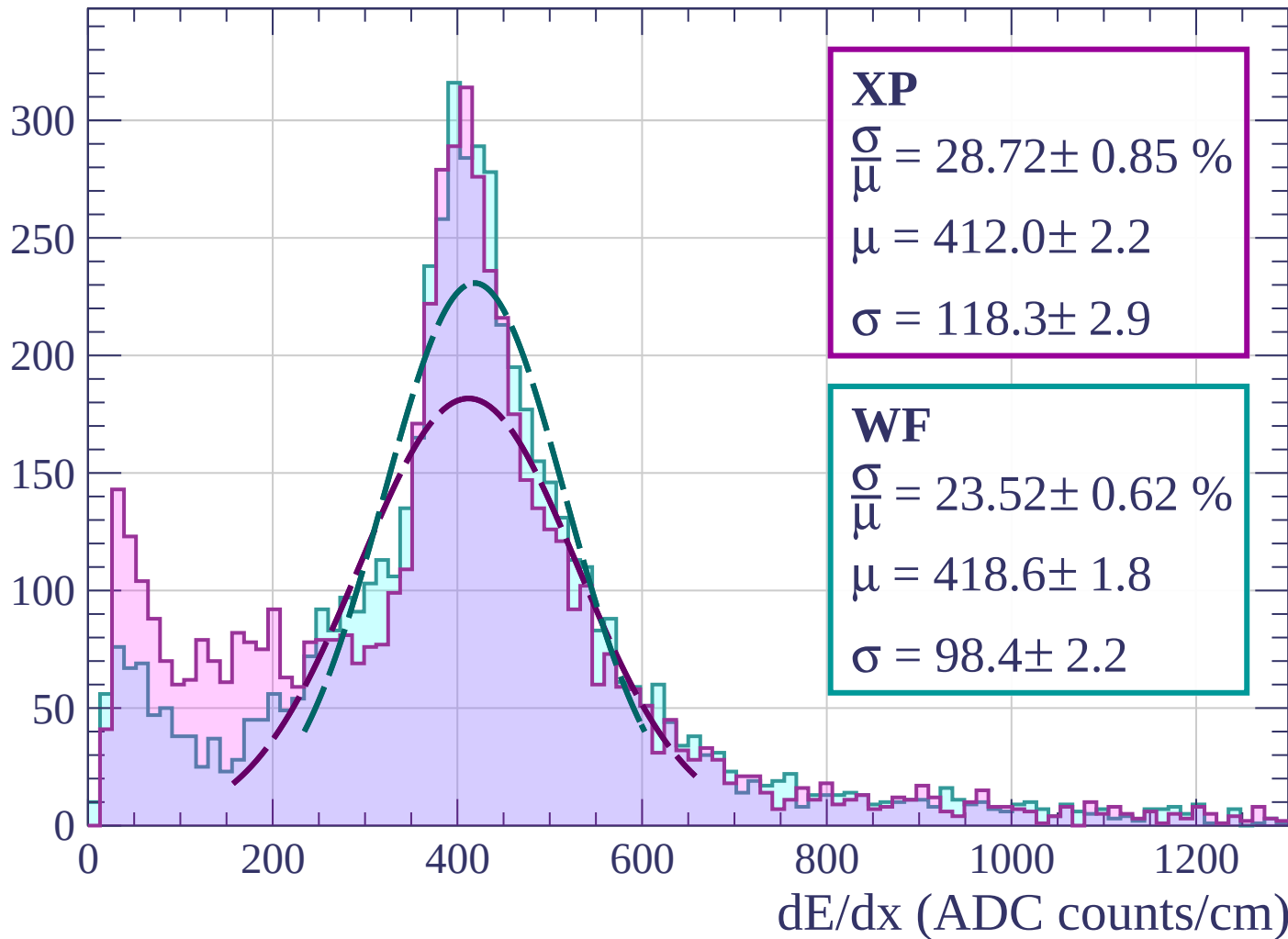
Energy loss | $48 < \theta < 51$

Count



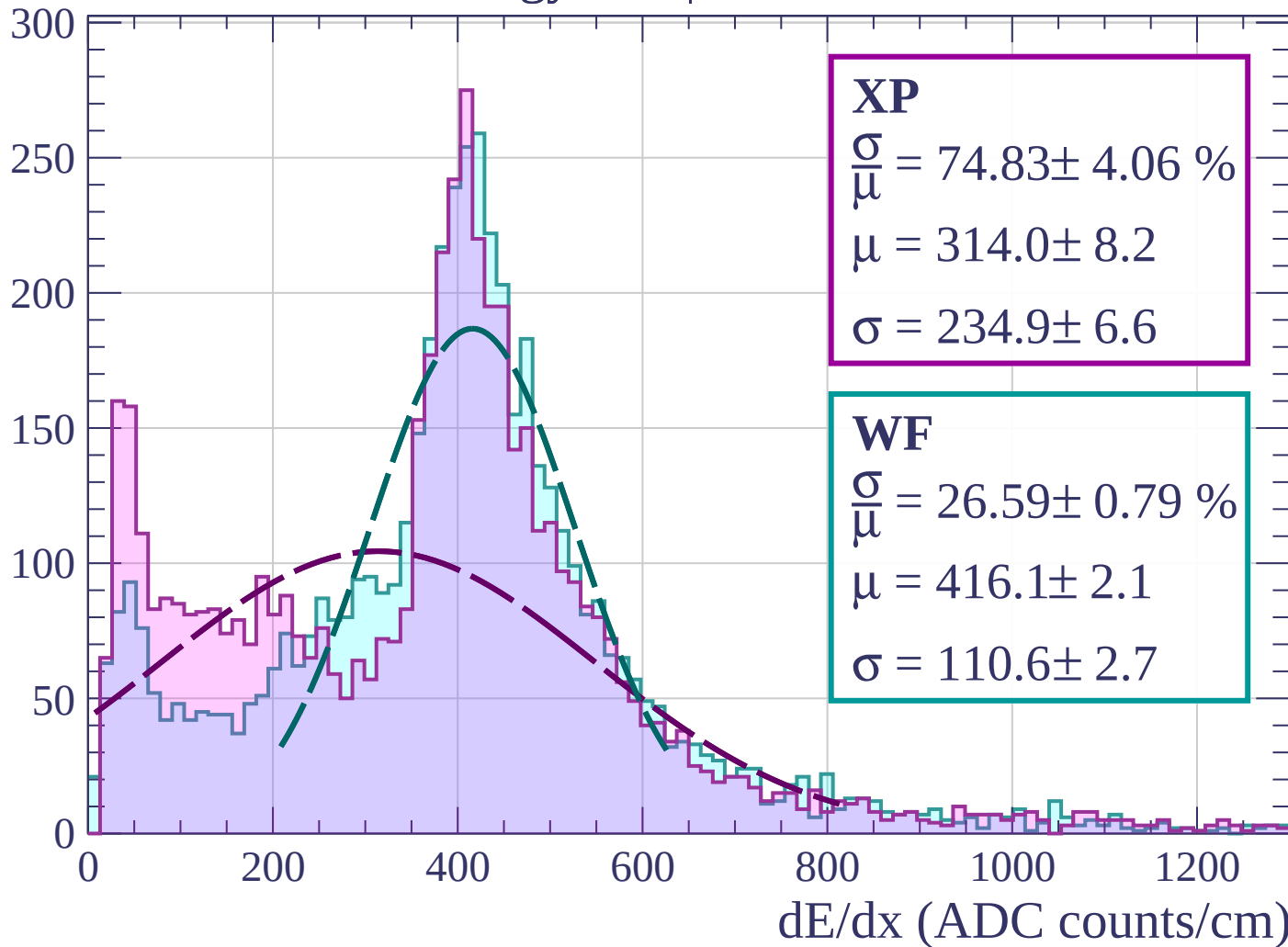
Energy loss | $51 < \theta < 54$

Count



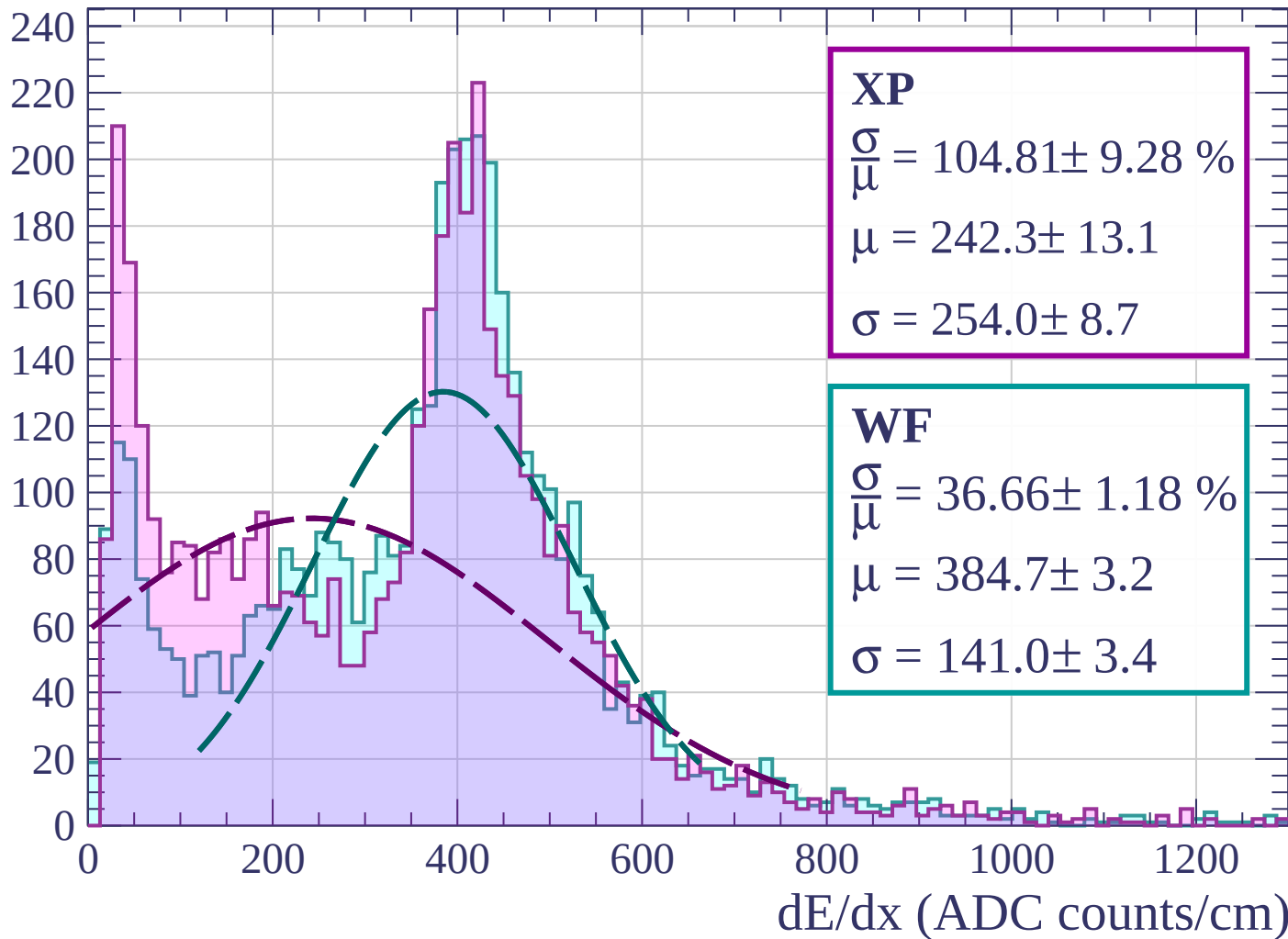
Energy loss | $54 < \theta < 57$

Count



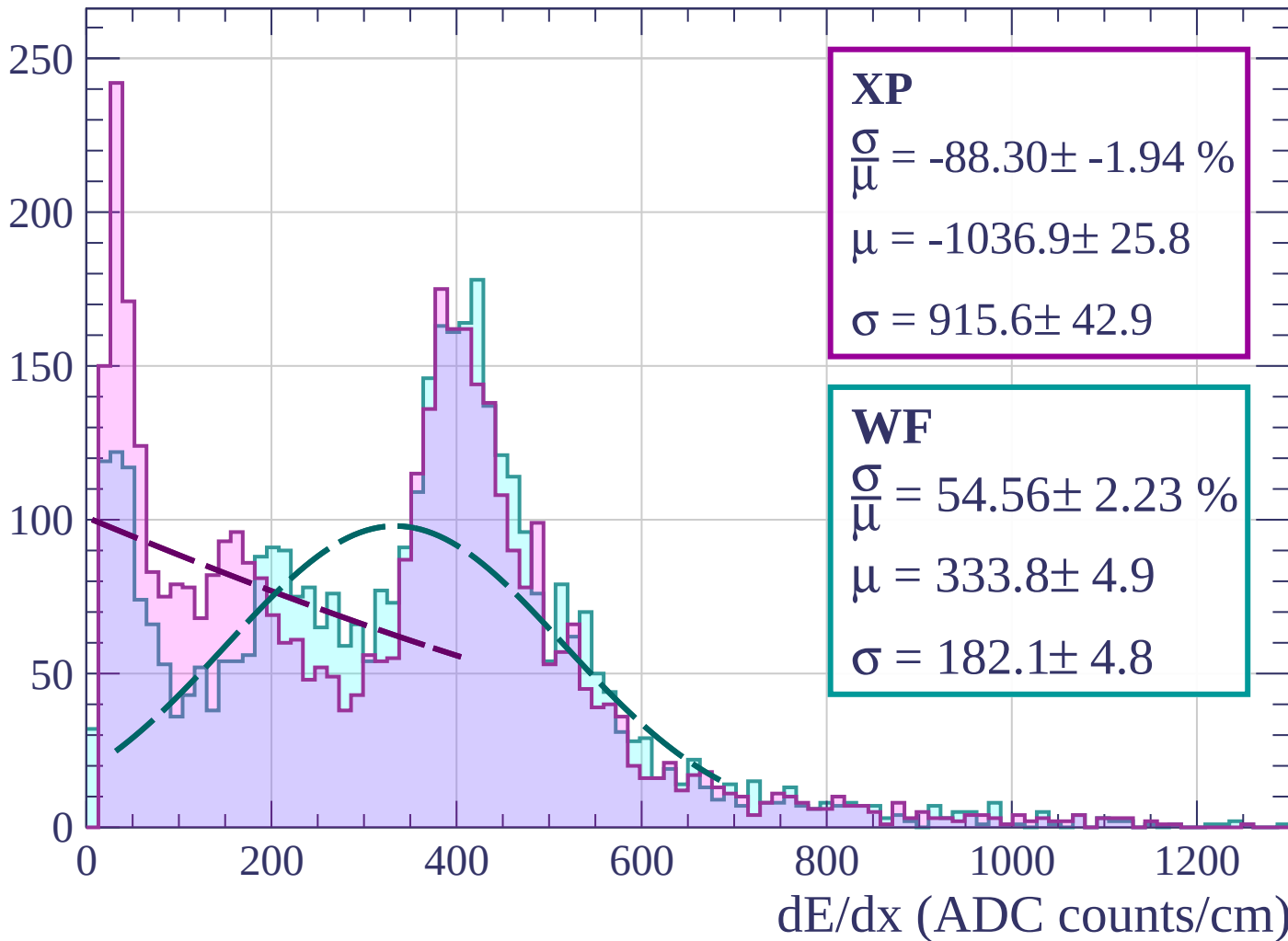
Energy loss | $57 < \theta < 60$

Count



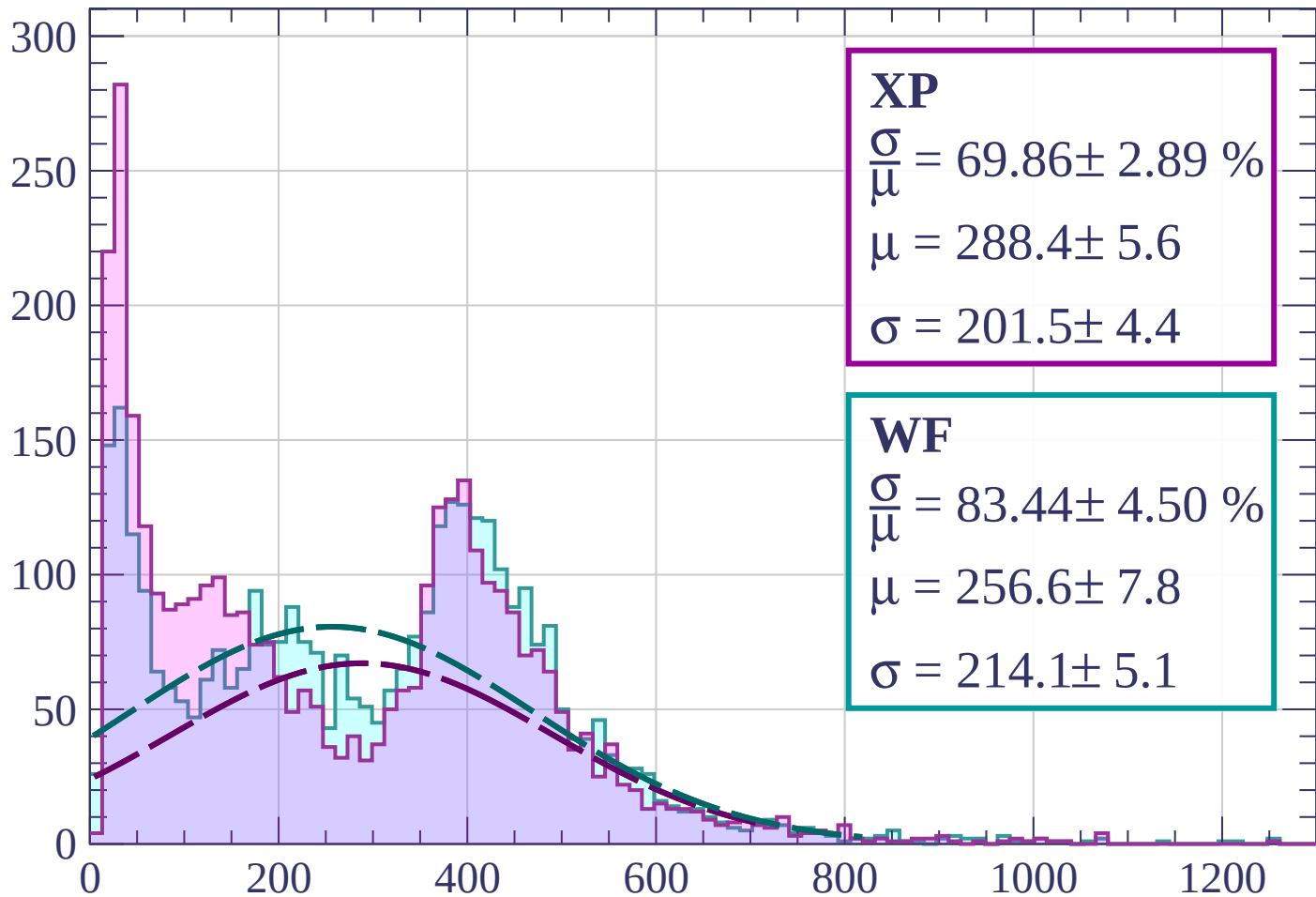
Energy loss | $60 < \theta < 63$

Count



Energy loss | $63 < \theta < 66$

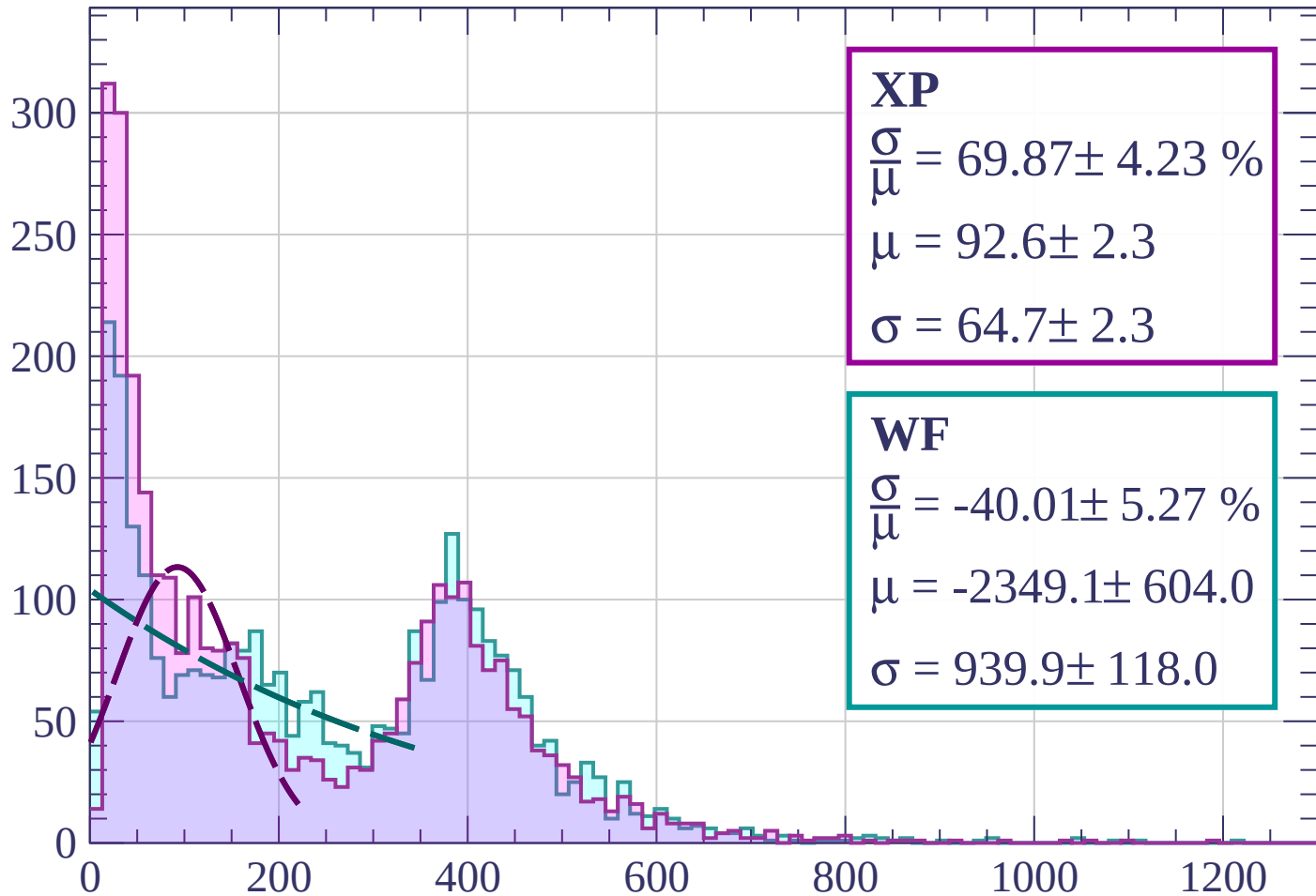
Count



dE/dx (ADC counts/cm)

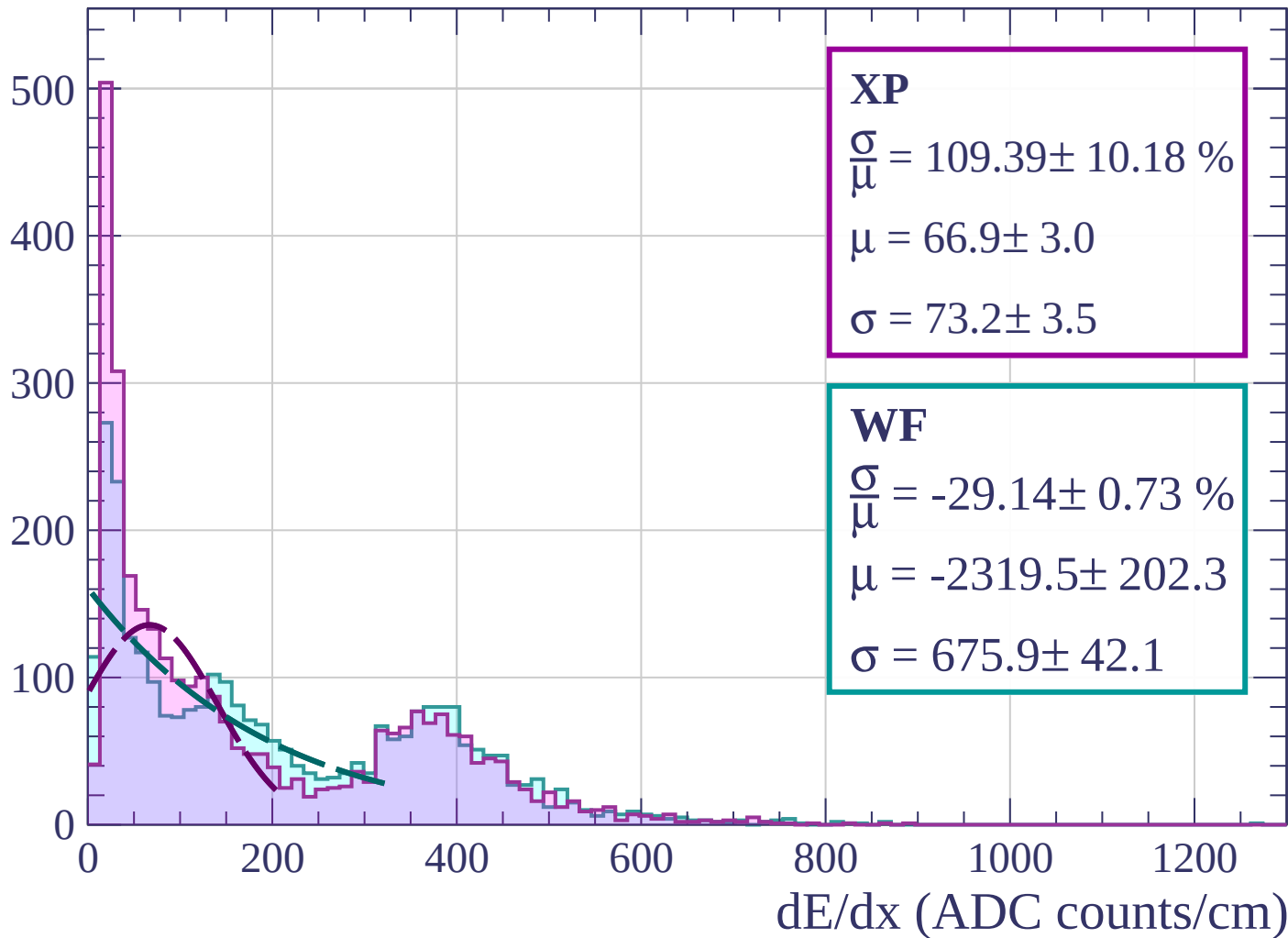
Energy loss | $66 < \theta < 69$

Count

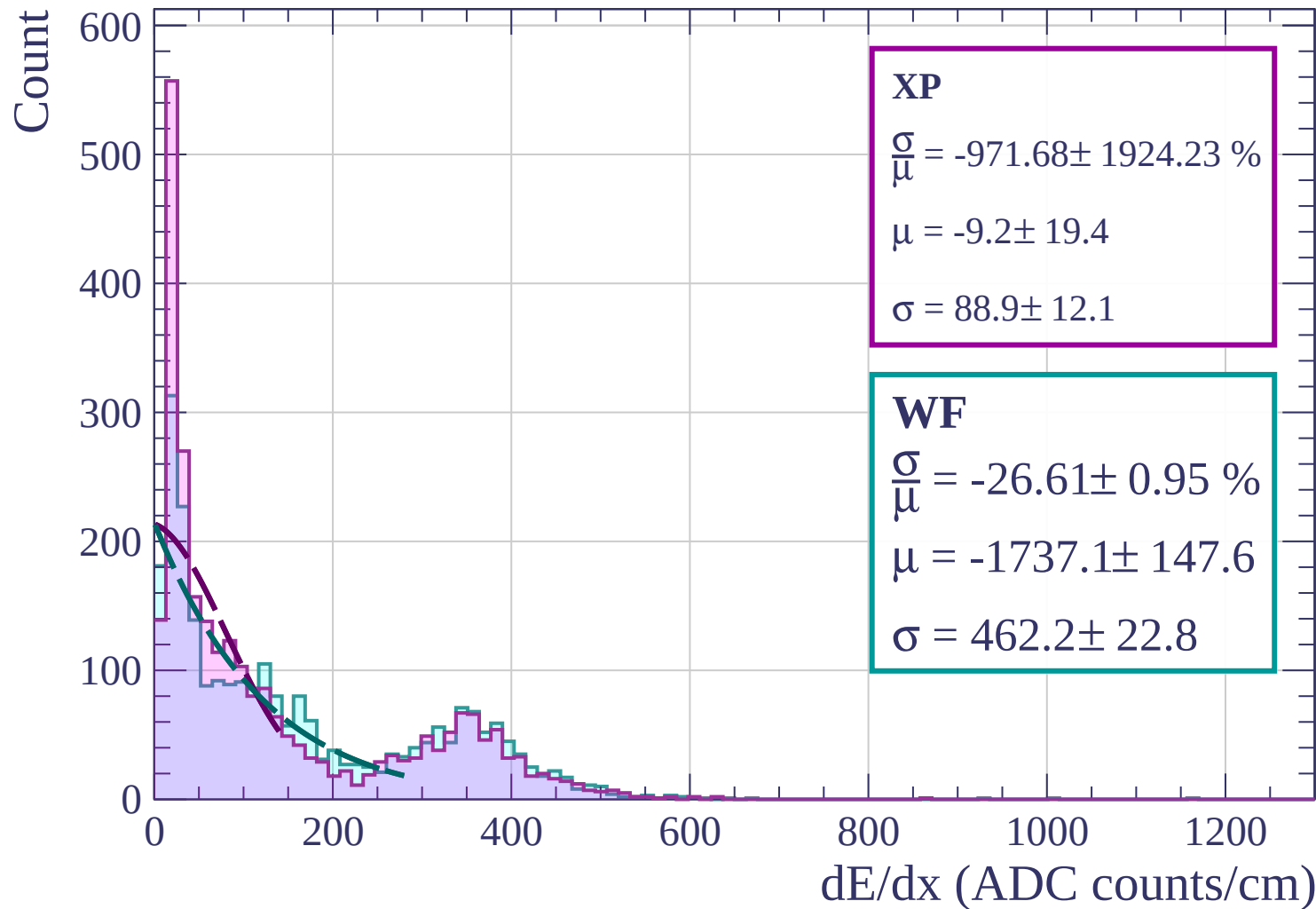


Energy loss | $69 < \theta < 72$

Count

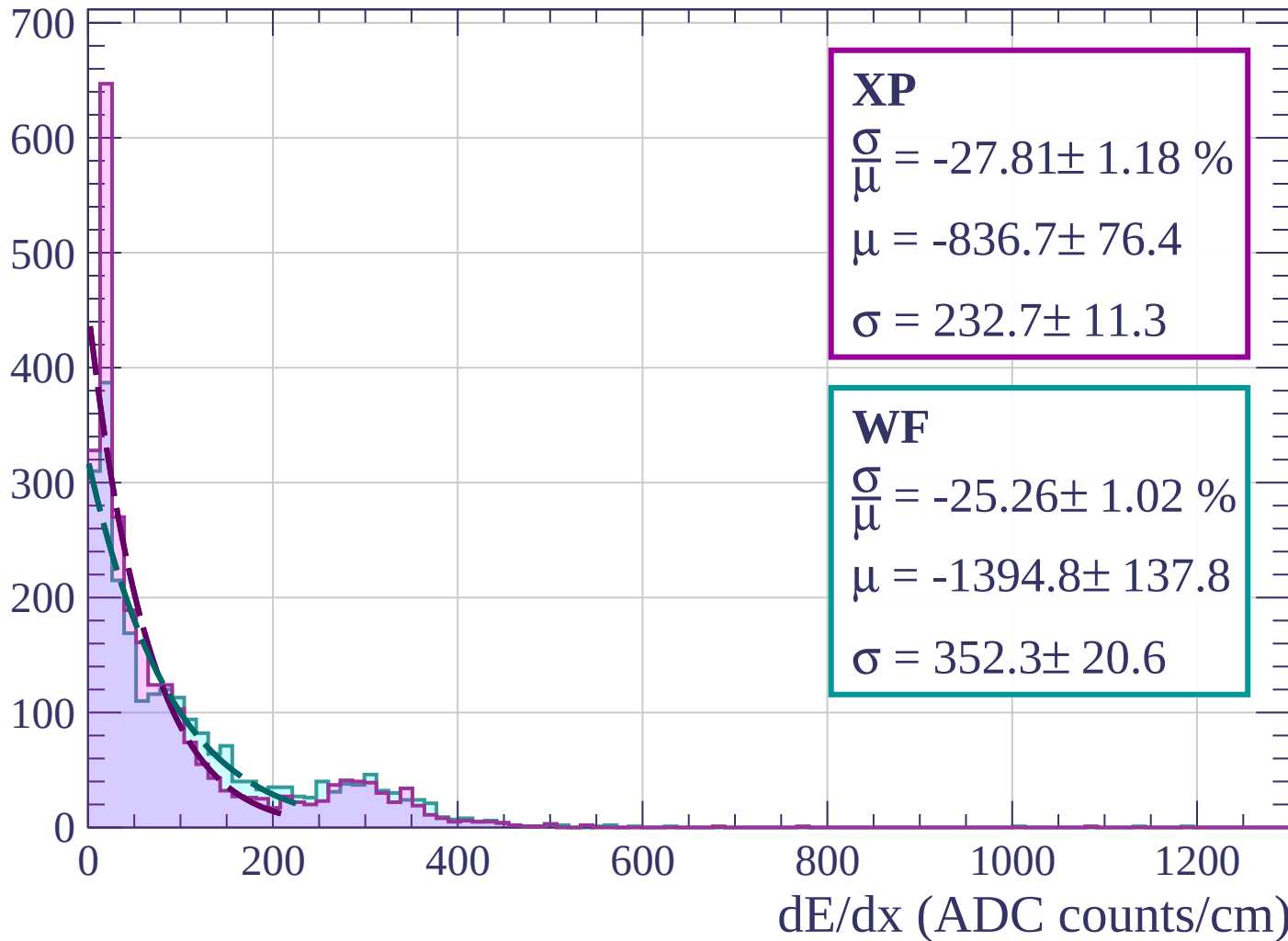


Energy loss | $72 < \theta < 75$



Energy loss | $75 < \theta < 78$

Count



Energy loss | $78 < \theta < 81$

Count

600

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = -24.44 \pm 0.42 \%$$

$$\mu = -662.9 \pm 34.5$$

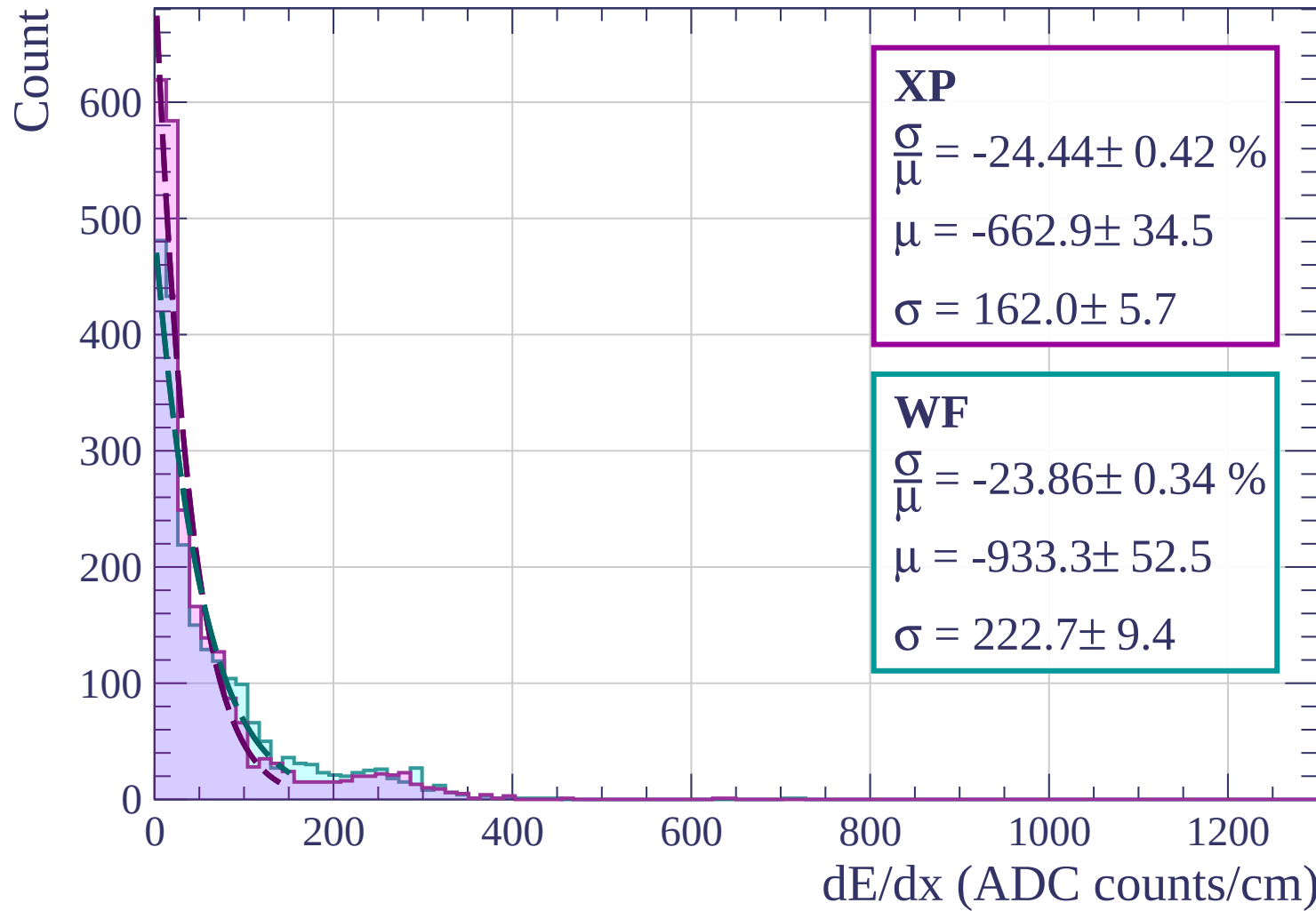
$$\sigma = 162.0 \pm 5.7$$

WF

$$\frac{\sigma}{\mu} = -23.86 \pm 0.34 \%$$

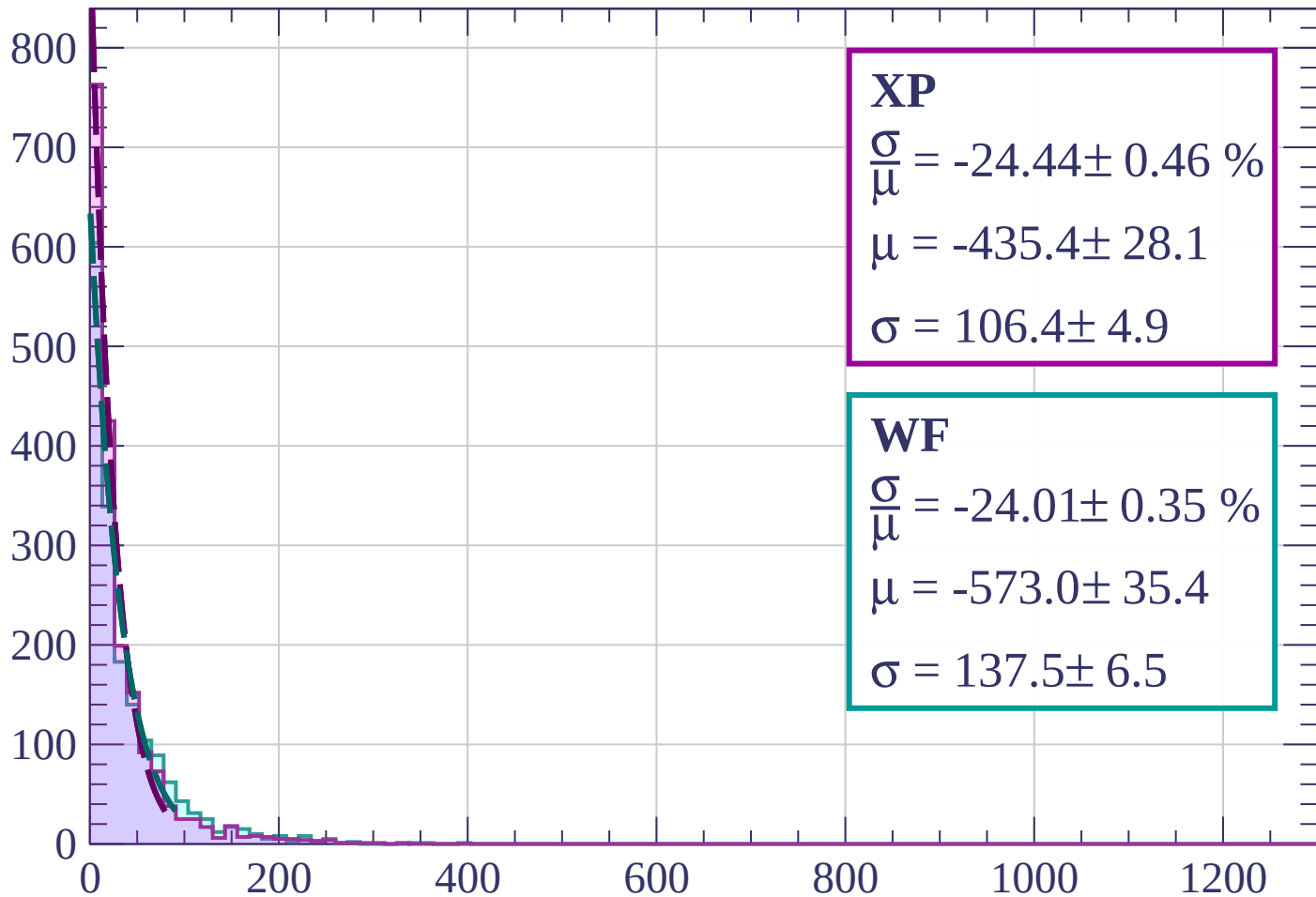
$$\mu = -933.3 \pm 52.5$$

$$\sigma = 222.7 \pm 9.4$$



Energy loss | $81 < \theta < 84$

Count



XP

$$\frac{\sigma}{\mu} = -24.44 \pm 0.46 \%$$

$$\mu = -435.4 \pm 28.1$$

$$\sigma = 106.4 \pm 4.9$$

WF

$$\frac{\sigma}{\mu} = -24.01 \pm 0.35 \%$$

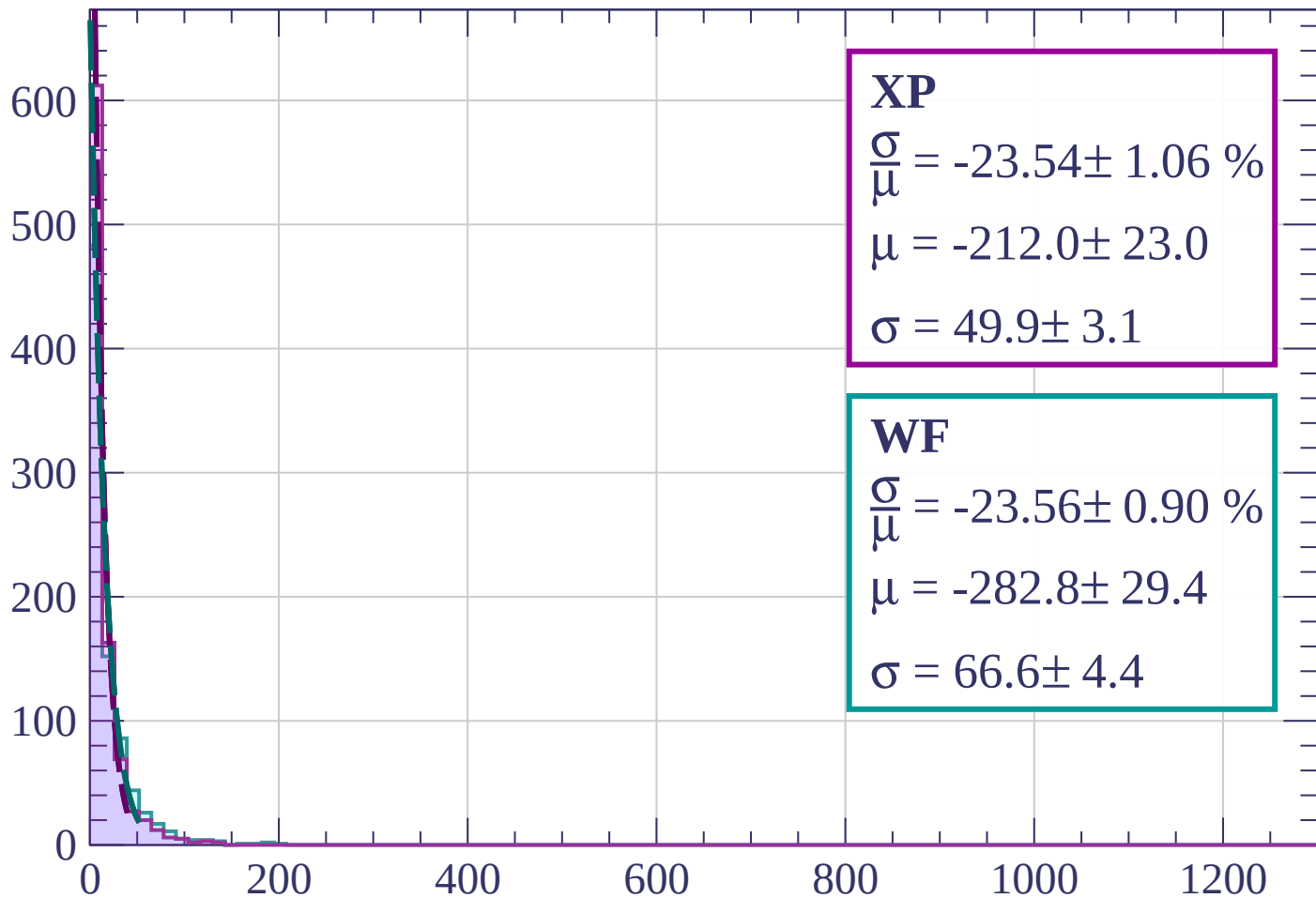
$$\mu = -573.0 \pm 35.4$$

$$\sigma = 137.5 \pm 6.5$$

dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

Count



XP

$$\frac{\sigma}{\mu} = -23.54 \pm 1.06 \%$$

$$\mu = -212.0 \pm 23.0$$

$$\sigma = 49.9 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = -23.56 \pm 0.90 \%$$

$$\mu = -282.8 \pm 29.4$$

$$\sigma = 66.6 \pm 4.4$$

dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

